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(19) **United States**(12) **Patent Application Publication**
CHEN et al.(10) **Pub. No.: US 2025/0286154 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SYSTEMS AND METHODS FOR
MINIMIZING AND PREVENTING
DENDRITE FORMATION IN
ELECTROCHEMICAL CELLS**provisional application No. 63/528,213, filed on Jul.
21, 2023, provisional application No. 63/546,980,
filed on Nov. 2, 2023.**Publication Classification**

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Woburn, MA (US); **Anthony
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Daniel SALAZAR, Brookline, MA
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- (51) **Int. Cl.**
H01M 10/42 (2006.01)
H01M 10/0562 (2010.01)
H01M 10/44 (2006.01)
H01M 10/48 (2006.01)
H01M 50/431 (2021.01)
H01M 50/449 (2021.01)
H02J 7/00 (2006.01)
- (52) **U.S. Cl.**
CPC ... **H01M 10/4235** (2013.01); **H01M 10/0562**
(2013.01); **H01M 10/425** (2013.01); **H01M**
10/44 (2013.01); **H01M 10/48** (2013.01);
H01M 50/431 (2021.01); **H01M 50/449**
(2021.01); **H02J 7/0047** (2013.01); **H01M**
2010/4271 (2013.01); **H01M 2200/00**
(2013.01); **H01M 2300/0068** (2013.01)

(21) Appl. No.: **19/210,930**(22) Filed: **May 16, 2025****Related U.S. Application Data**

- (63) Continuation of application No. 18/741,181, filed on
Jun. 12, 2024, now Pat. No. 12,334,518, which is a
continuation of application No. 18/650,789, filed on
Apr. 30, 2024, now Pat. No. 12,119,458, which is a
continuation of application No. 18/543,515, filed on
Dec. 18, 2023, now Pat. No. 11,984,564.
- (60) Provisional application No. 63/433,269, filed on Dec.
16, 2022, provisional application No. 63/450,208,
filed on Mar. 6, 2023, provisional application No.
63/461,506, filed on Apr. 24, 2023, provisional ap-
plication No. 63/470,679, filed on Jun. 2, 2023,

(57) **ABSTRACT**

Embodiments described herein relate to electrochemical cells with dendrite prevention mechanisms. In some aspects, an electrochemical cell can include an anode disposed on an anode current collector, a cathode disposed on a cathode current collector, the cathode having a first thickness at a proximal end of the cathode and a second thickness at a distal end of the cathode, the second thickness greater than the first thickness, a first separator disposed on the anode, a second separator disposed on the cathode, an interlayer disposed between the first separator and the second separator, the interlayer including electroactive material and having a proximal end and a distal end, and a power source electrically connected to the proximal end of the cathode and the proximal end of the interlayer, the power source configured to maintain a voltage difference between the cathode and the interlayer below a threshold value.

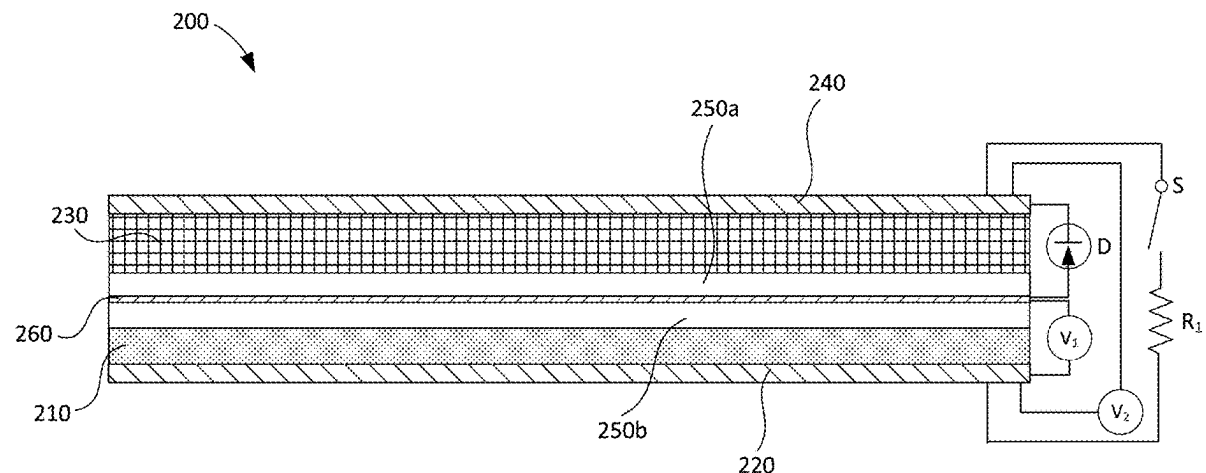


FIG. 1

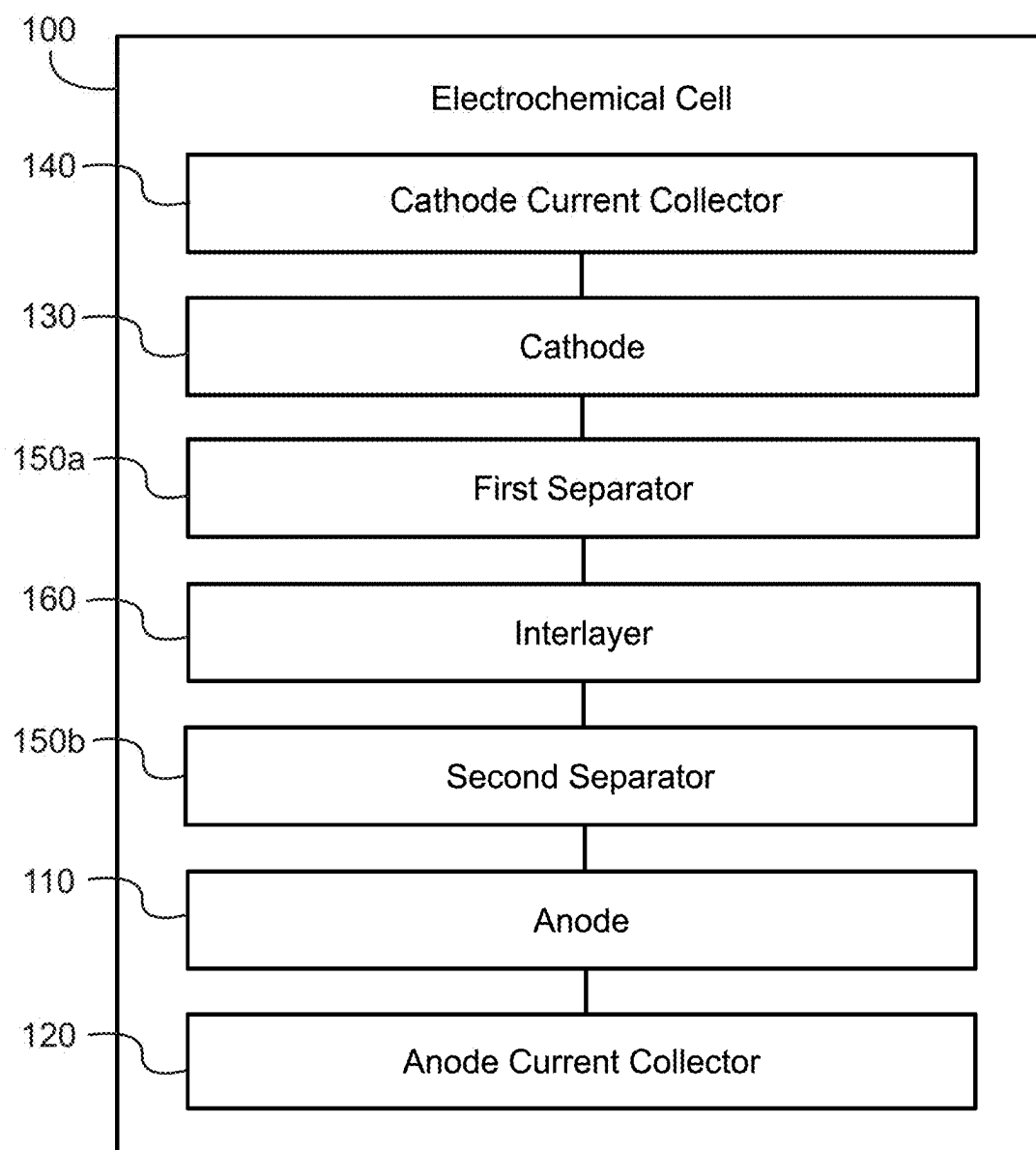


FIG. 2A

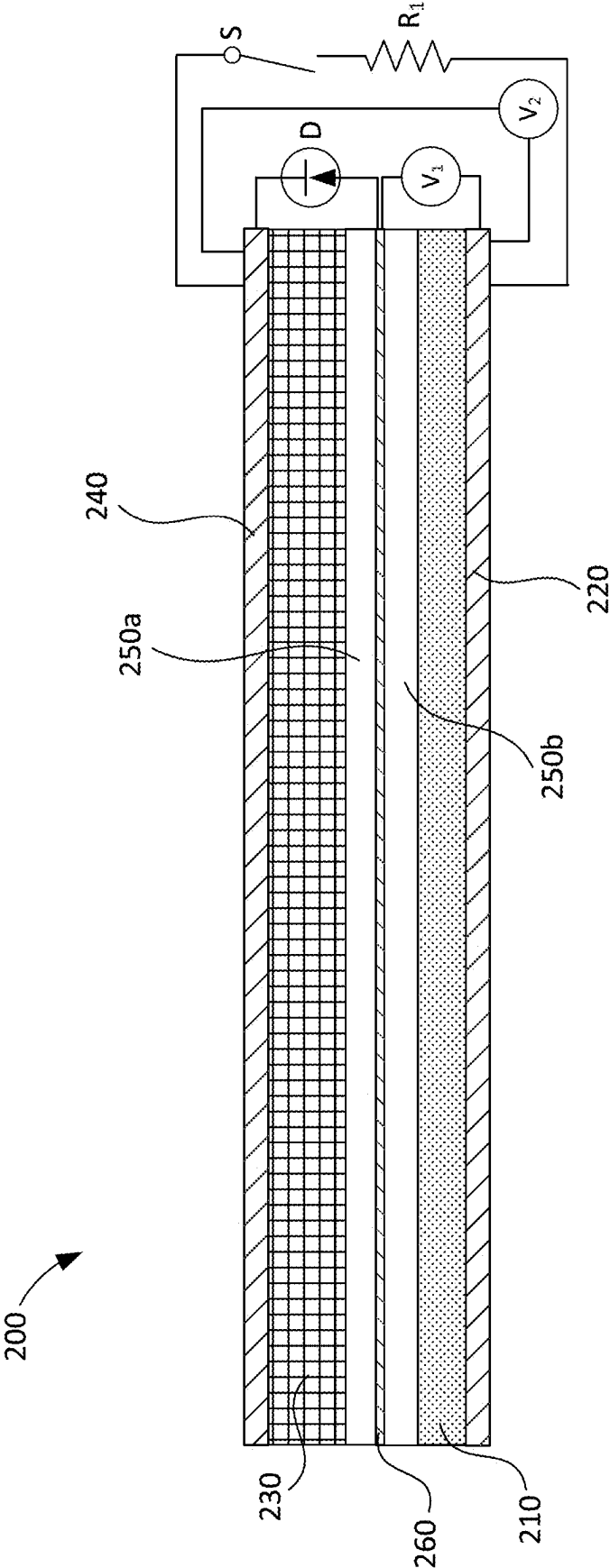


FIG. 2B

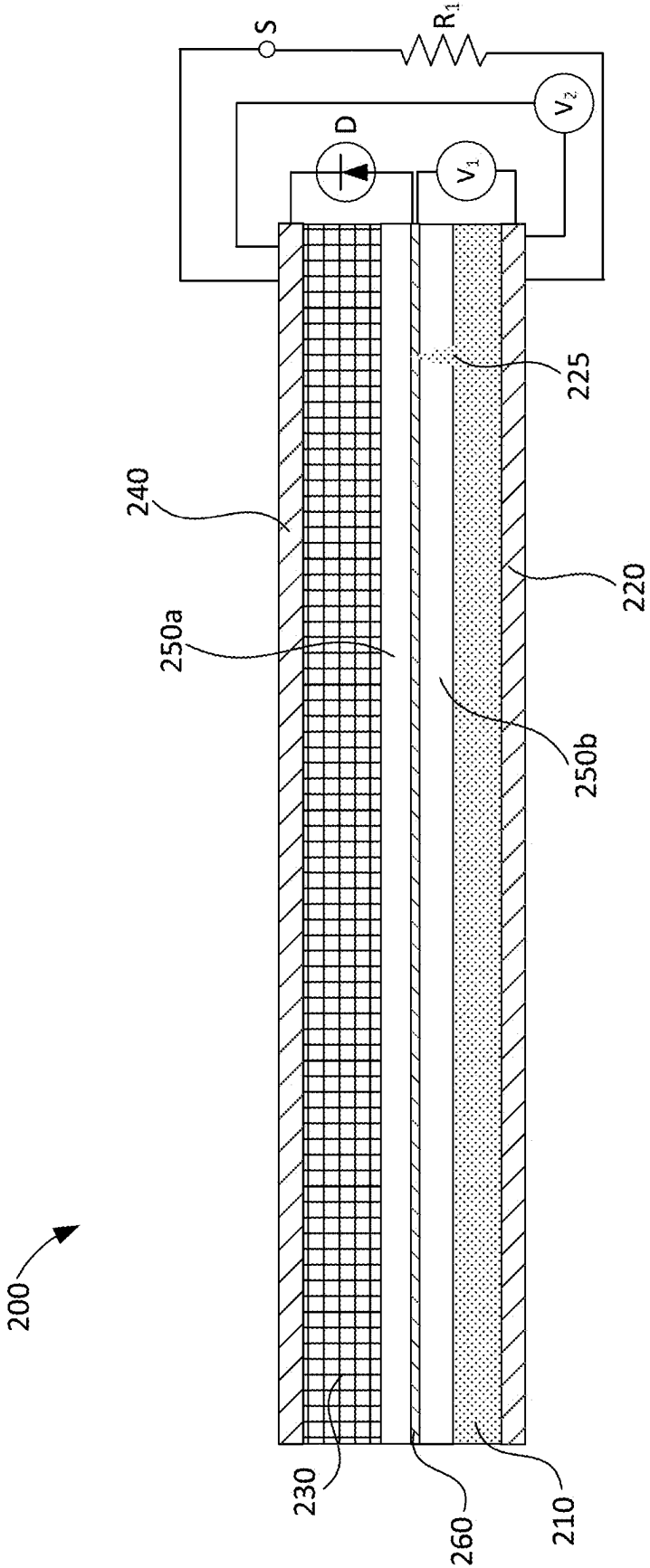


FIG. 3A

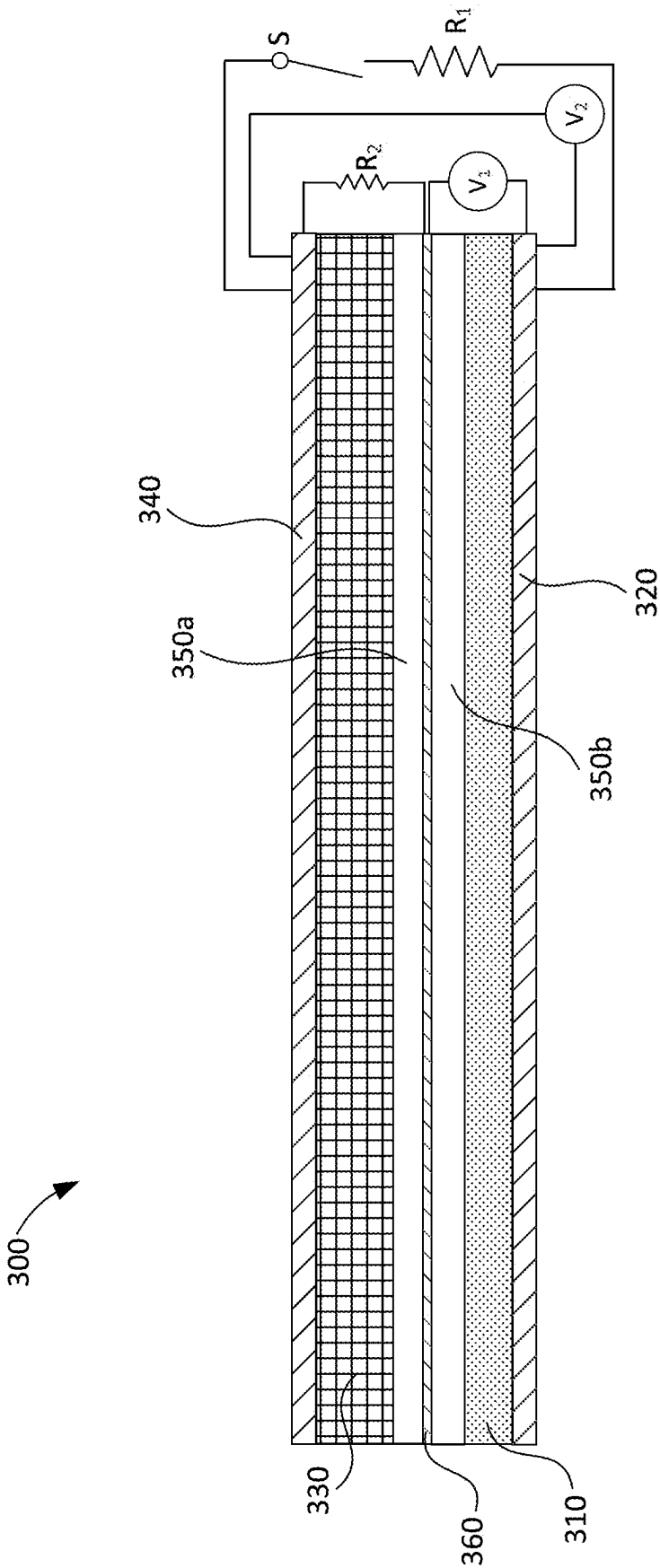


FIG. 3B

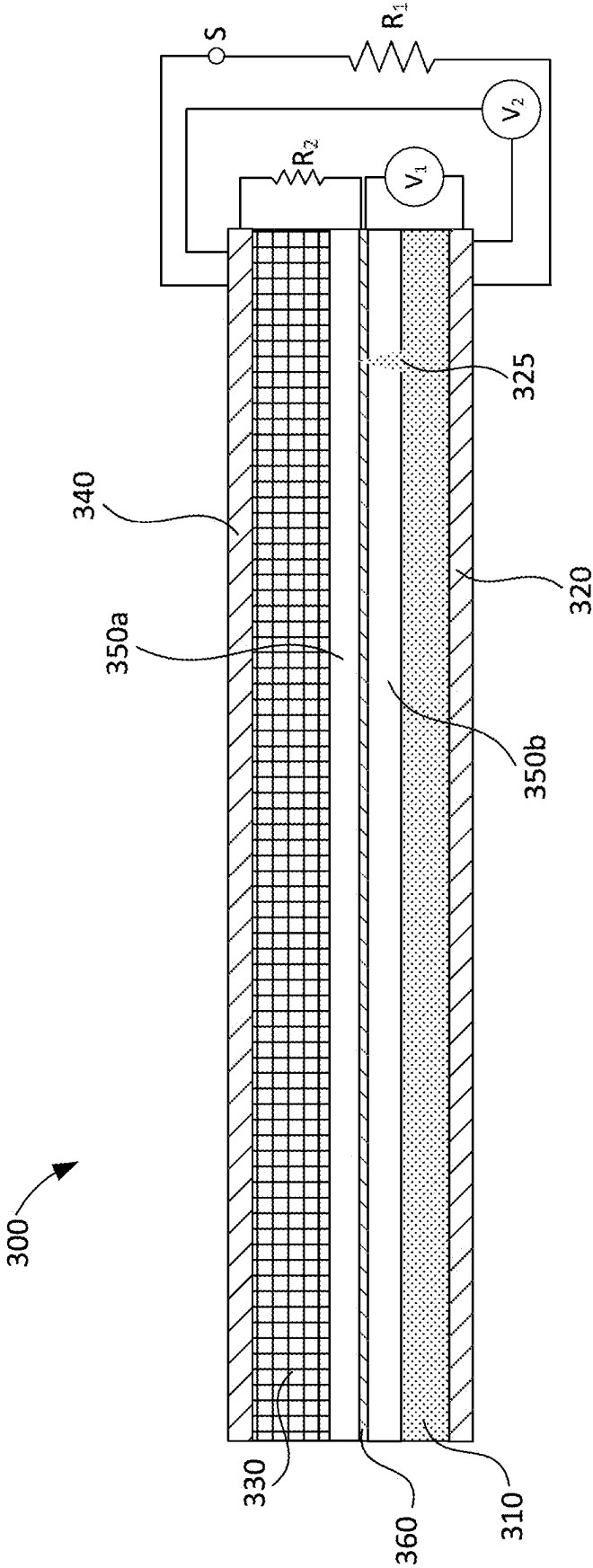


FIG. 4A

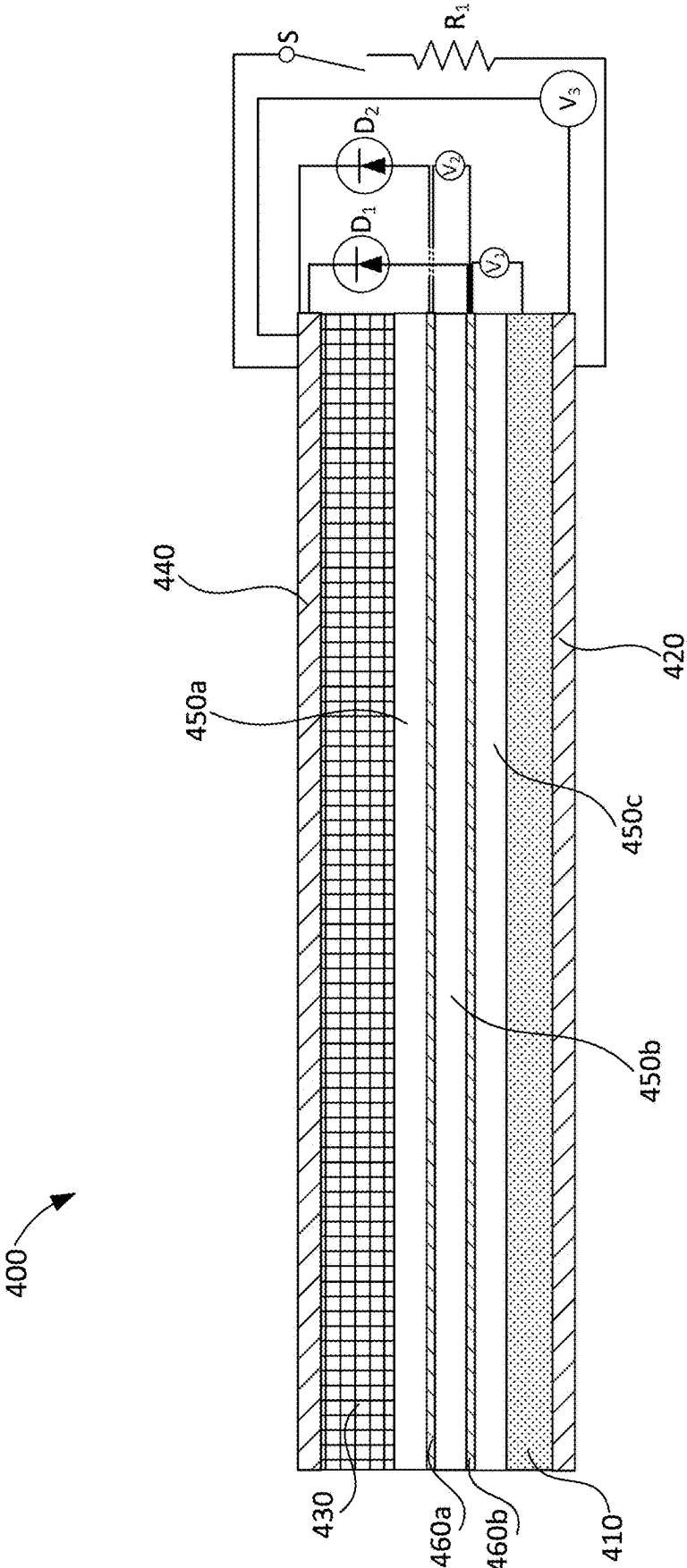


FIG. 4B

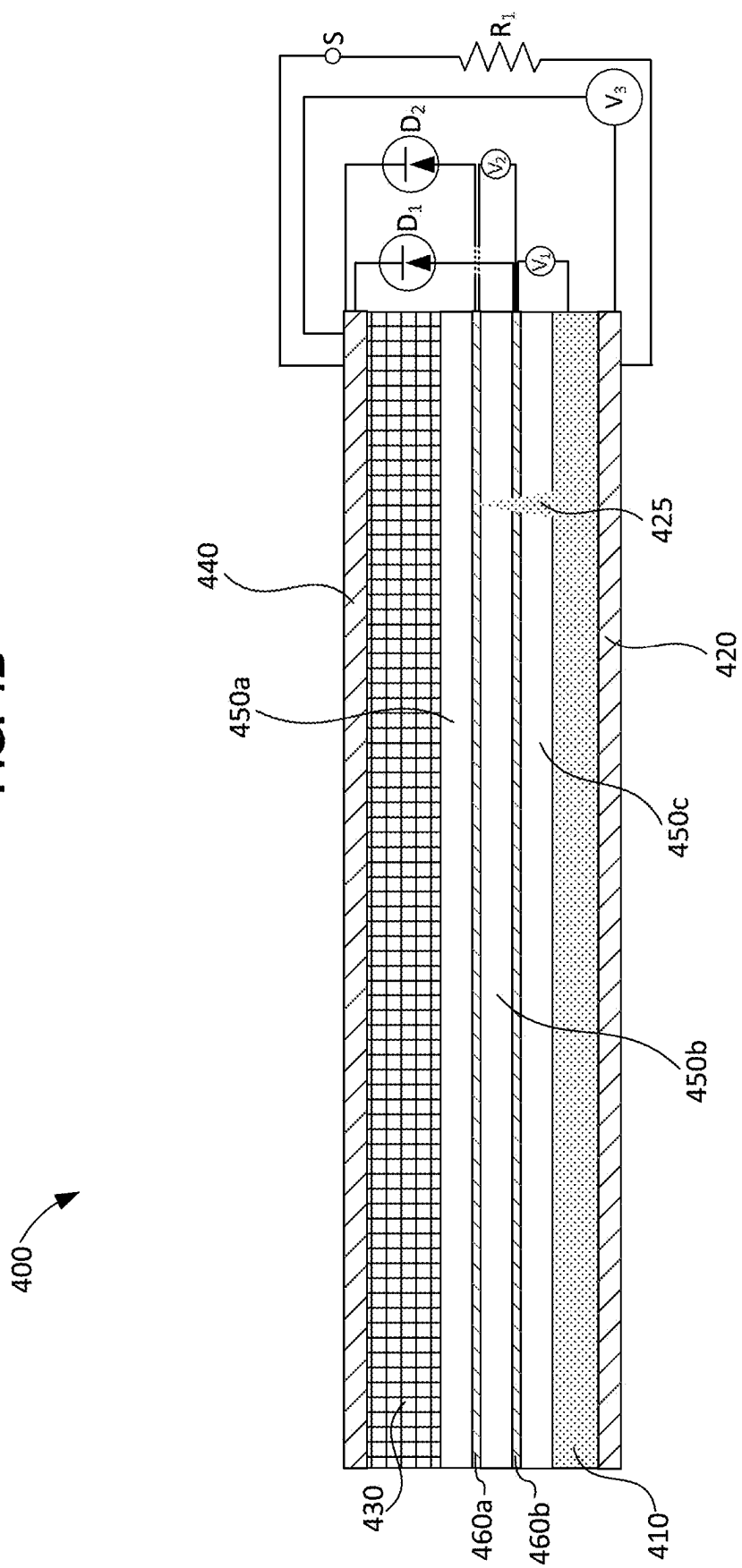


FIG. 5

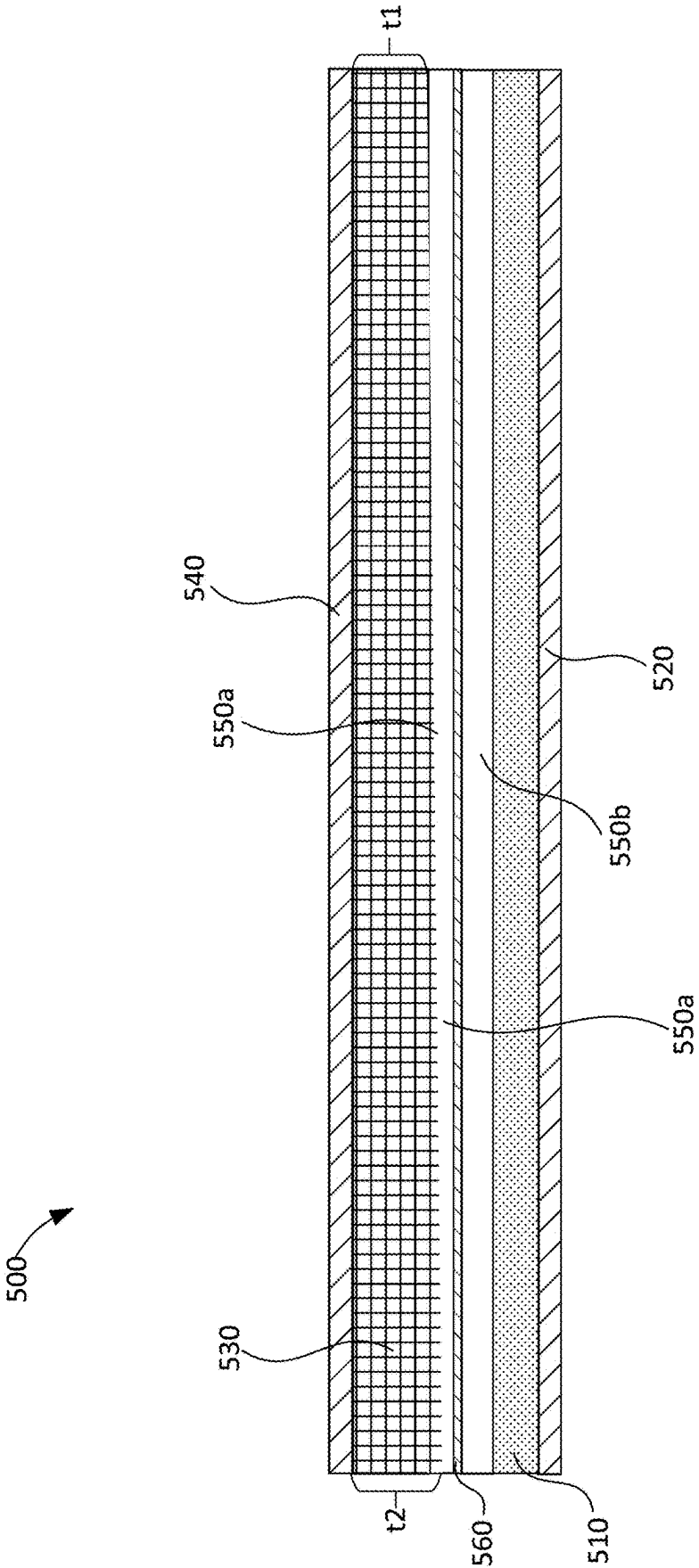


FIG. 6

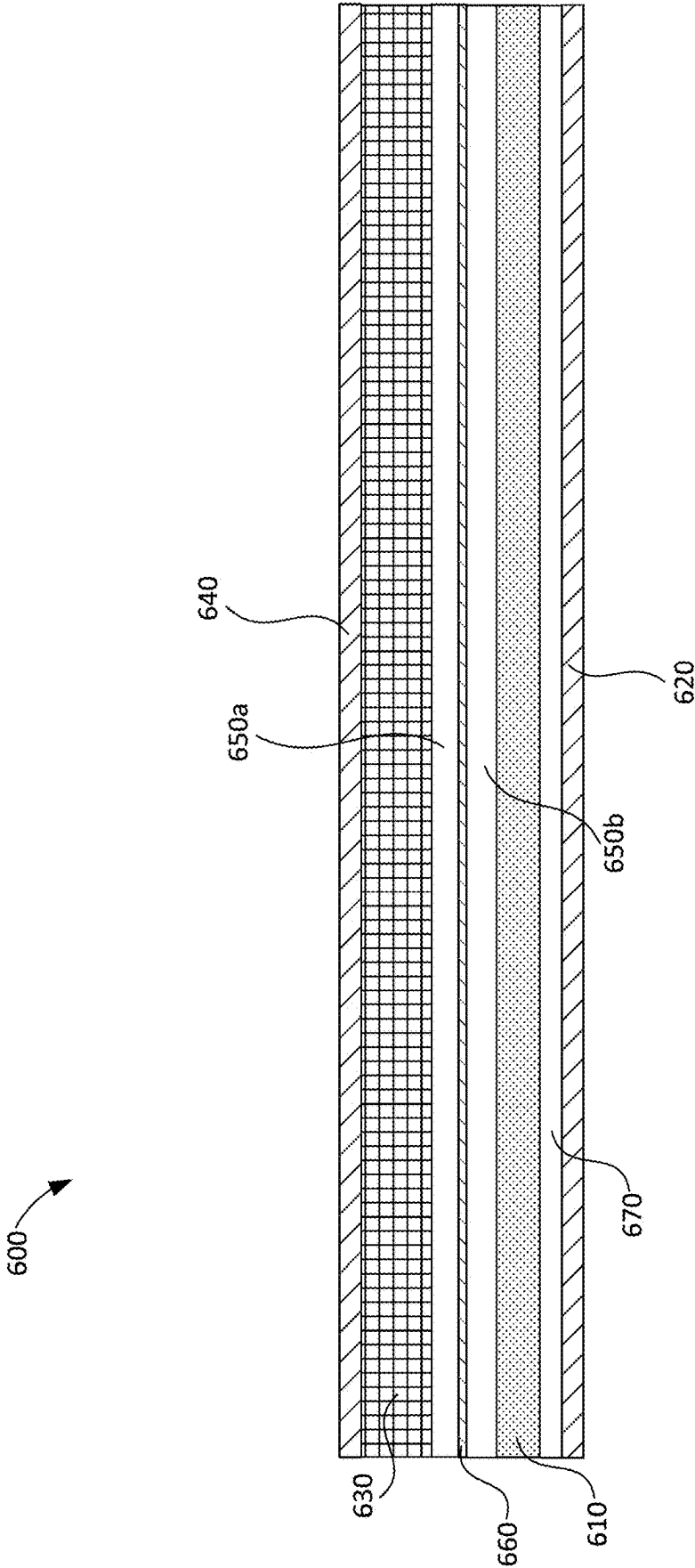


FIG. 7A

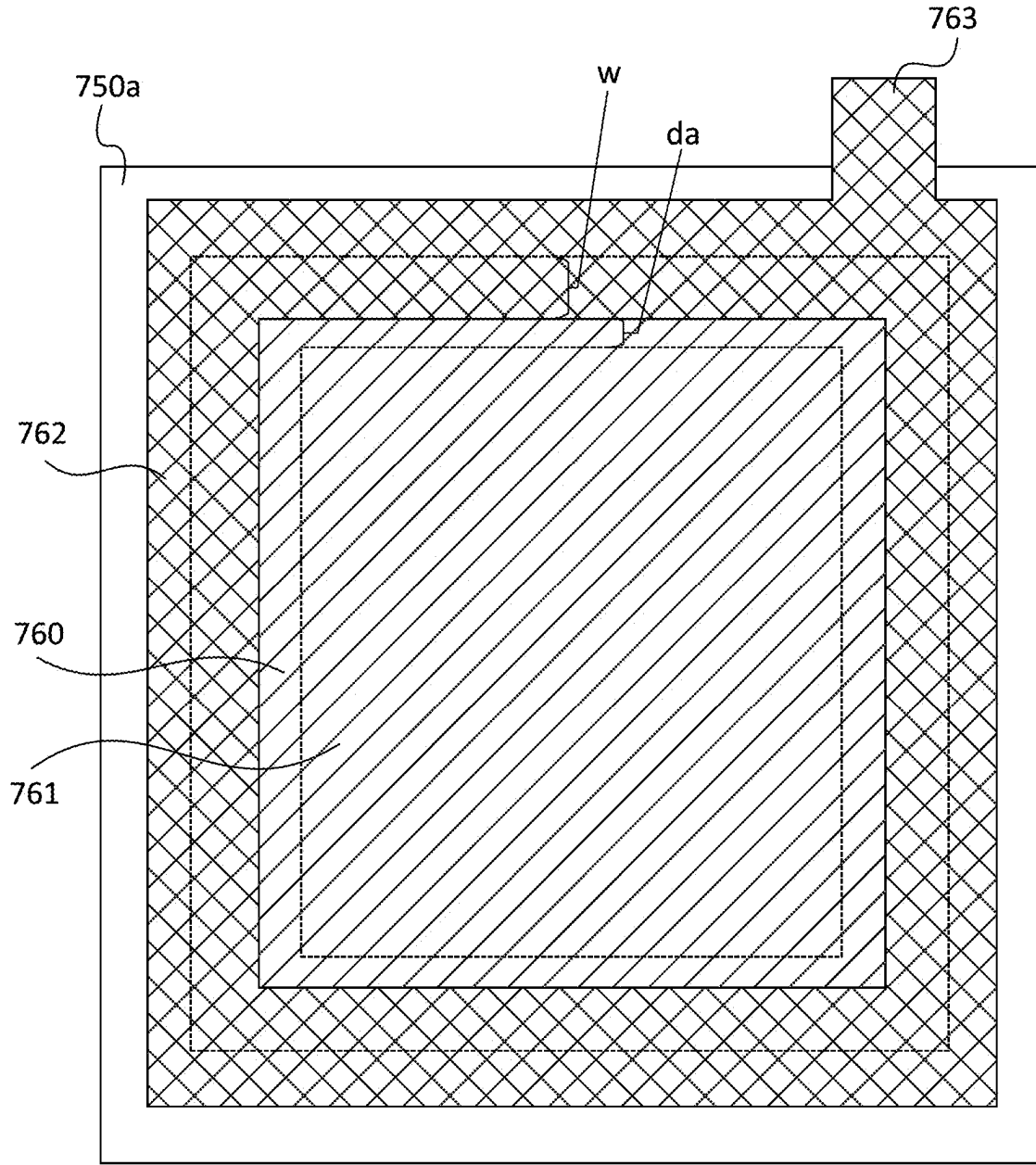


FIG. 7B

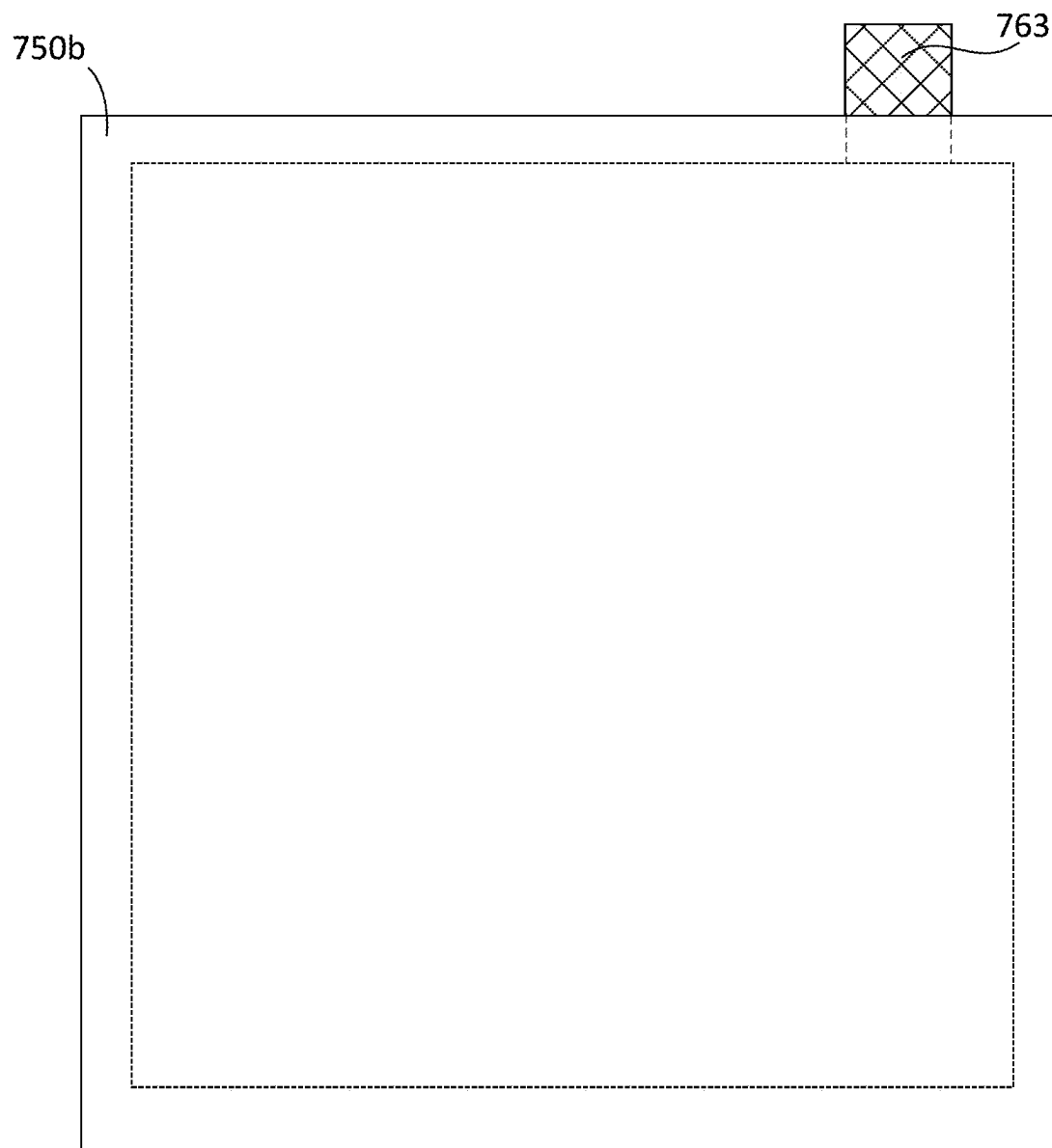


FIG. 8A

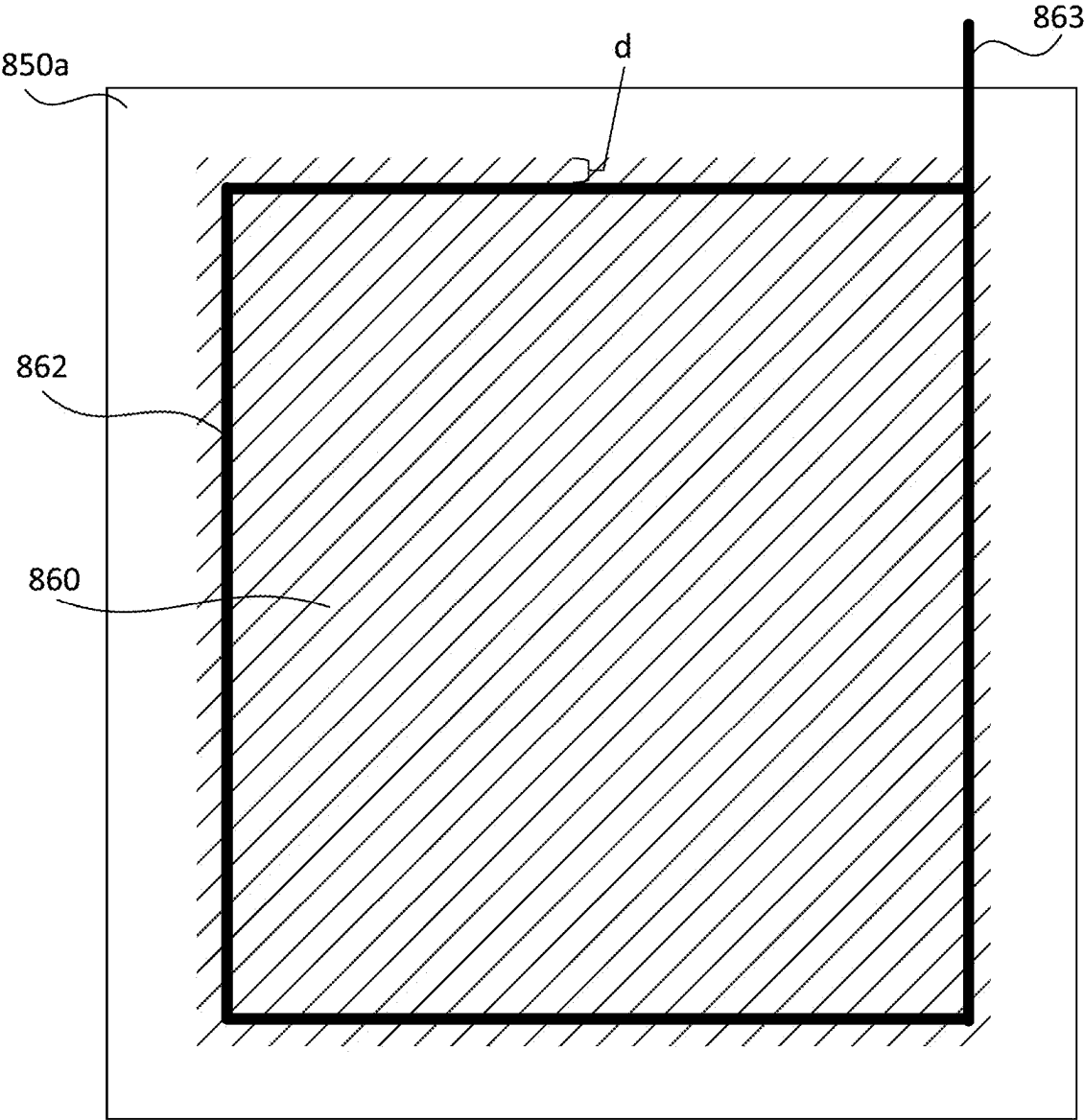


FIG. 8B

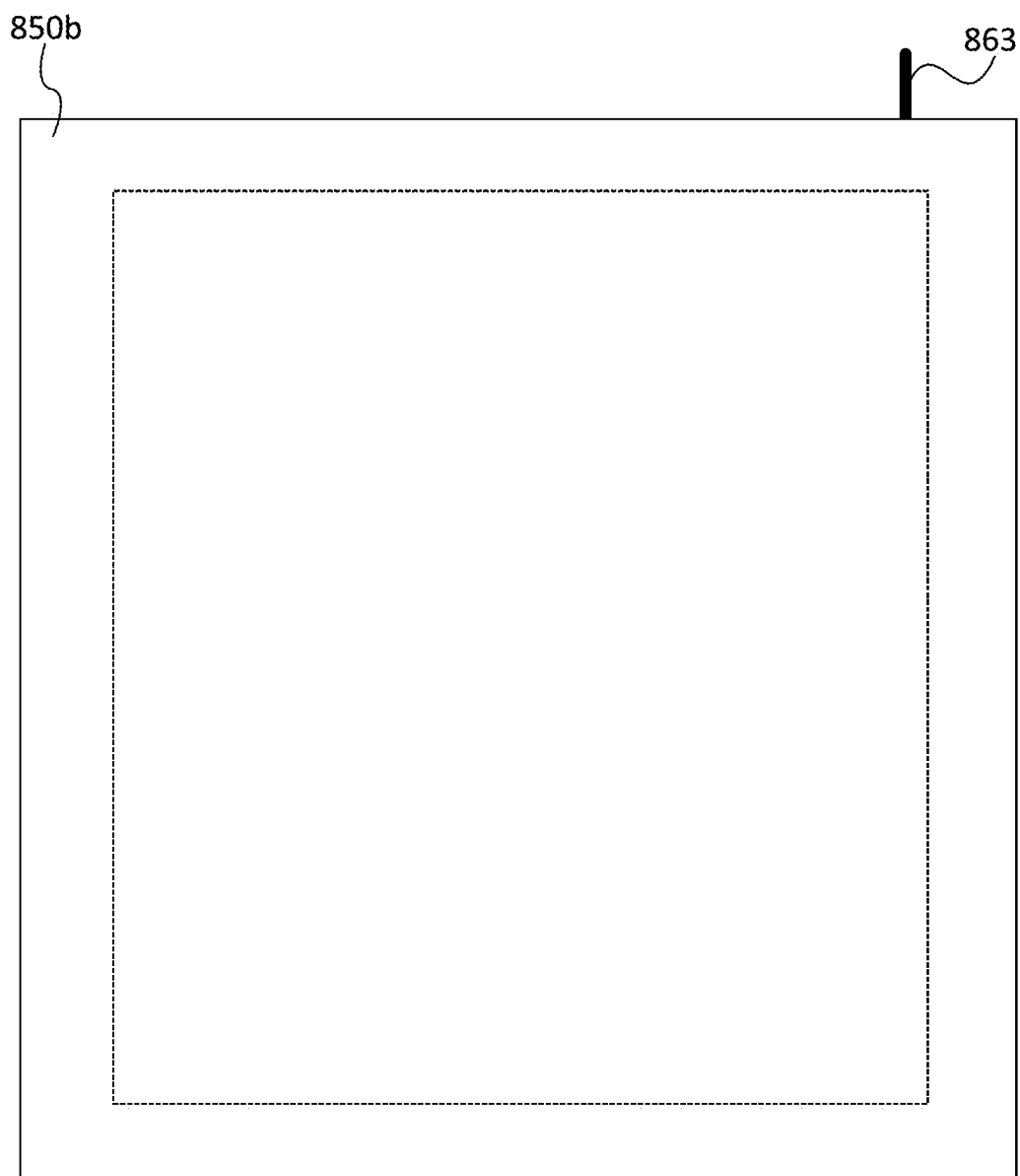


FIG. 9A

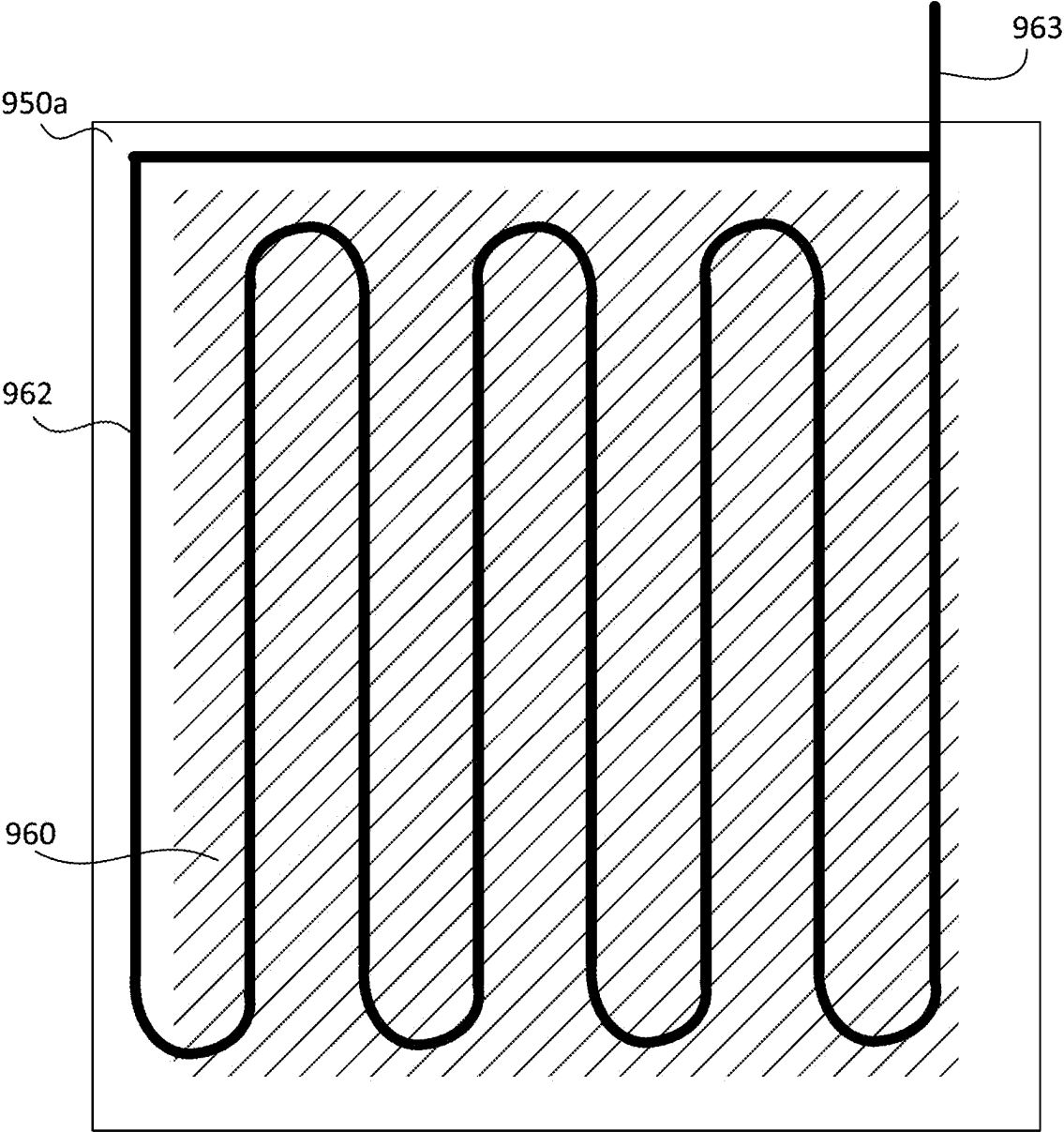


FIG. 9B

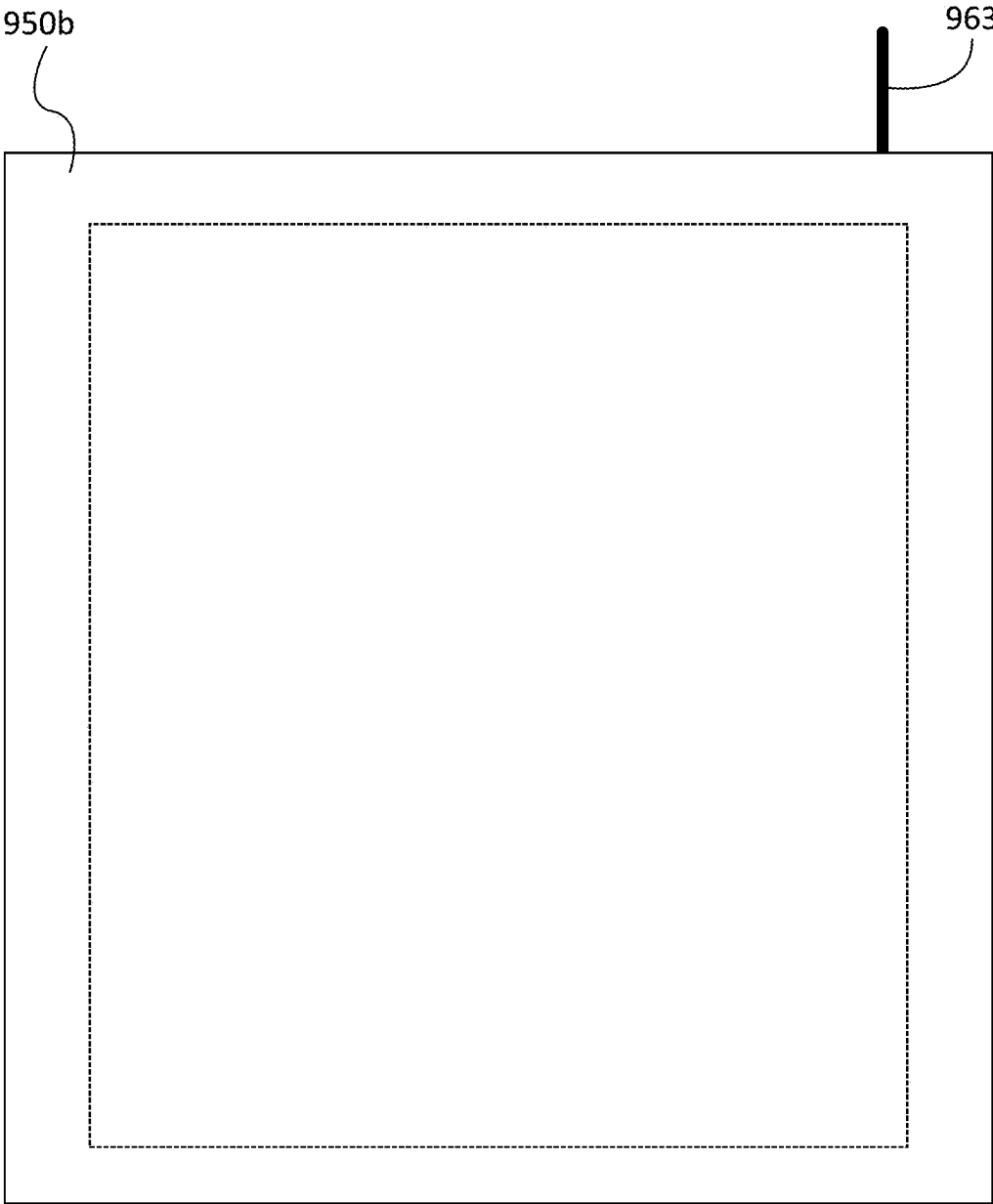


FIG. 10A

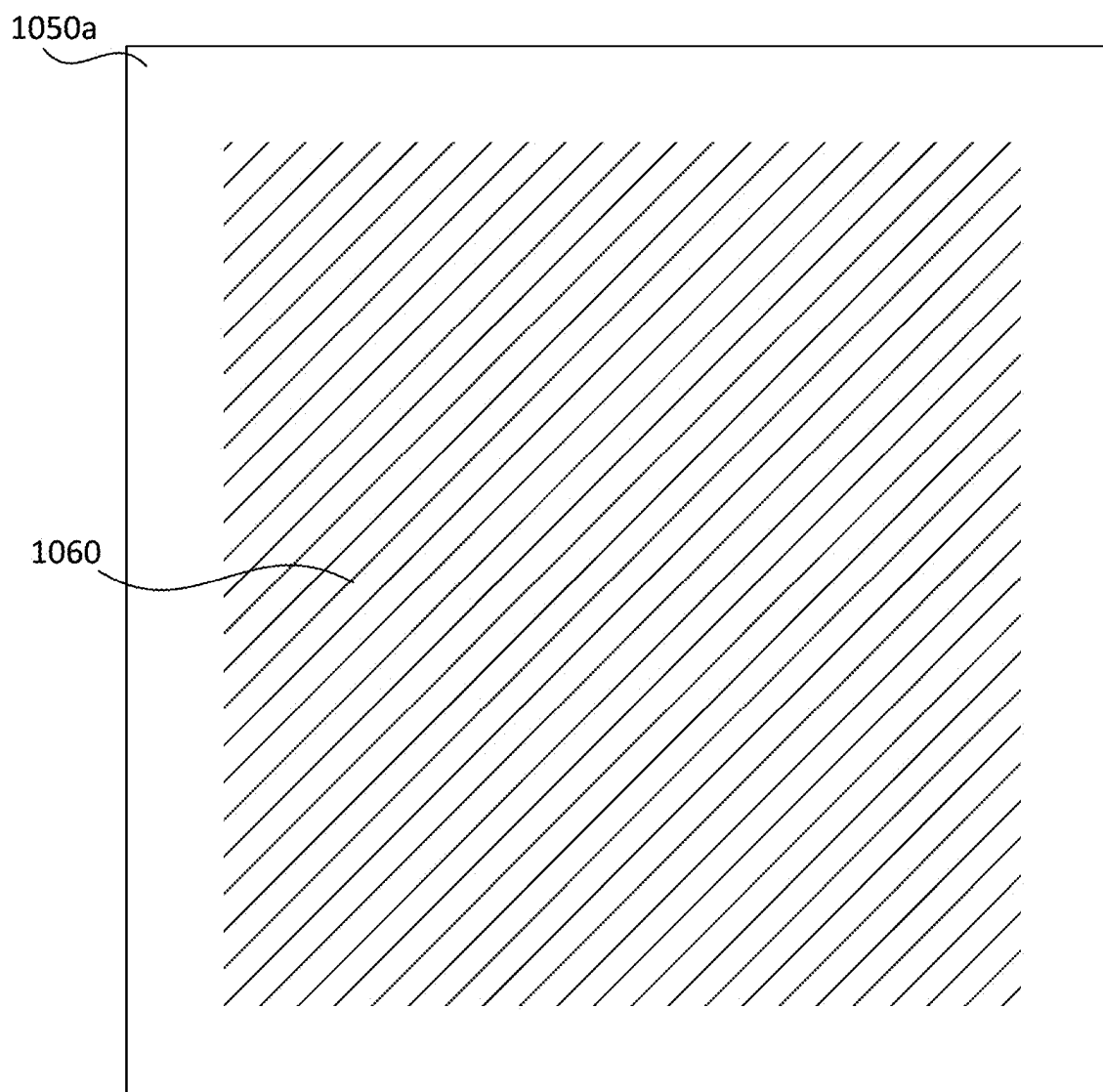


FIG. 10B

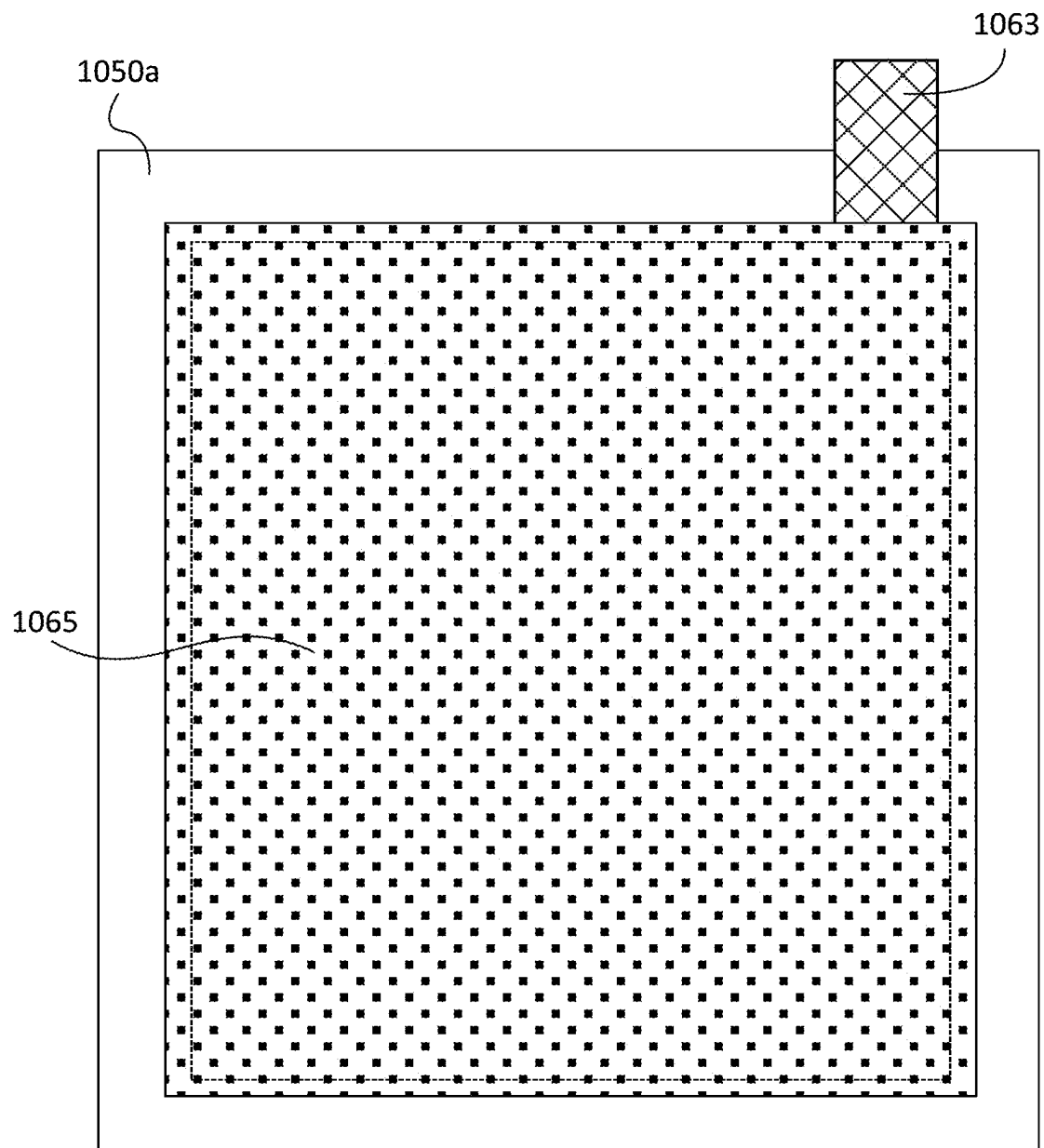


FIG. 10C

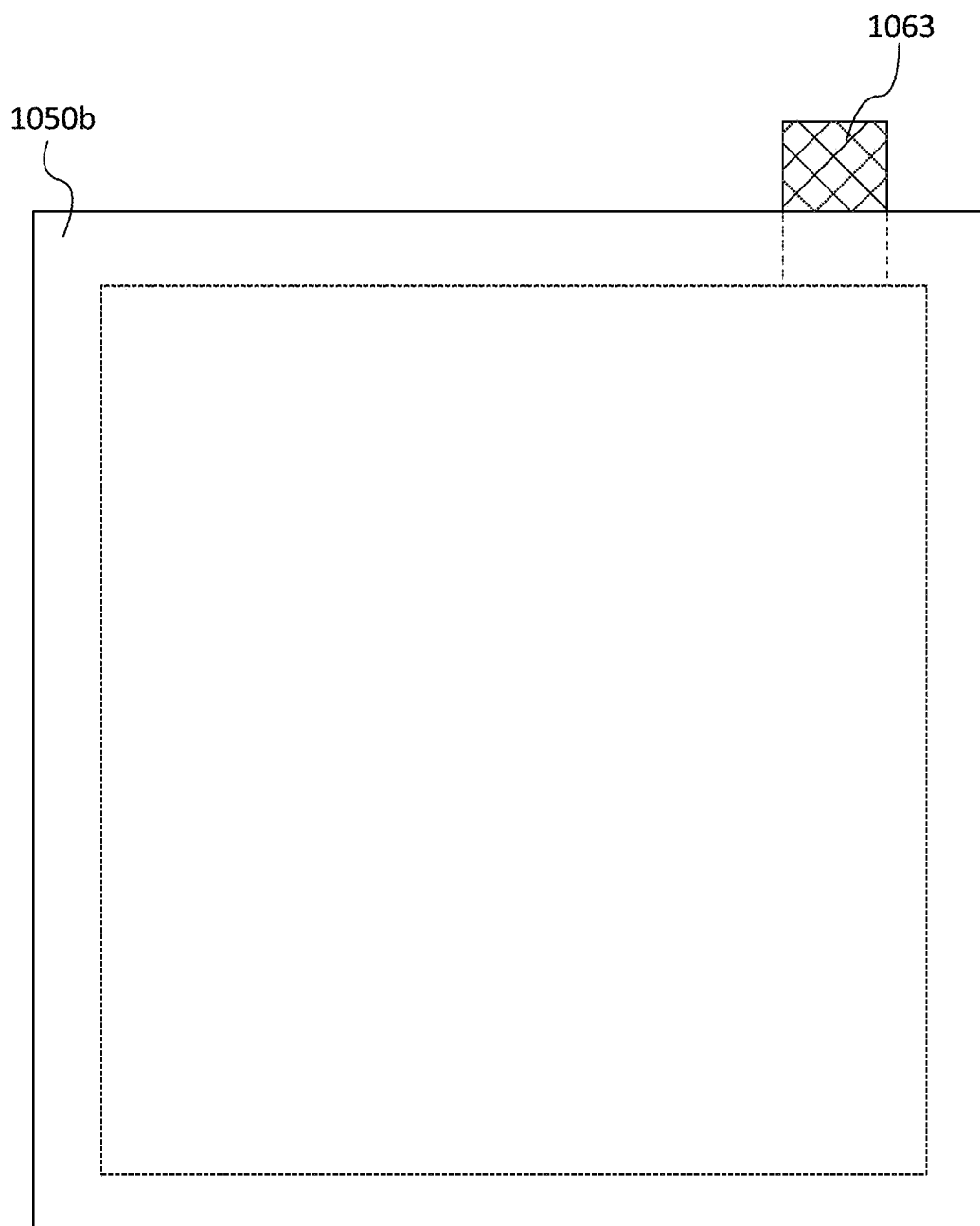


FIG. 11A

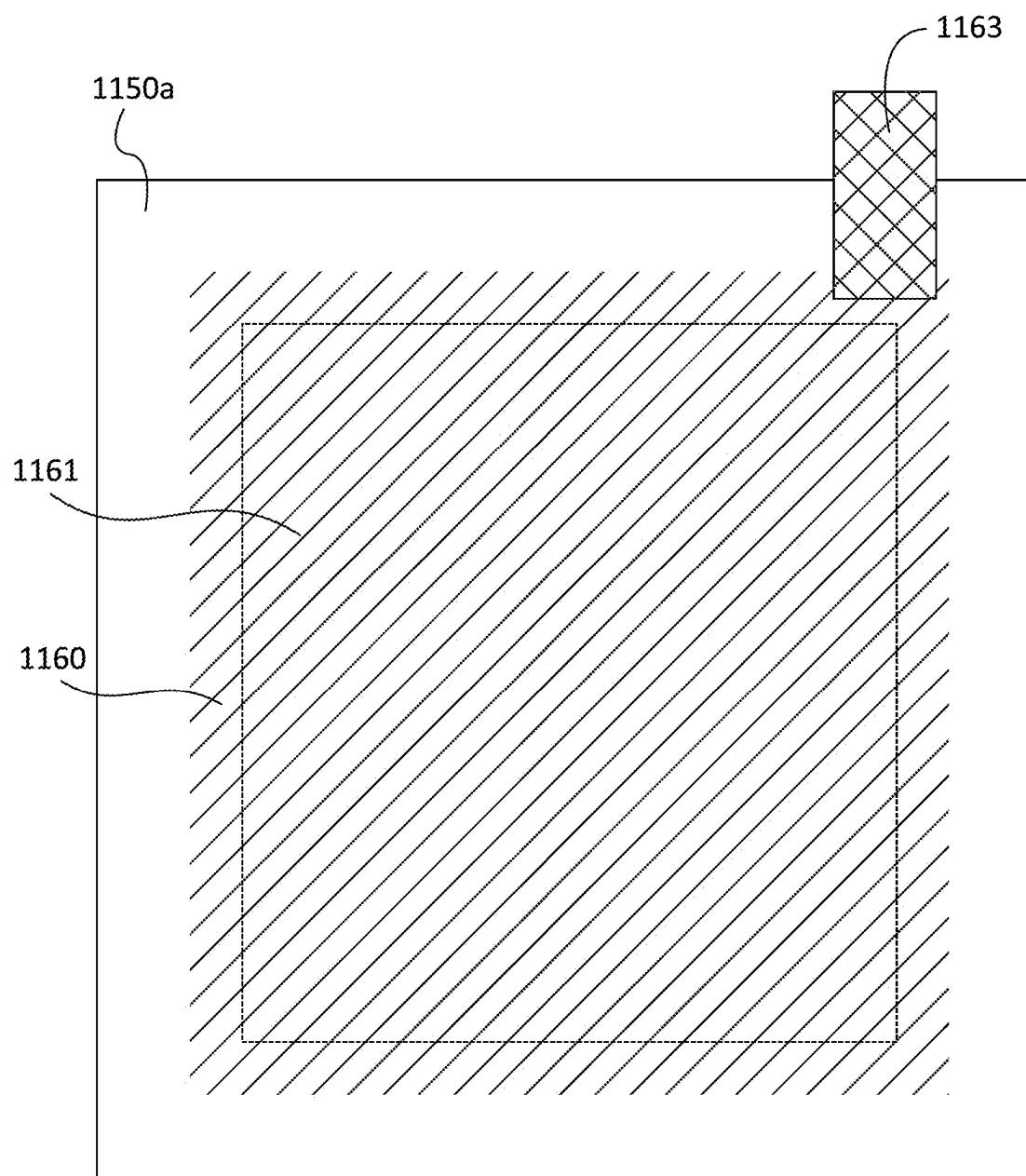


FIG. 11B

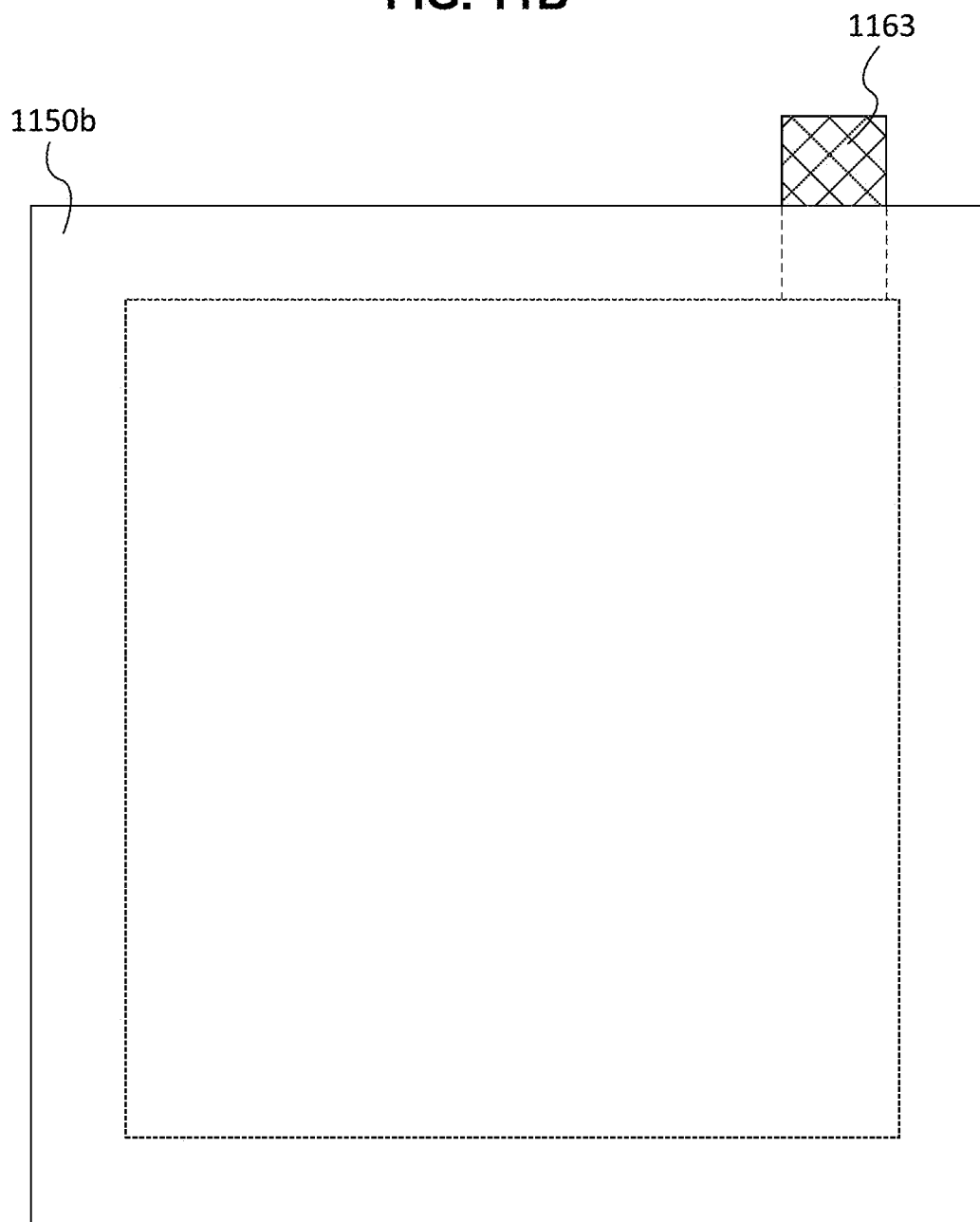


FIG. 12

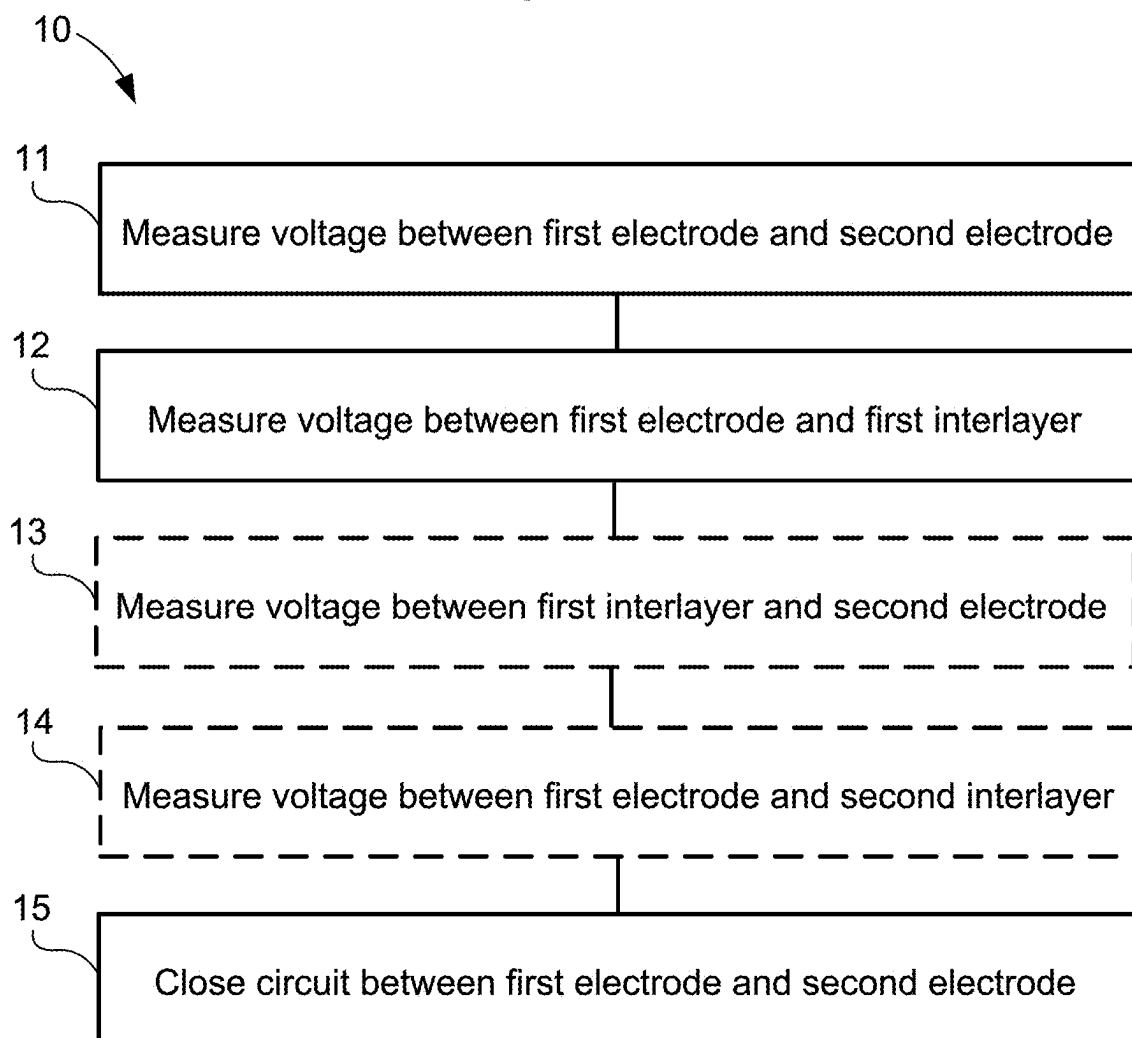


FIG. 13

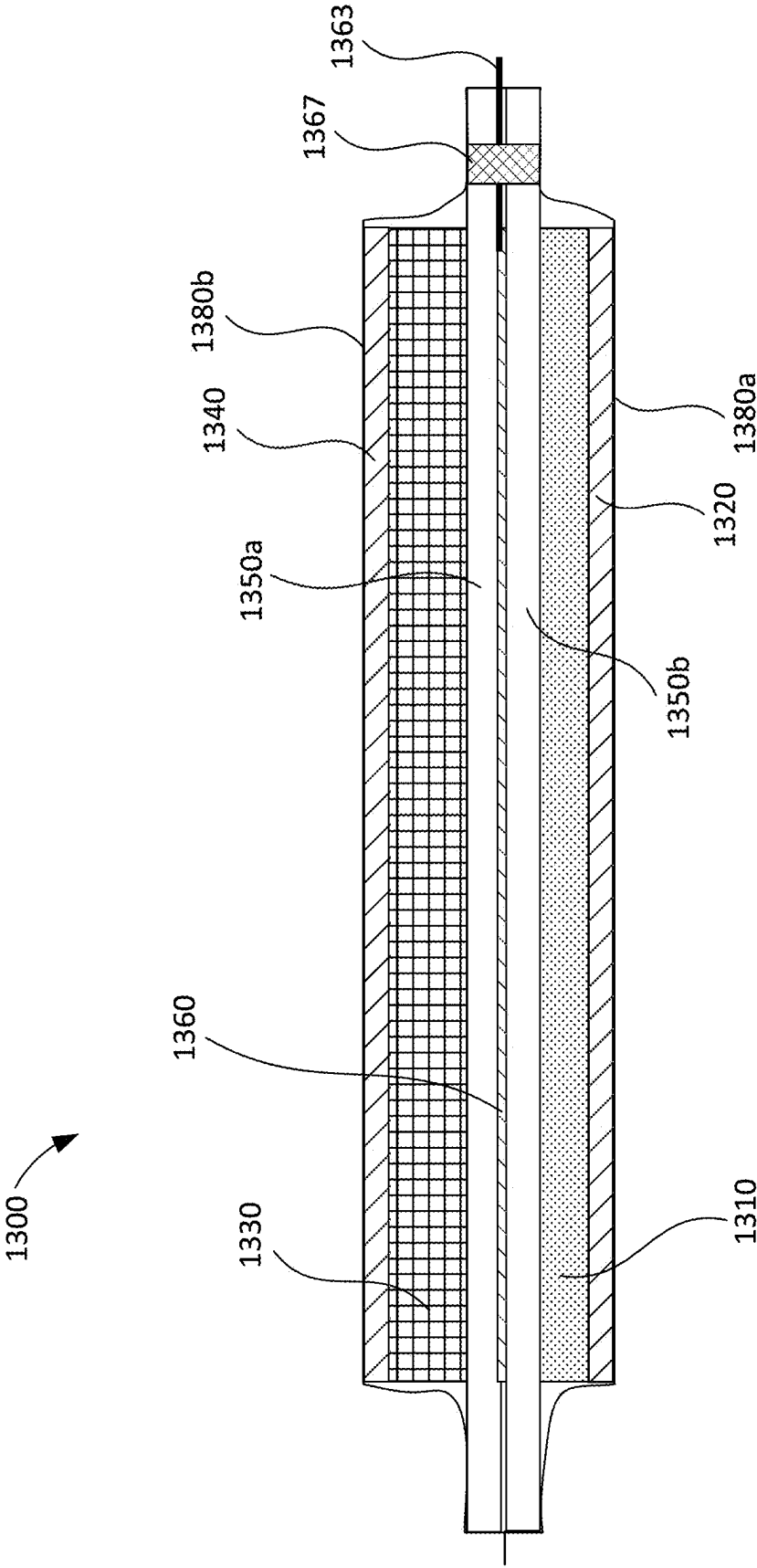


FIG. 14

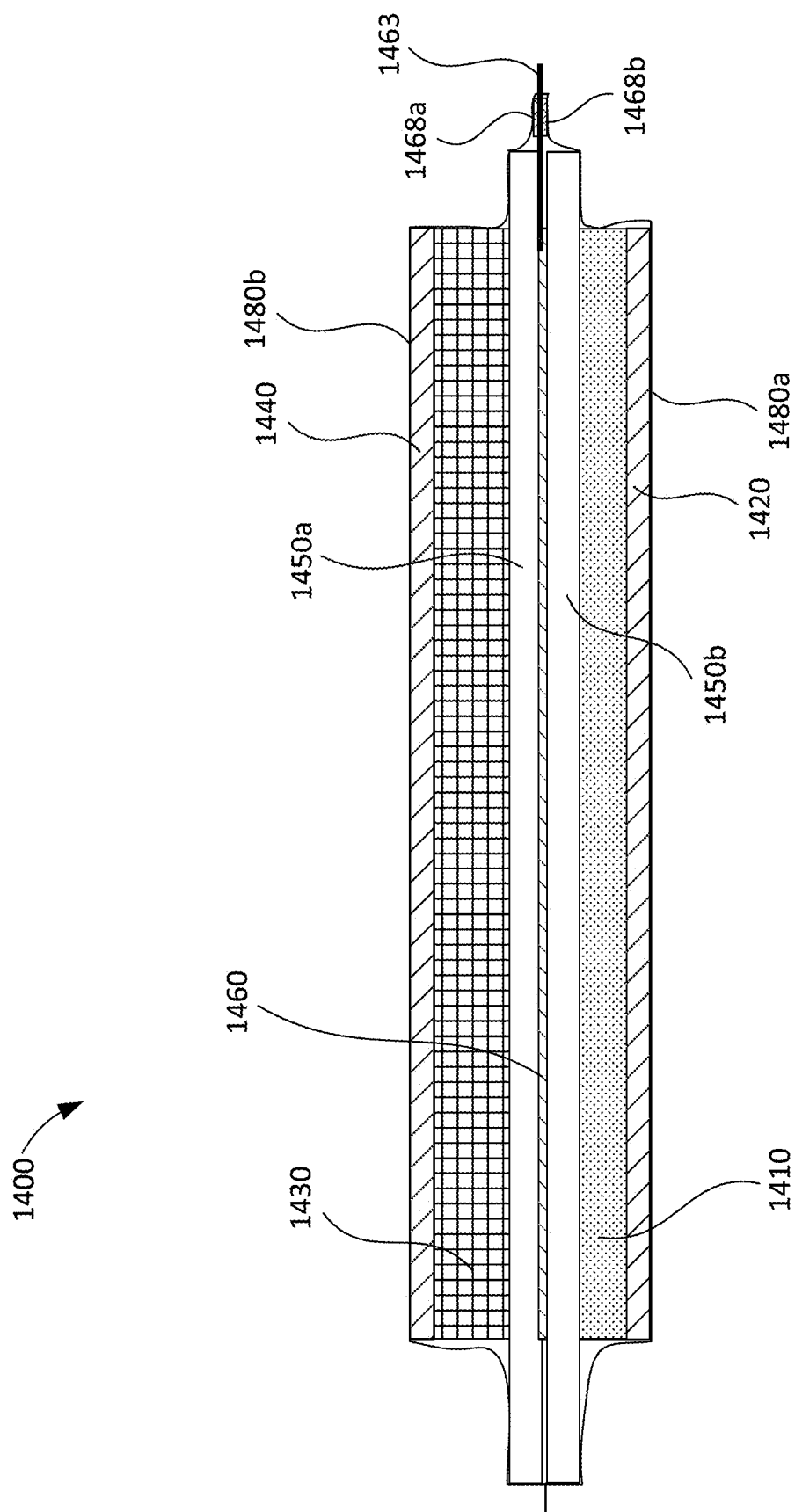


FIG. 15A

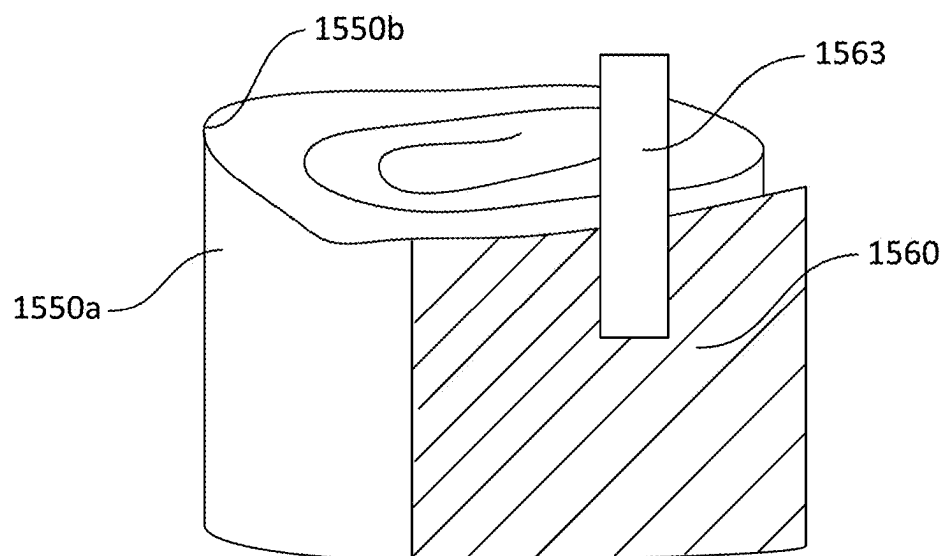


FIG. 15B

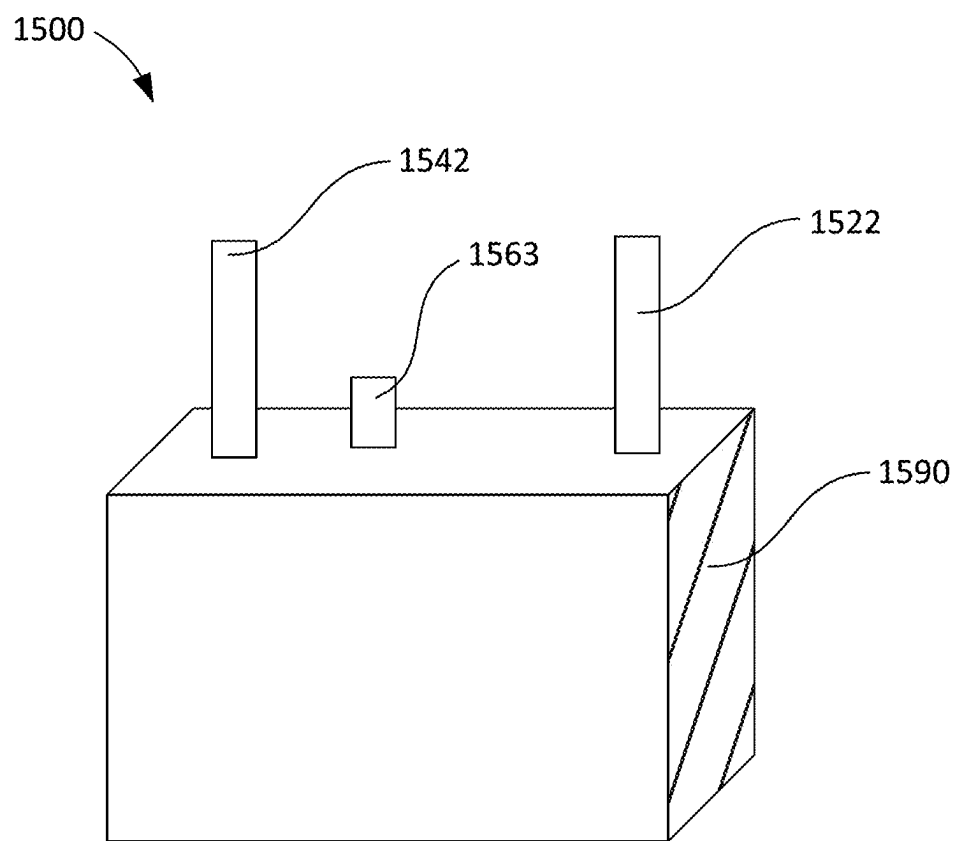


FIG. 16A

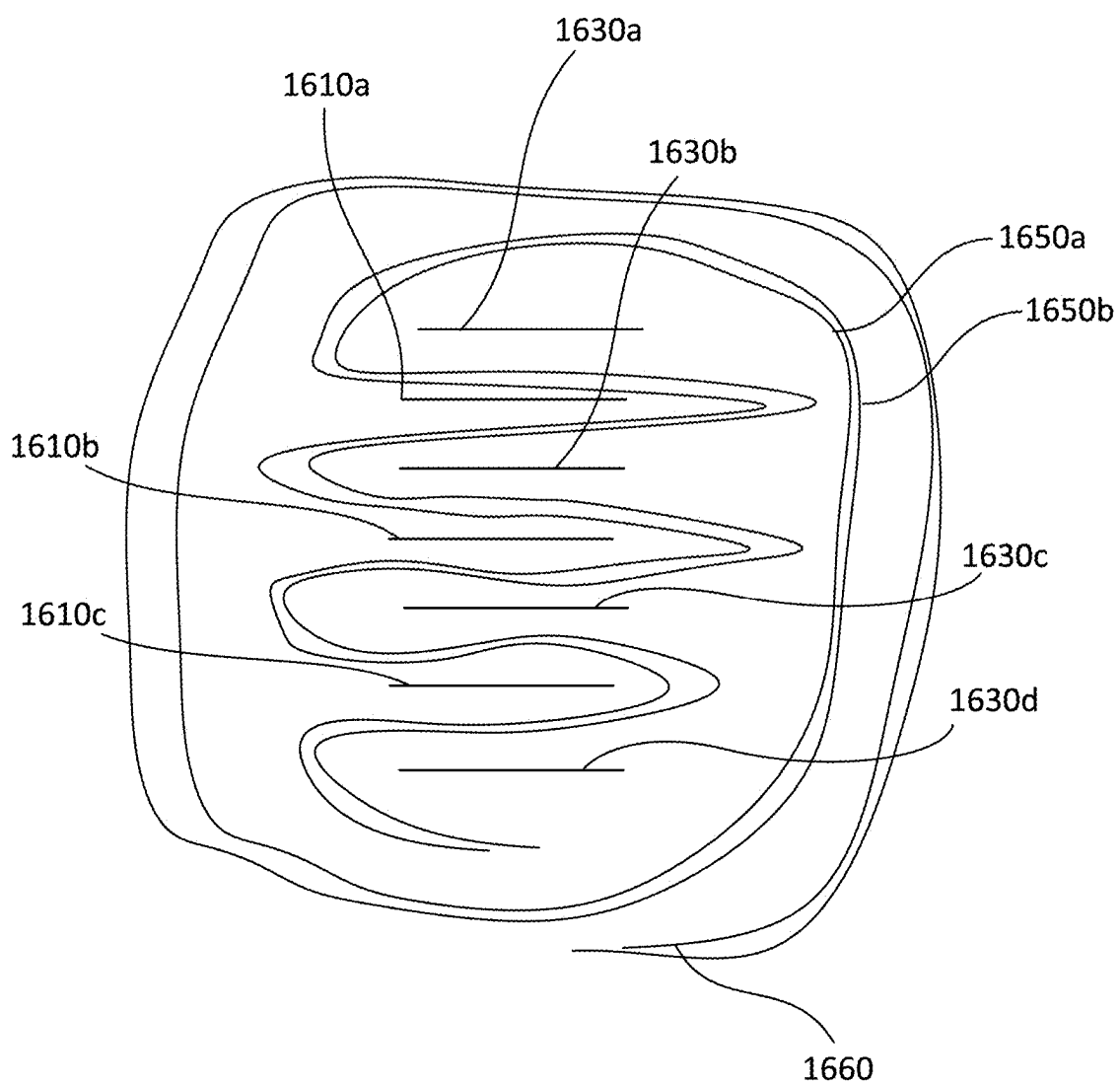


FIG. 16B

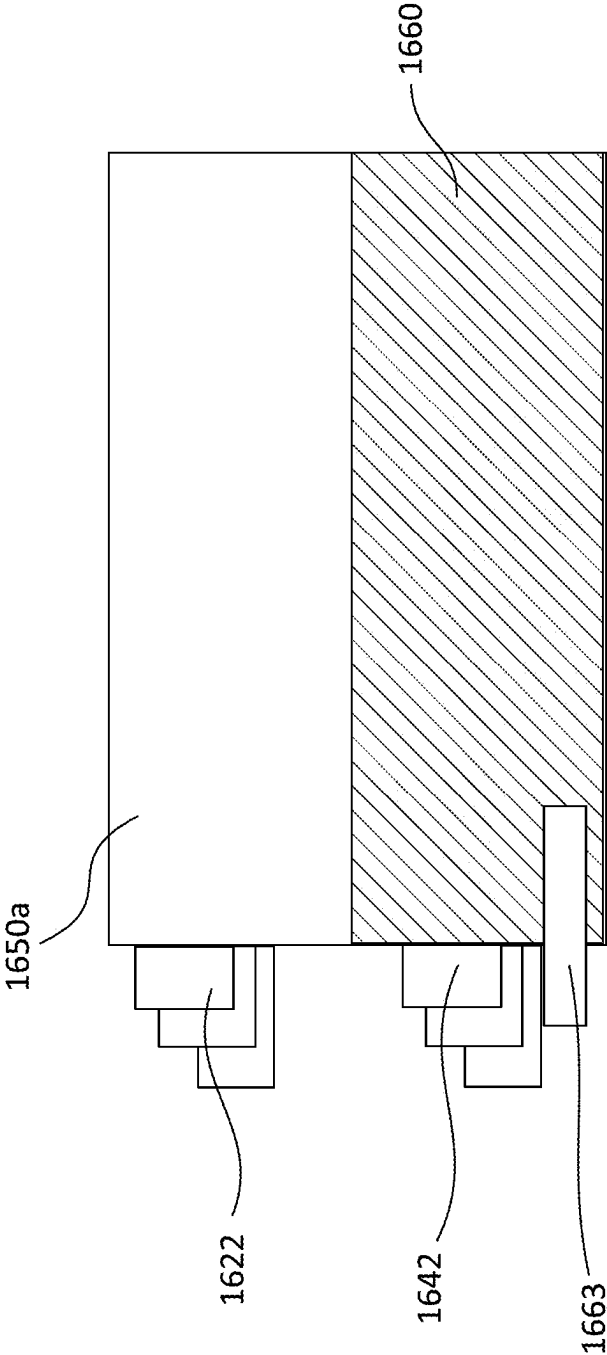


FIG. 17A

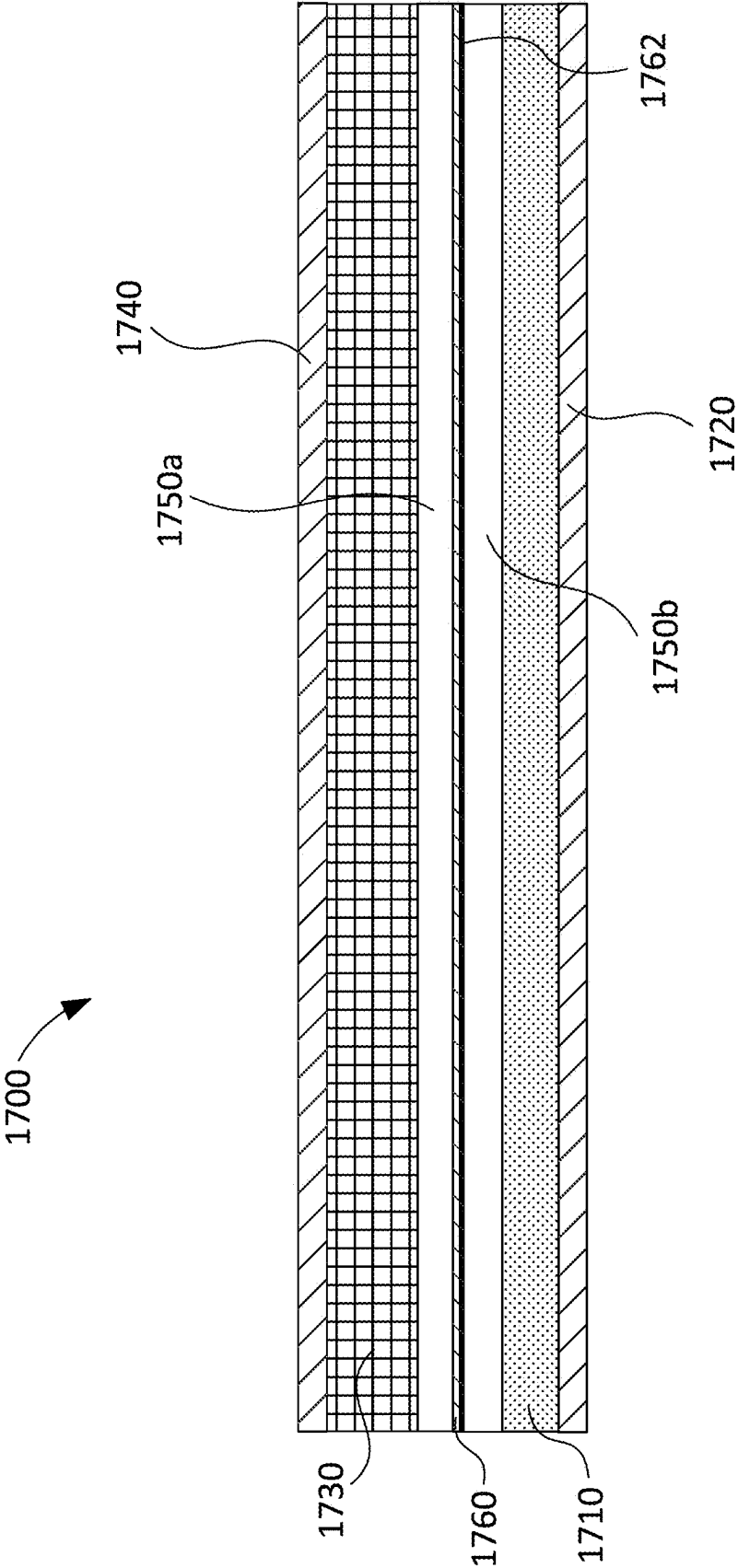


FIG. 17B

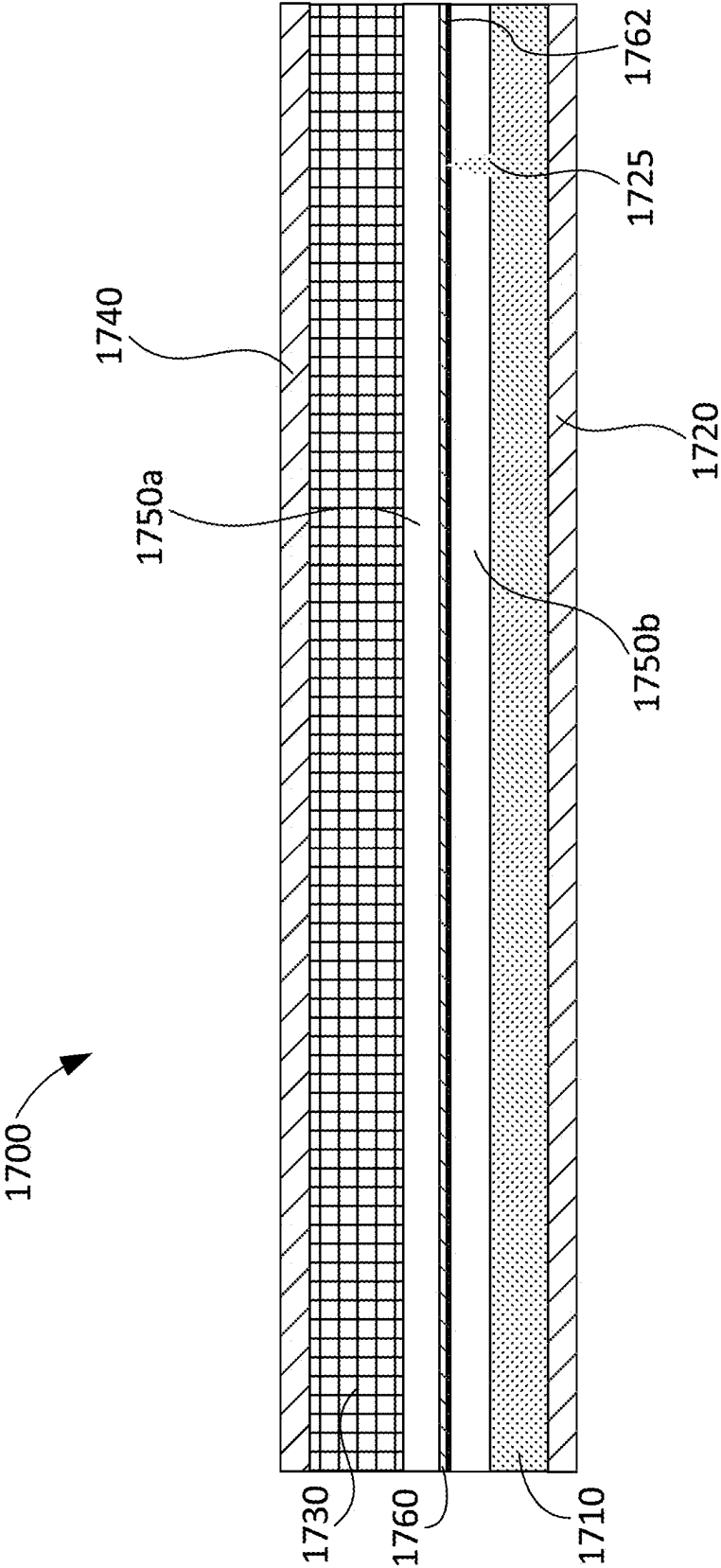


FIG. 18

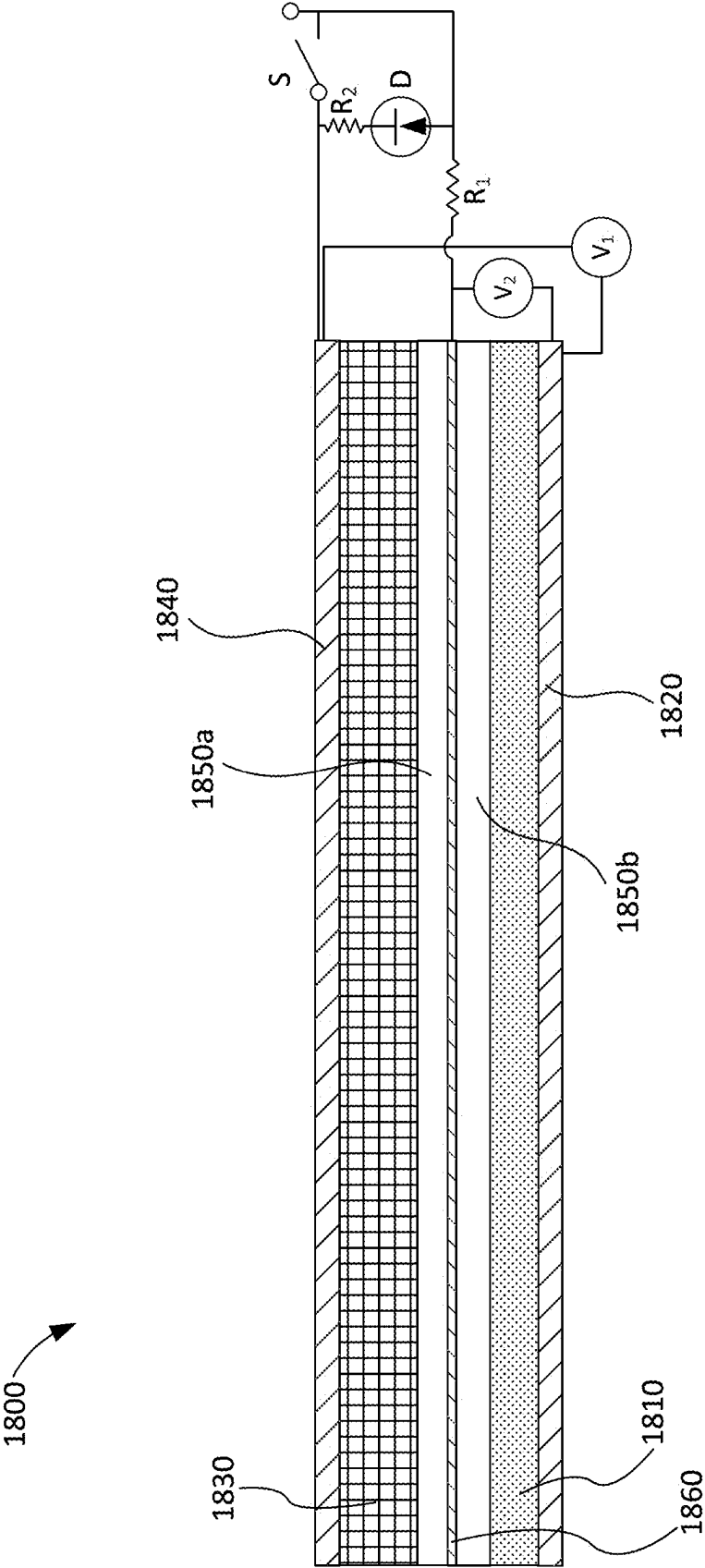


FIG. 19

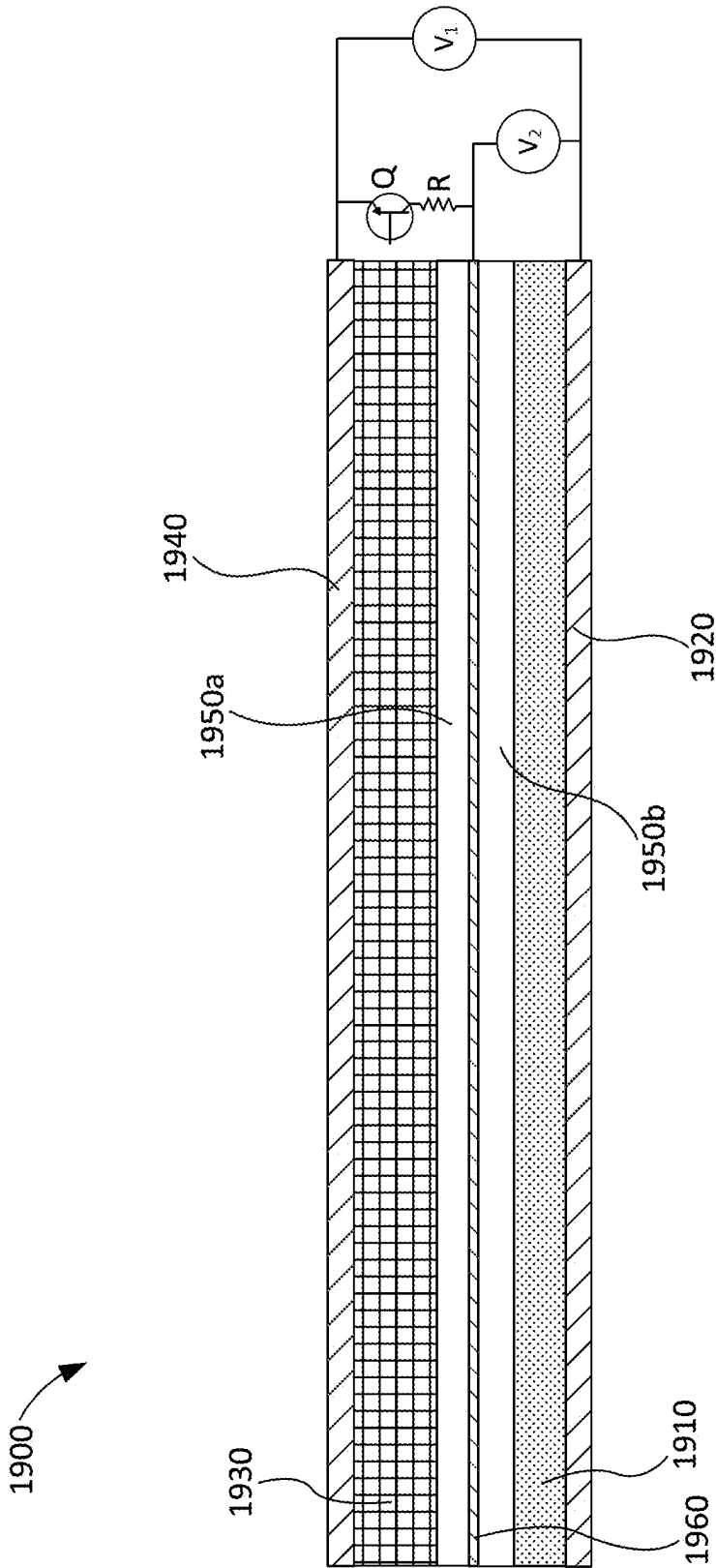


FIG. 20

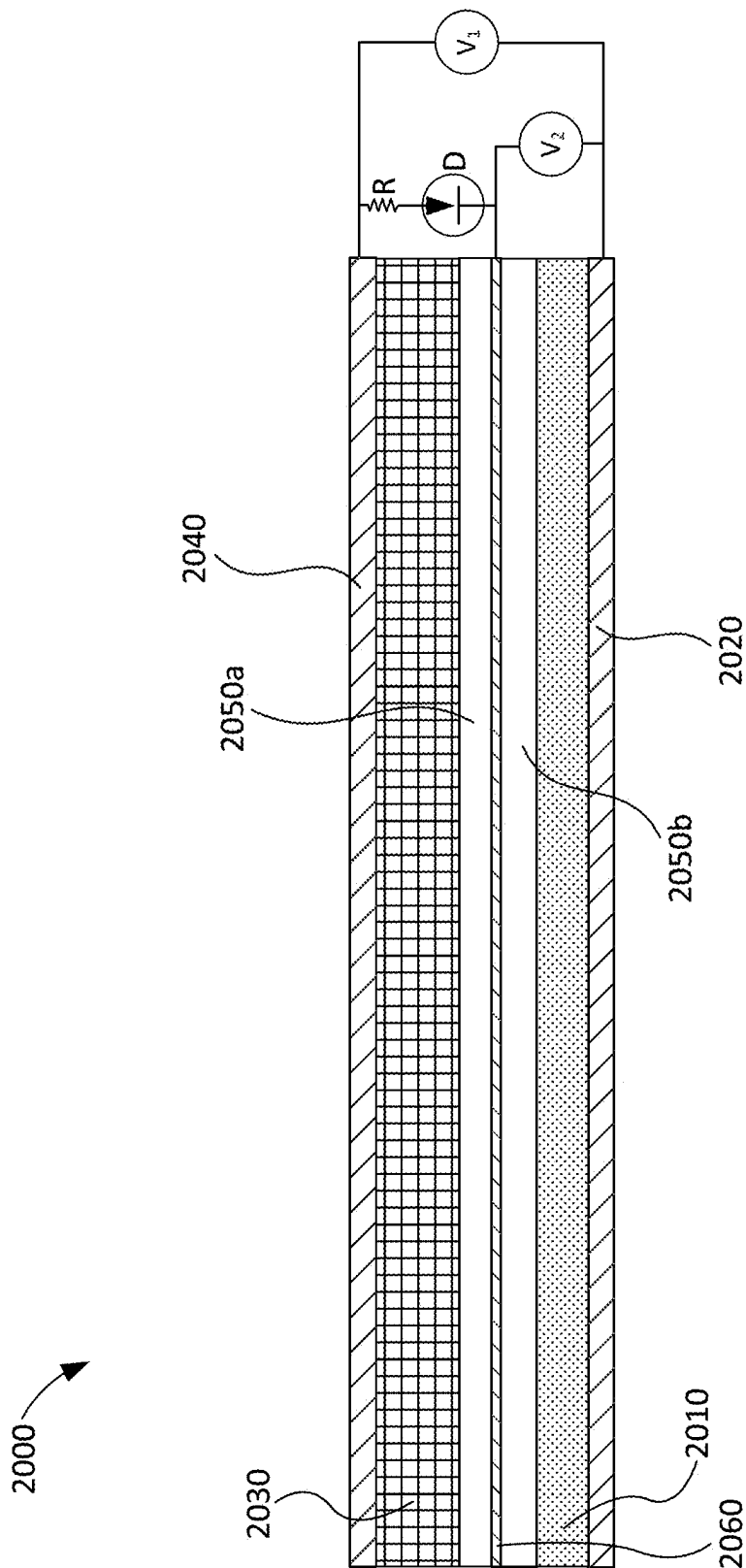


FIG. 21

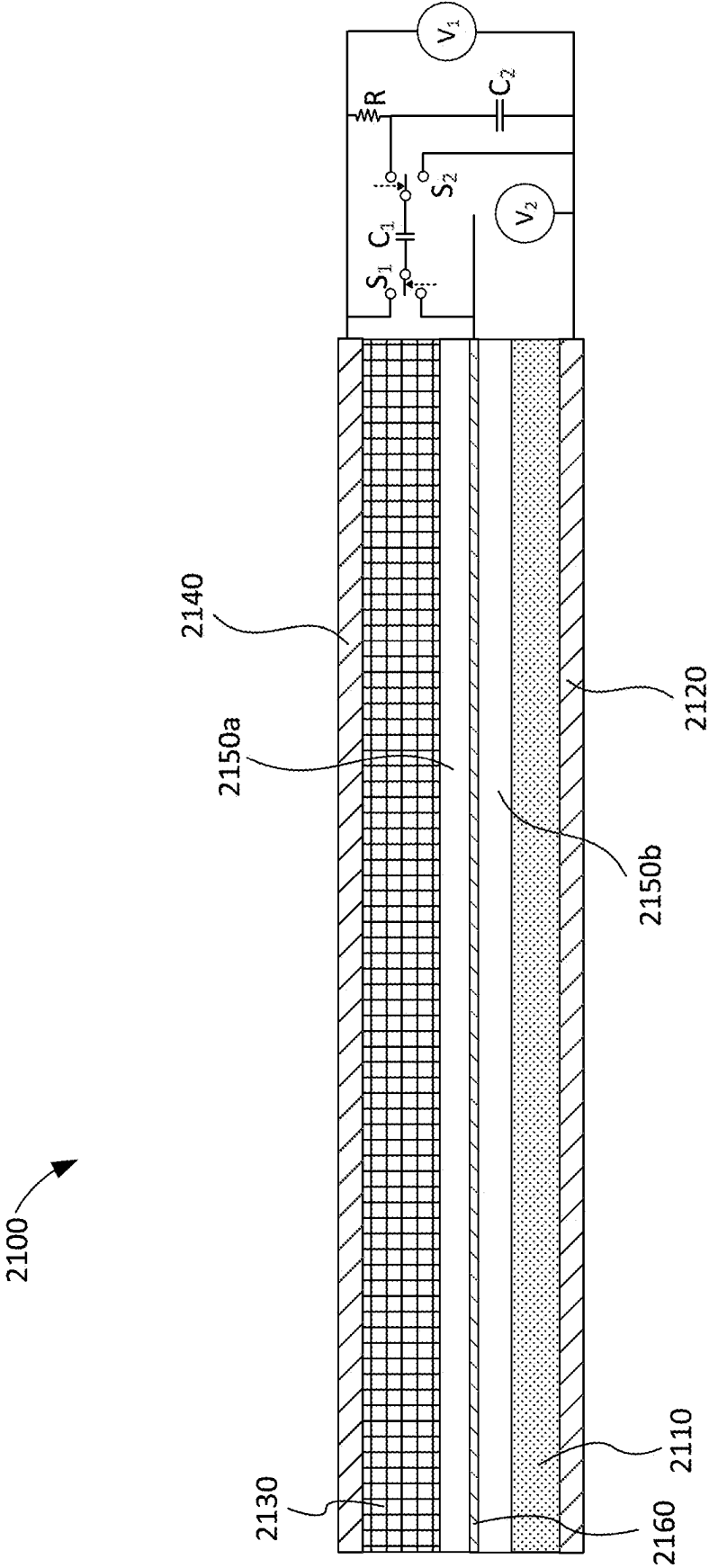


FIG. 22

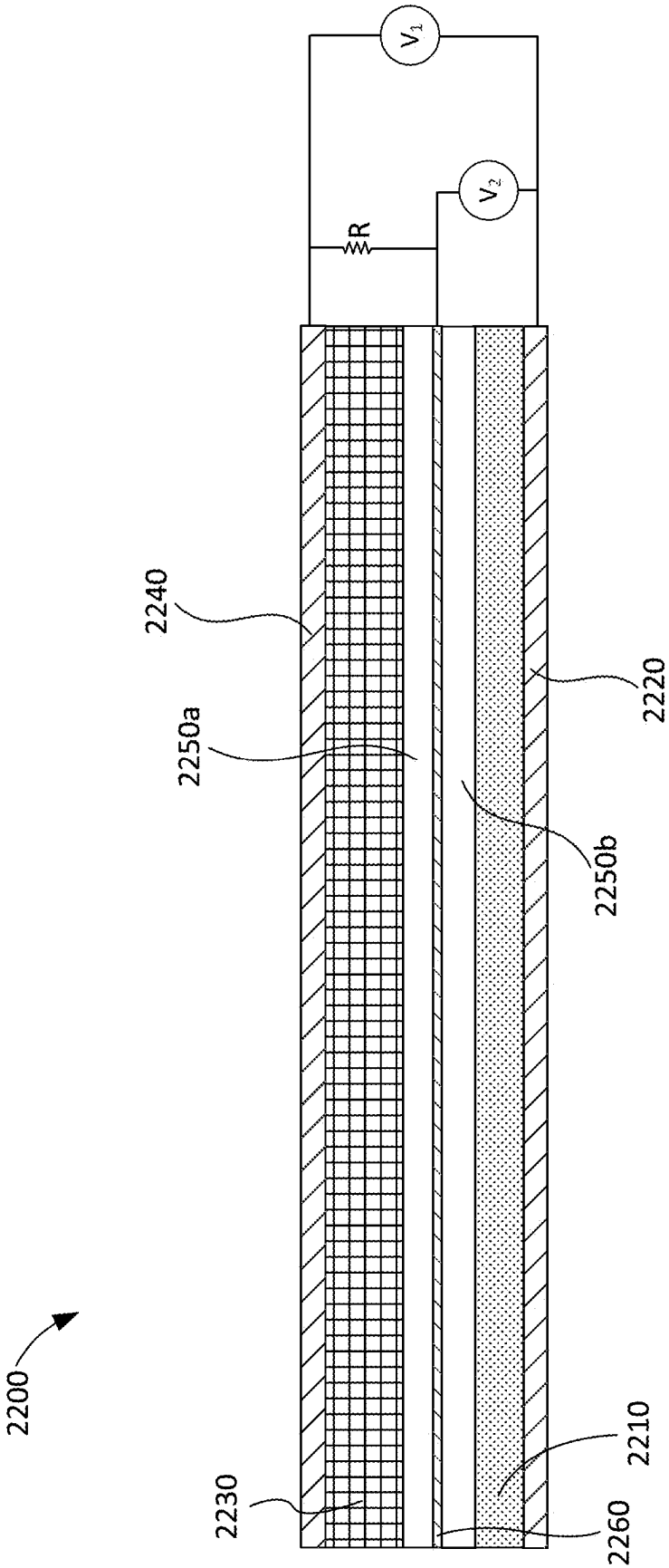


FIG. 23

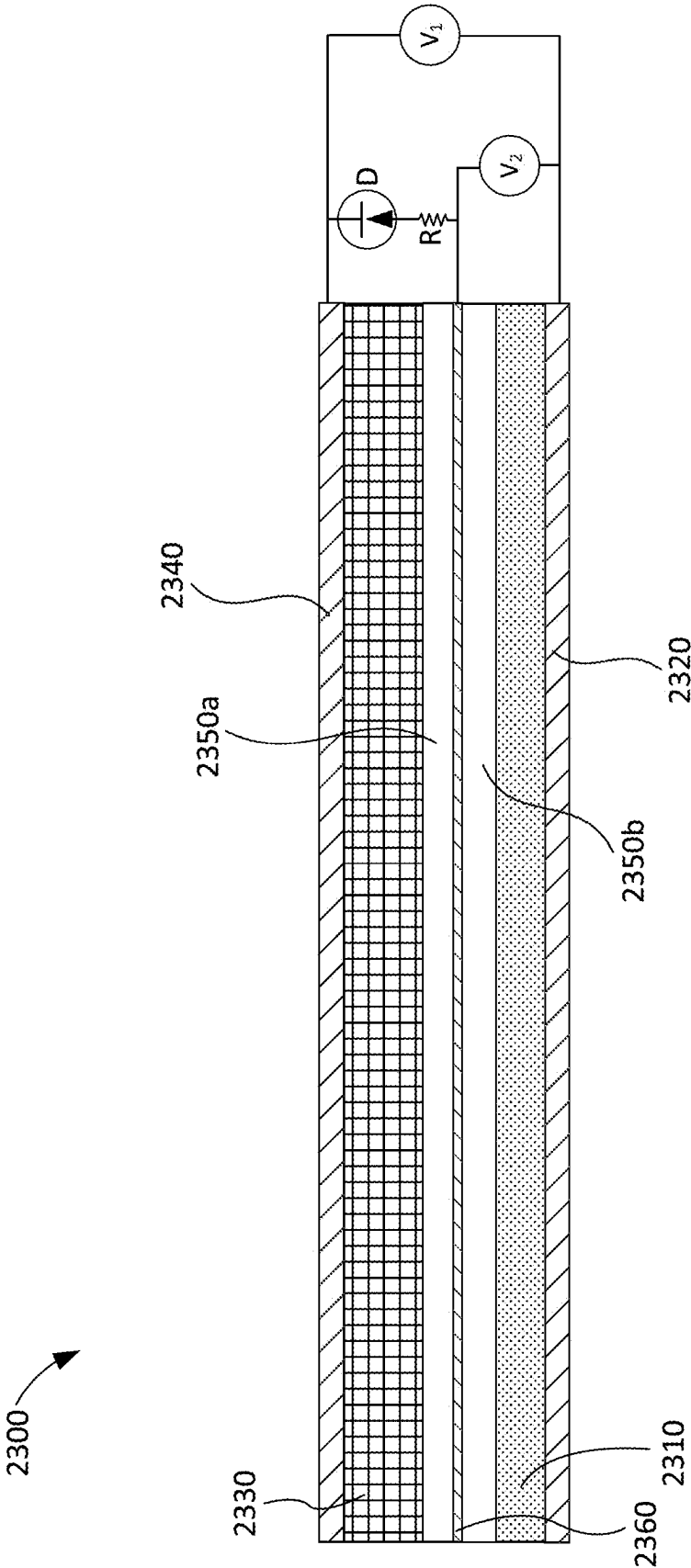


FIG. 24

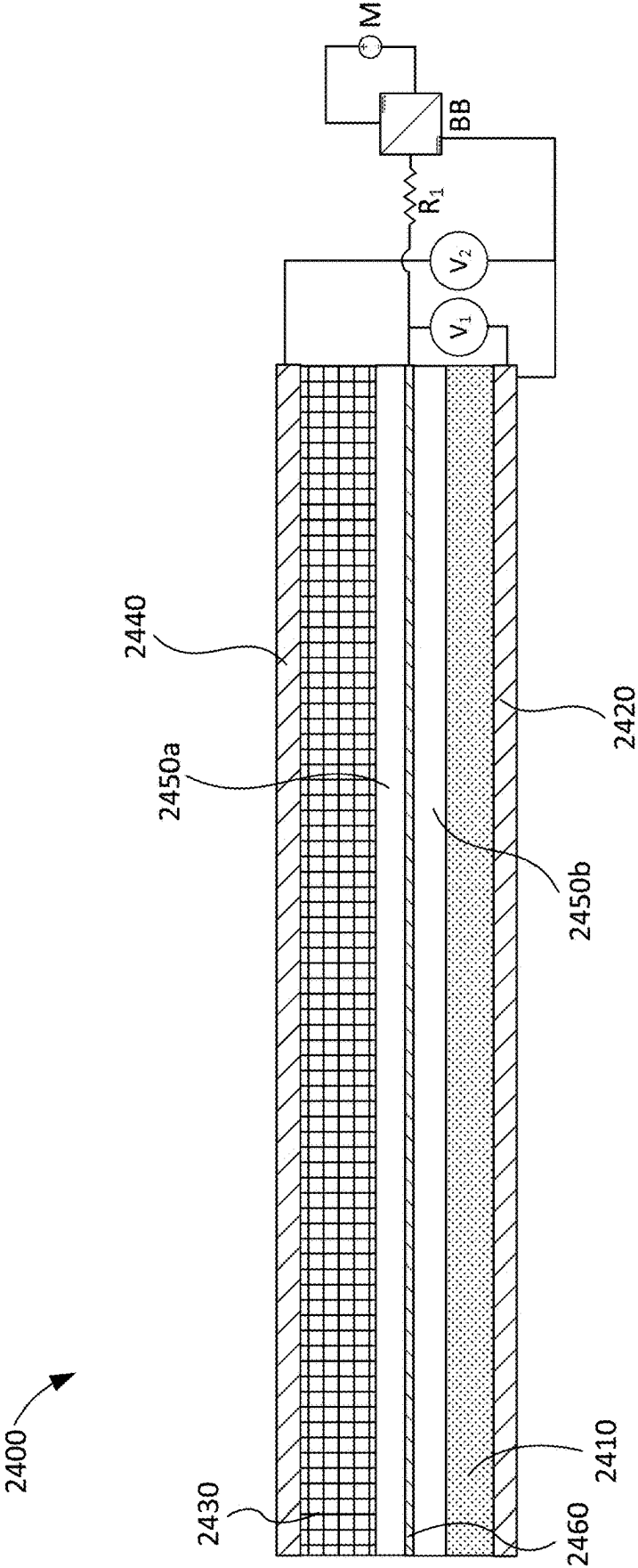


FIG. 25

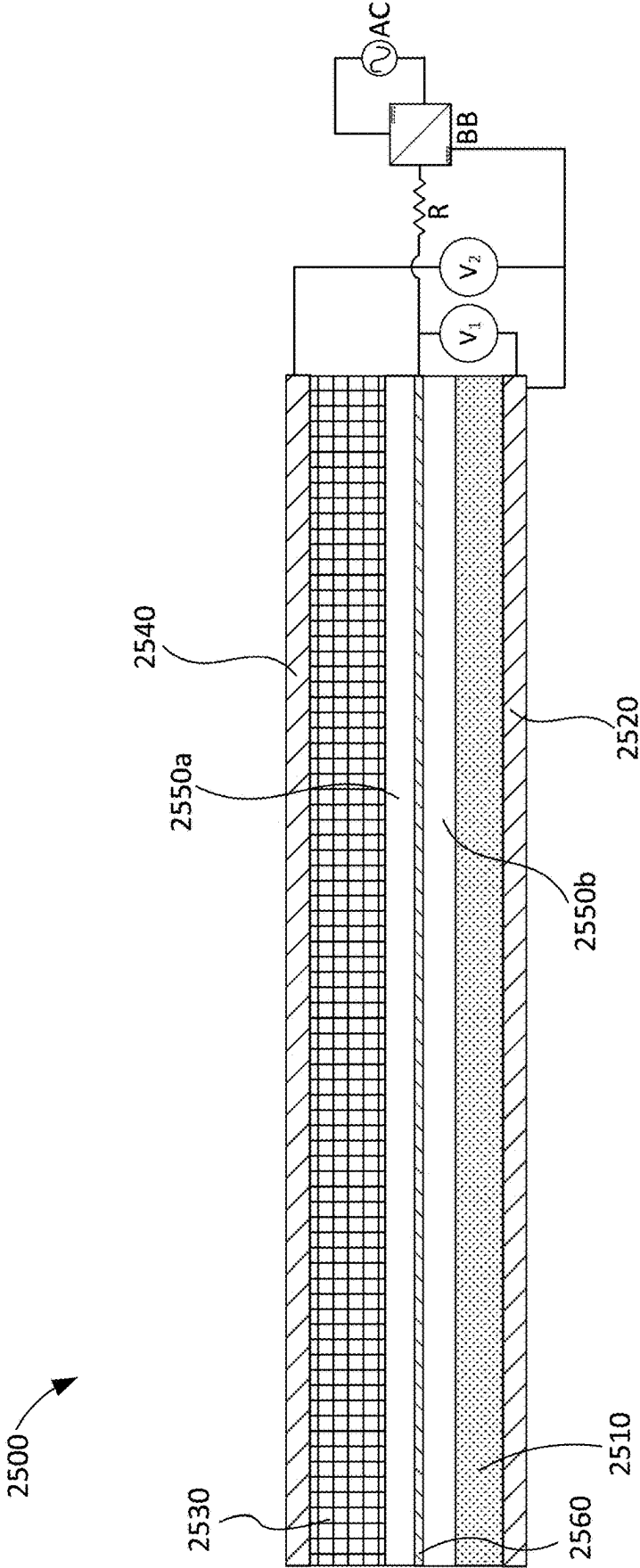


FIG. 26

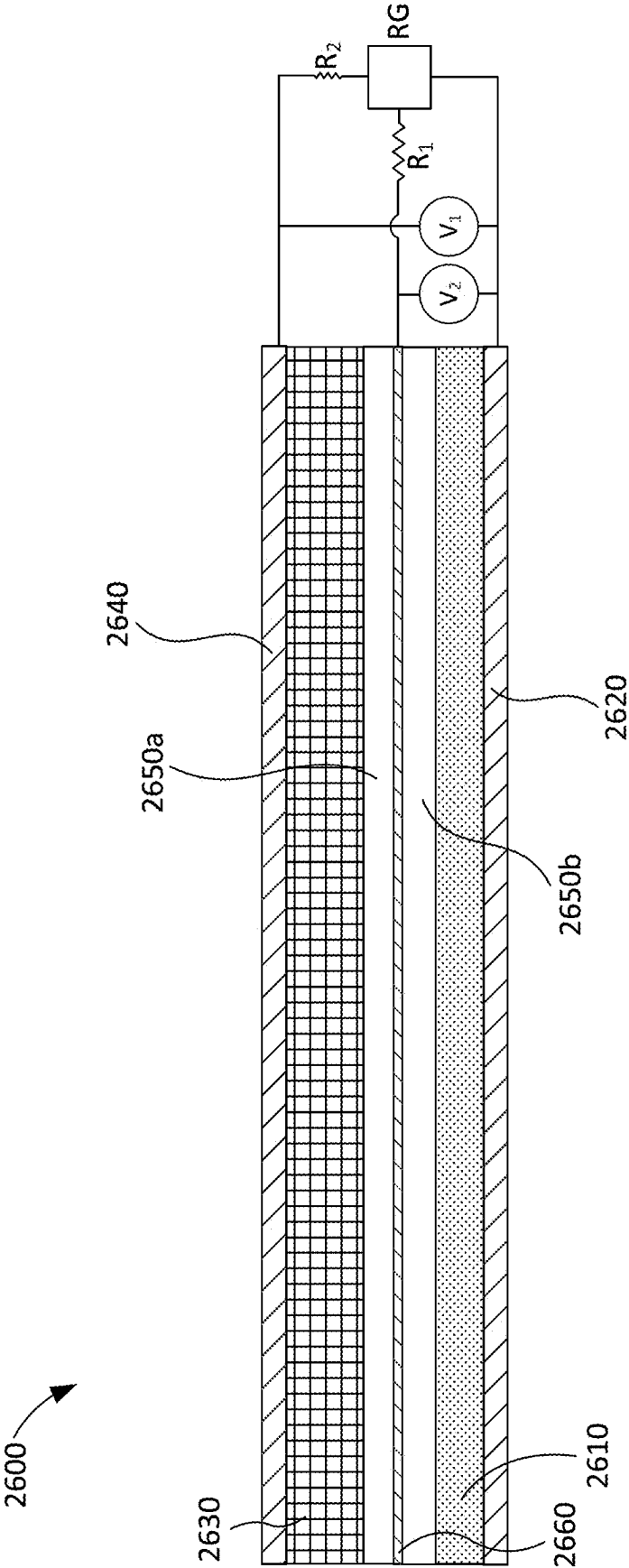


FIG. 27

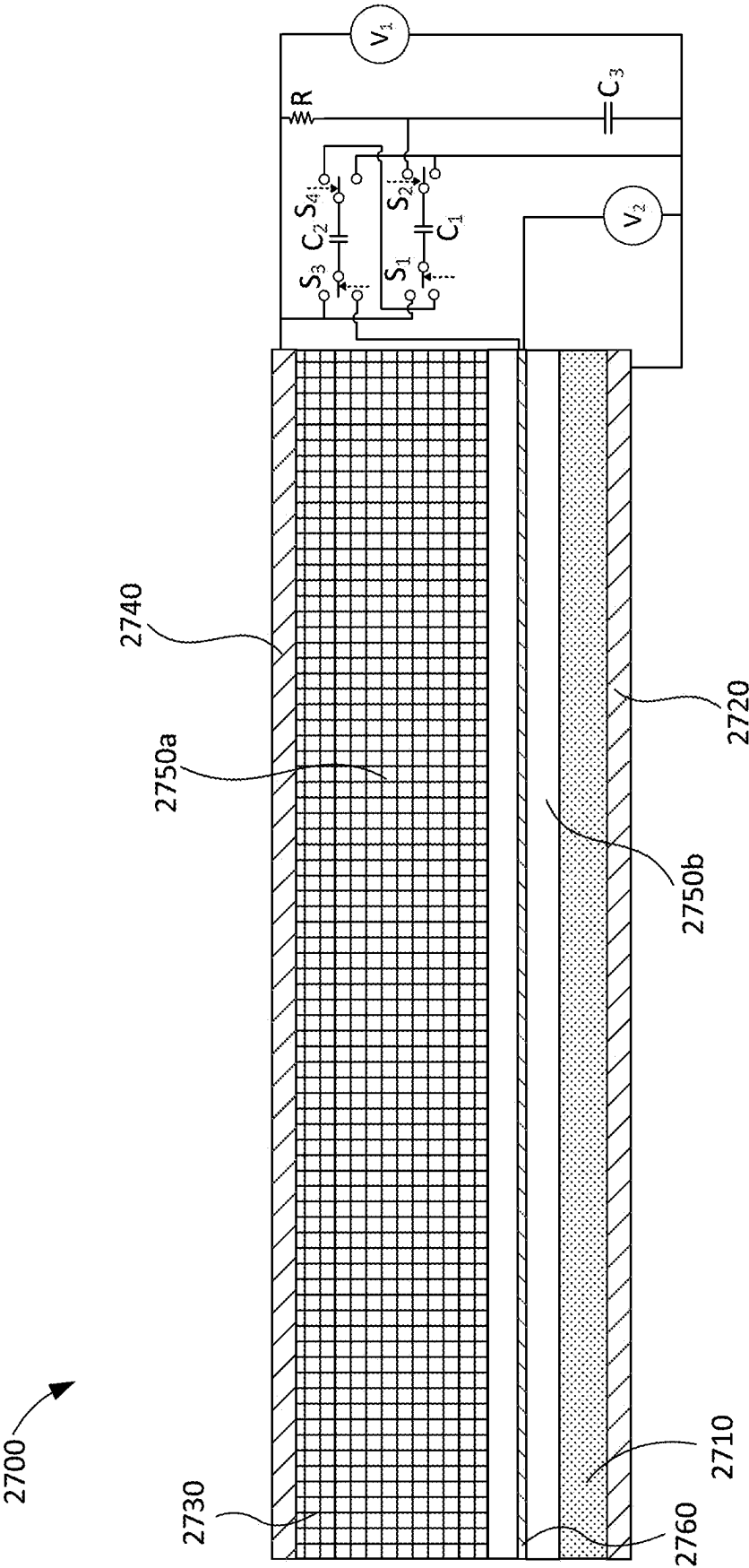


FIG. 28

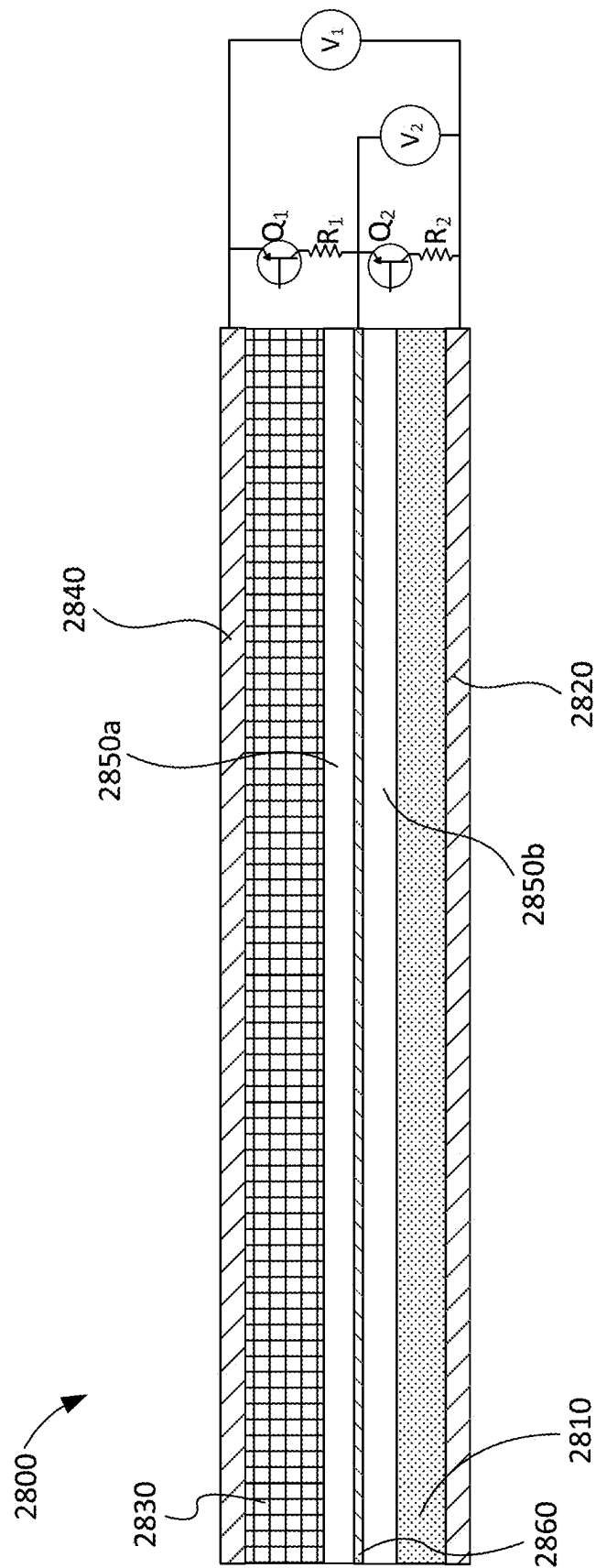


FIG. 29

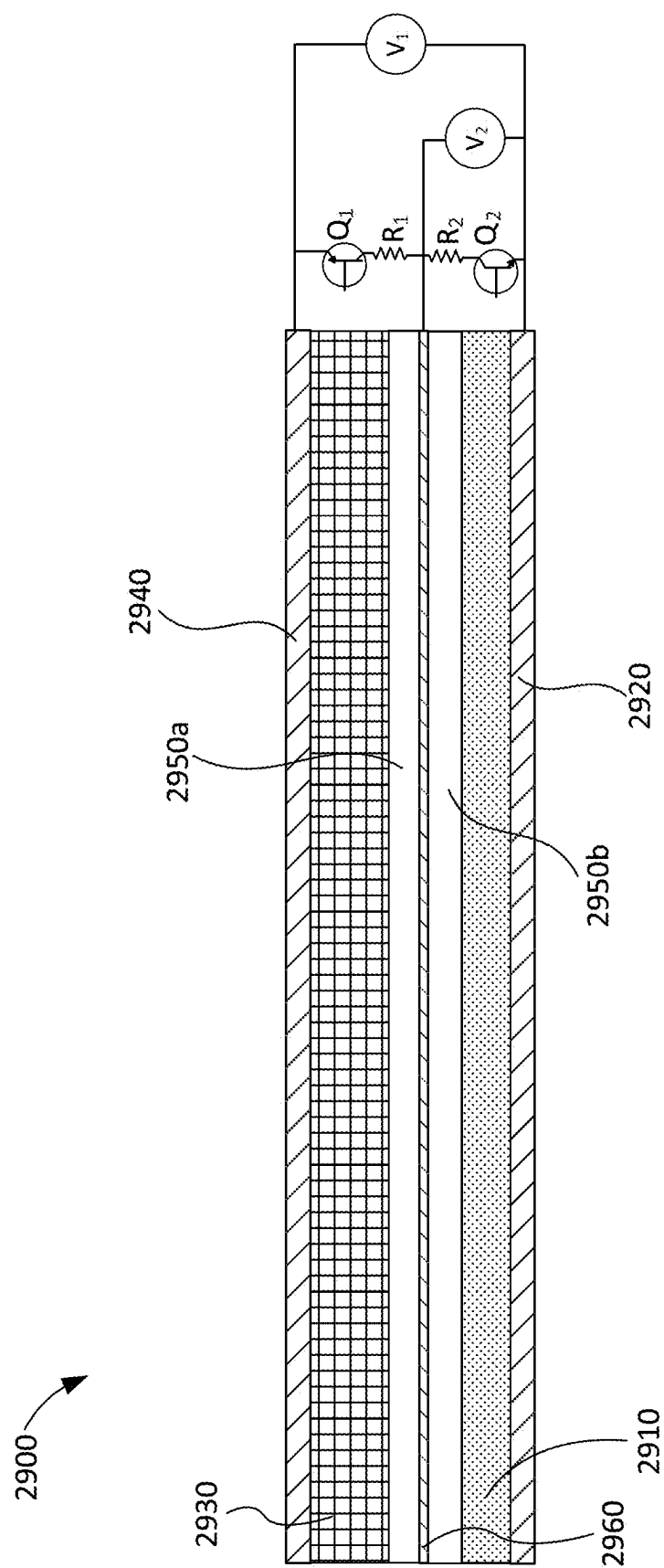


FIG. 30

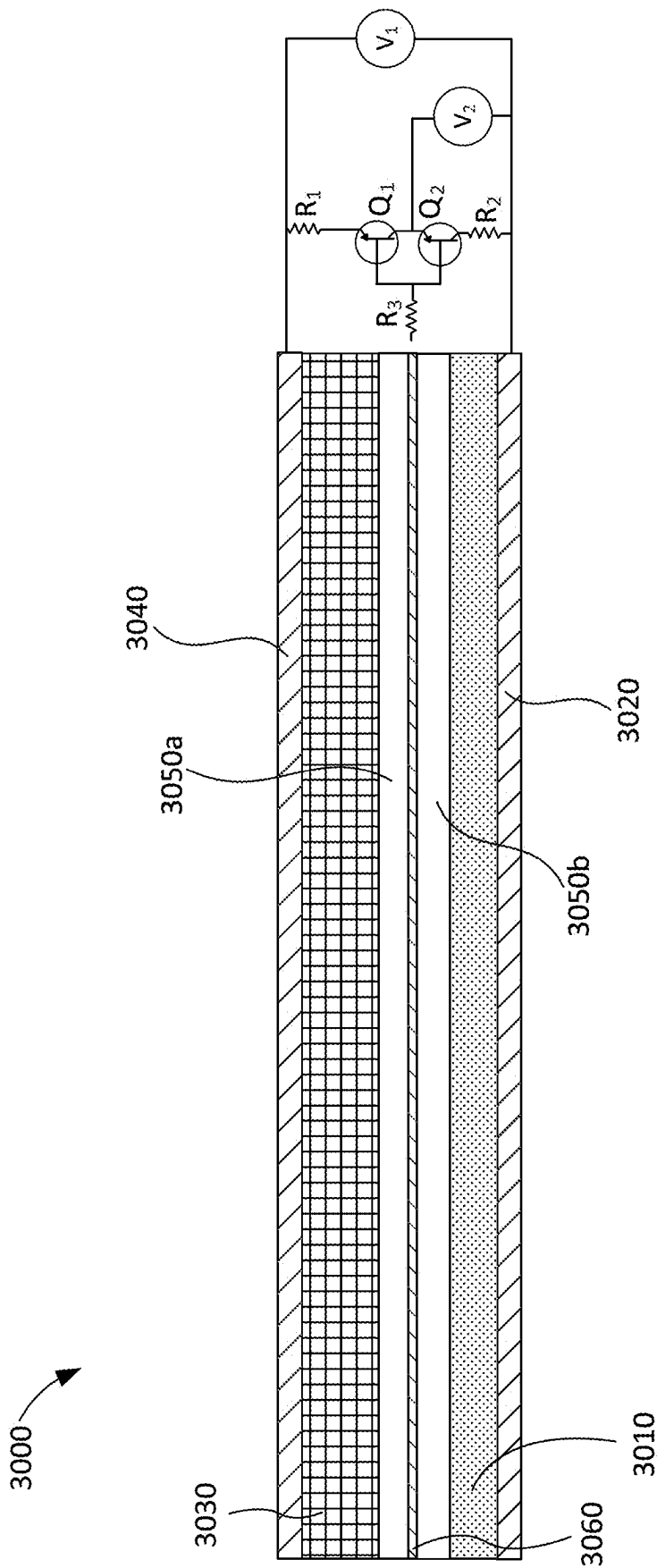


FIG. 31

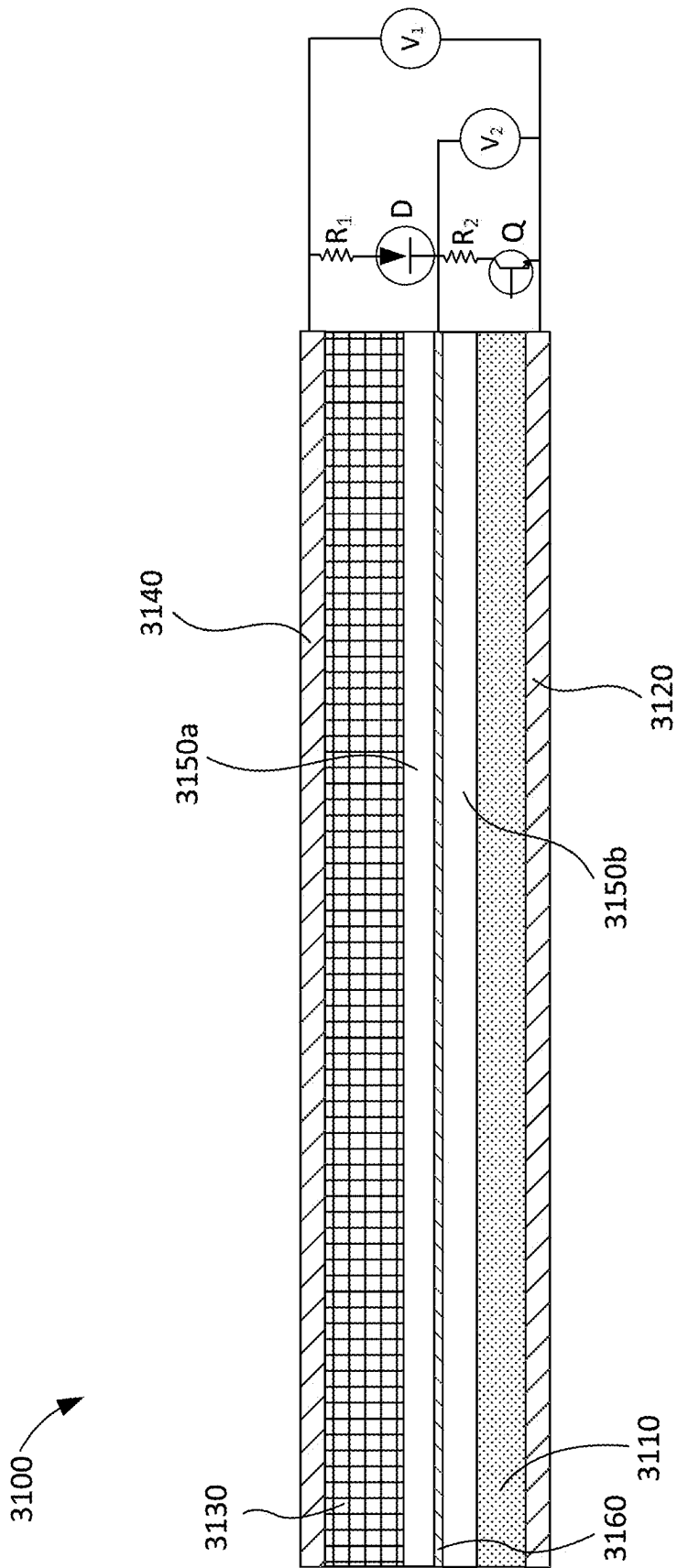


FIG. 32

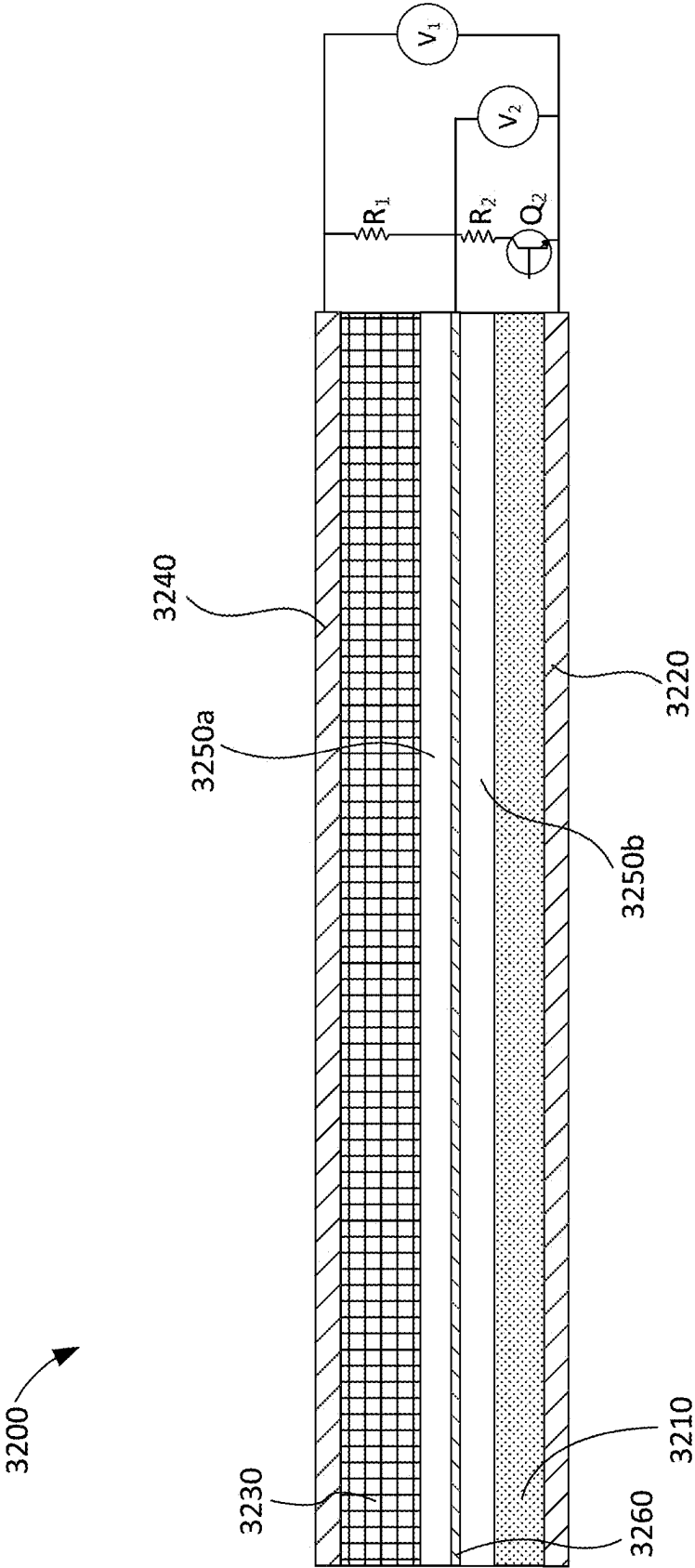


FIG. 33

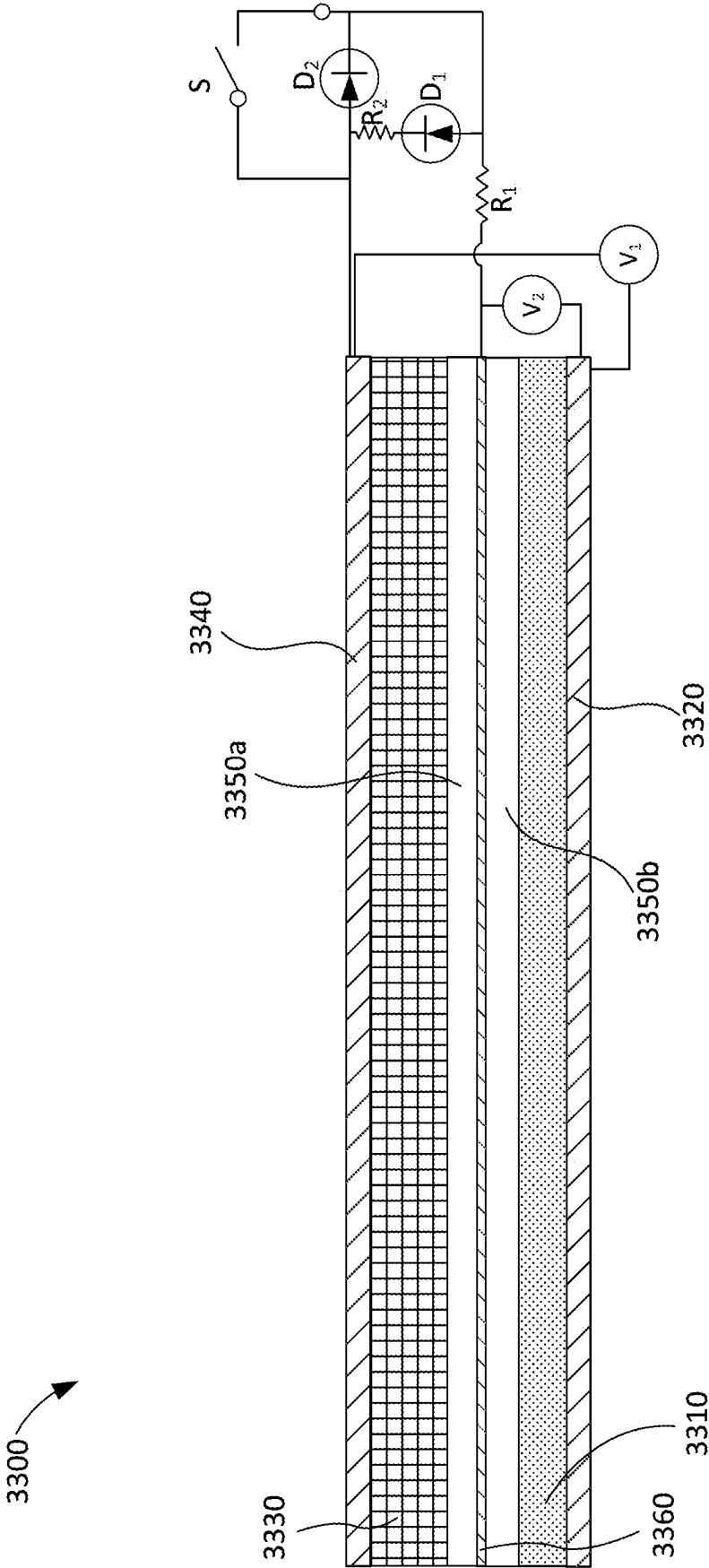


FIG. 34

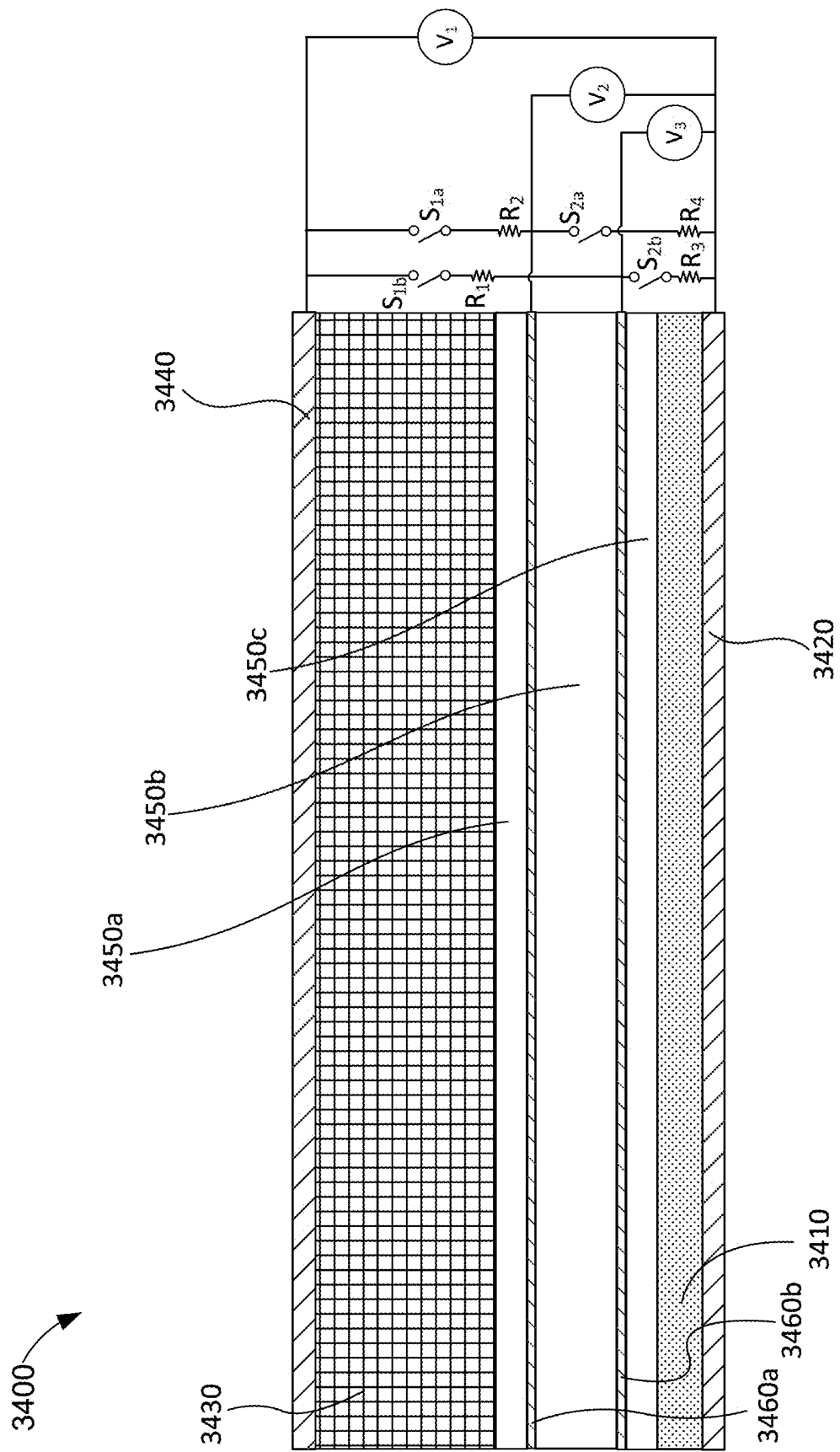


FIG. 35

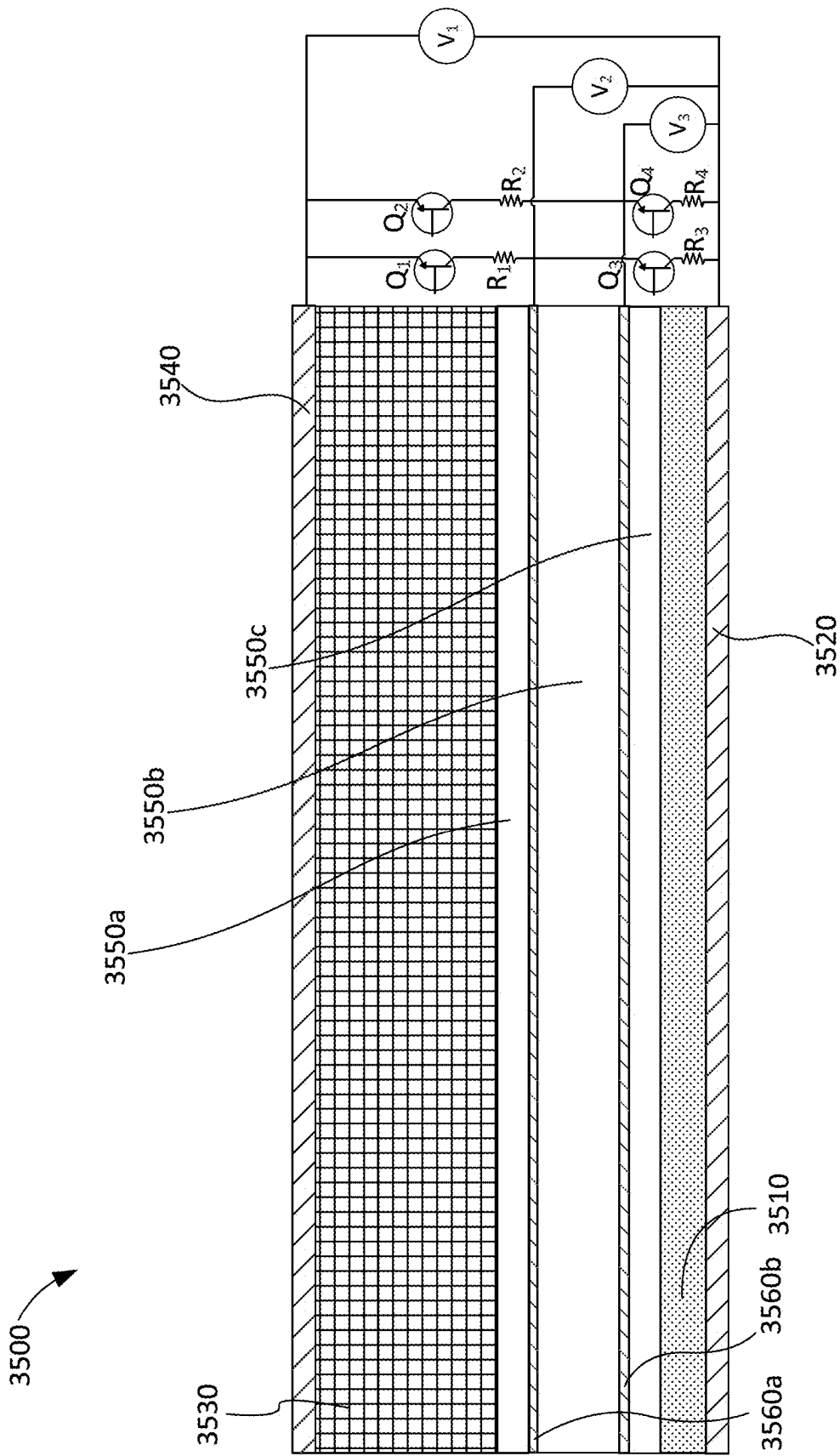


FIG. 36

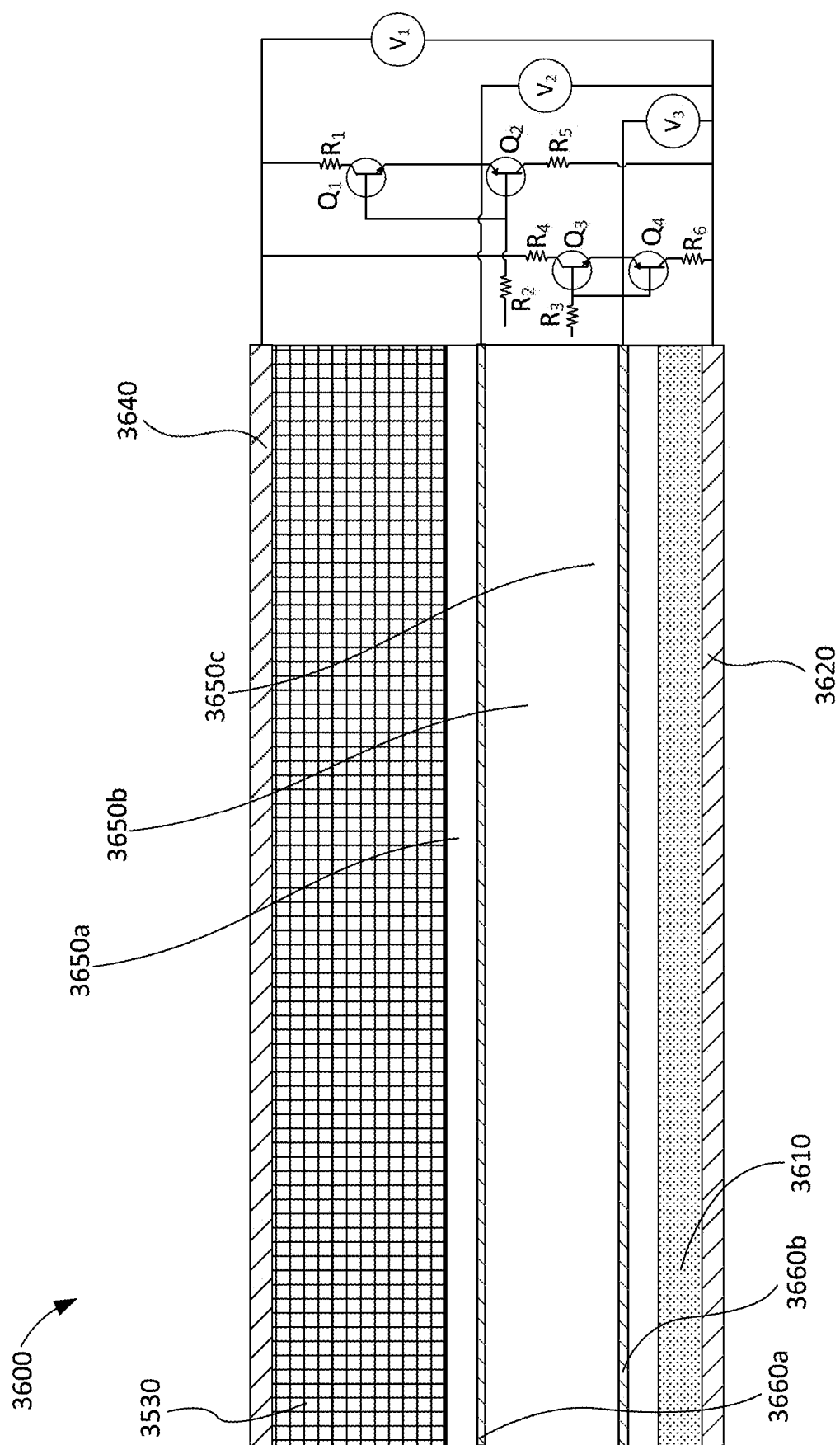


FIG. 37

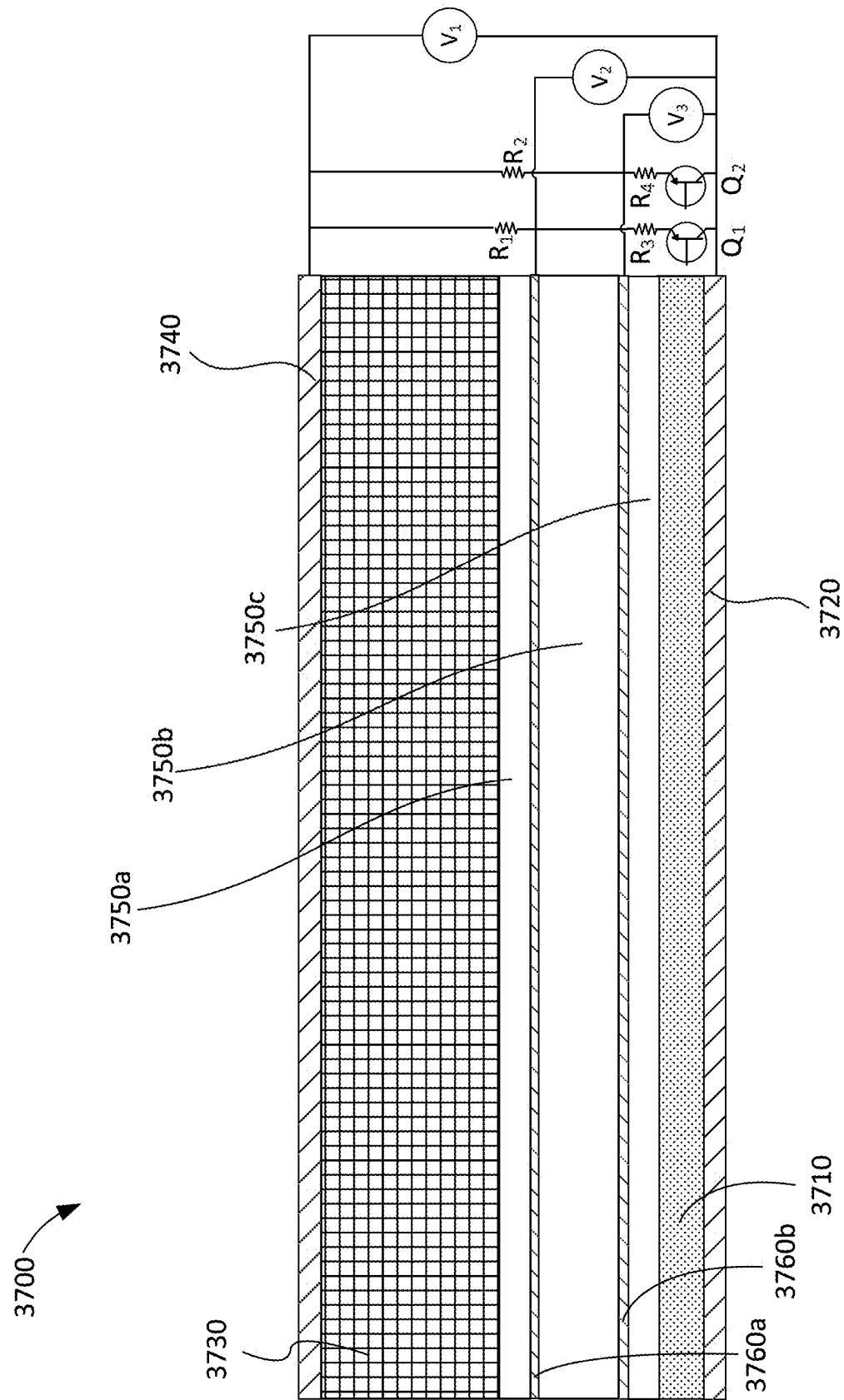


FIG. 38

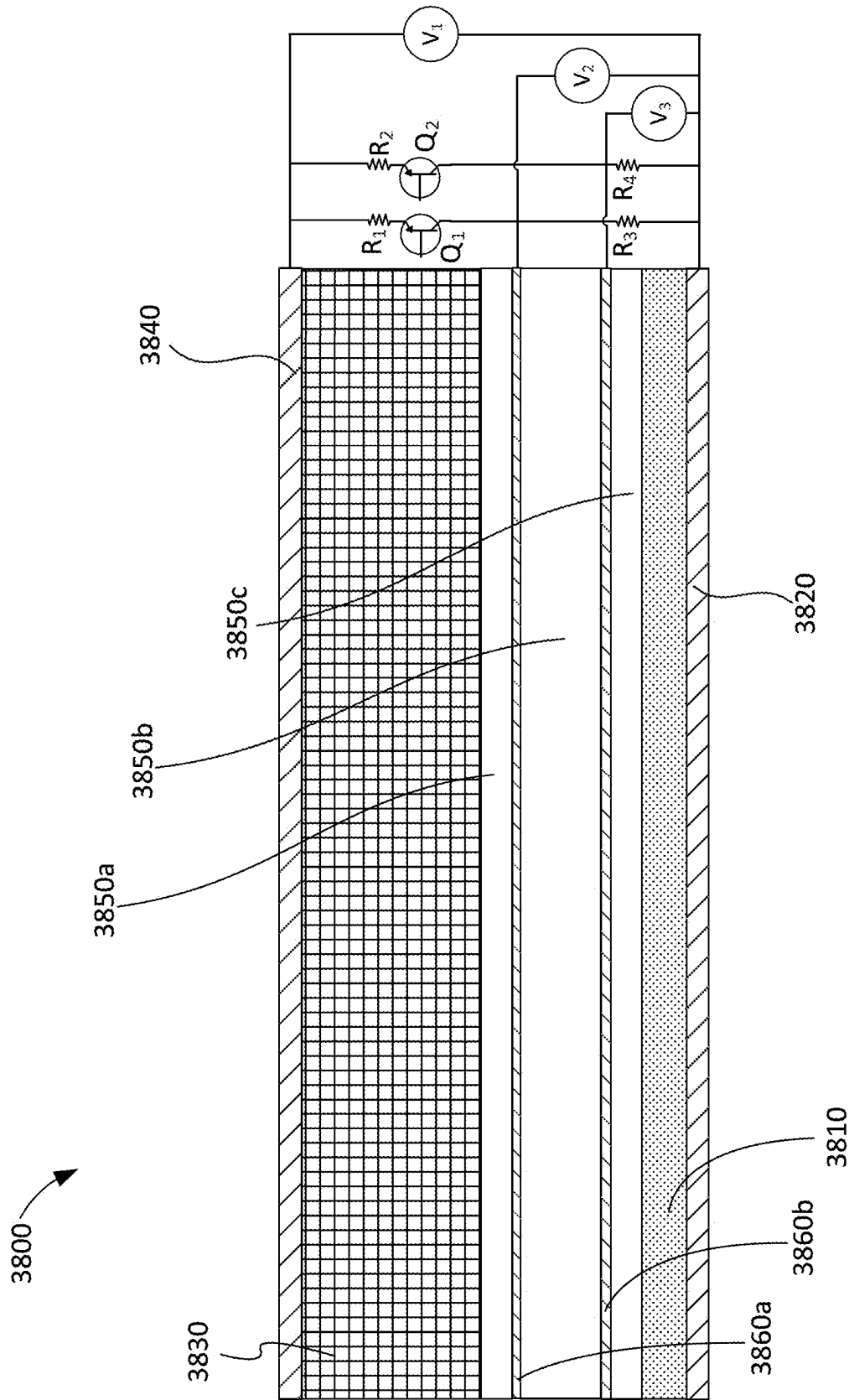


FIG. 39

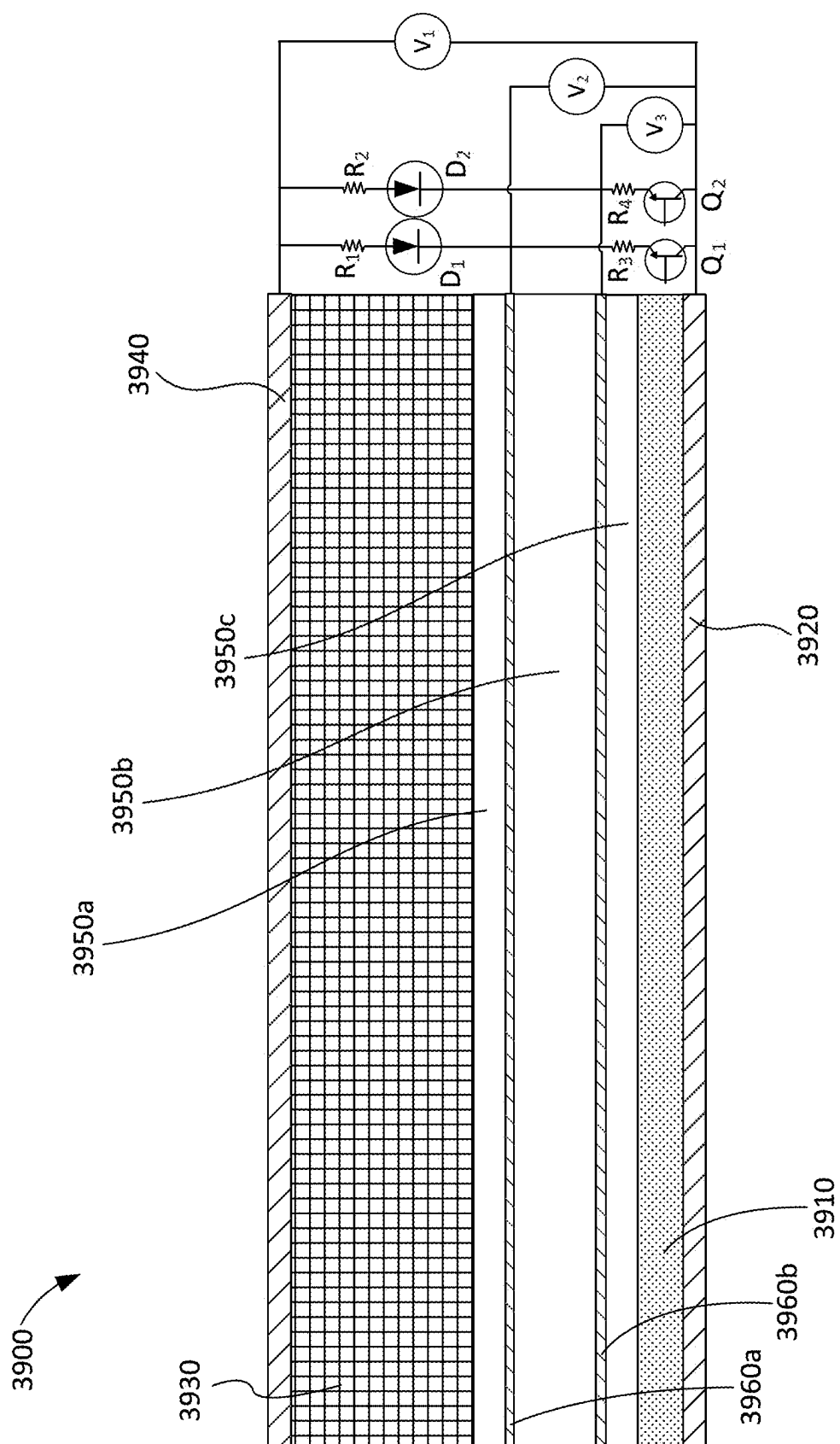


FIG. 40

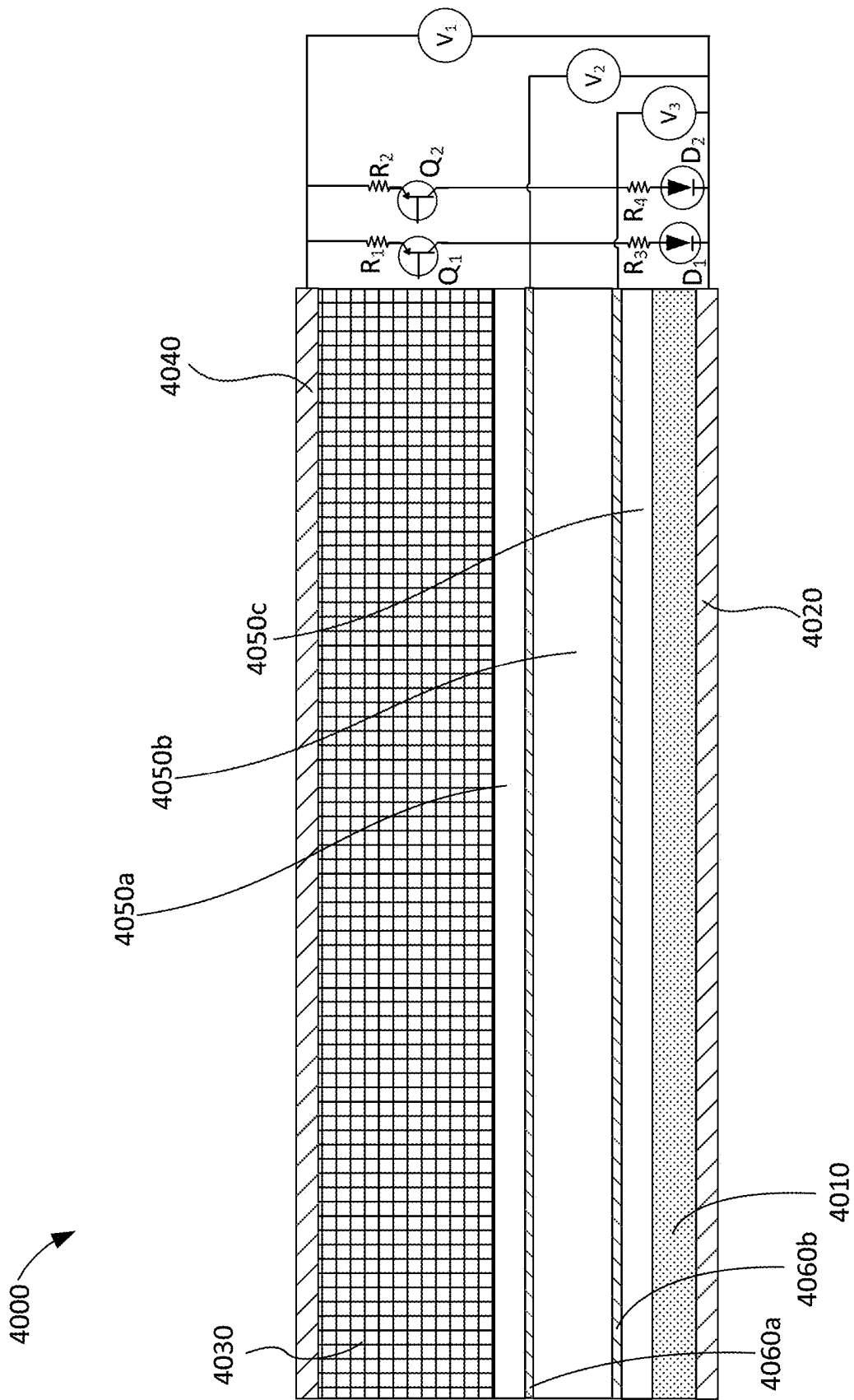


FIG. 41

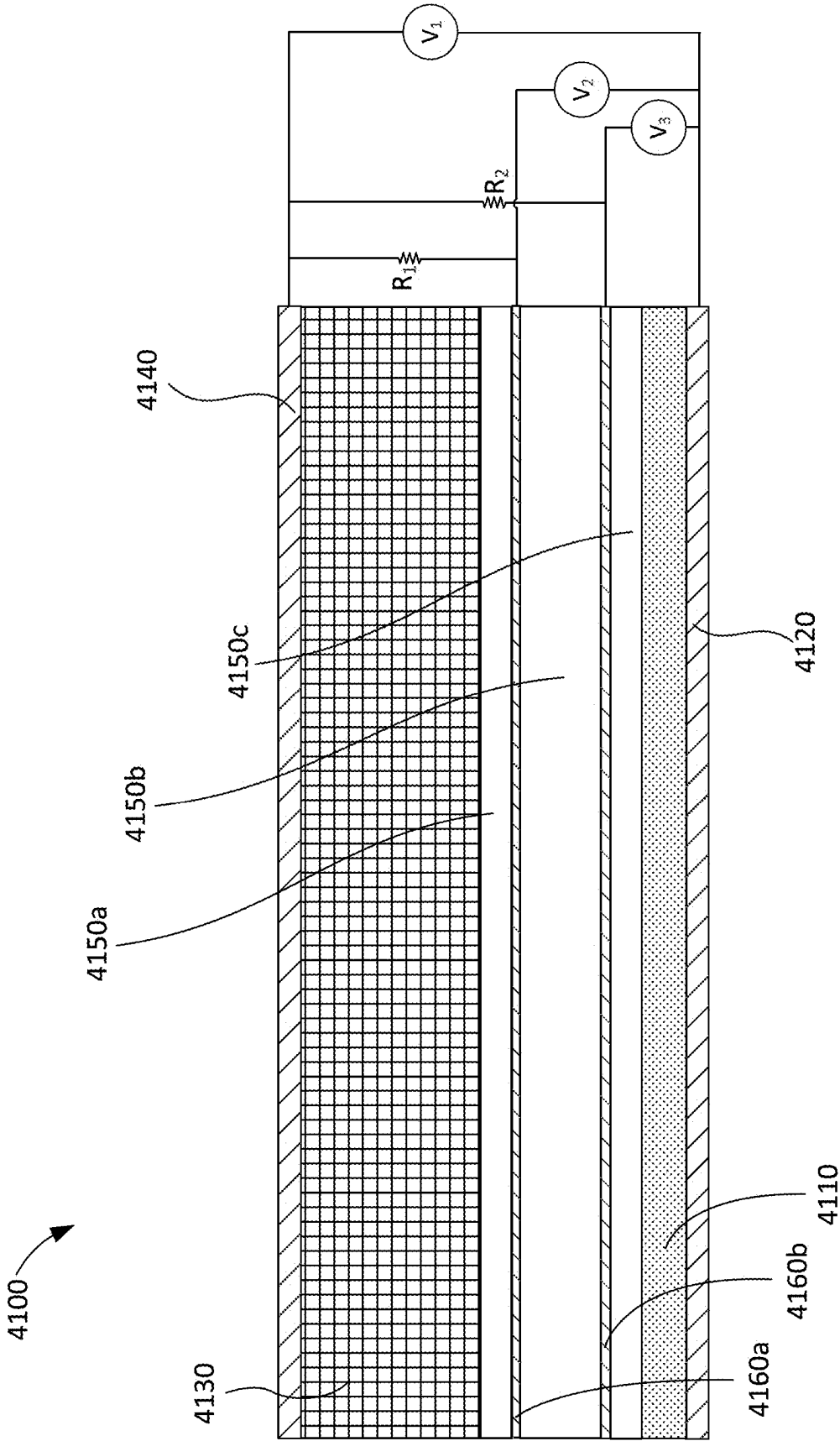


FIG. 42

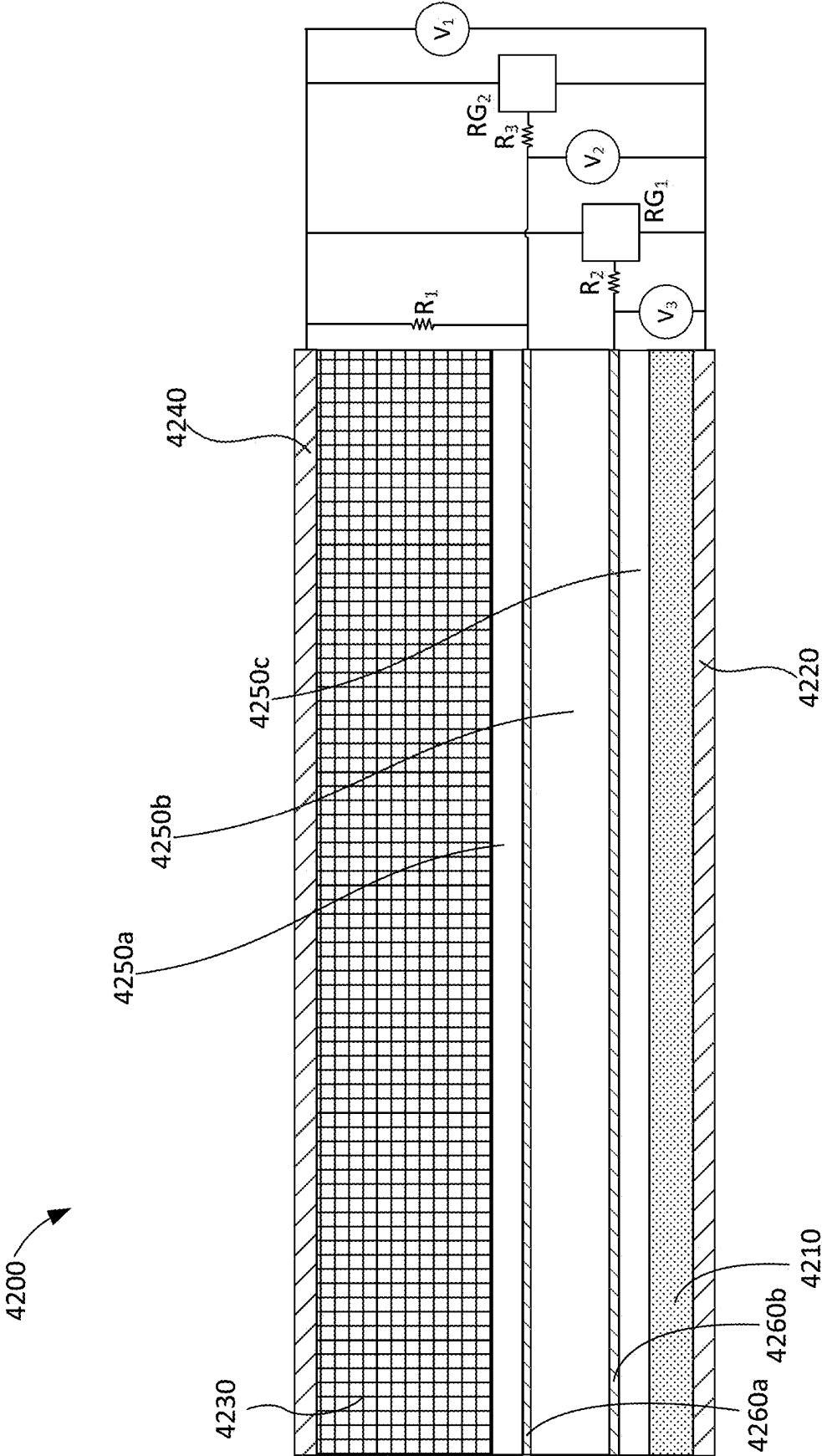


FIG. 43

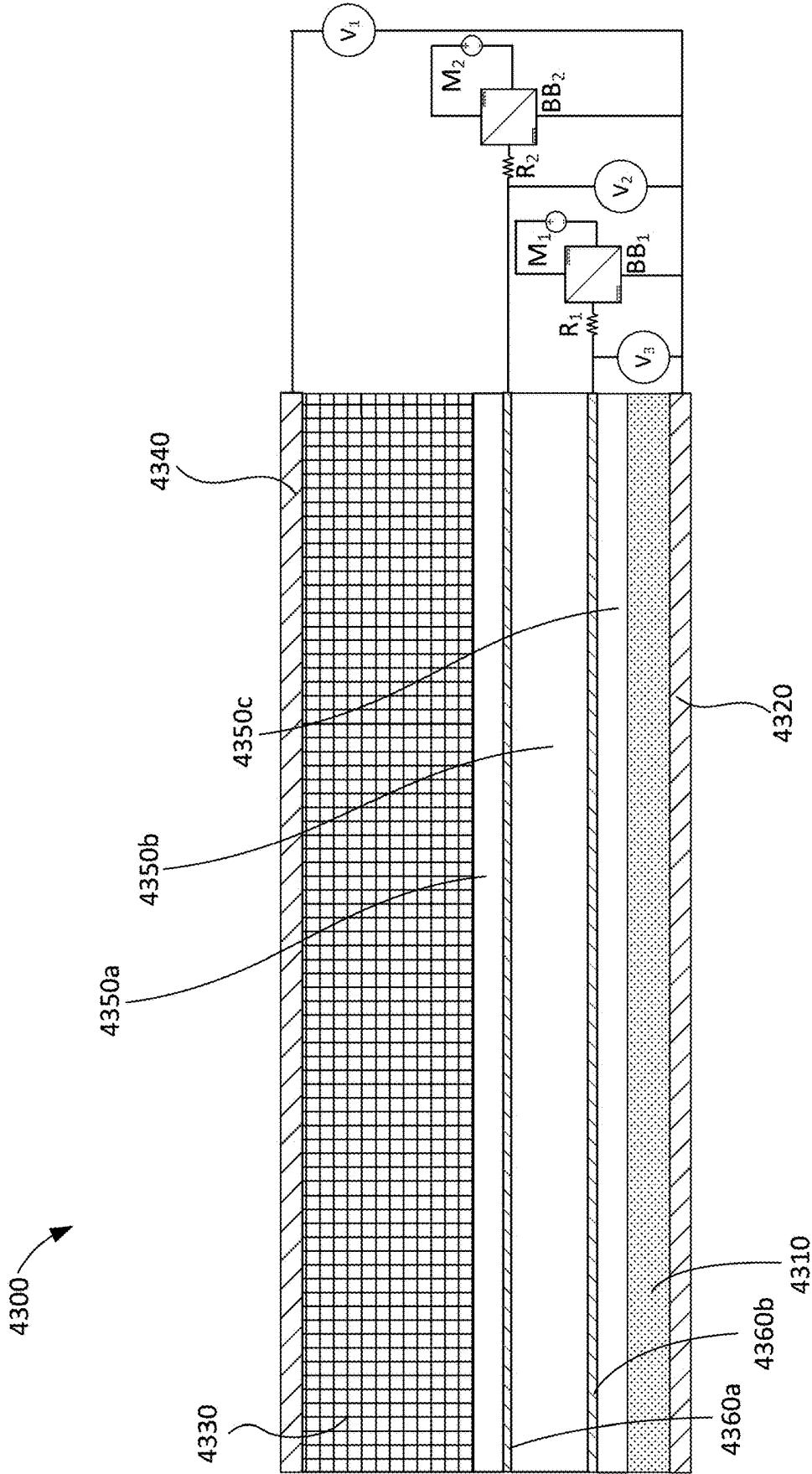


FIG. 44

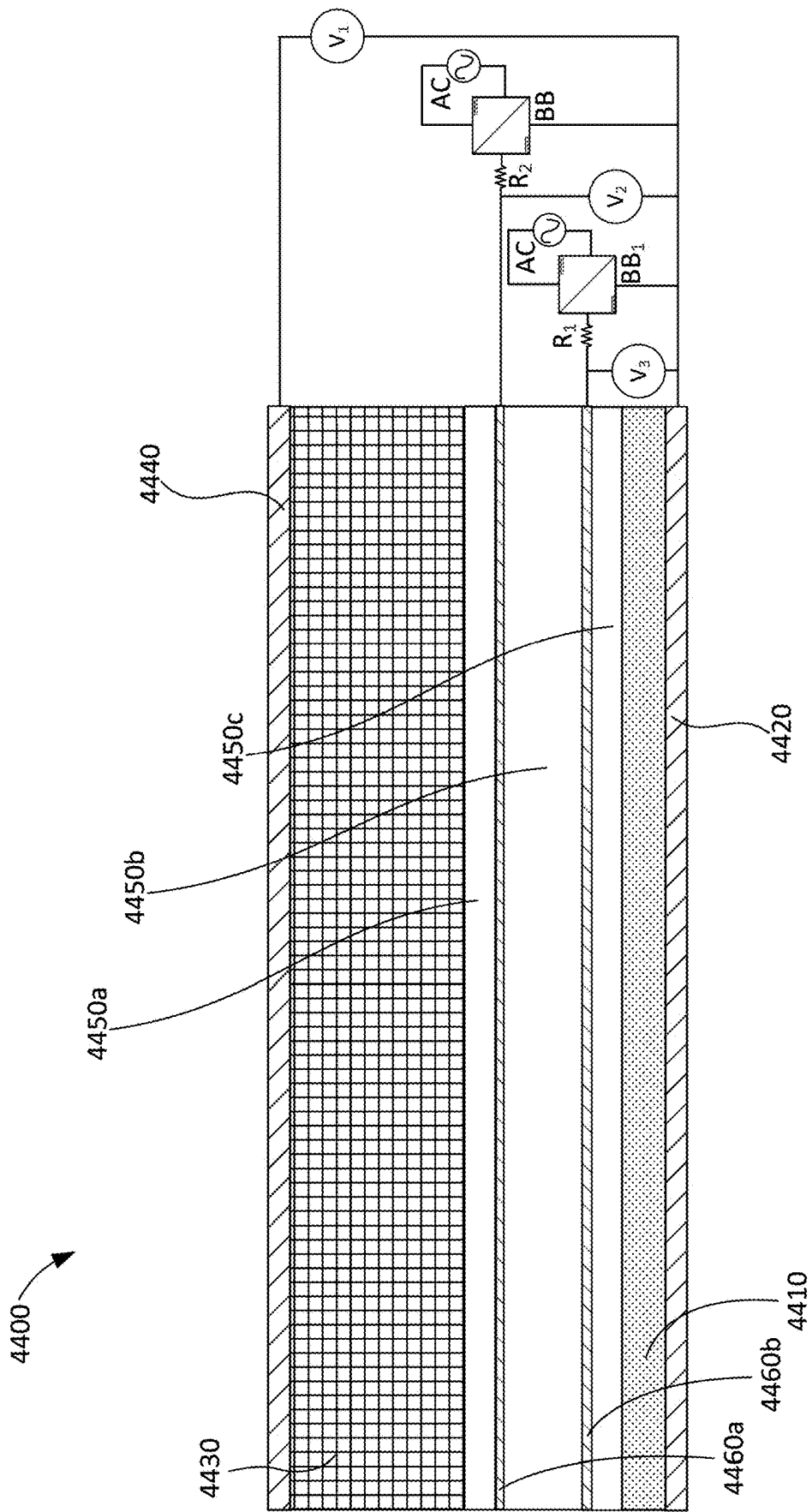


FIG. 45

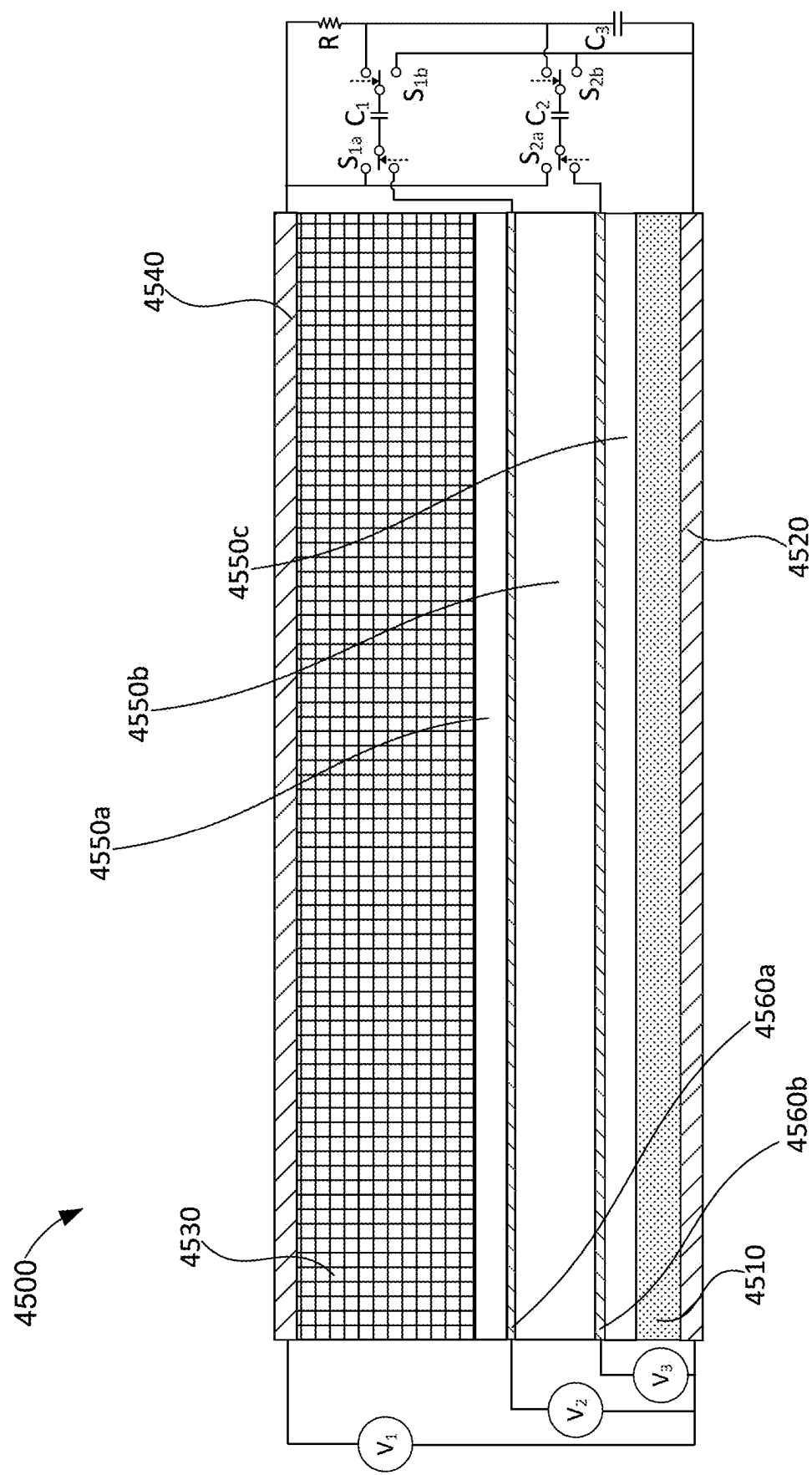


FIG. 46

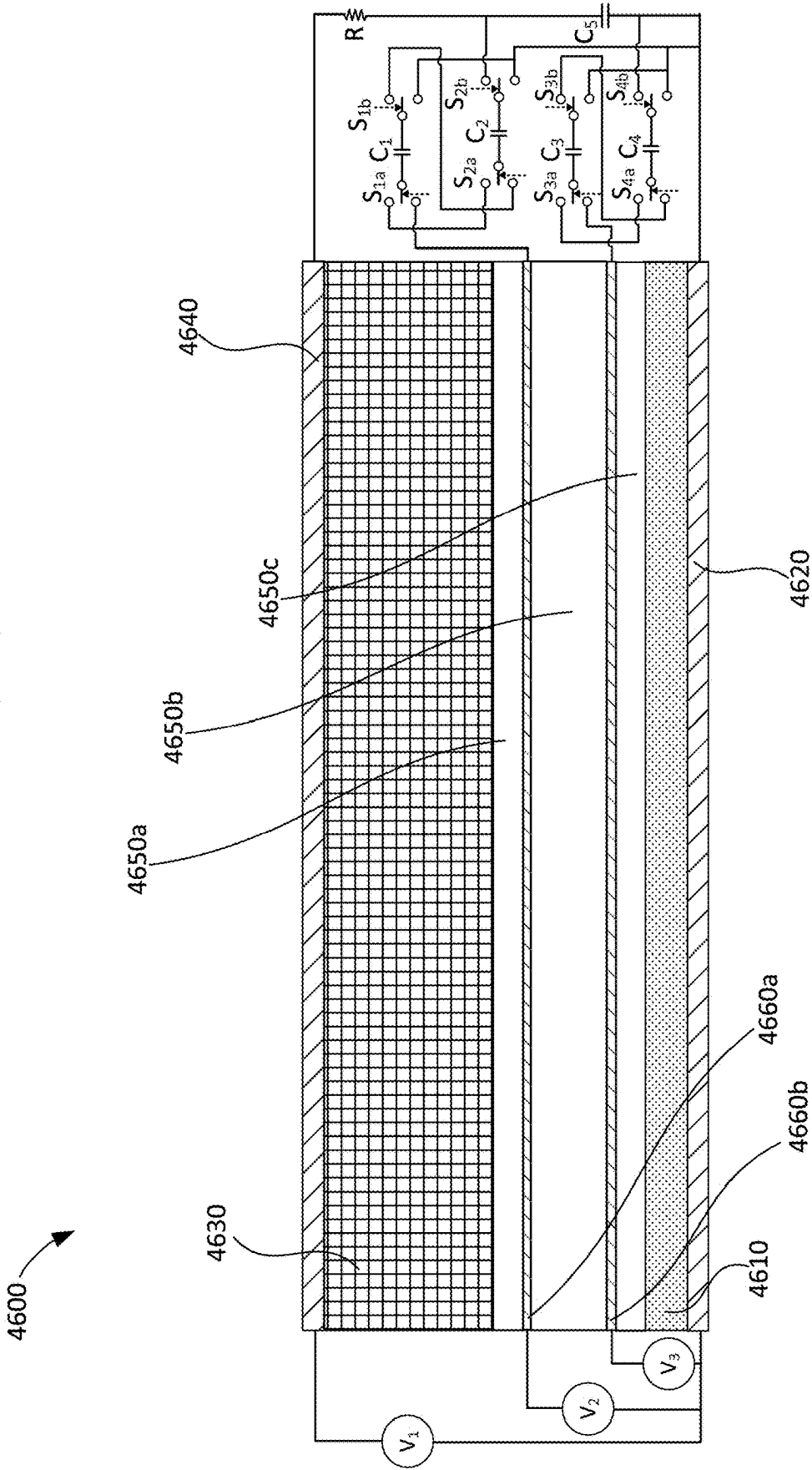


FIG. 47

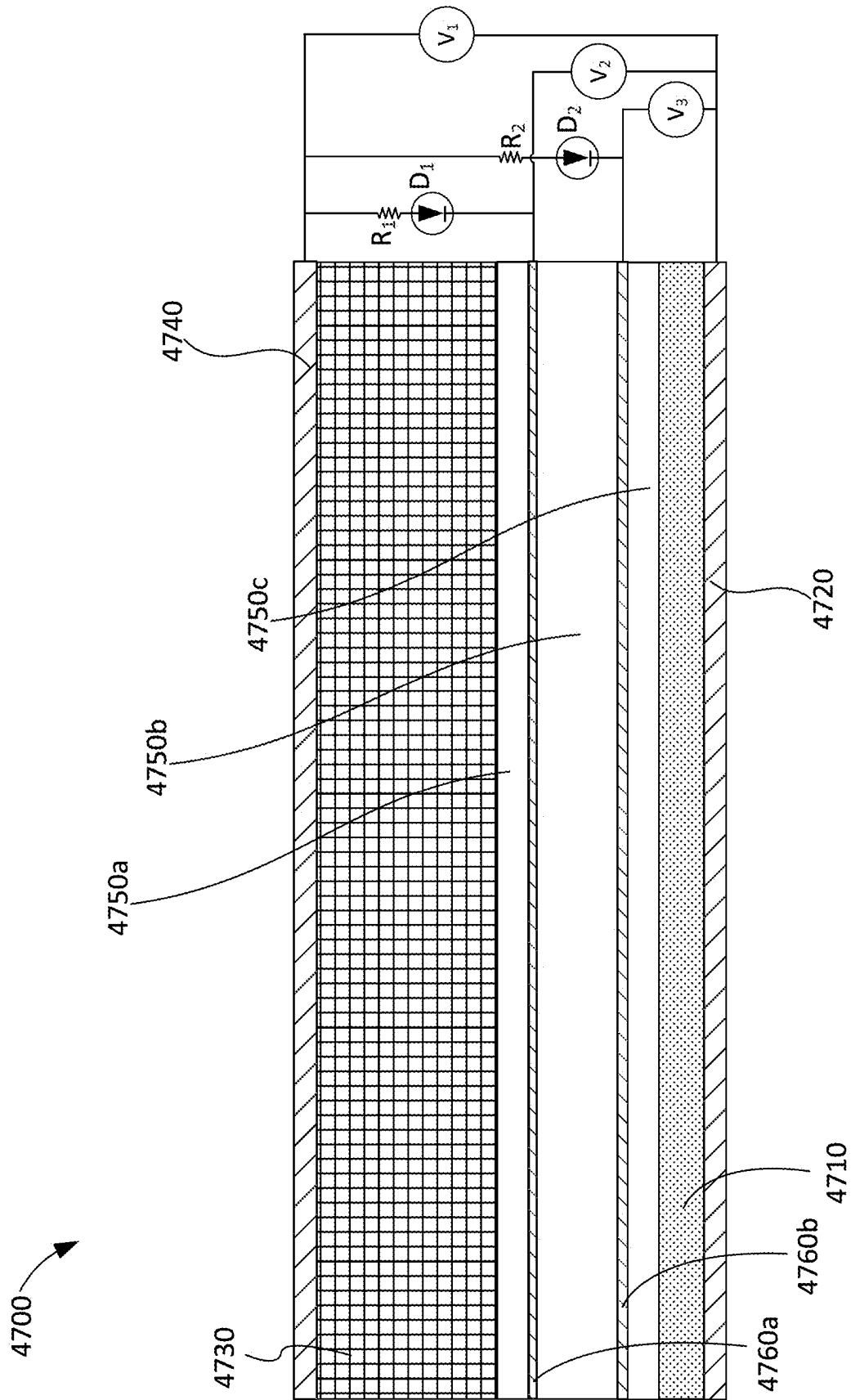


FIG. 48

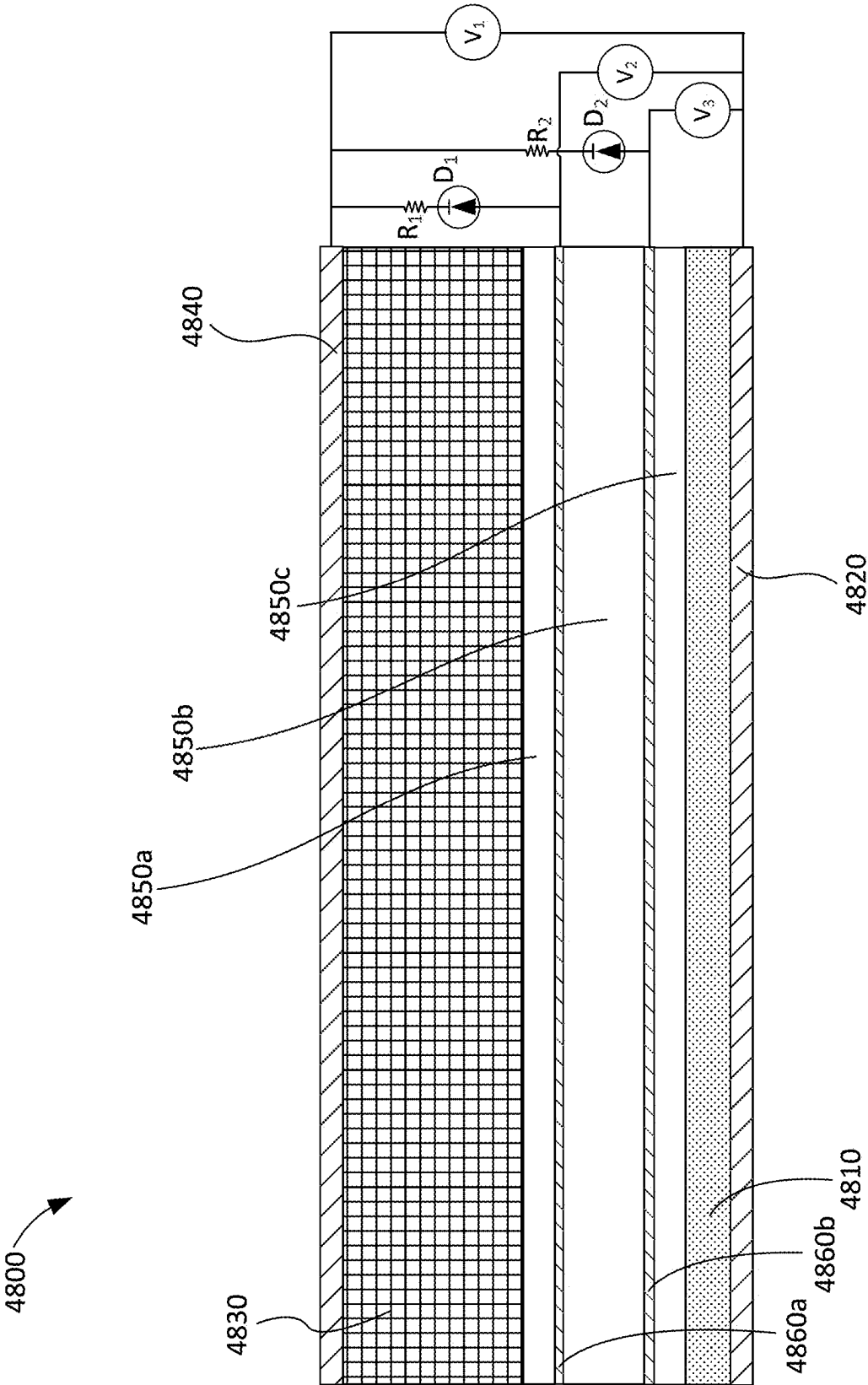


FIG. 49

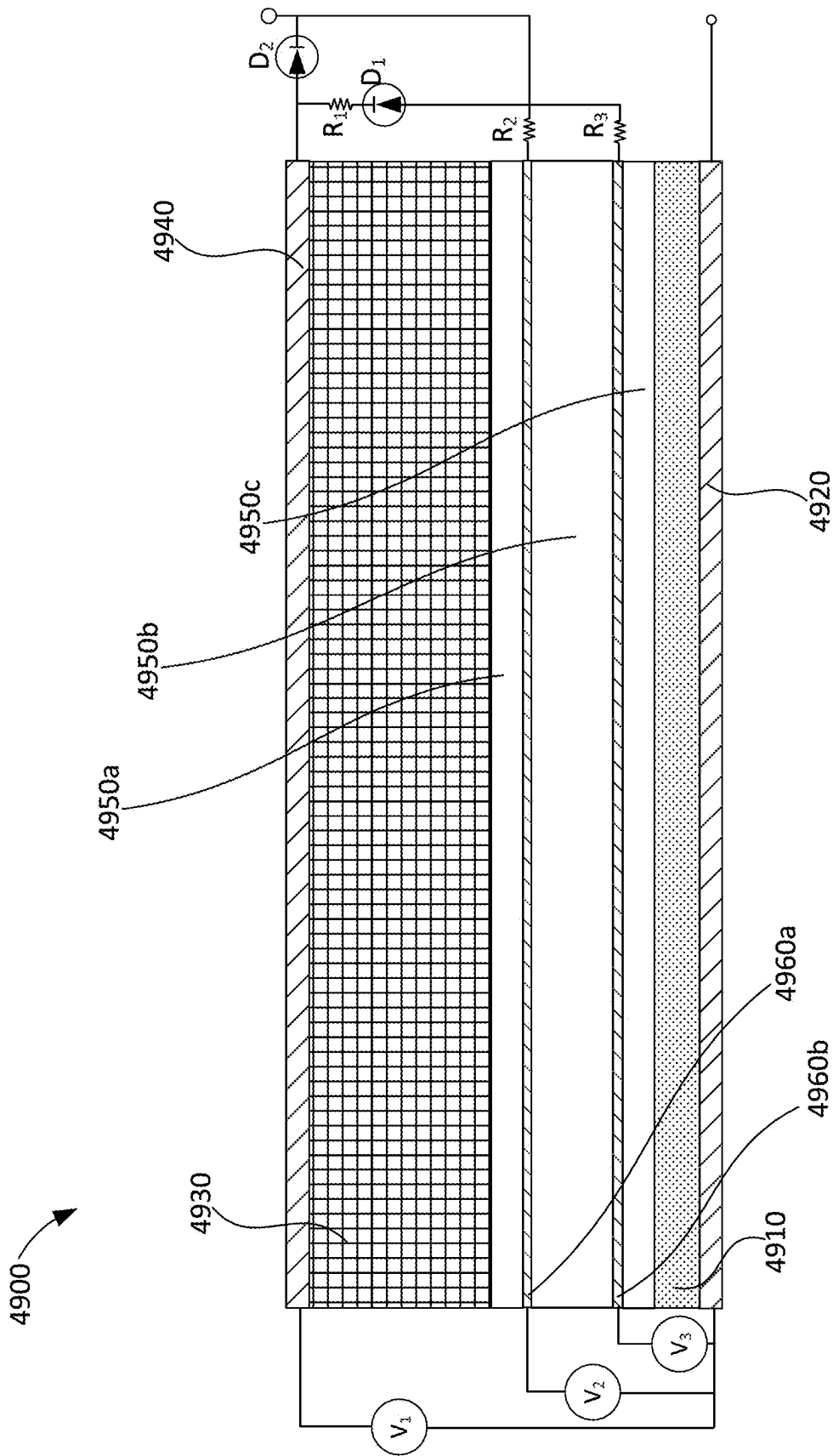


FIG. 50

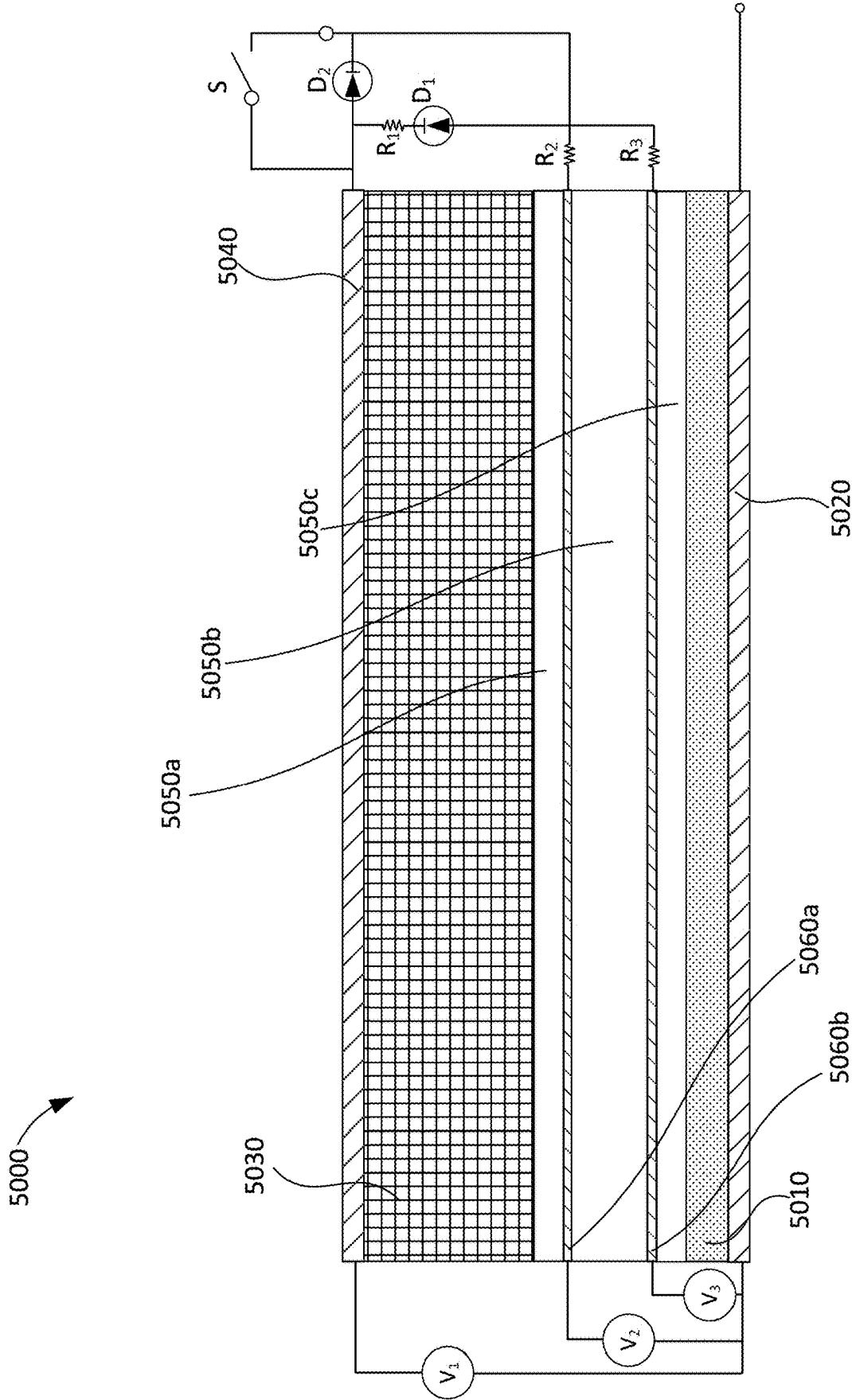


FIG. 51A

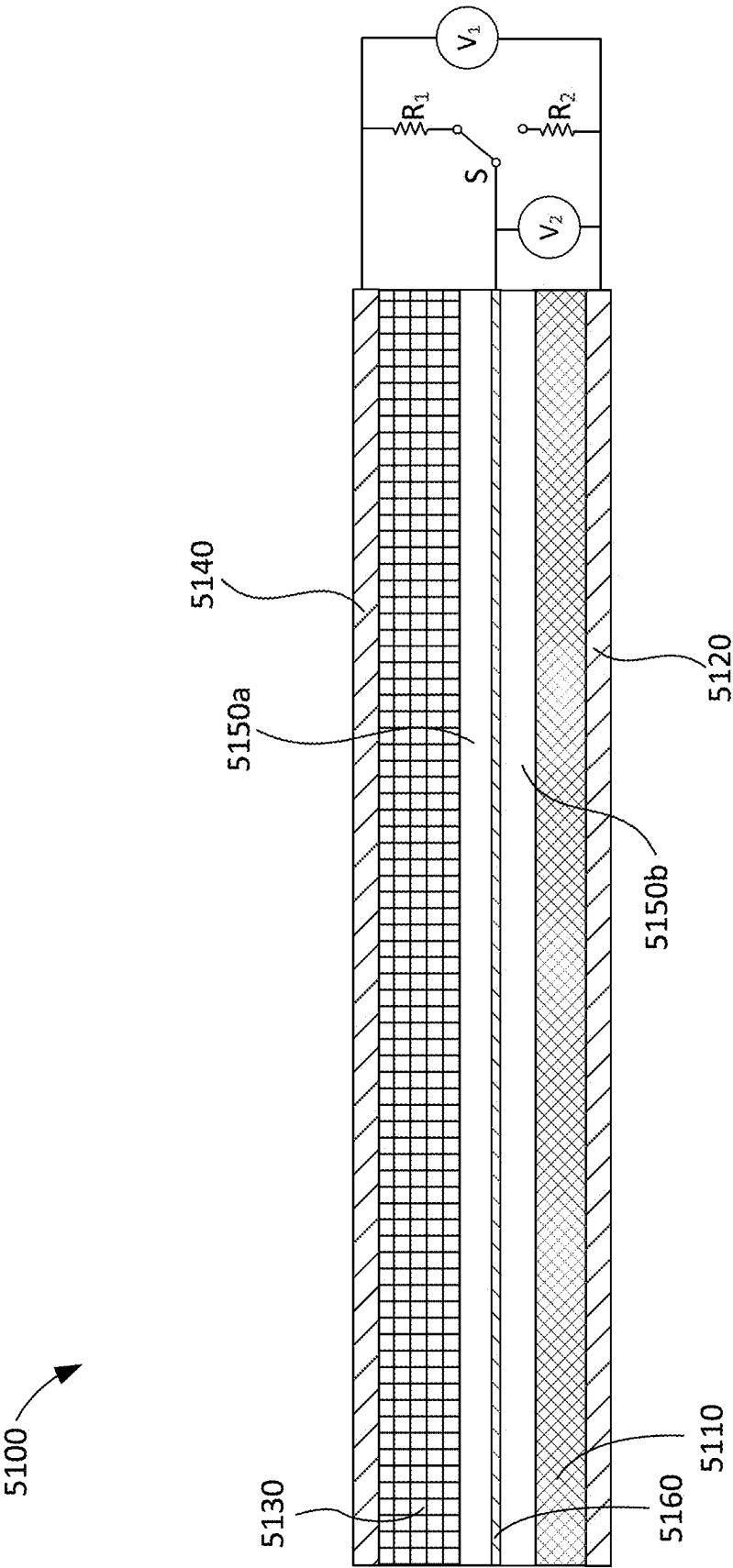


FIG. 51B

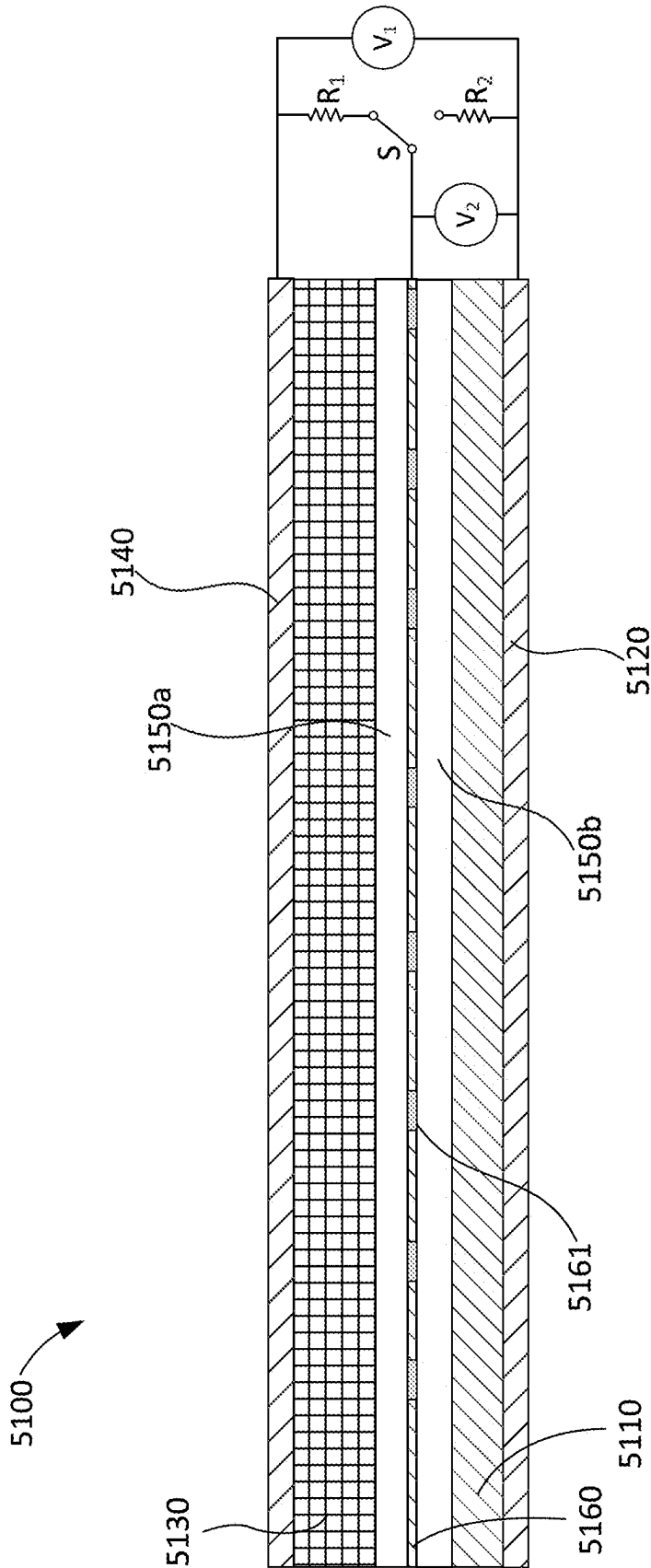


FIG. 51C

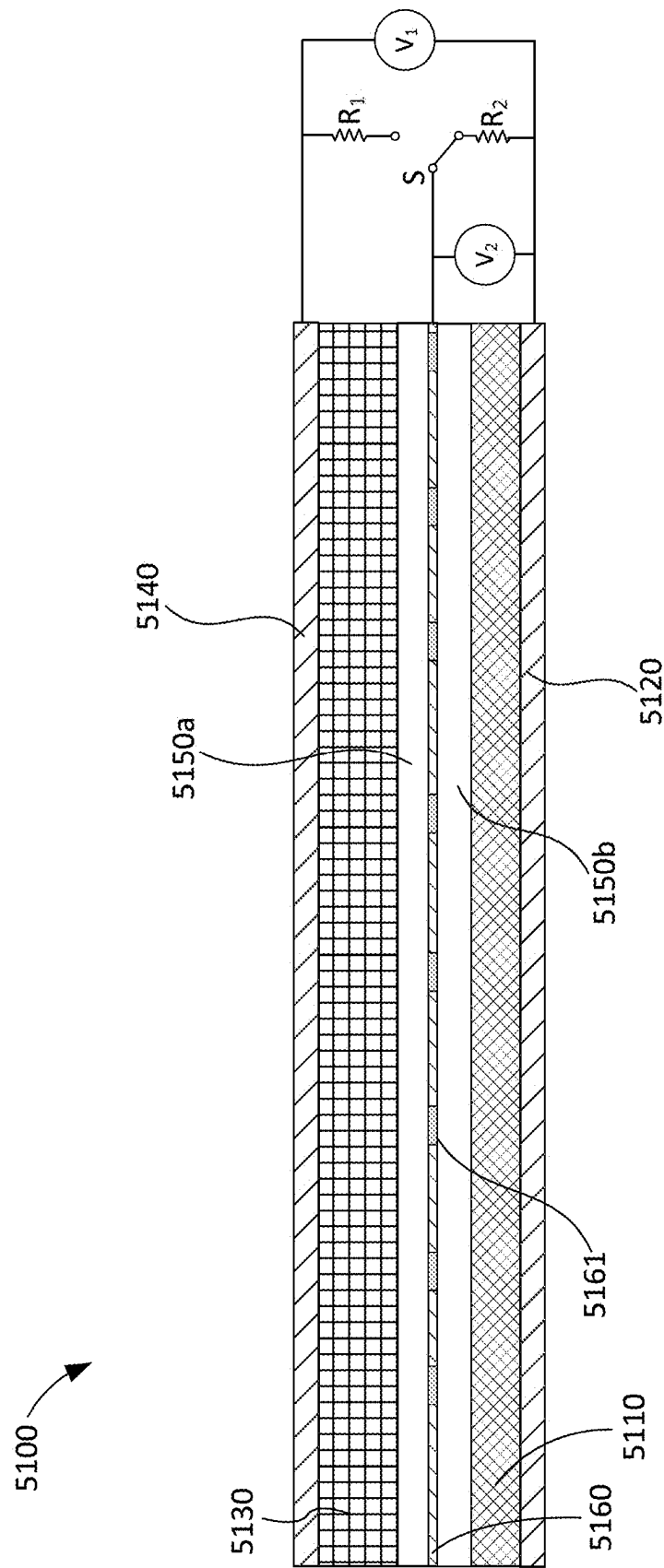


FIG. 52A

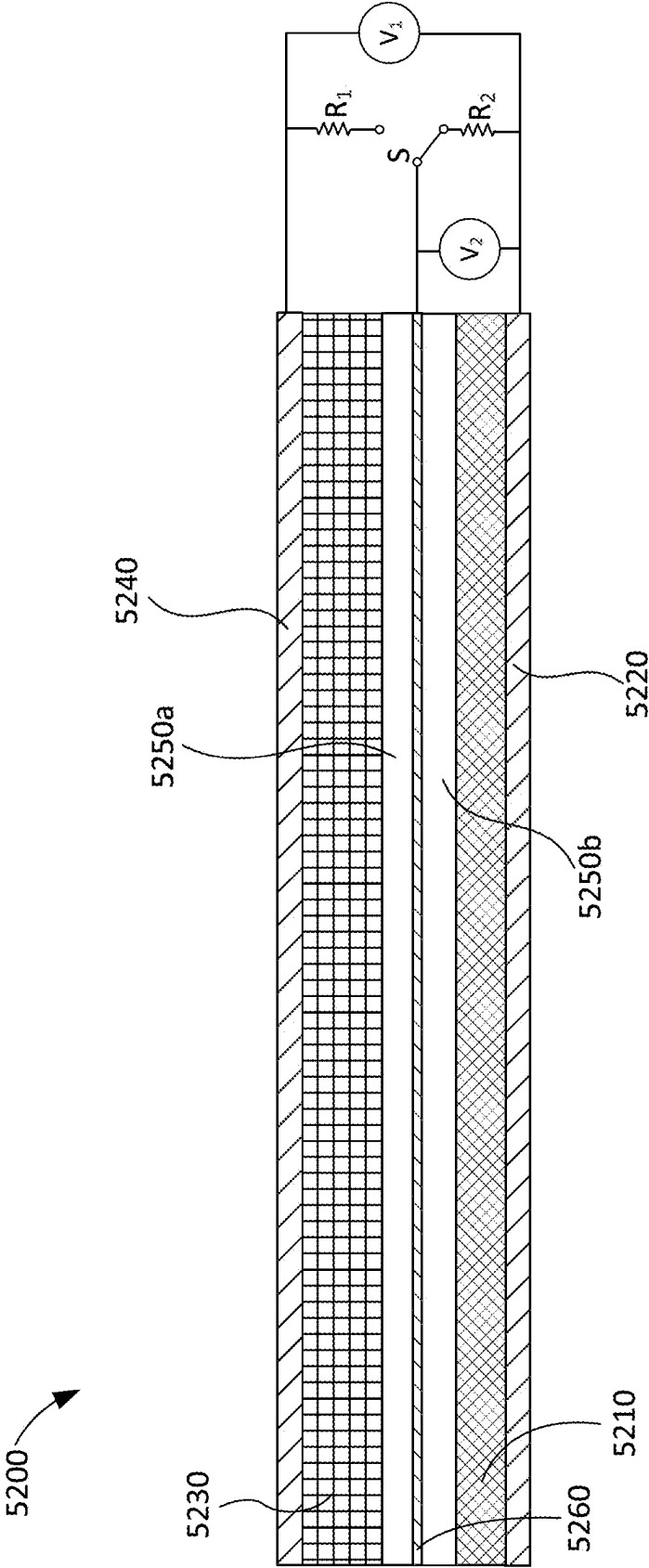


FIG. 52B

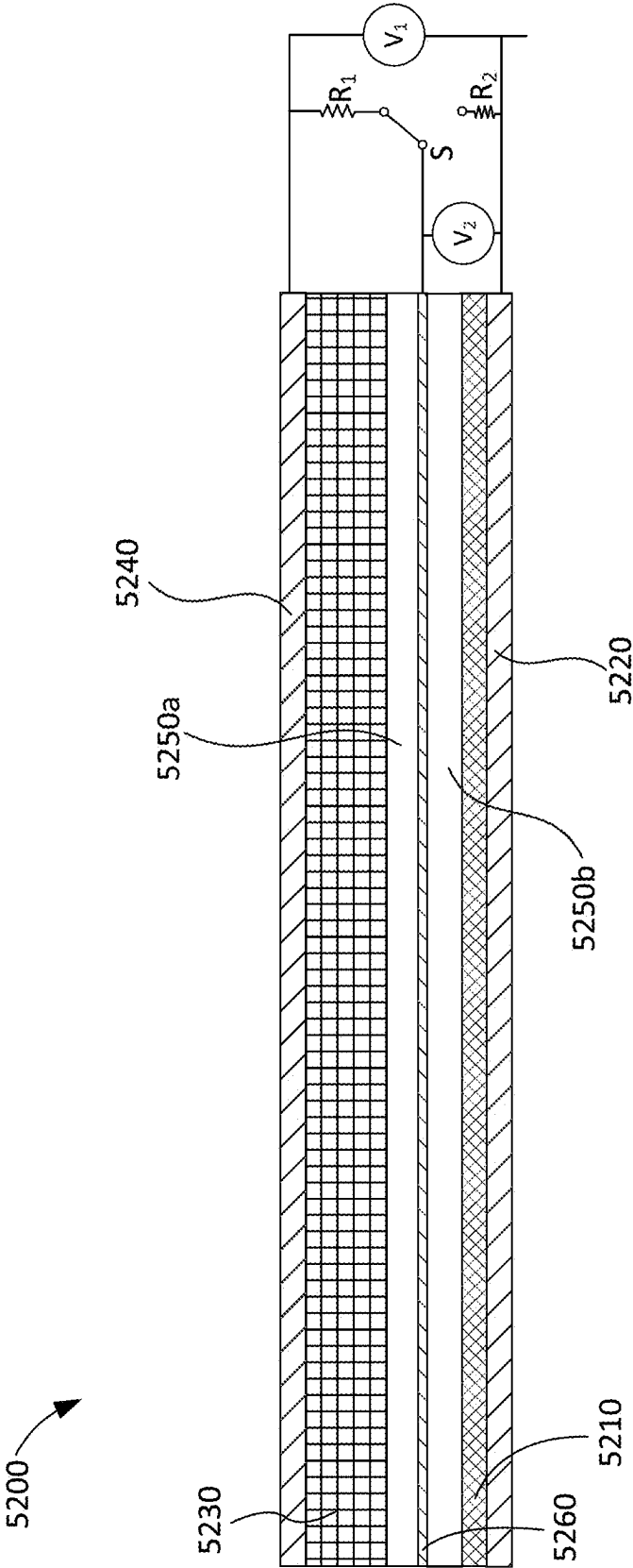


FIG. 52C

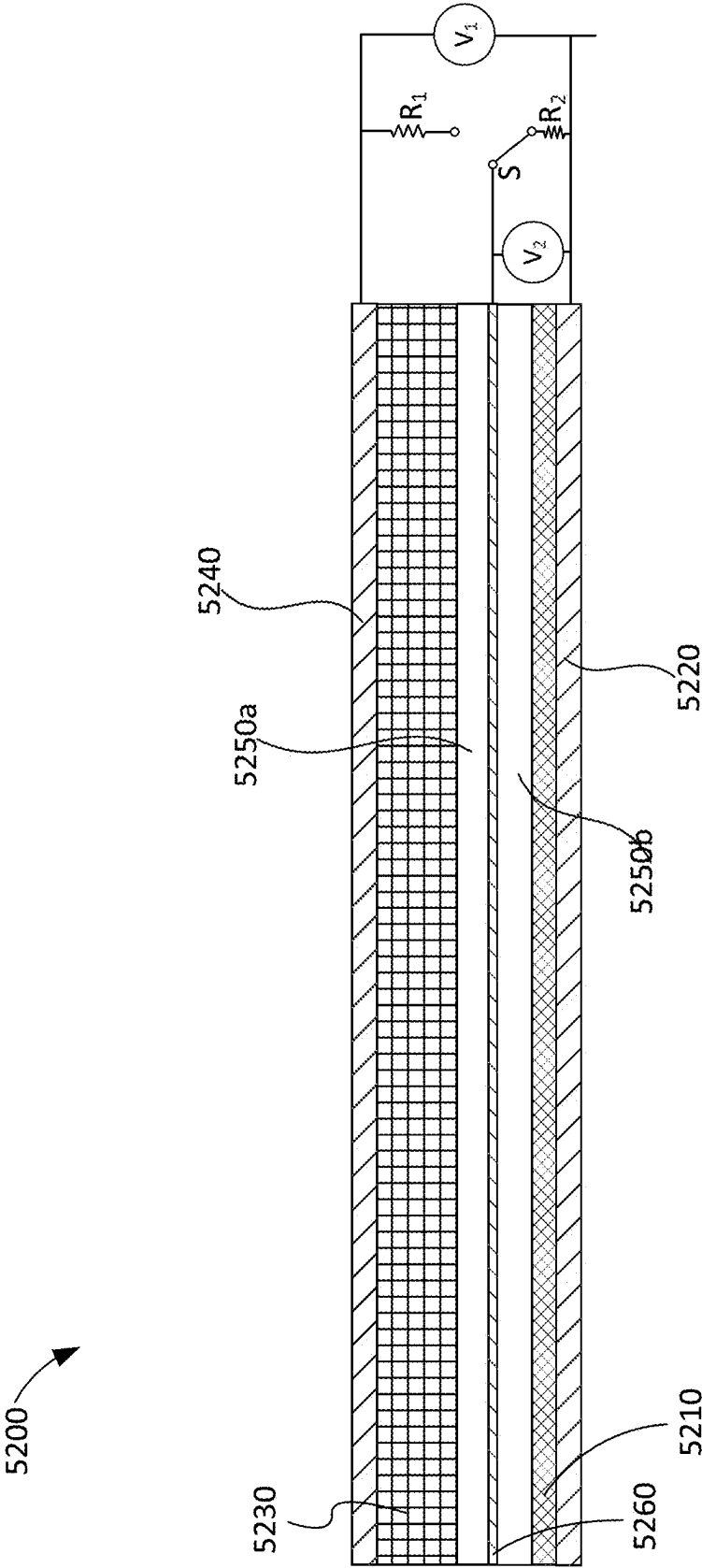


FIG. 53

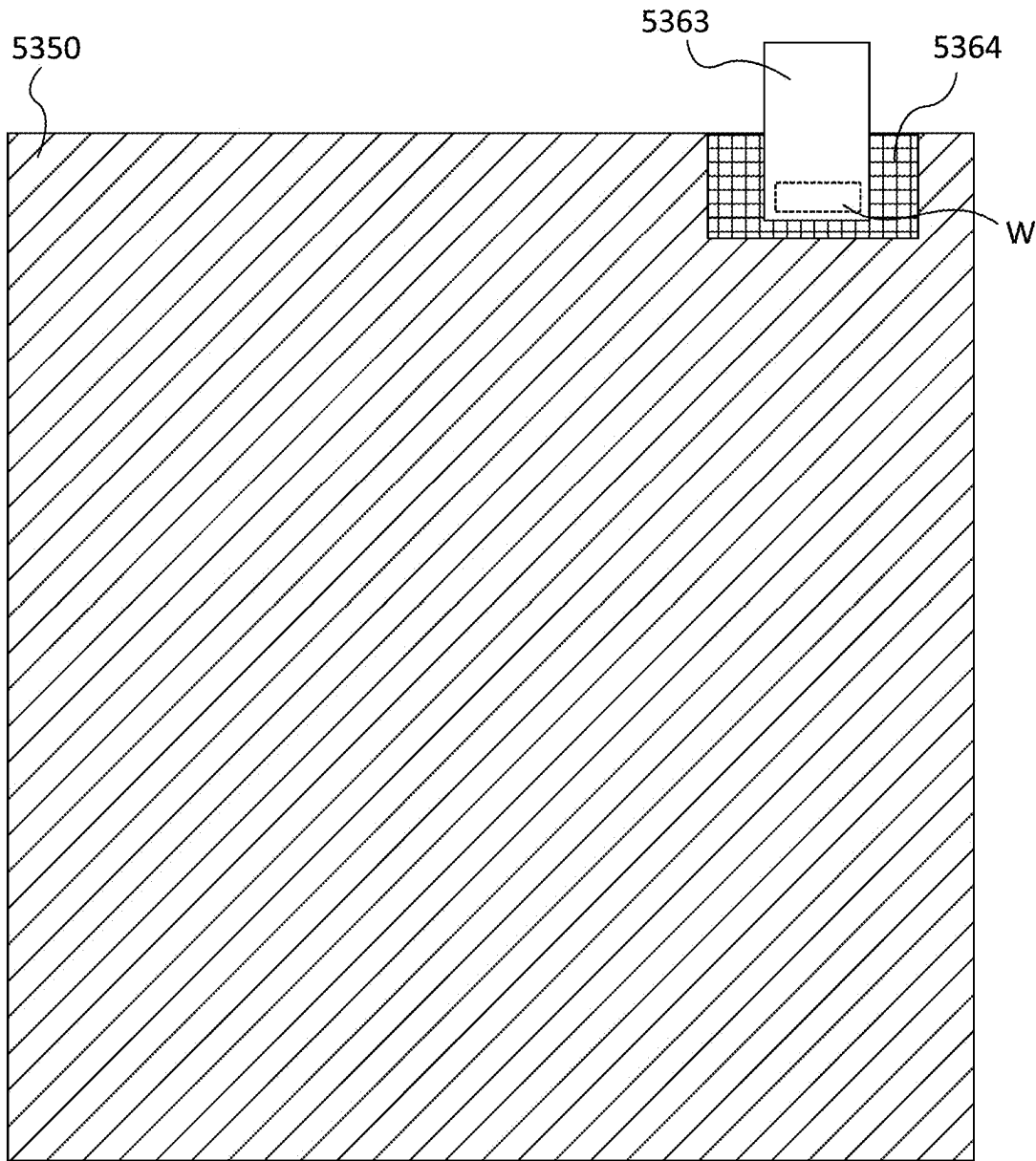


FIG. 54

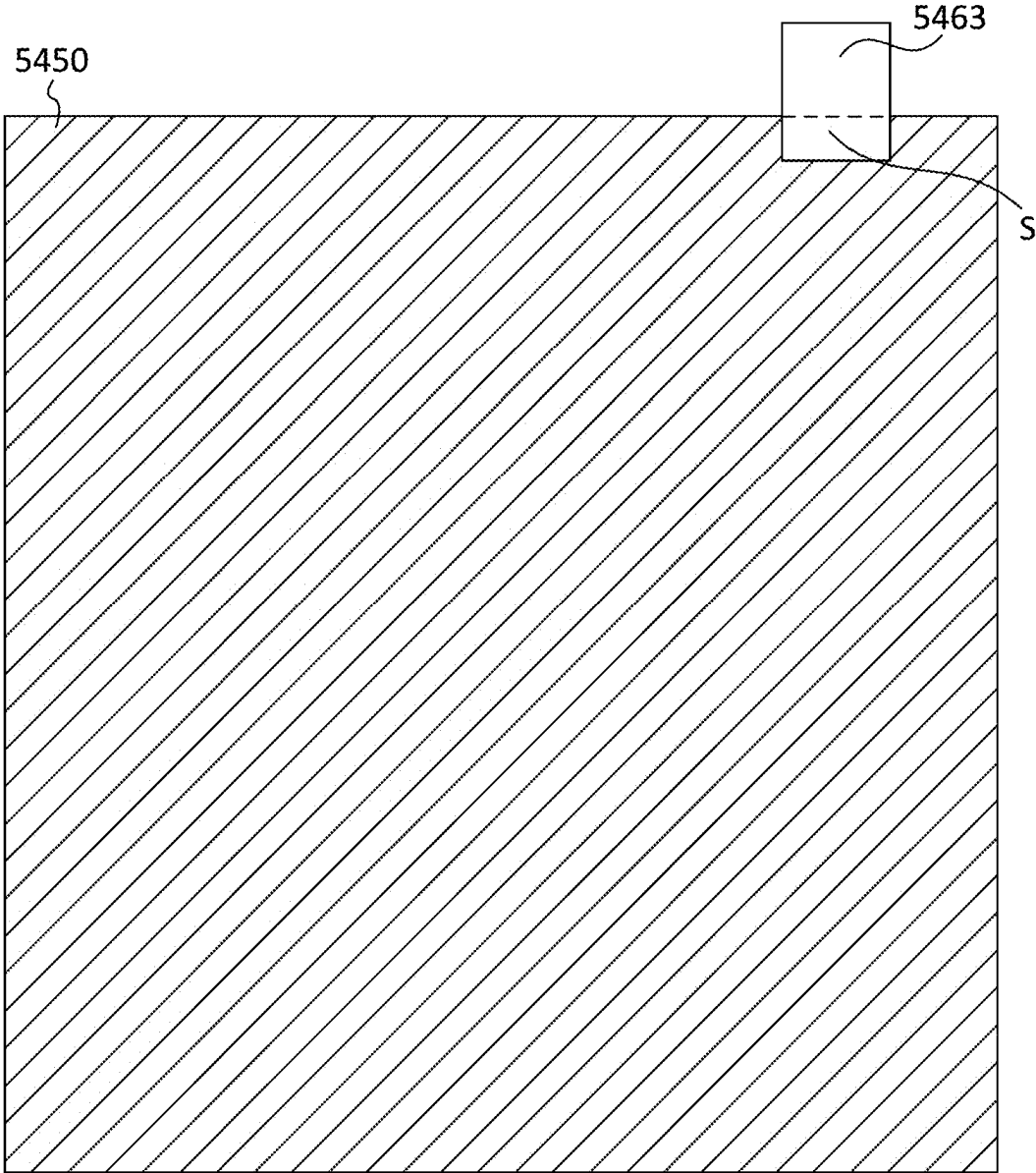


FIG. 55

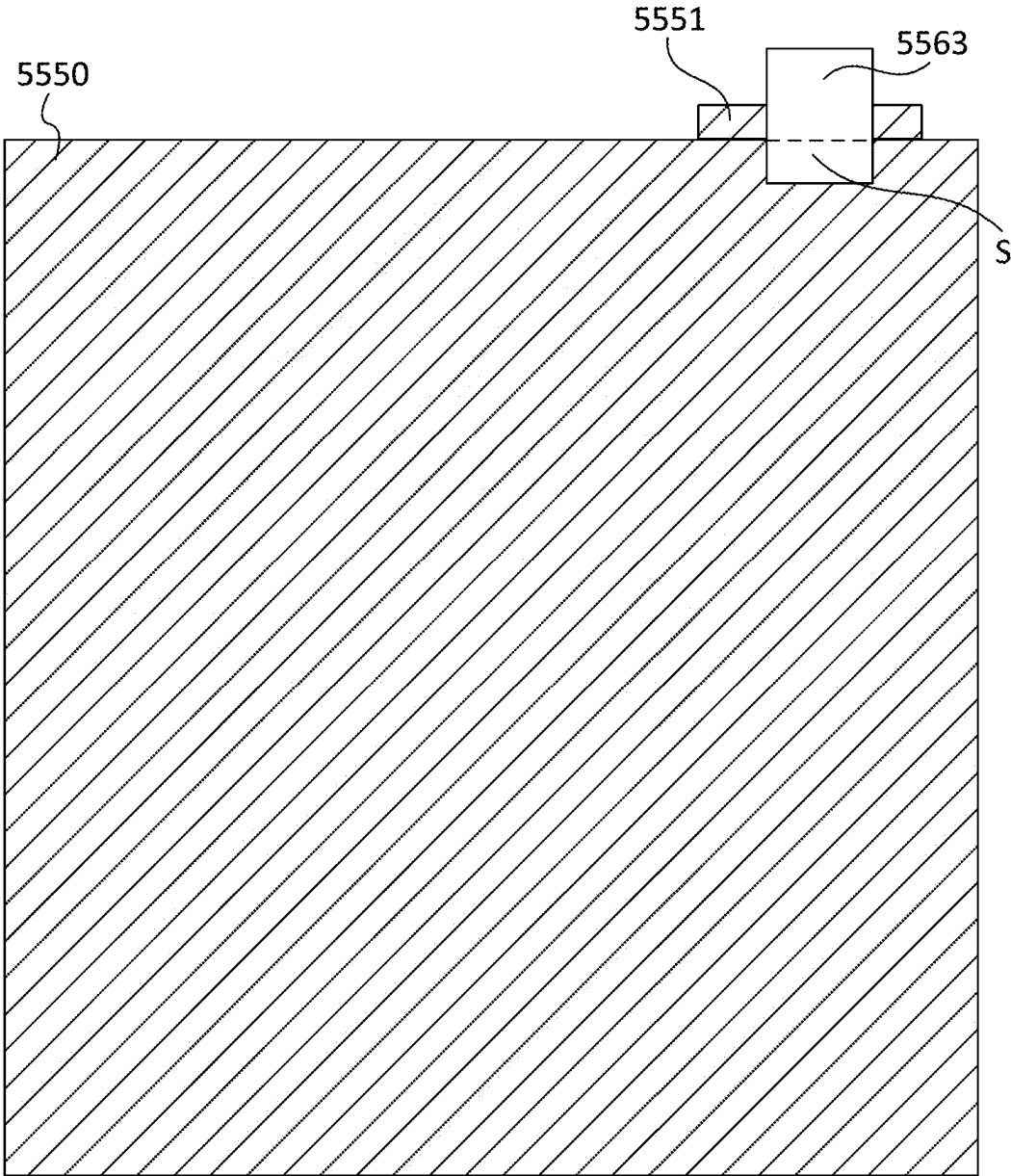


FIG. 56

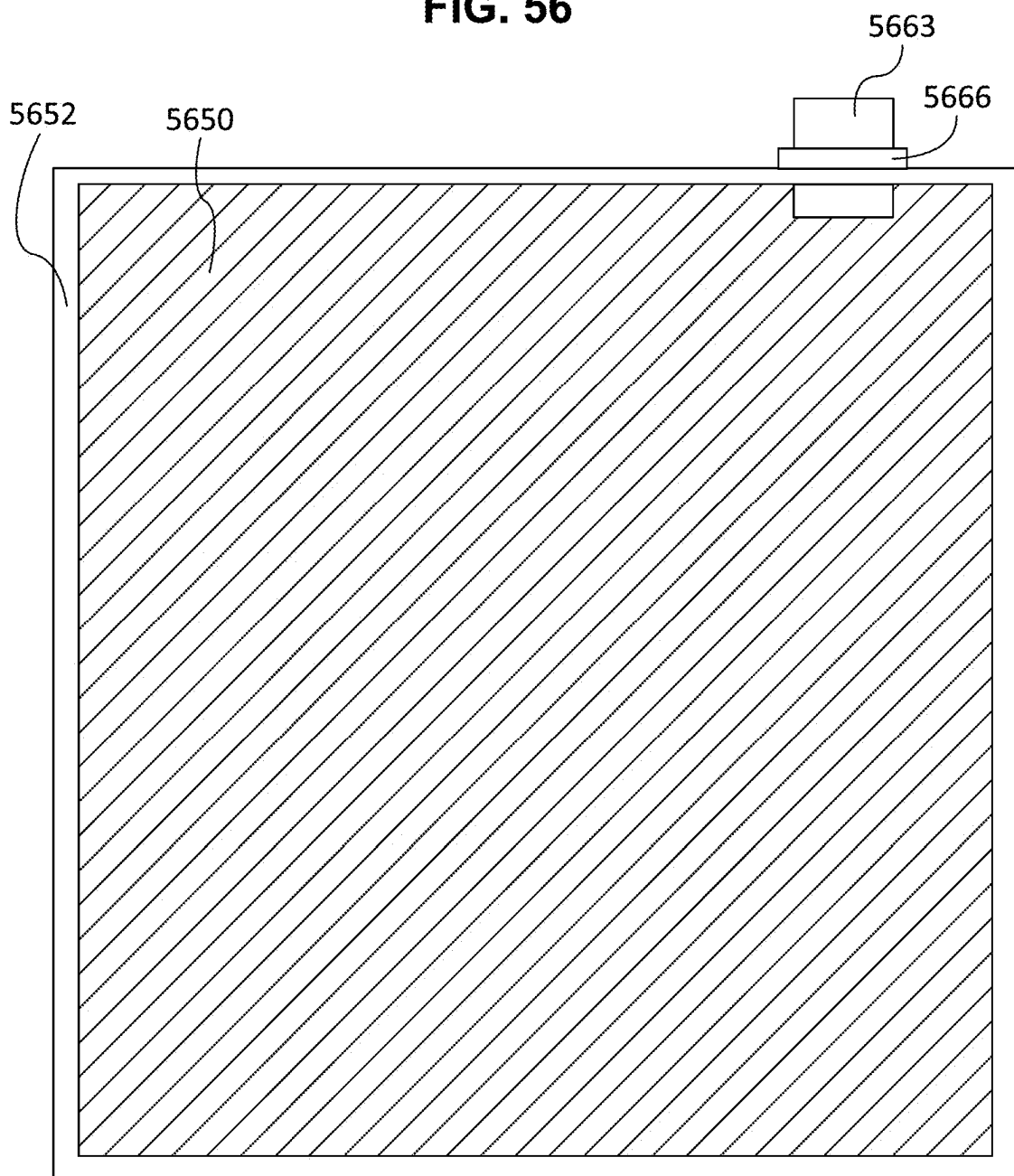


FIG. 57

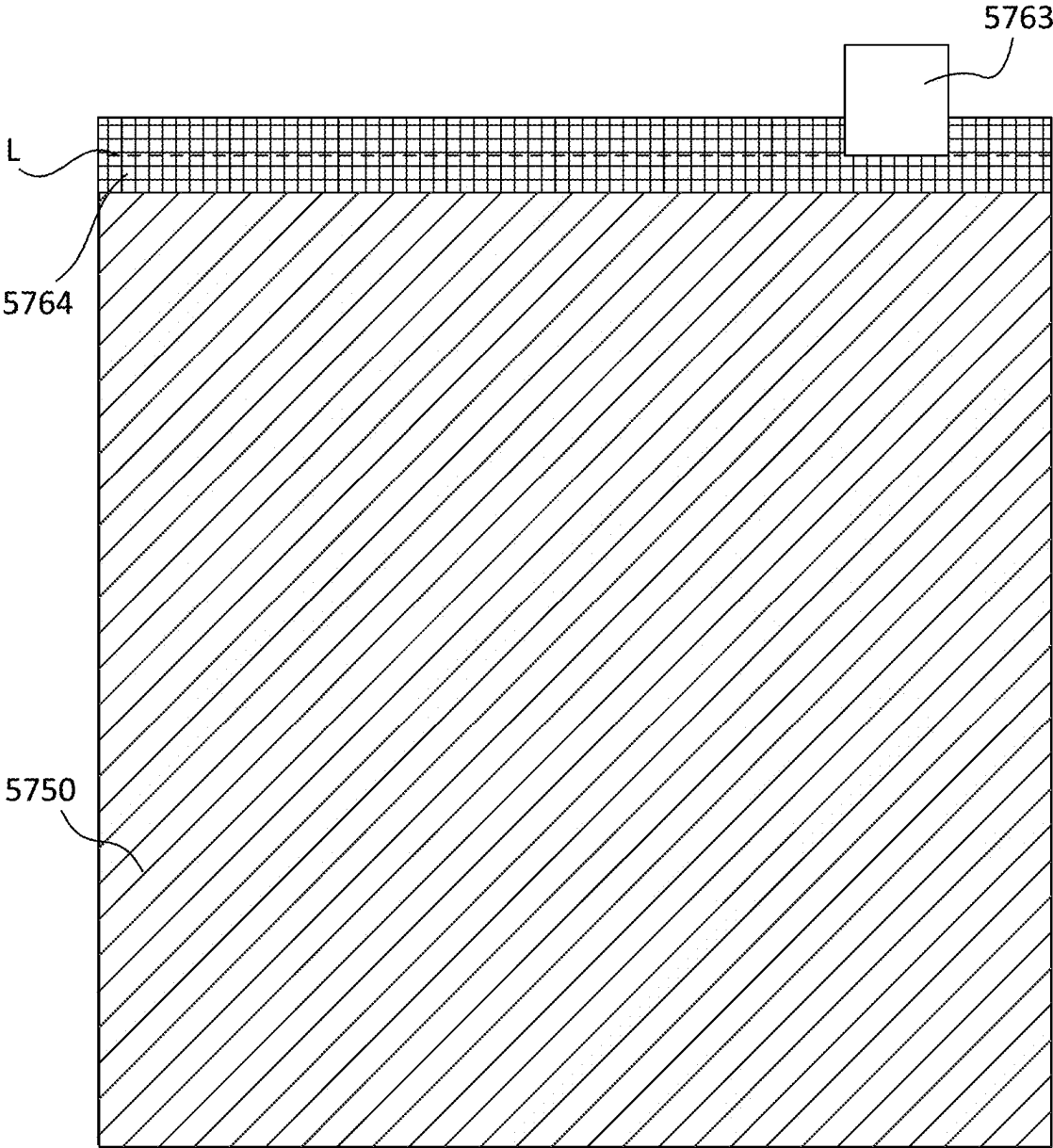


FIG. 58

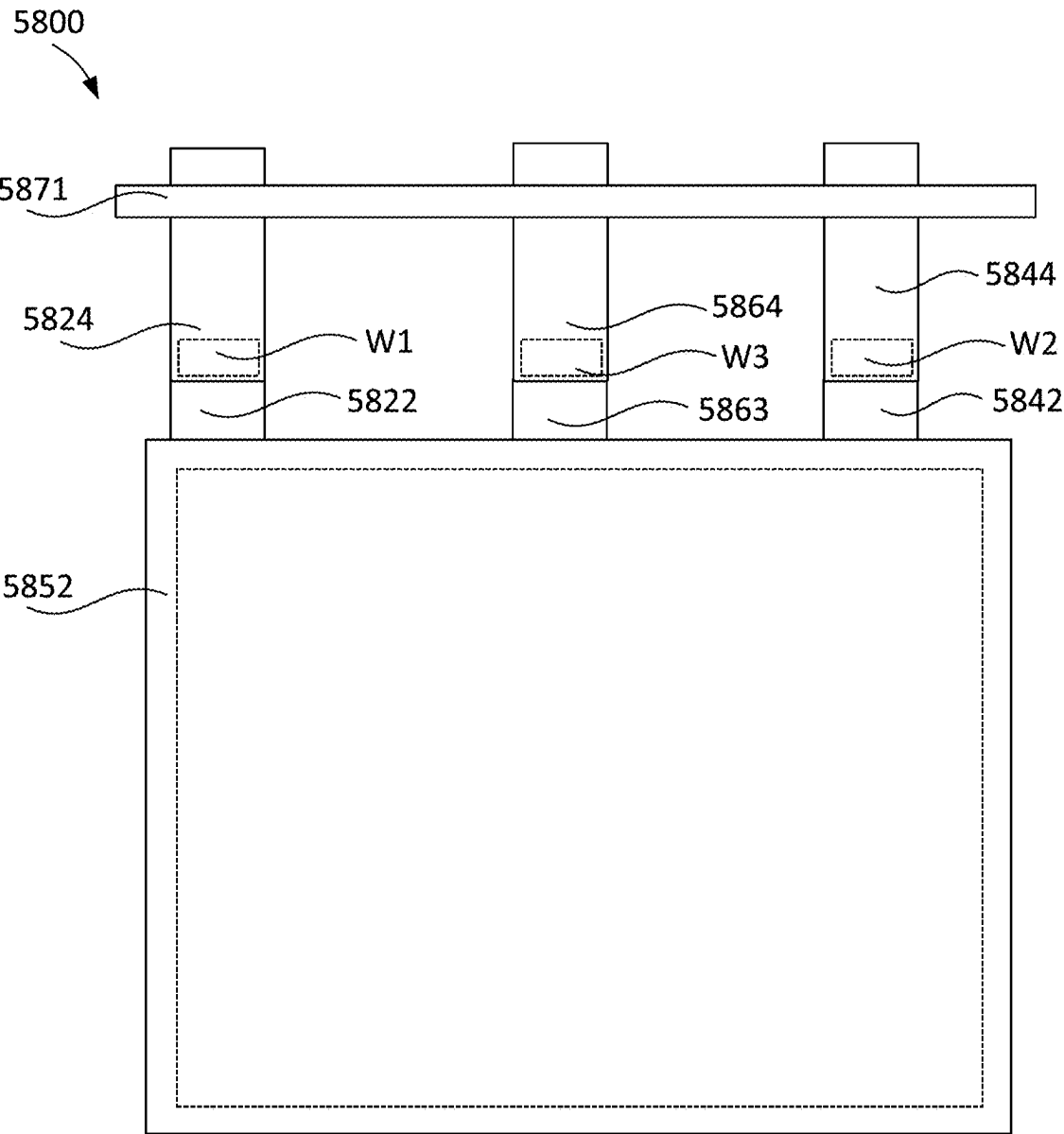


FIG. 59A

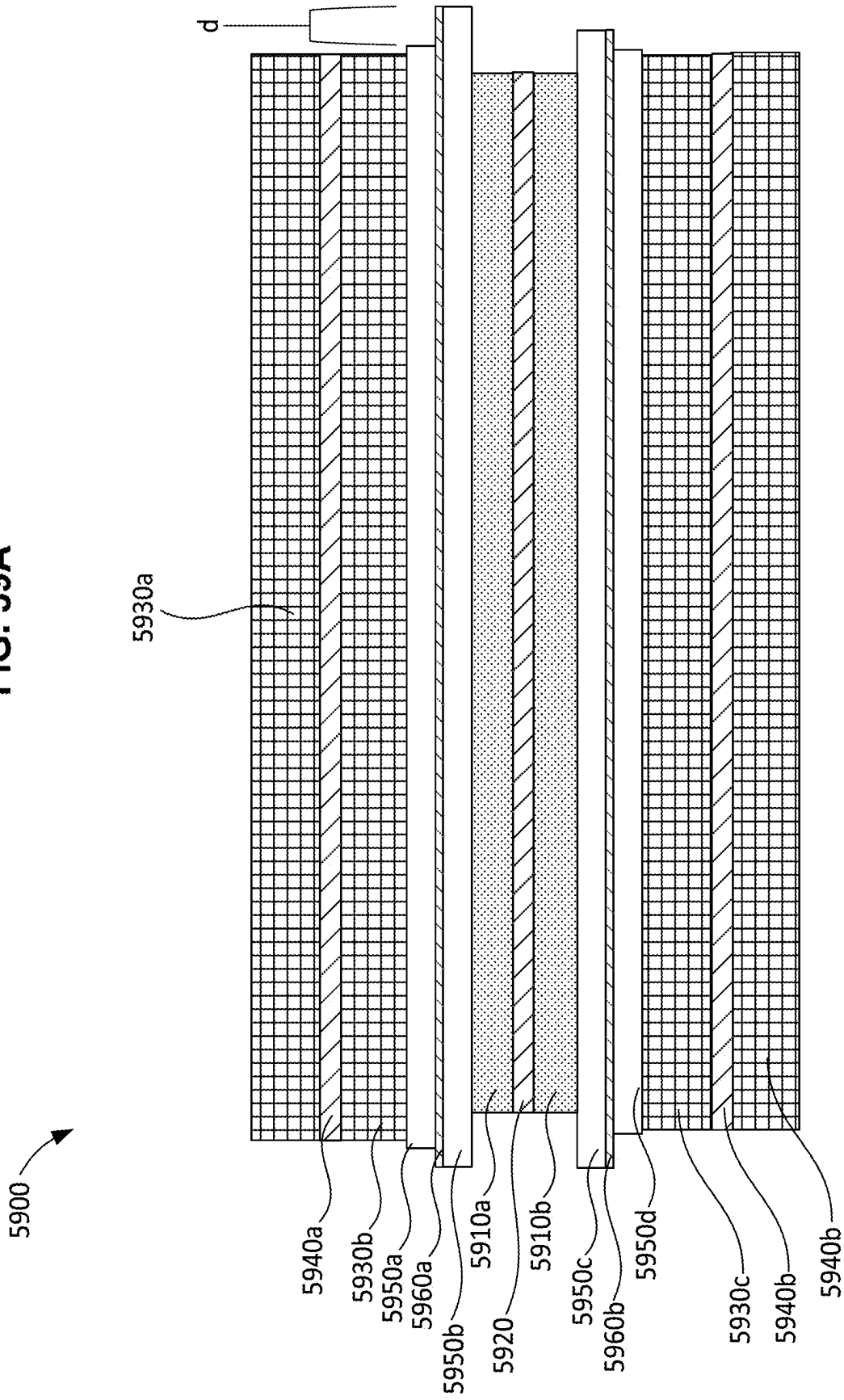


FIG. 59B

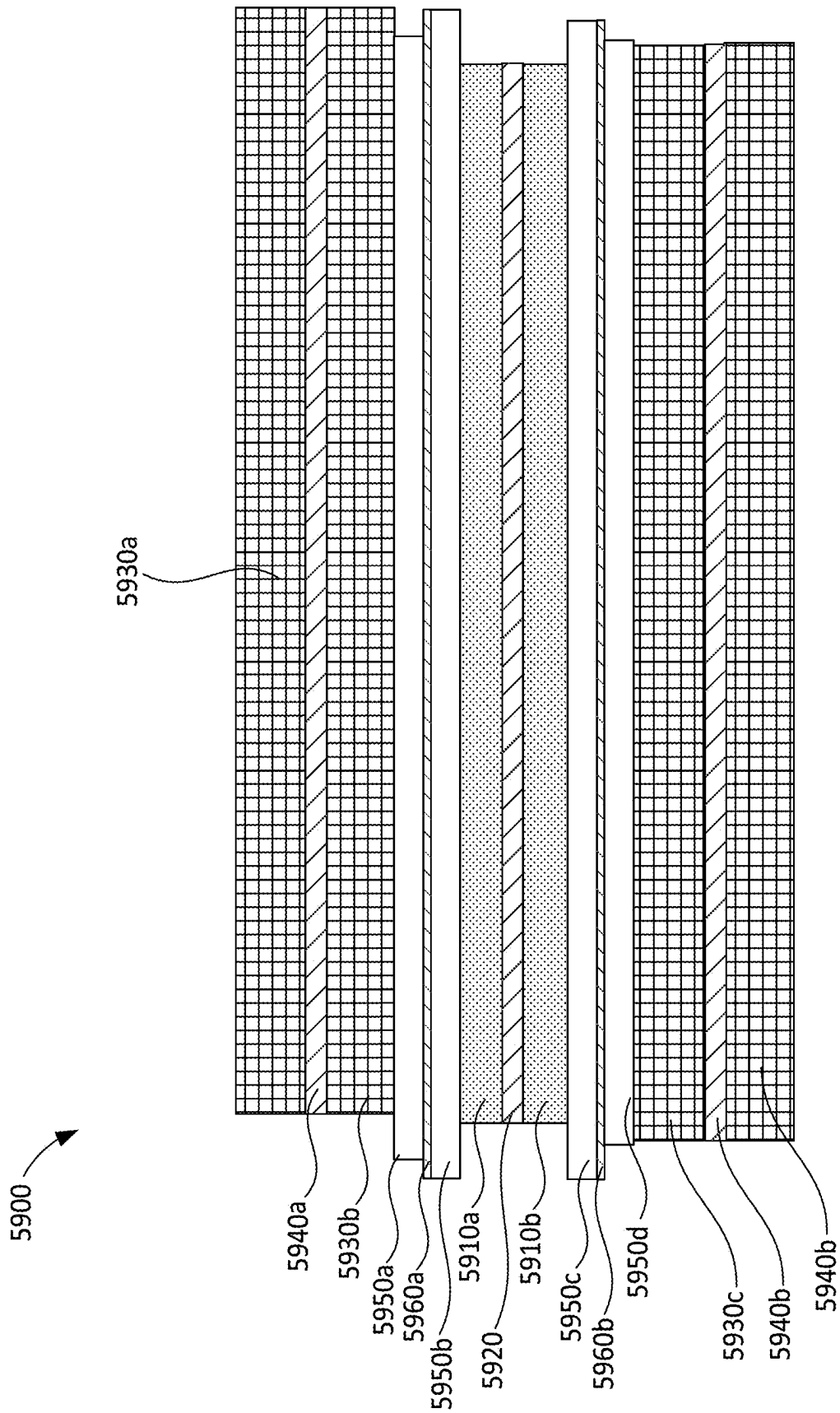


FIG. 59C

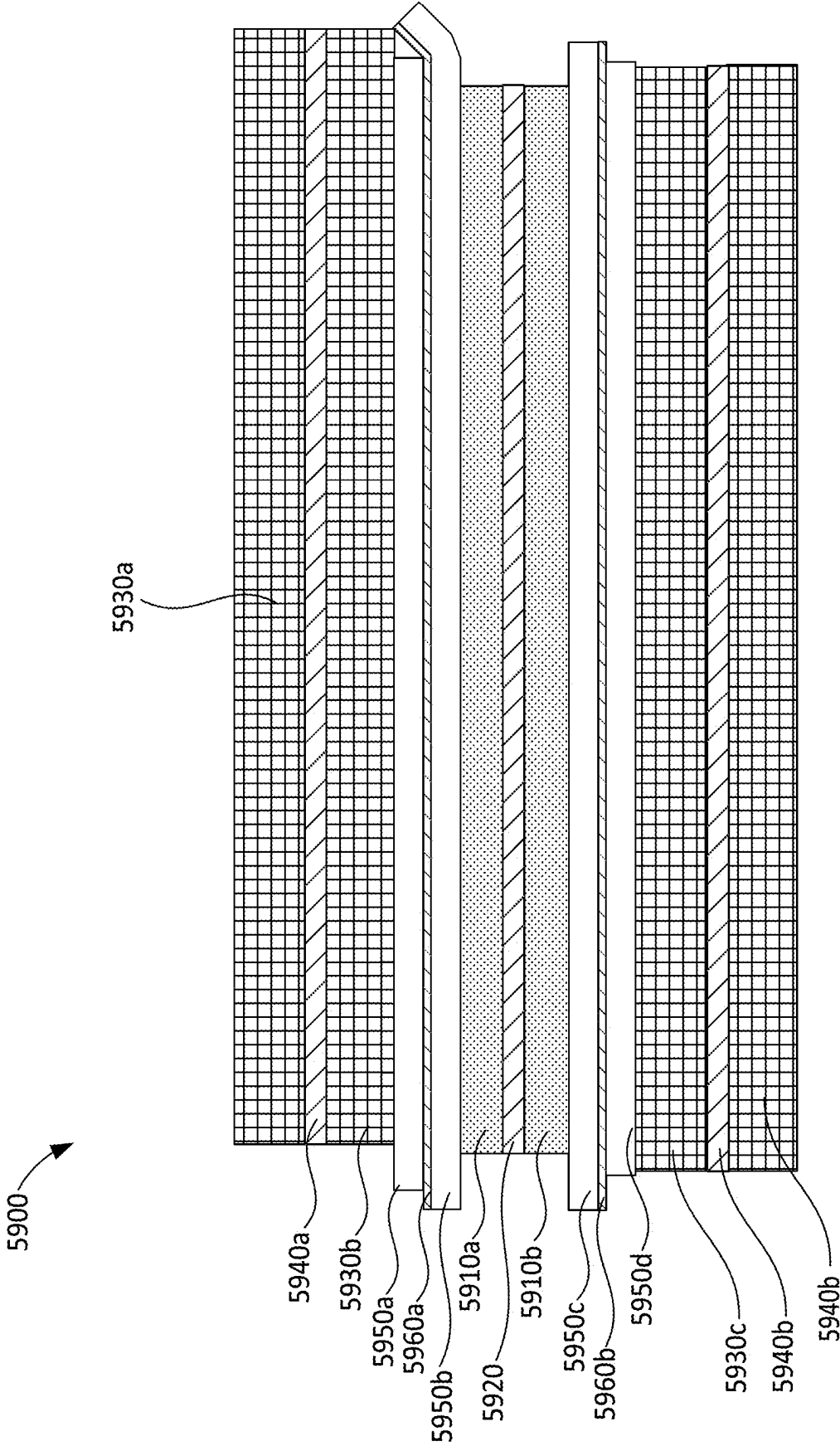


FIG. 60

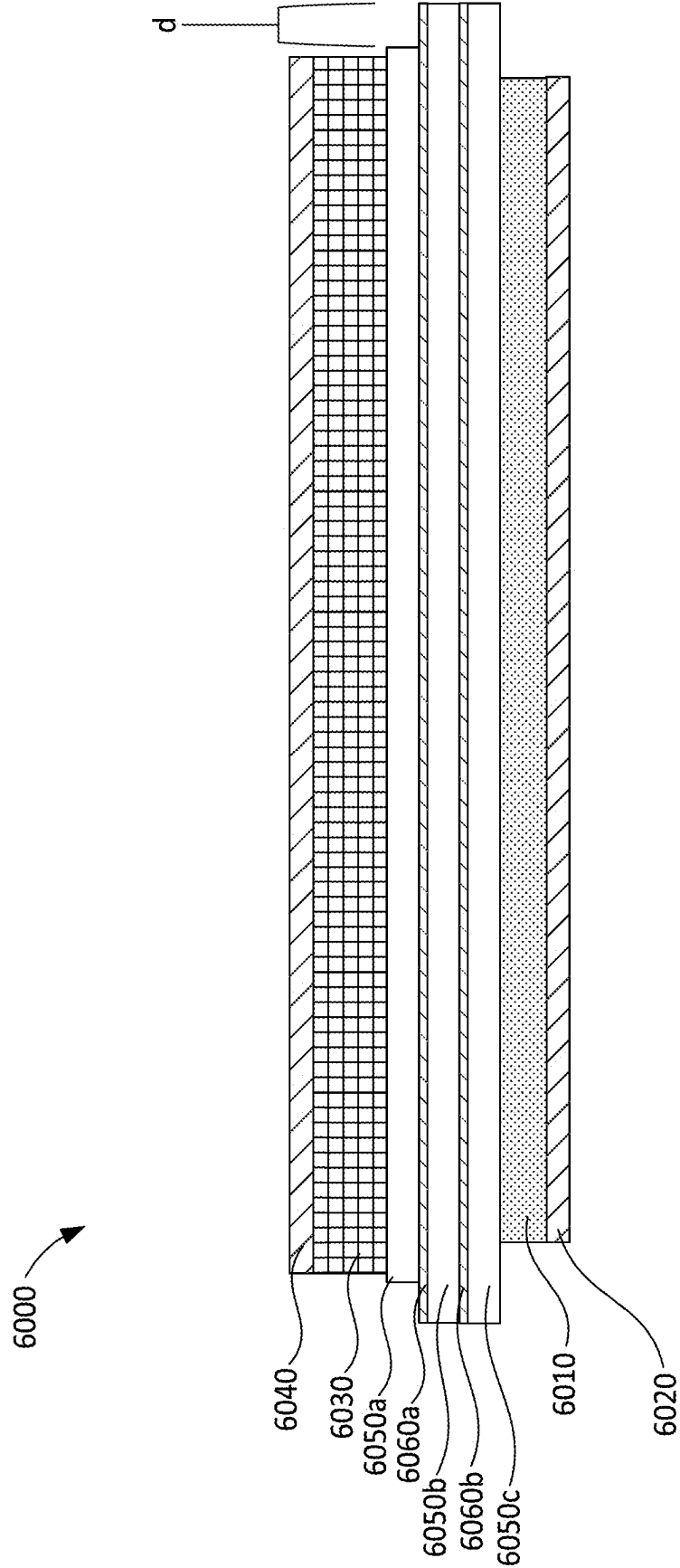


FIG. 62A

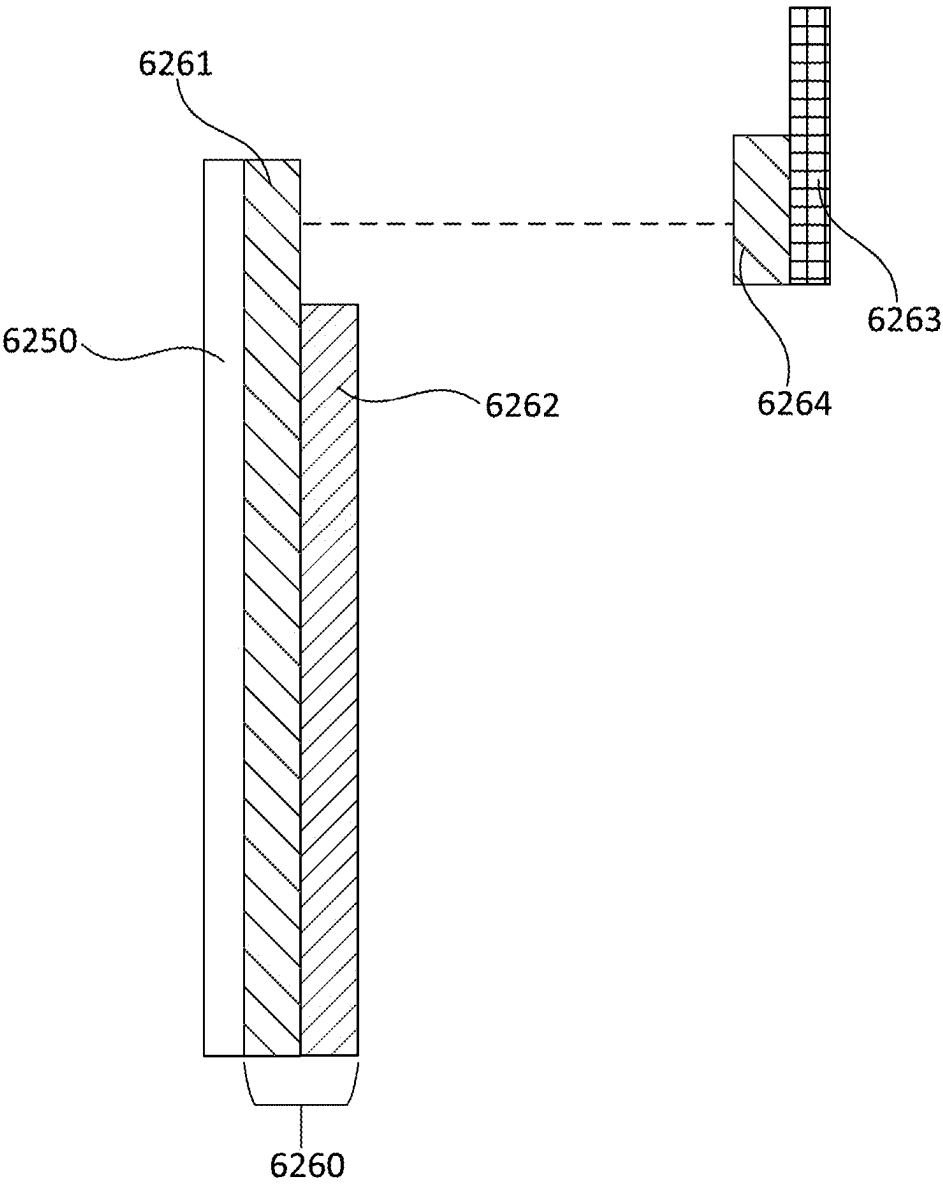


FIG. 62B

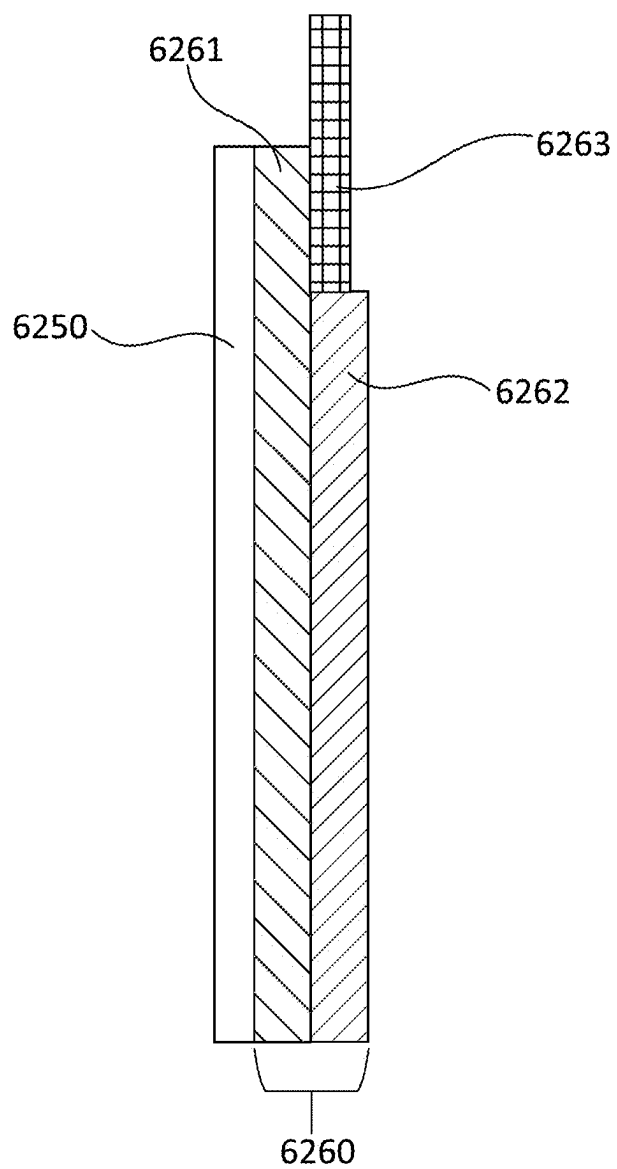


FIG. 63

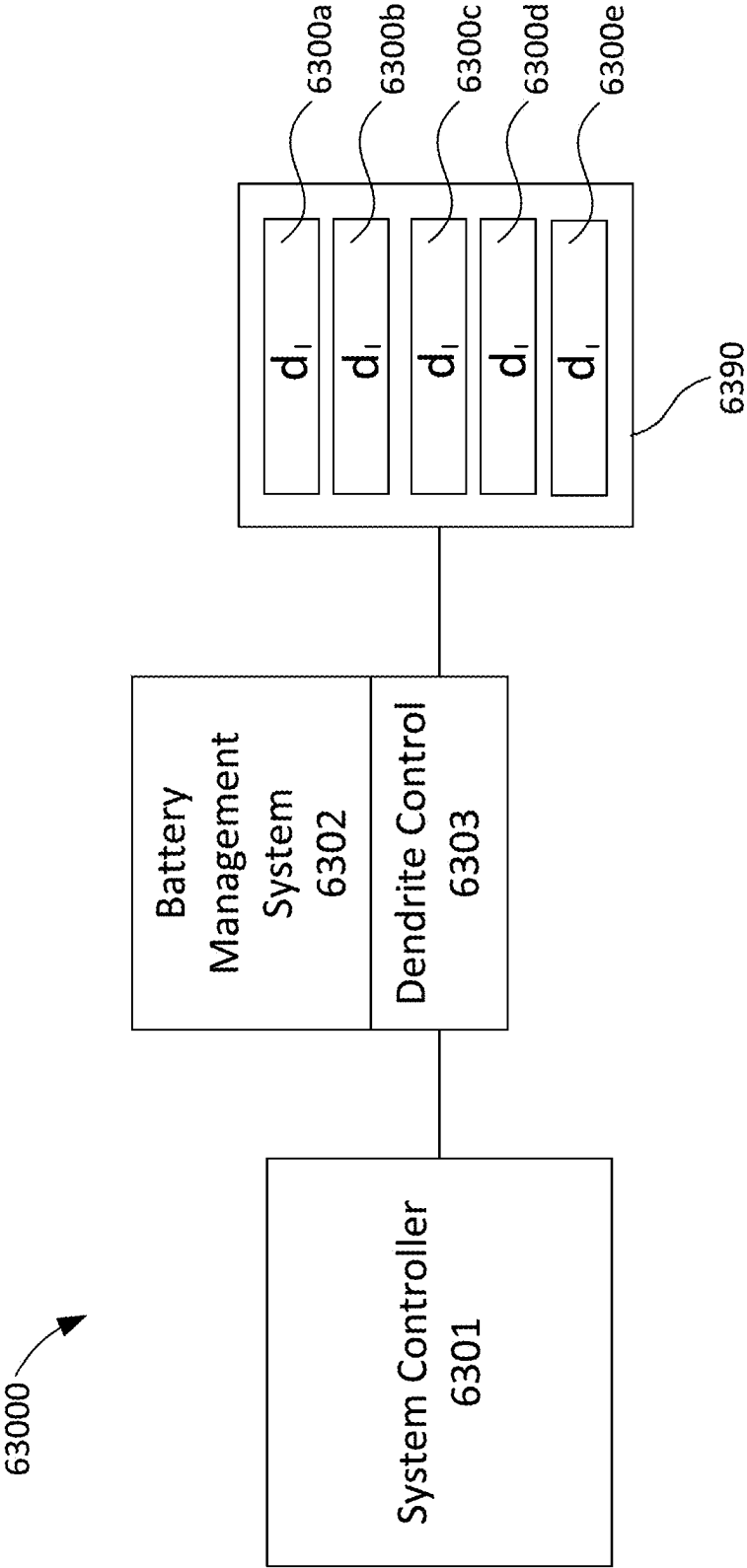


FIG. 64

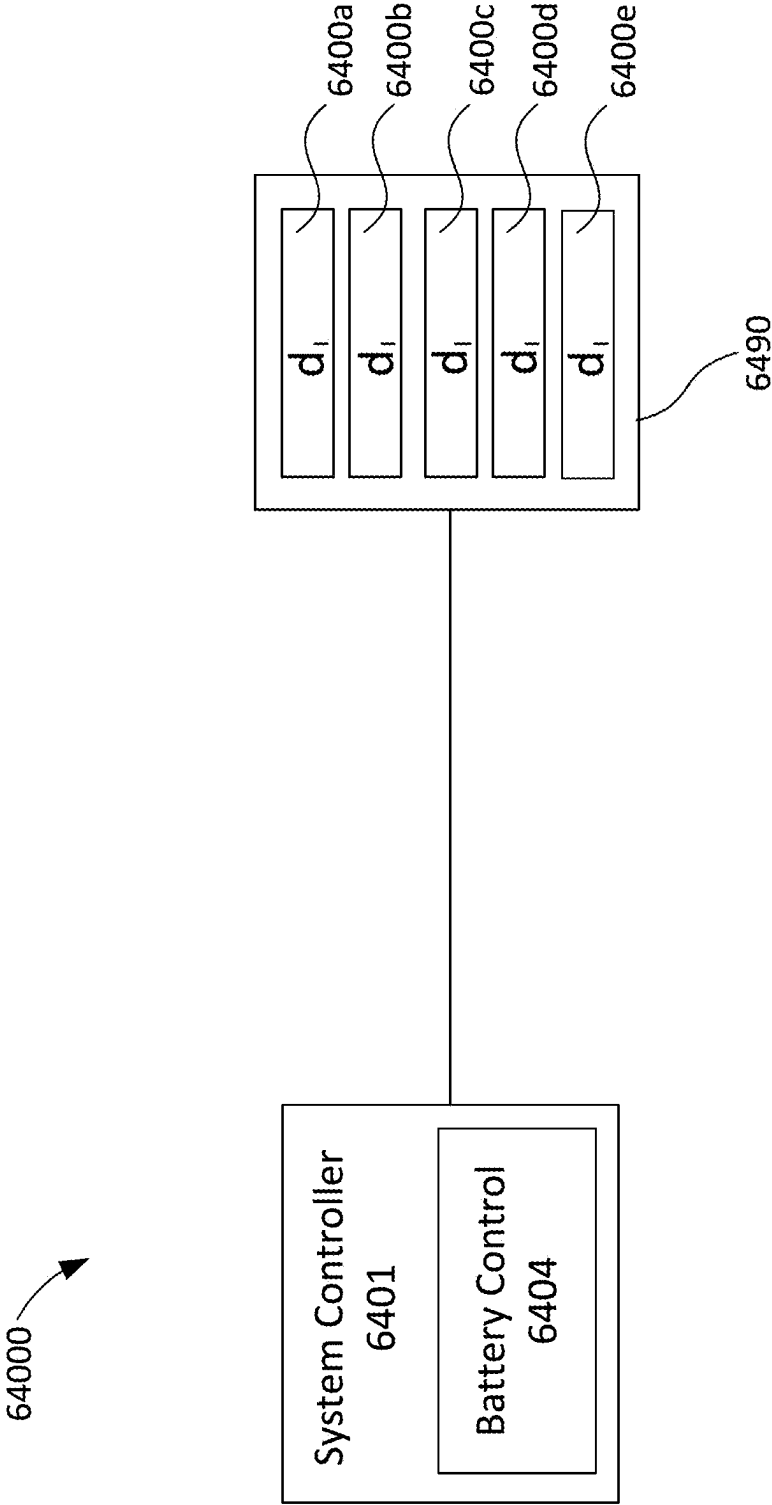


FIG. 65

65000

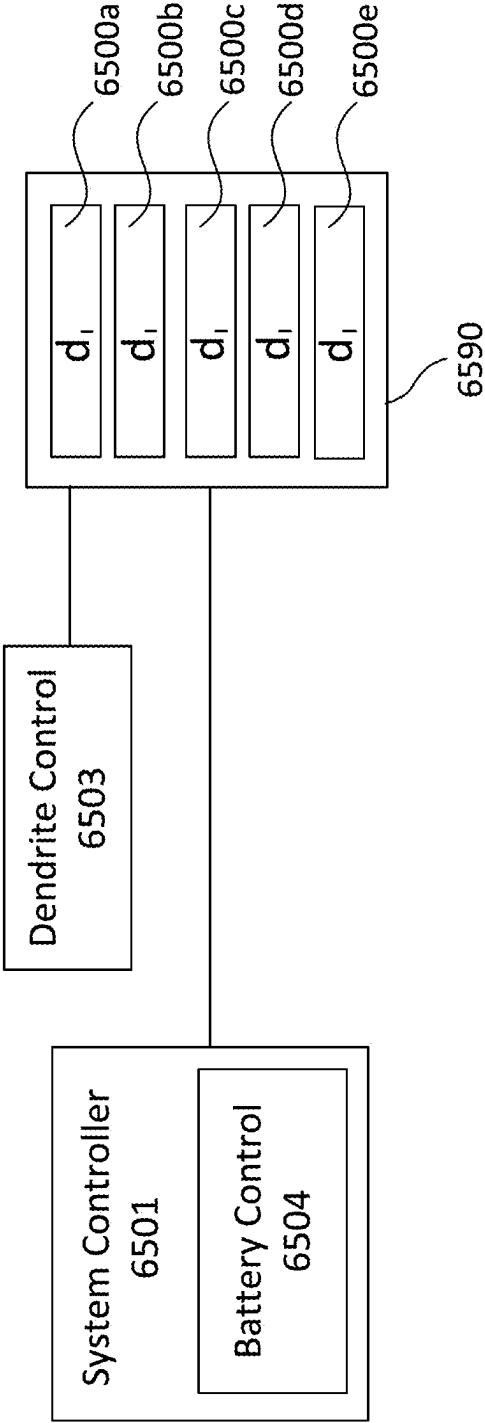


FIG. 66

66000

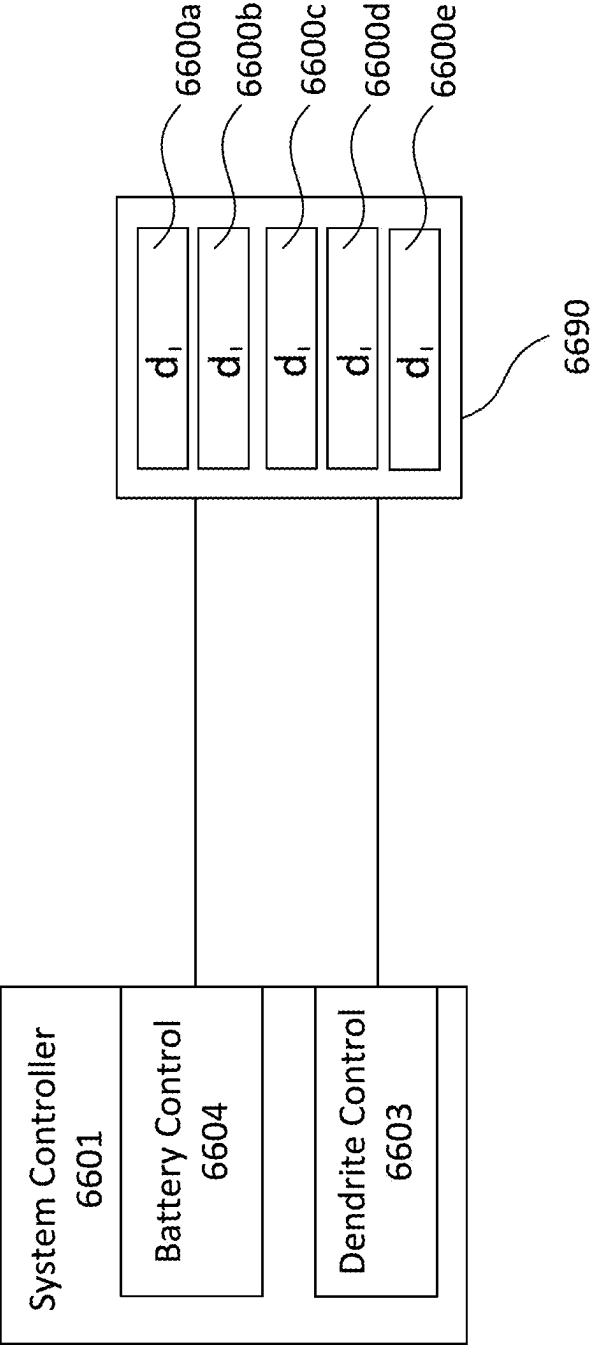


FIG. 67

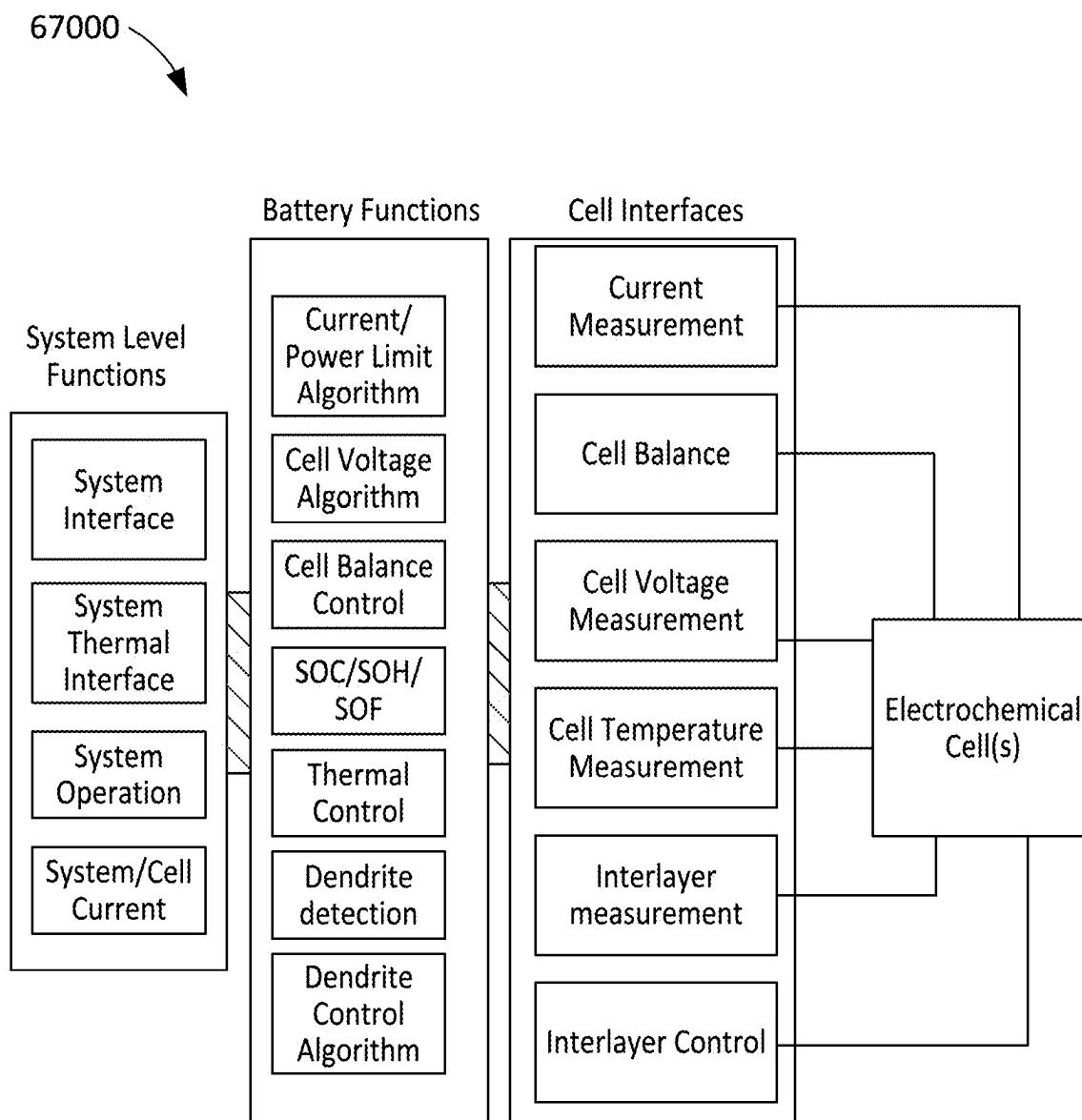


FIG. 68

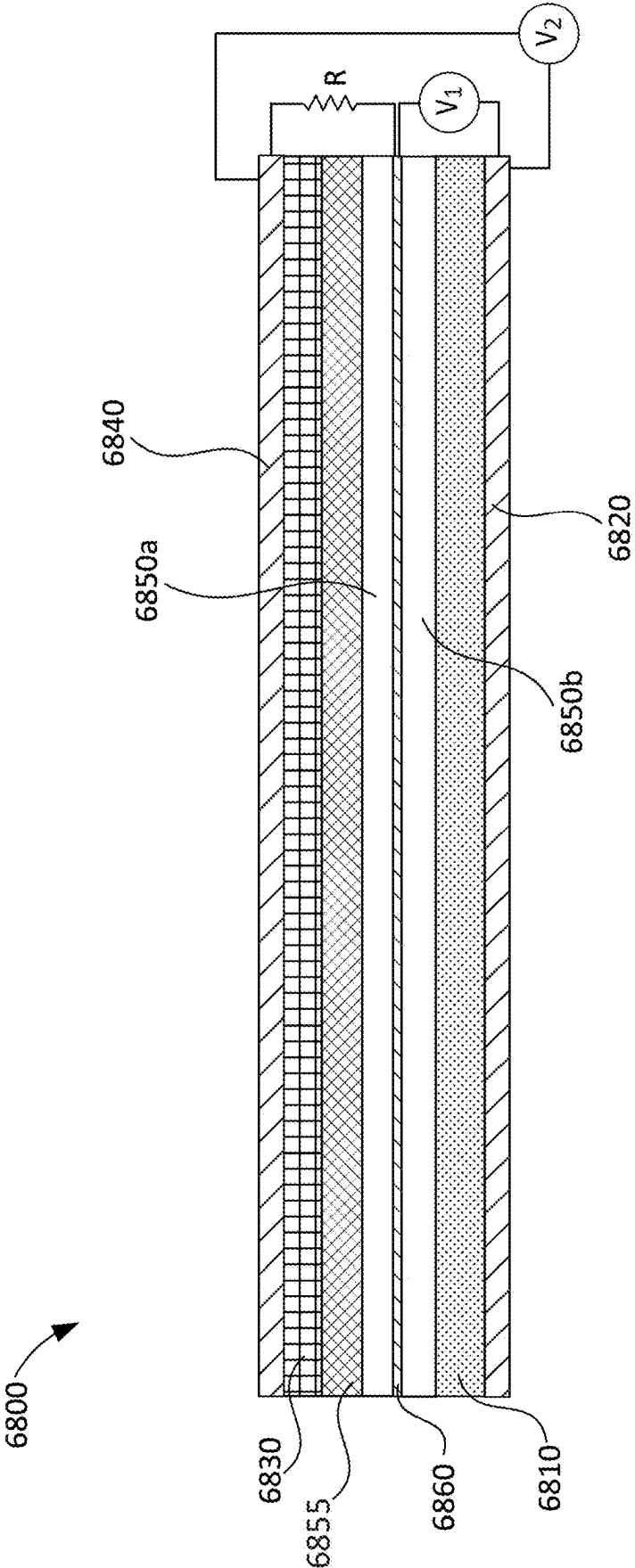


FIG. 69

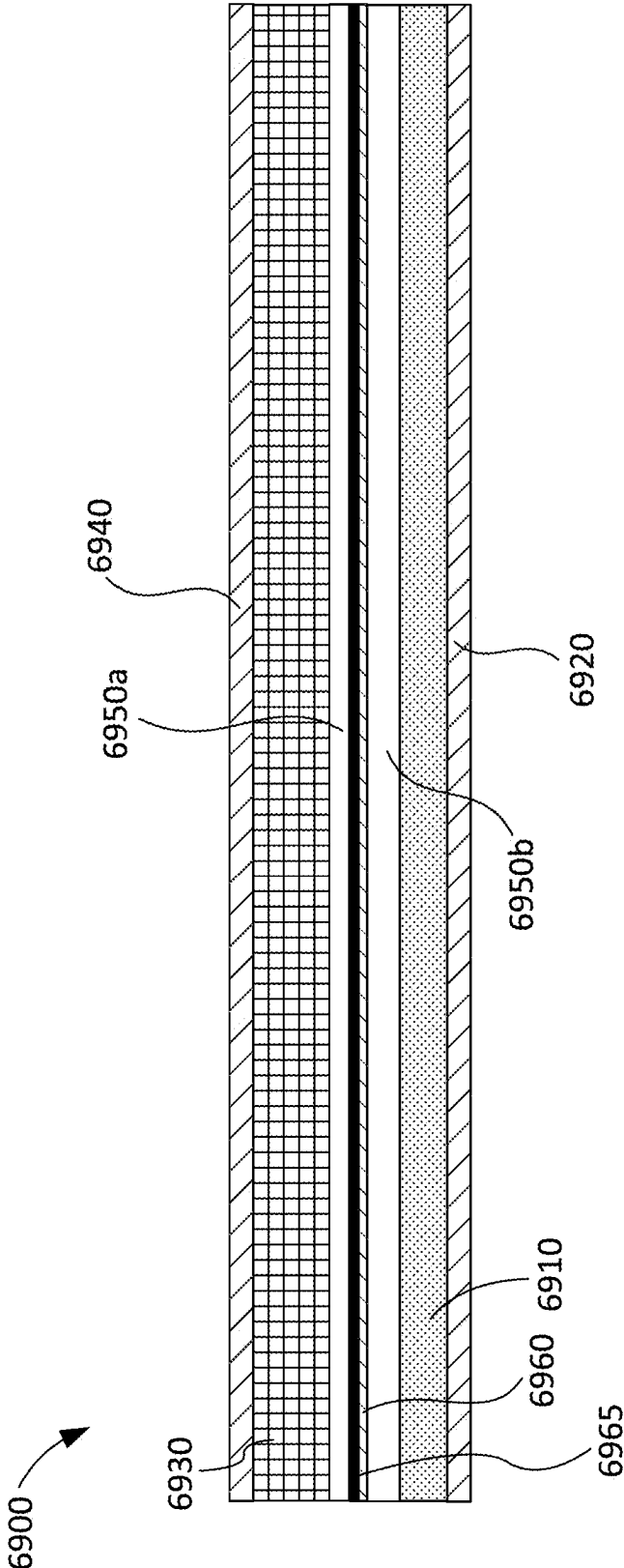


FIG. 70A

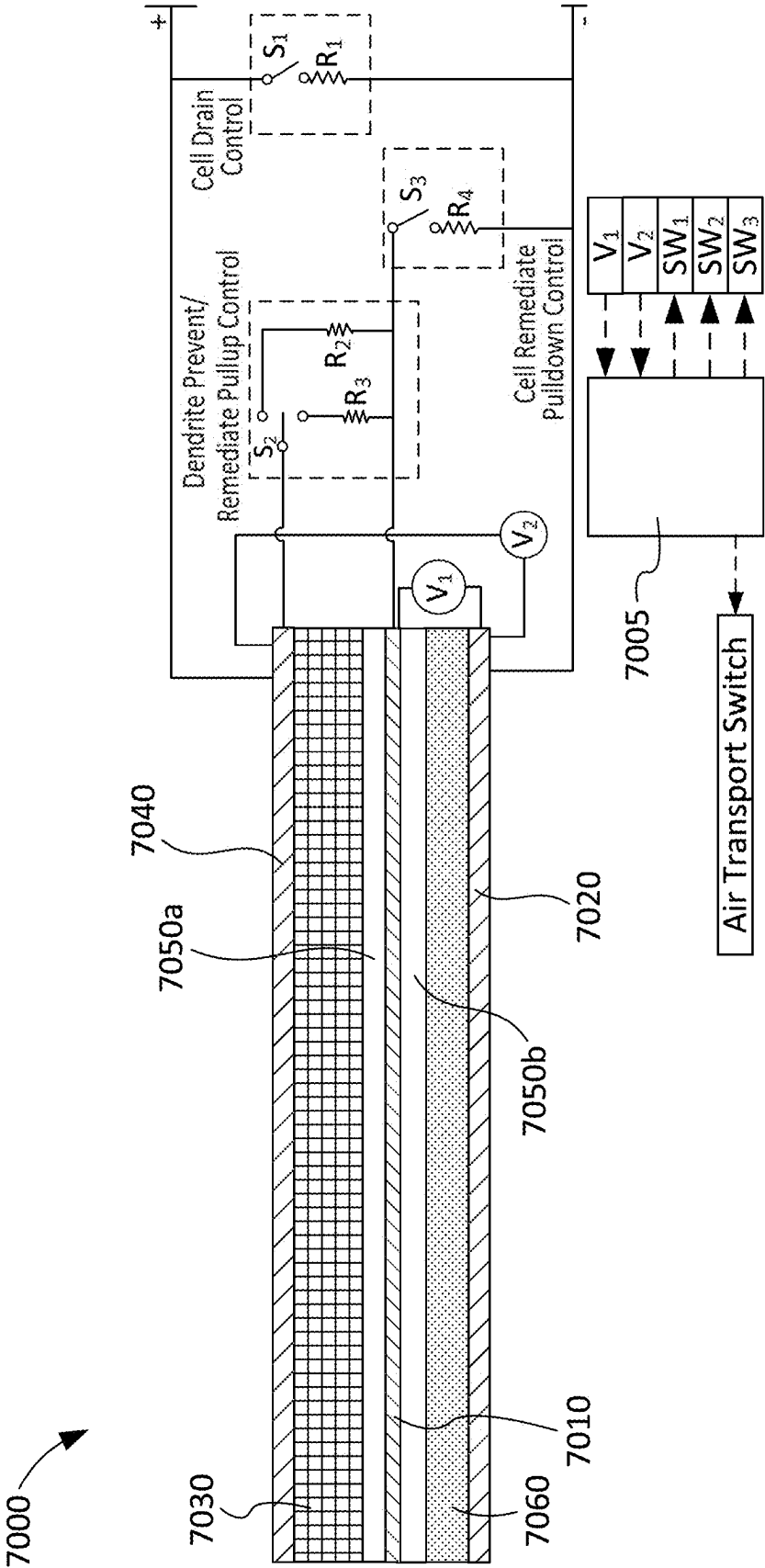


FIG. 70B

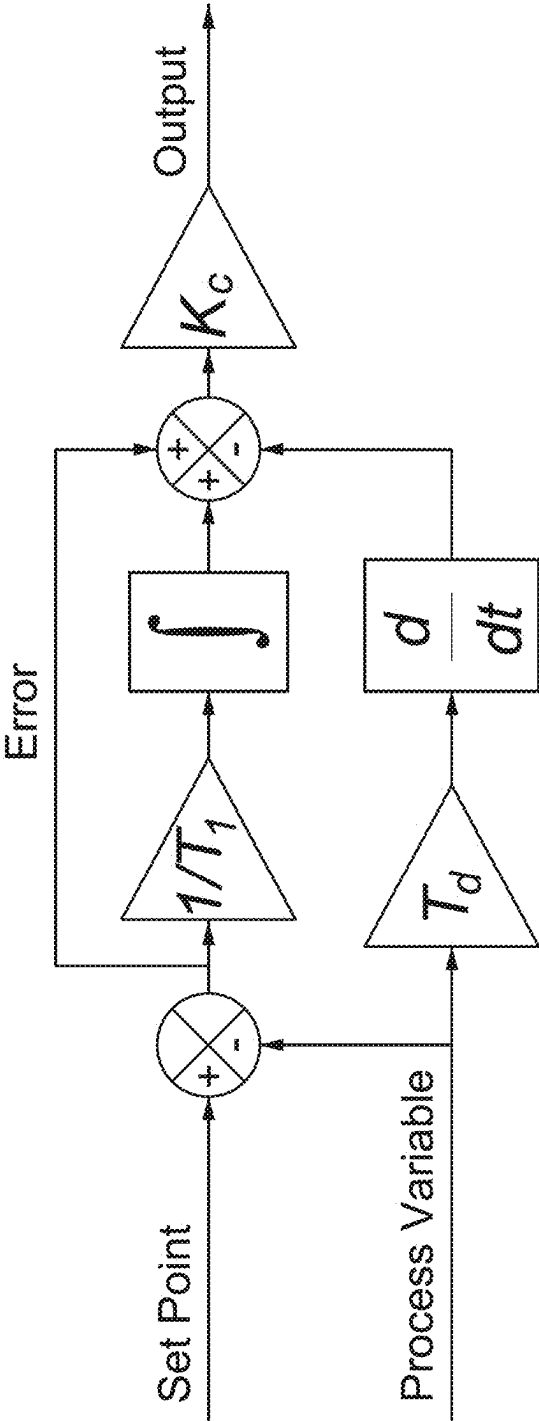


FIG. 71

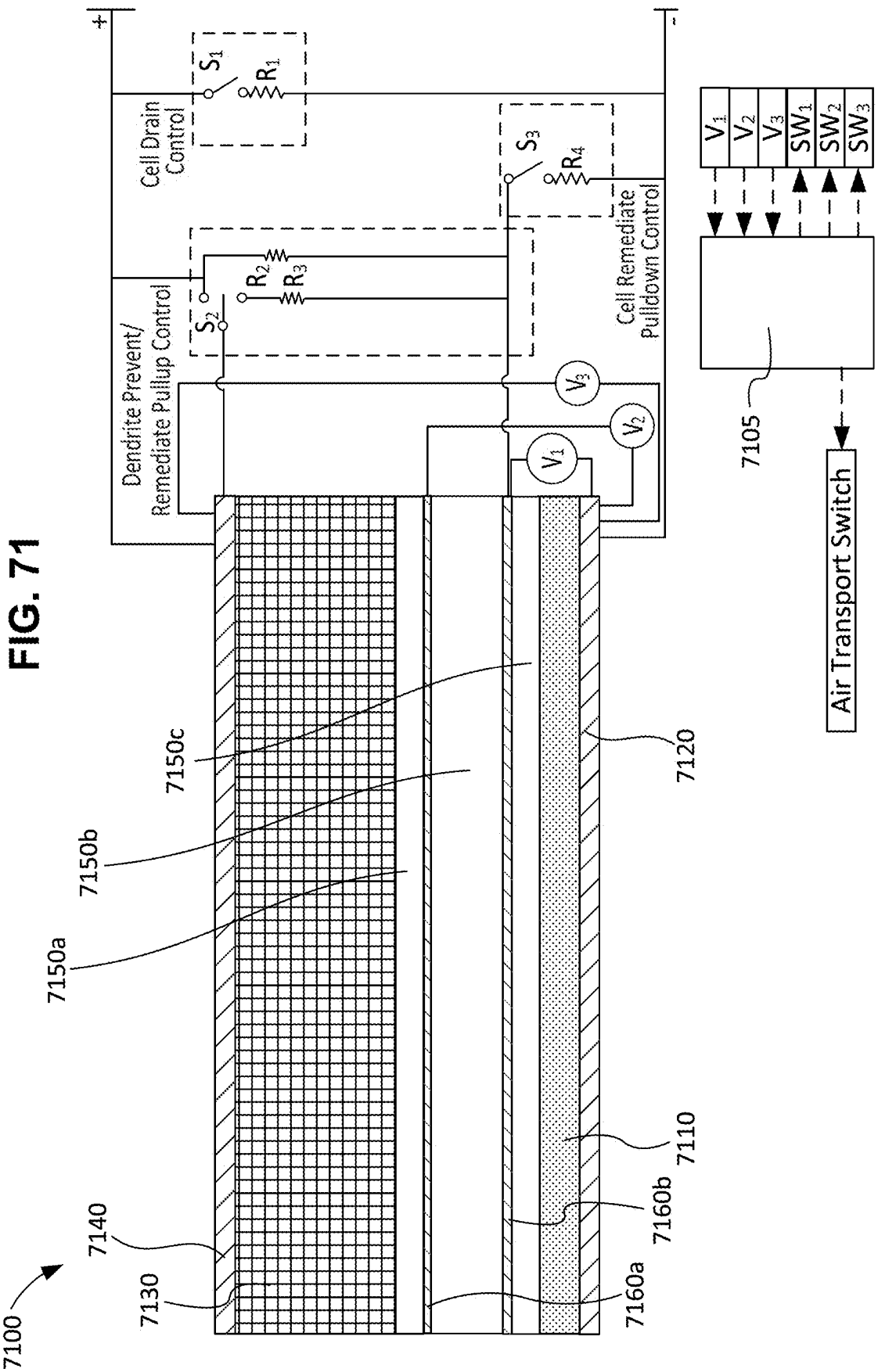


FIG. 72

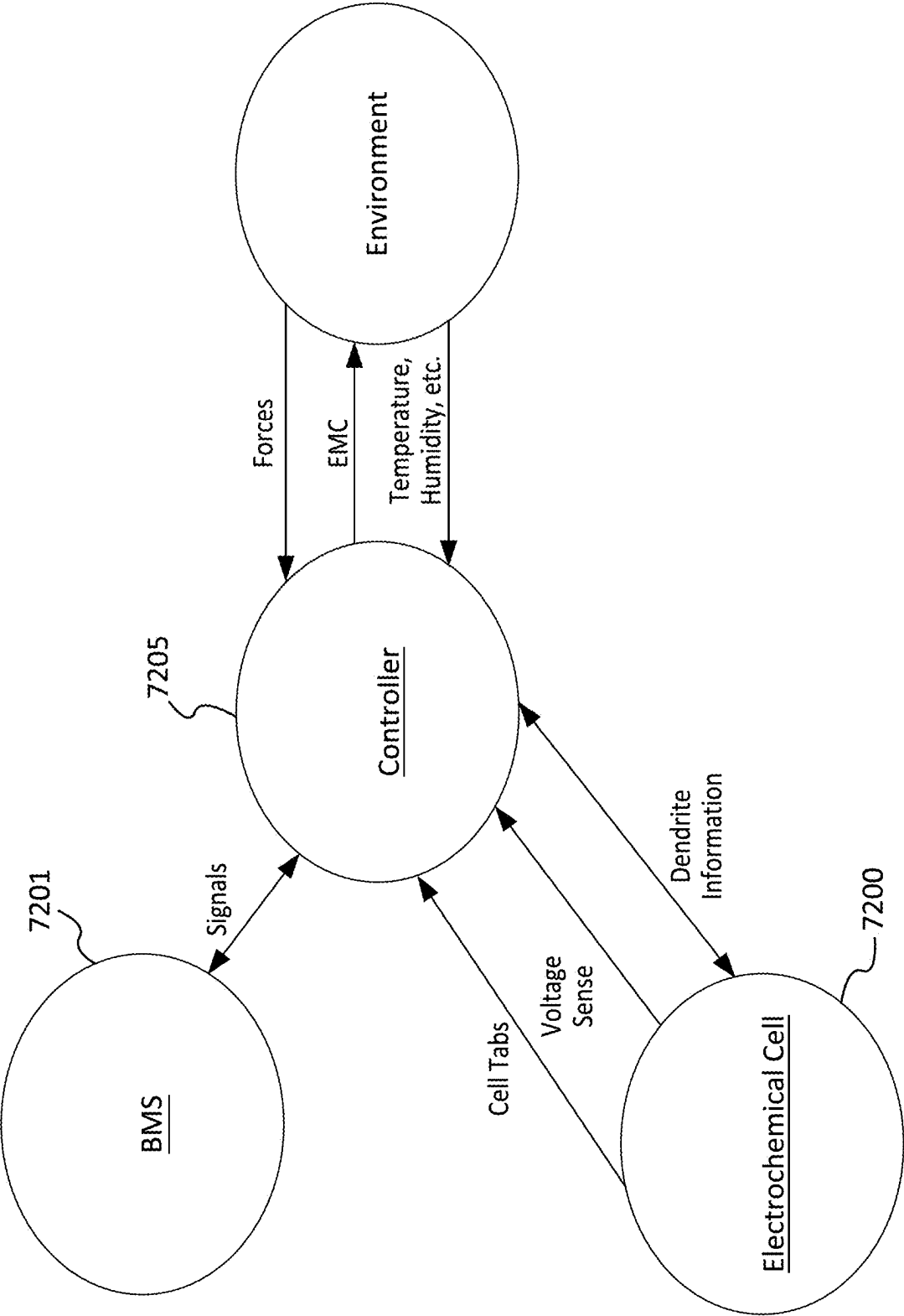


FIG. 73

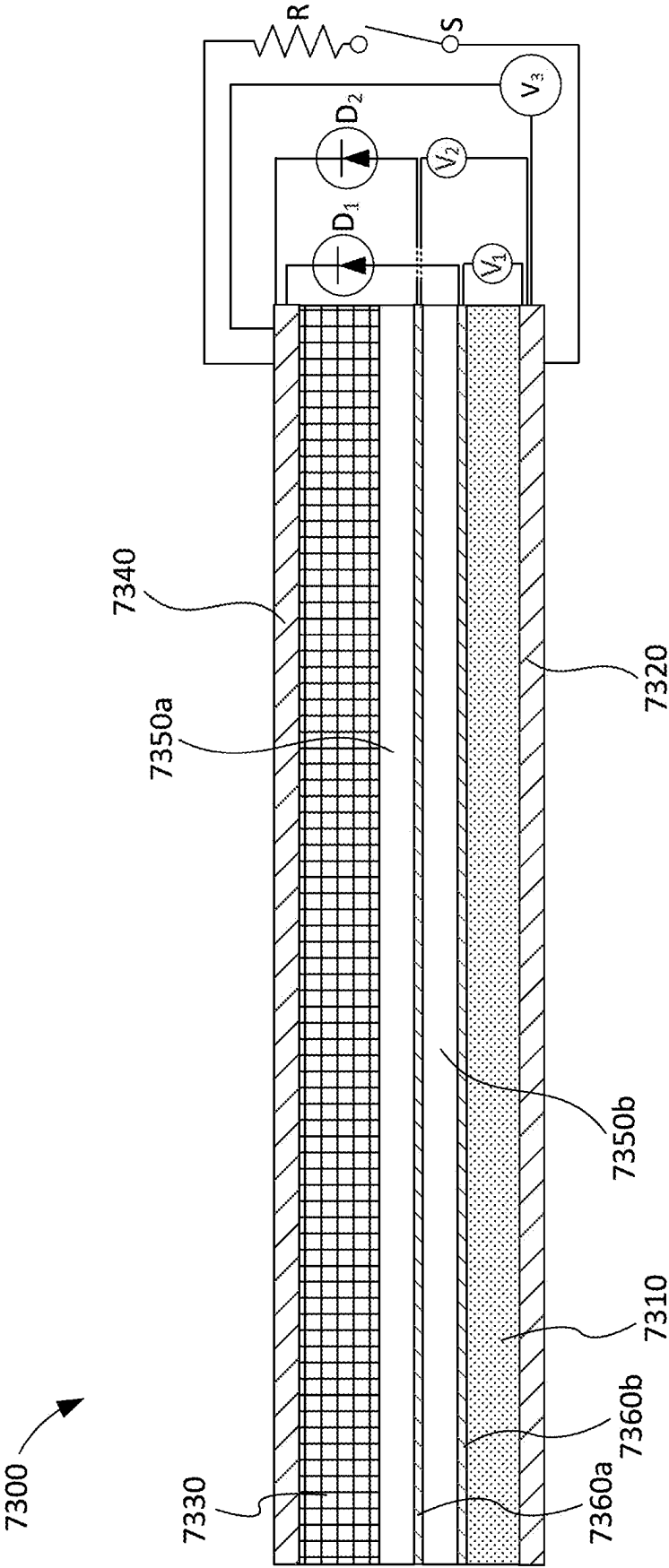


FIG. 74

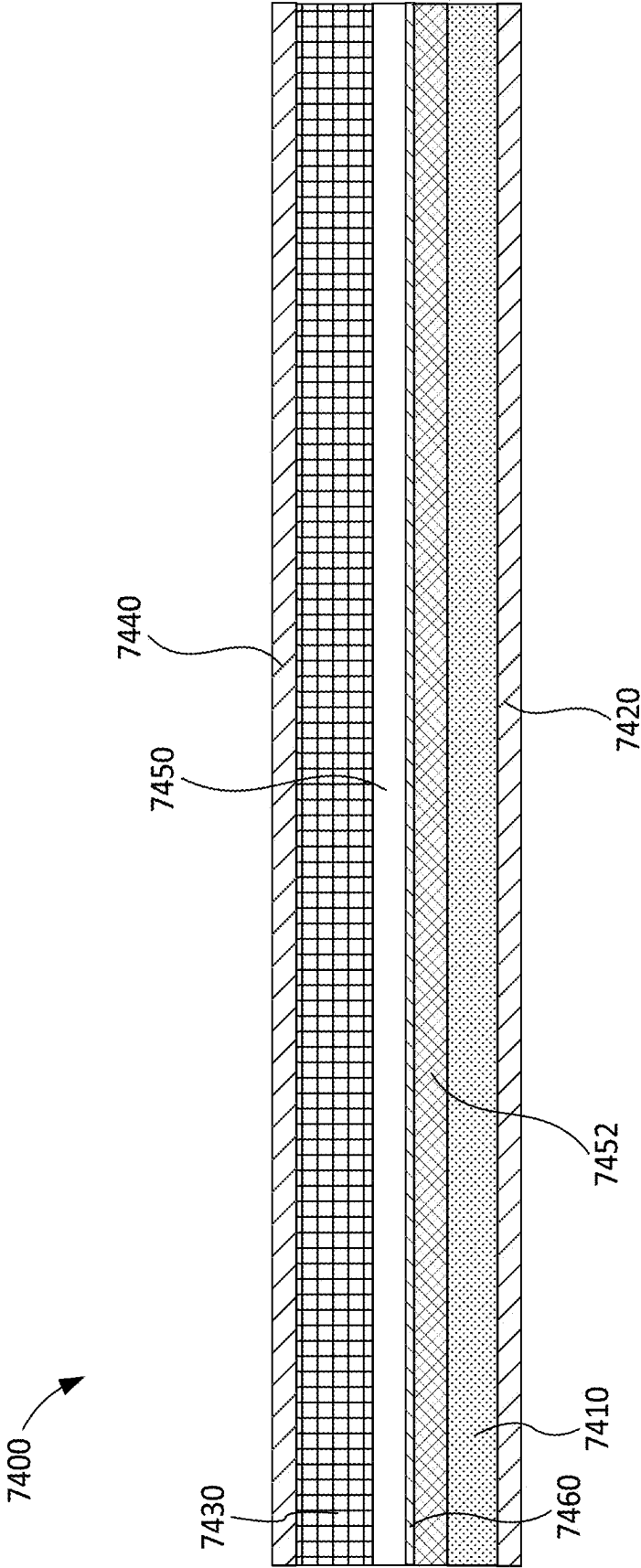


FIG. 75A

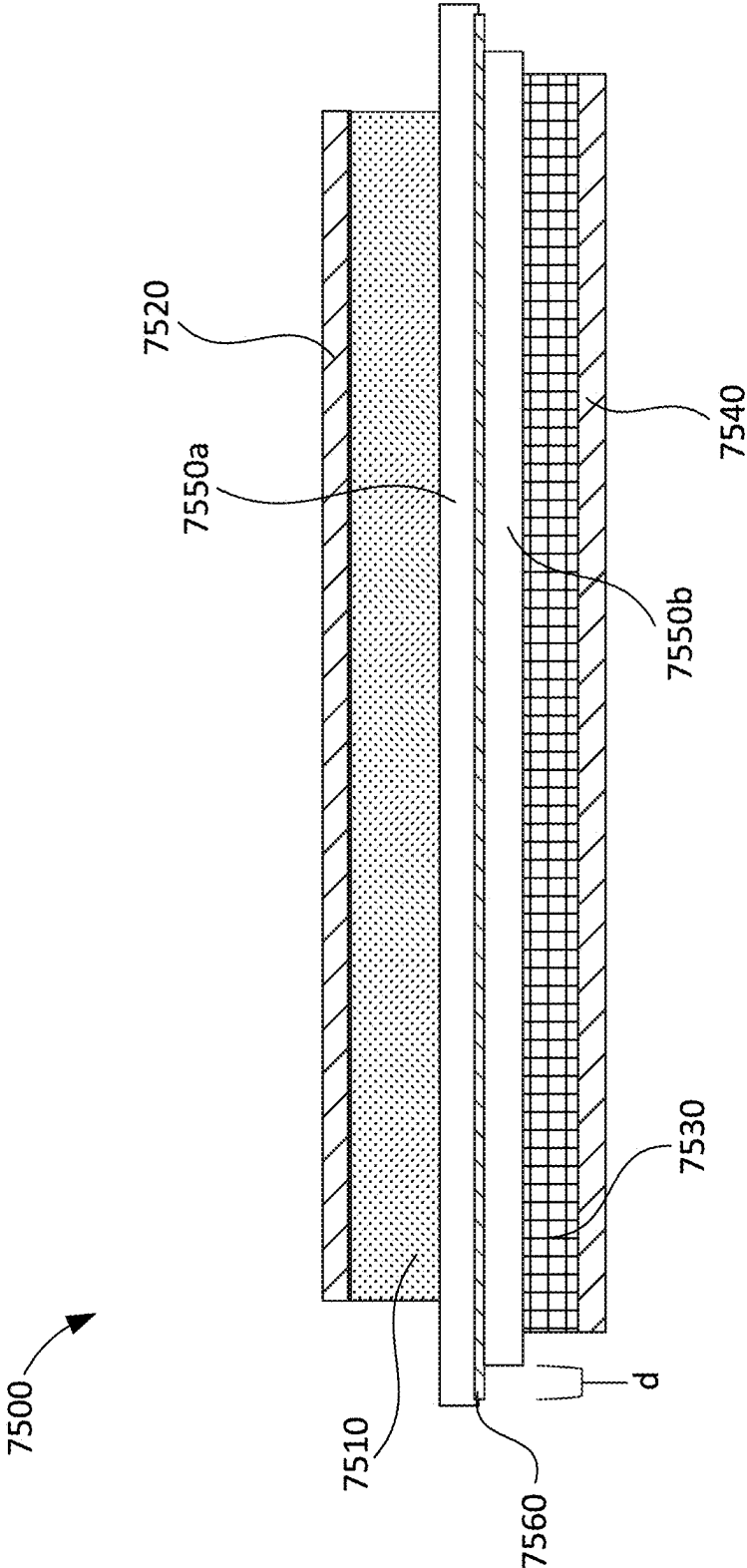


FIG. 75B

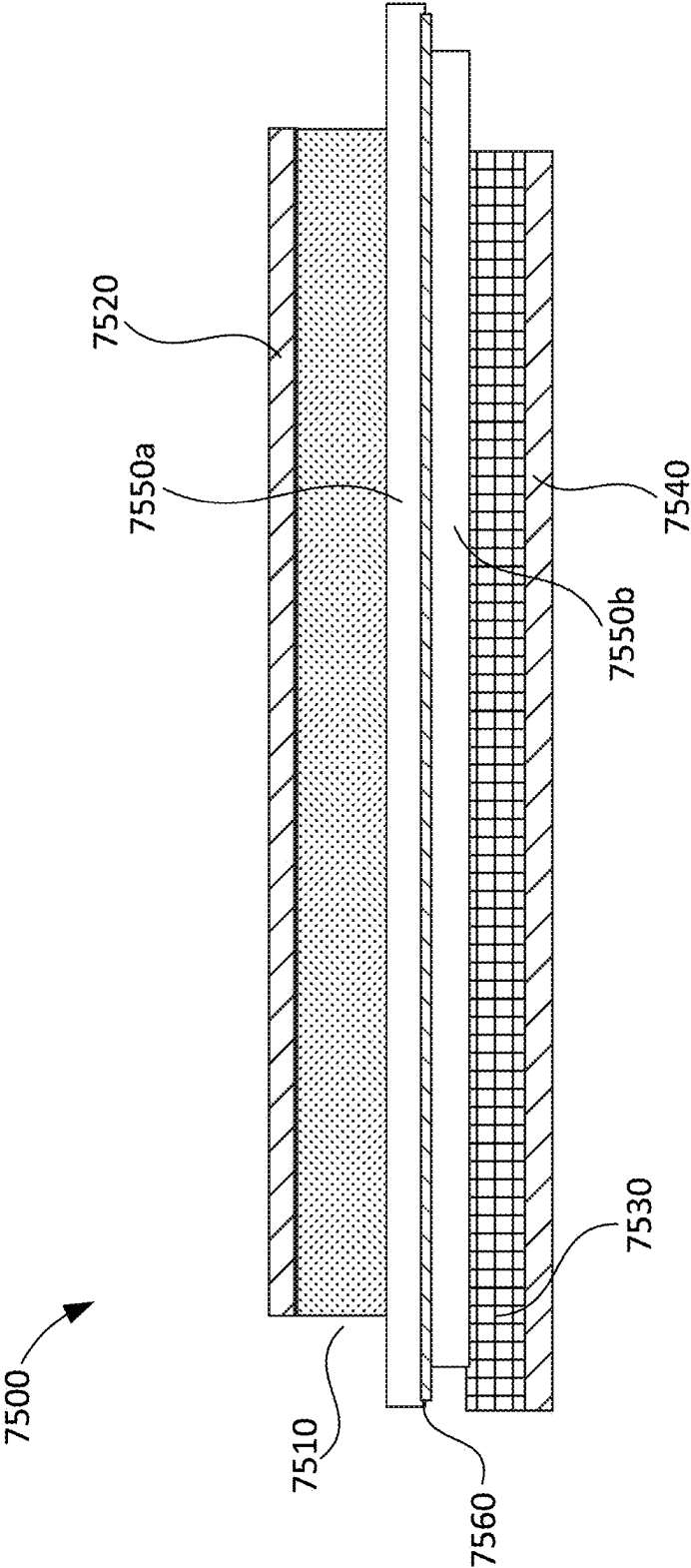


FIG. 75C

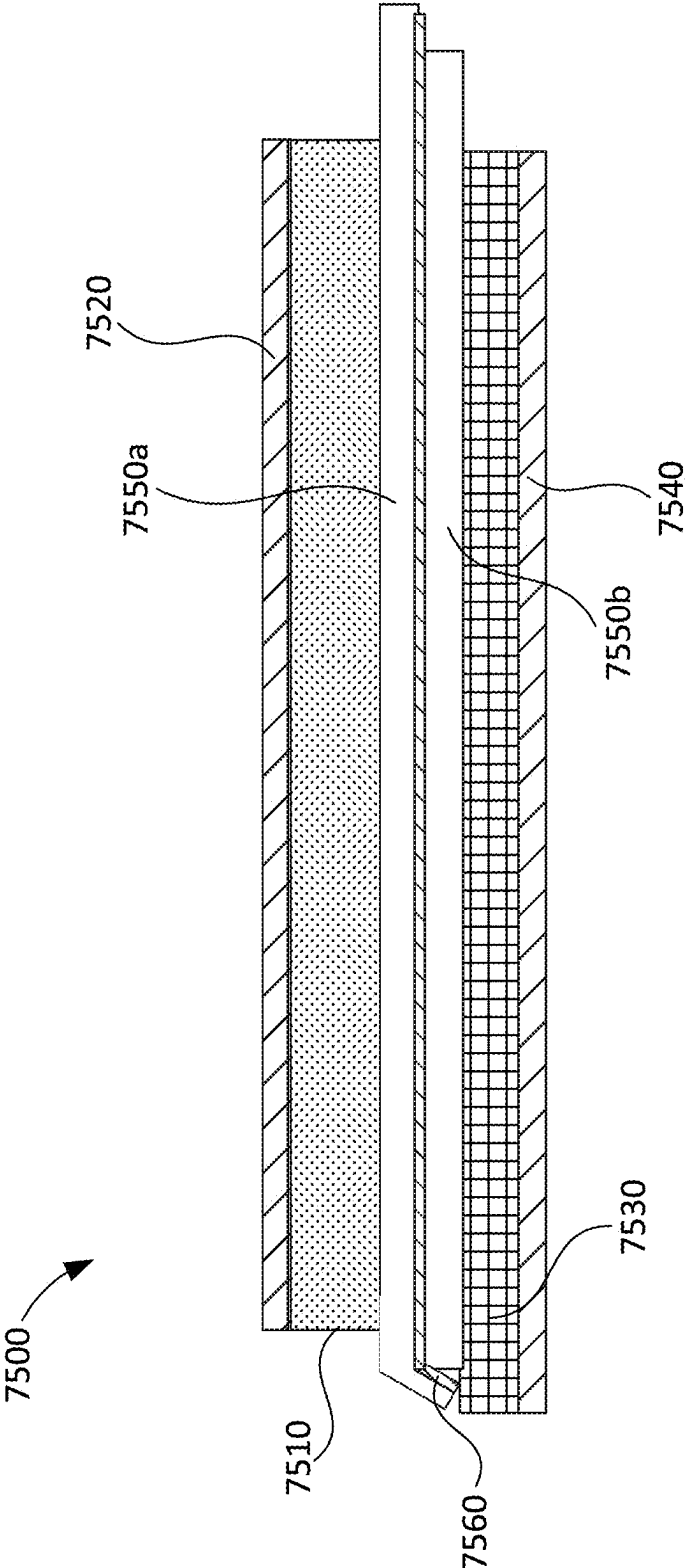


FIG. 76A

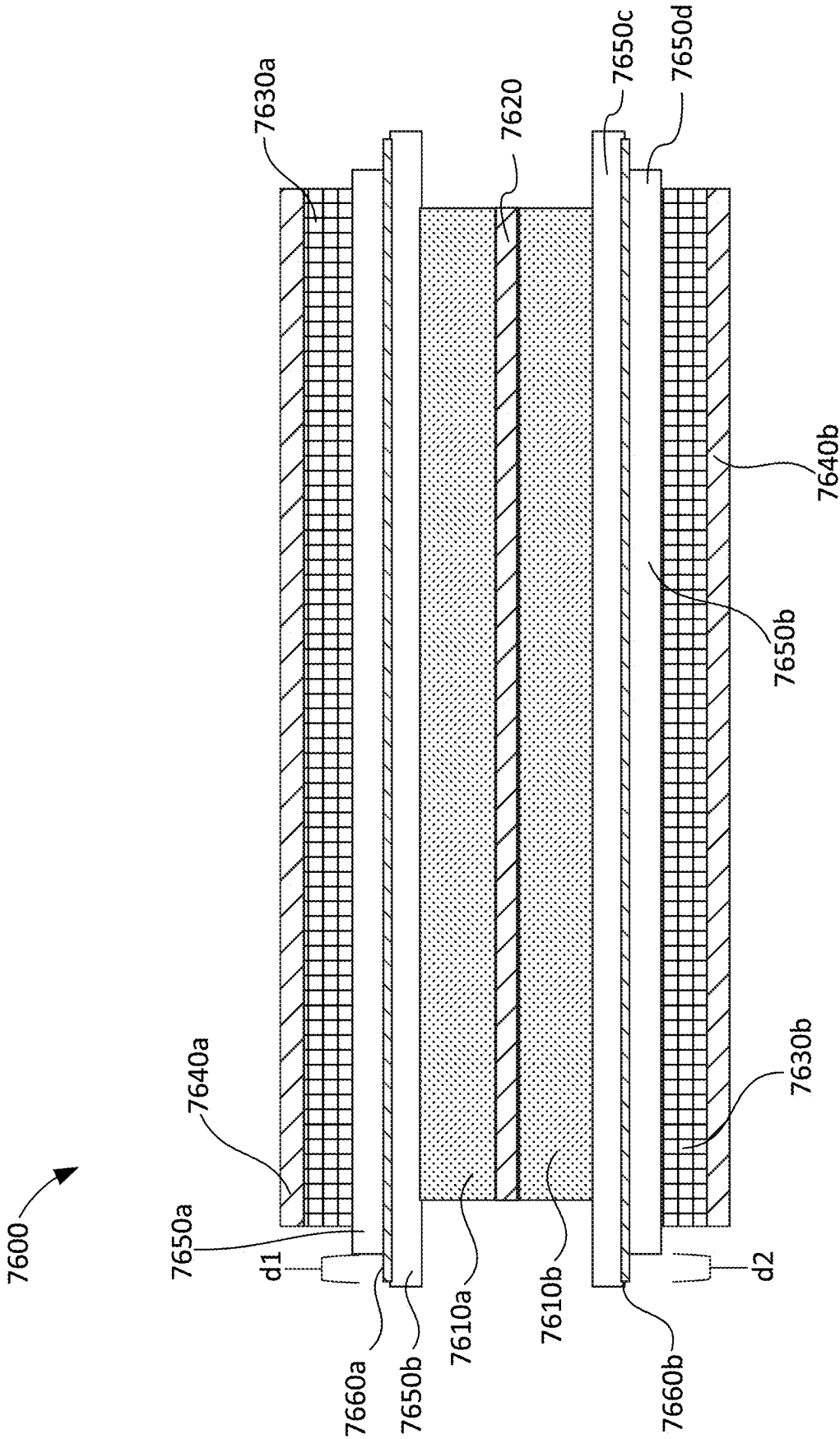


FIG. 76B

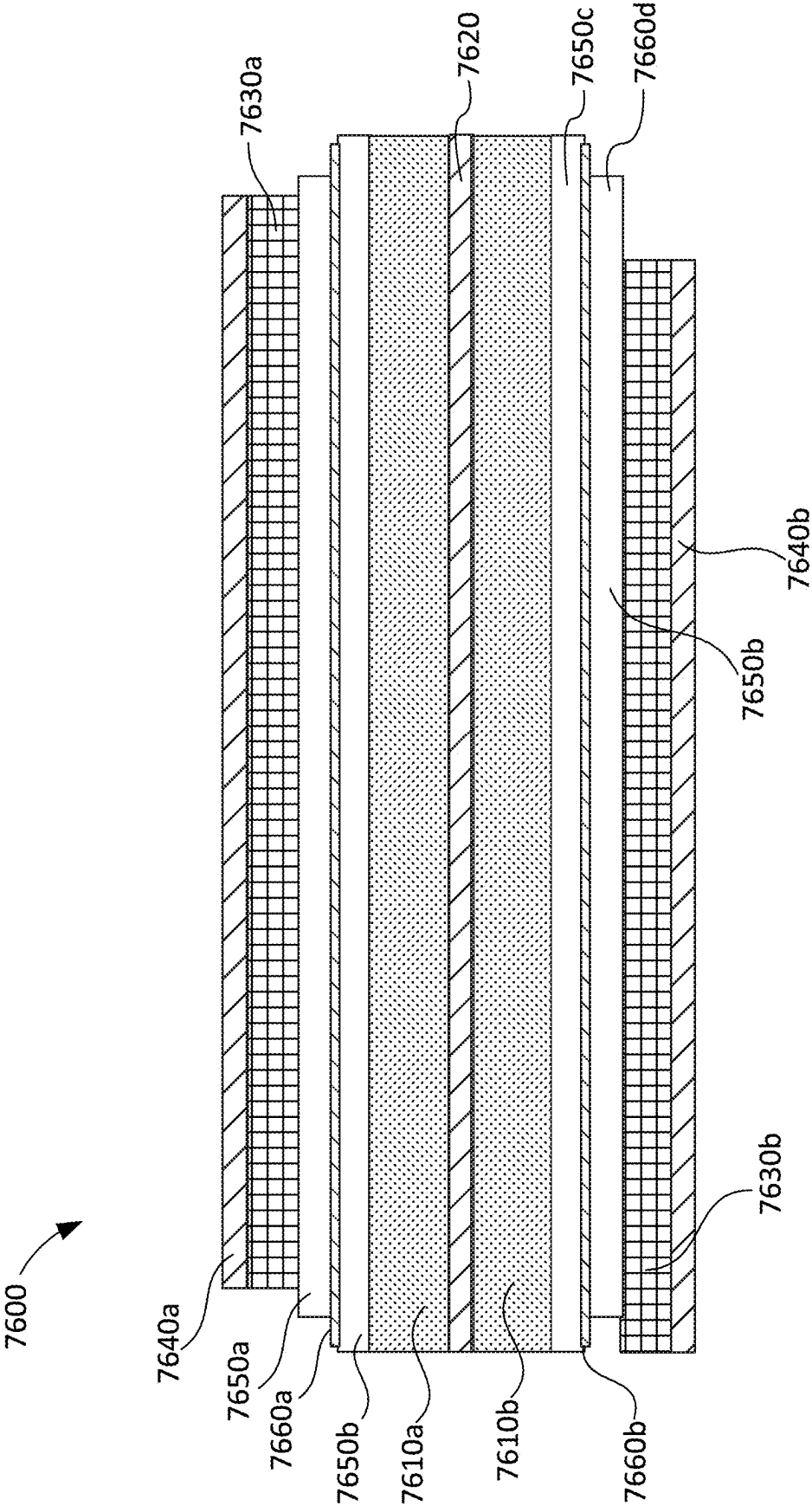


FIG. 76C

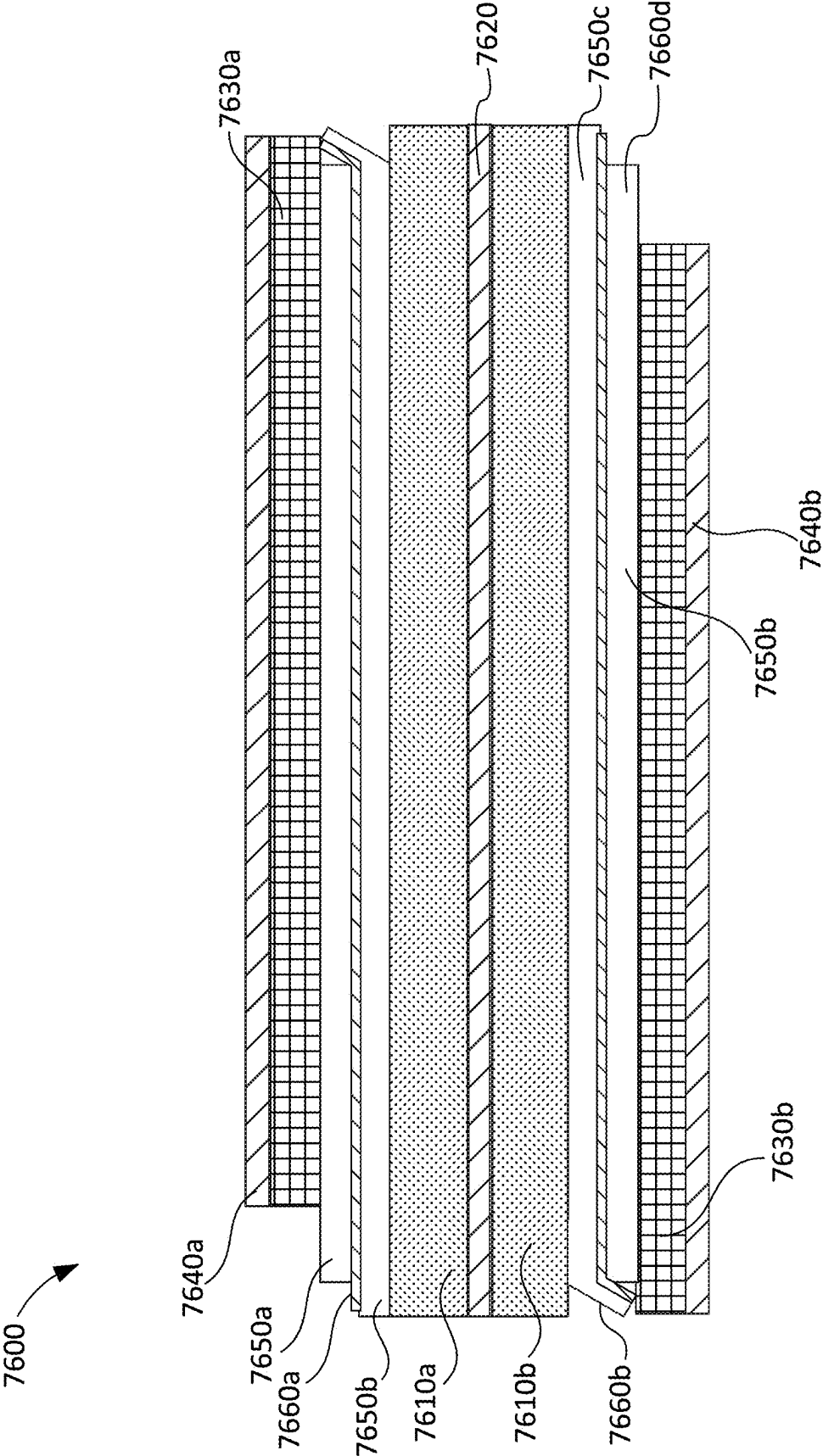


FIG. 77A

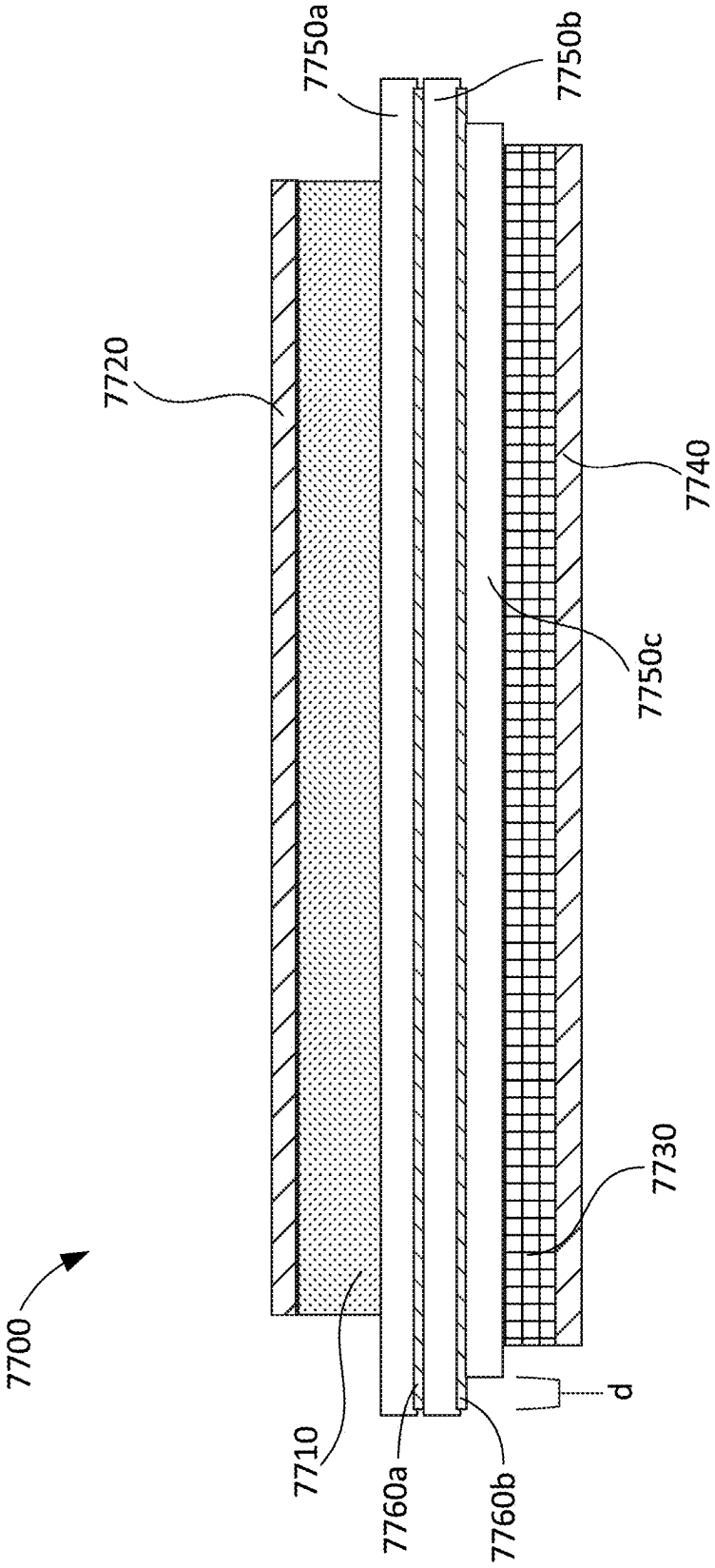


FIG. 77B

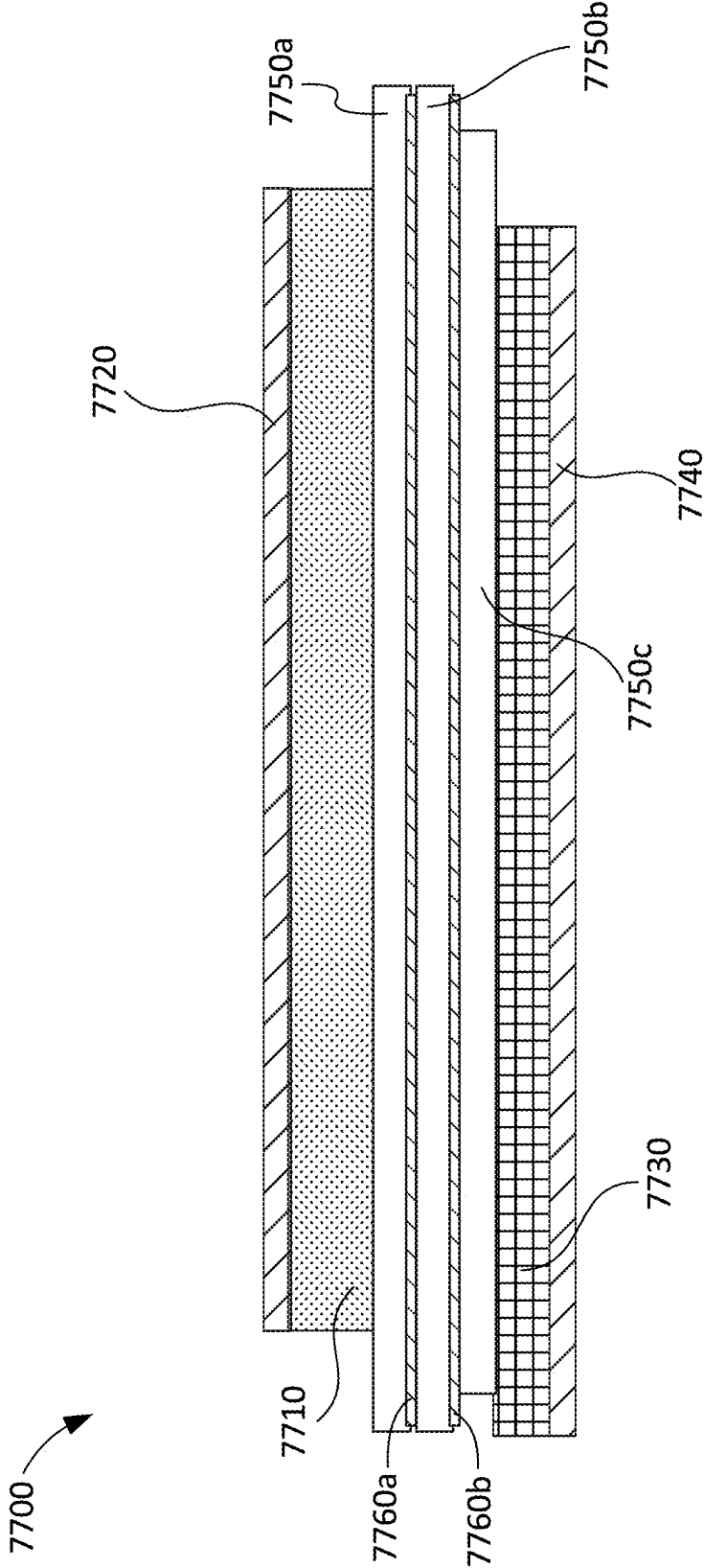


FIG. 77C

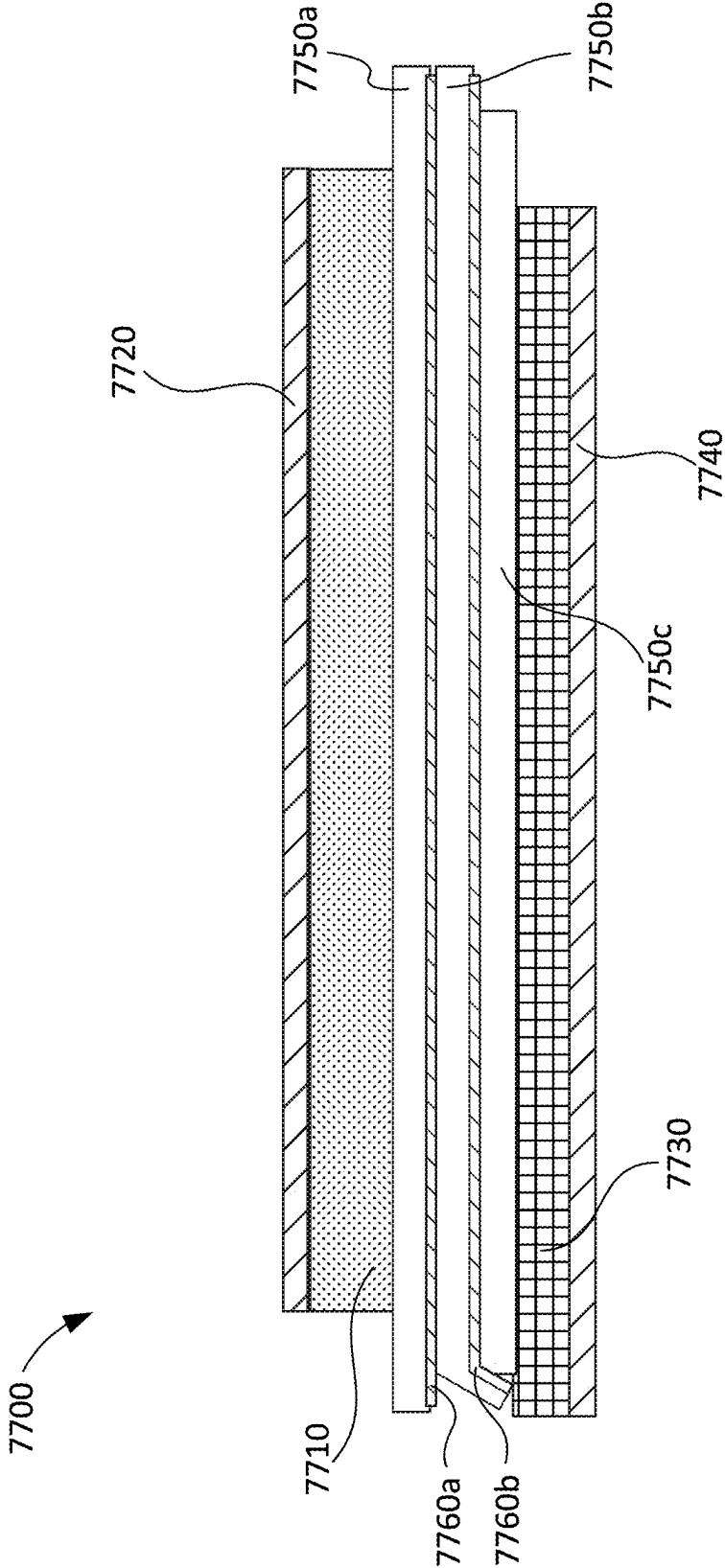


FIG. 78A

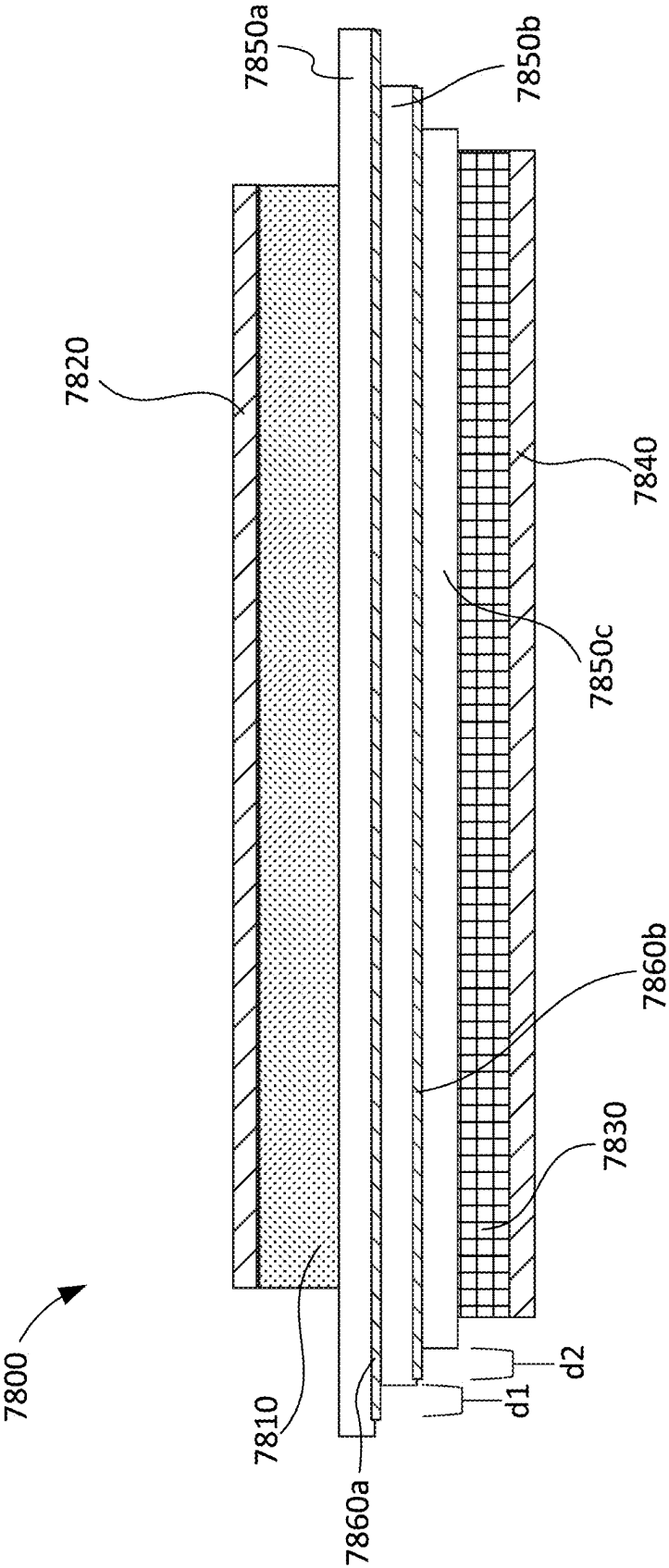


FIG. 78B

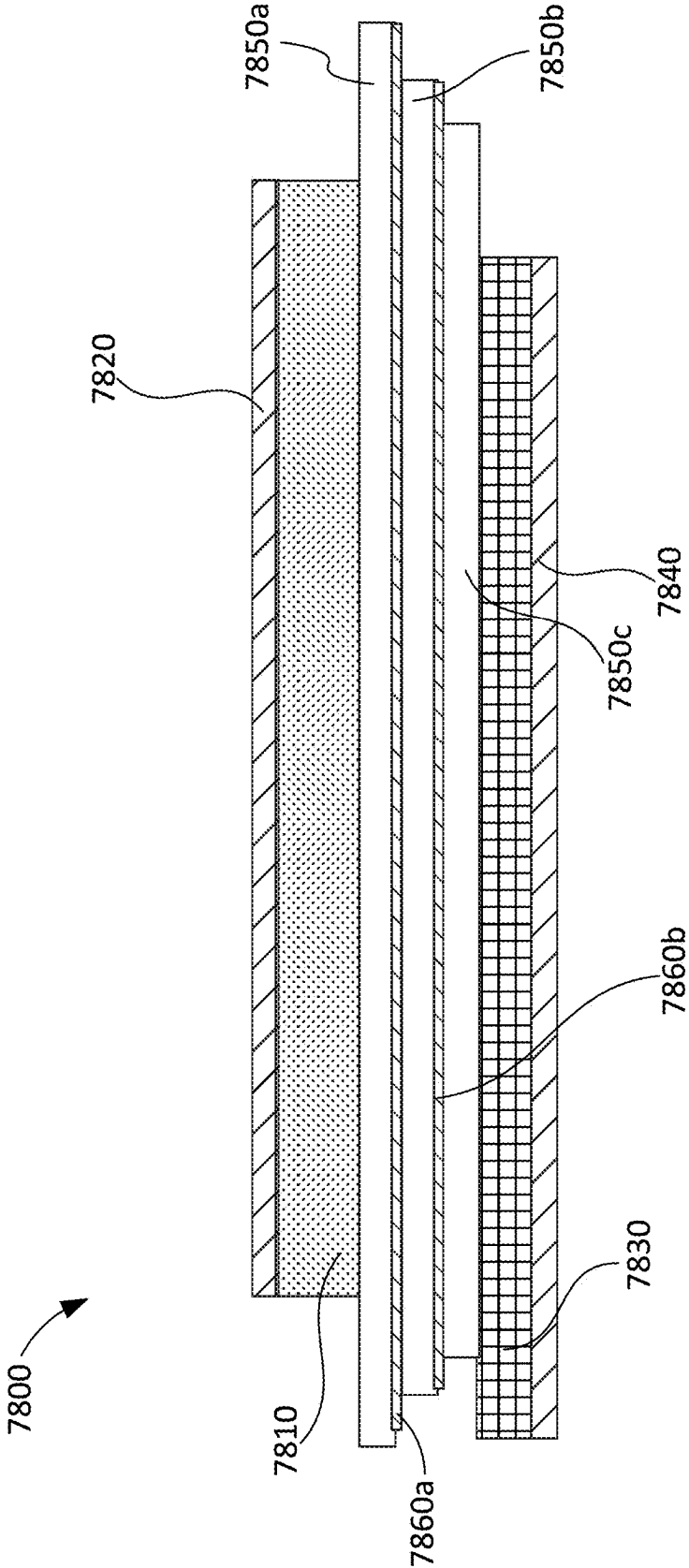


FIG. 78C

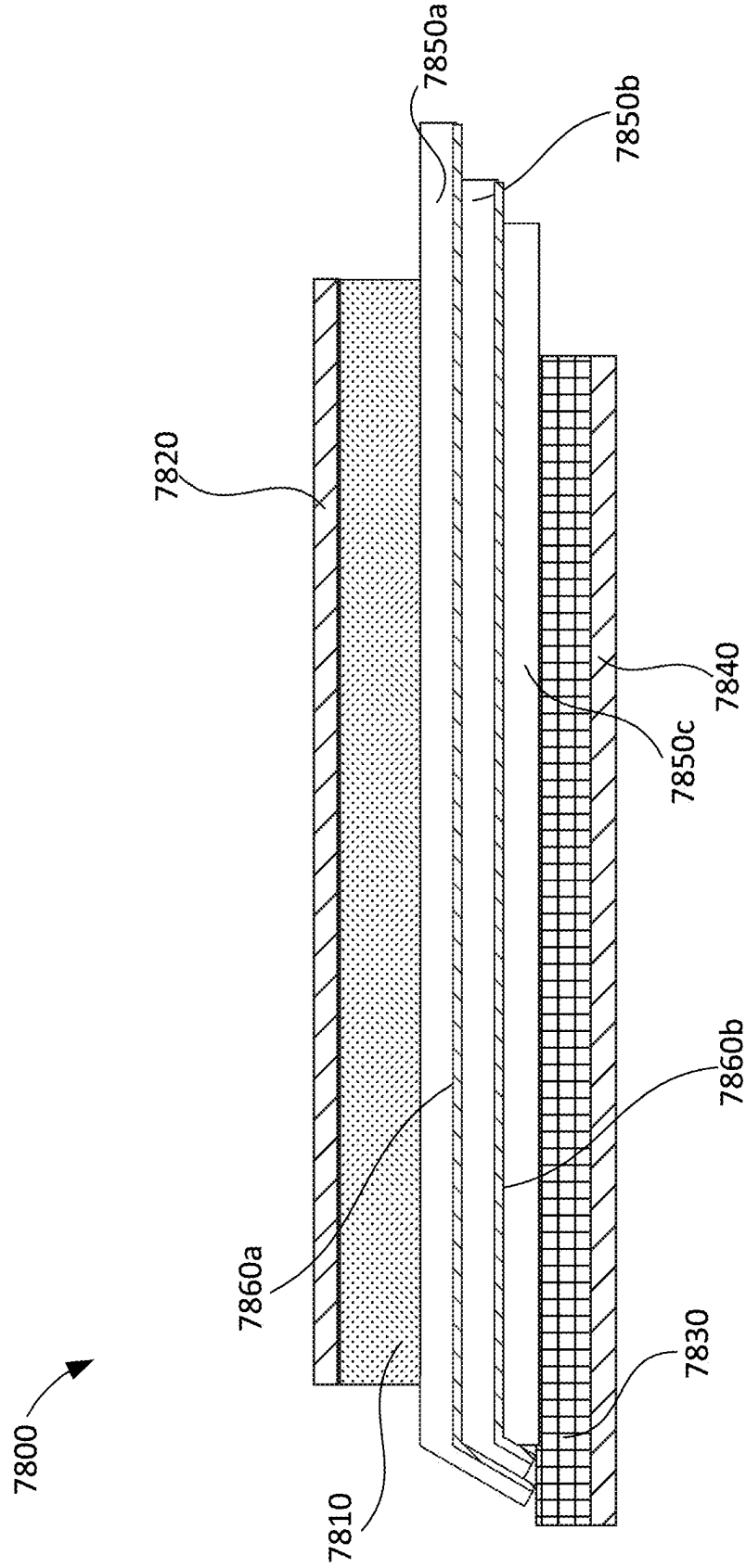
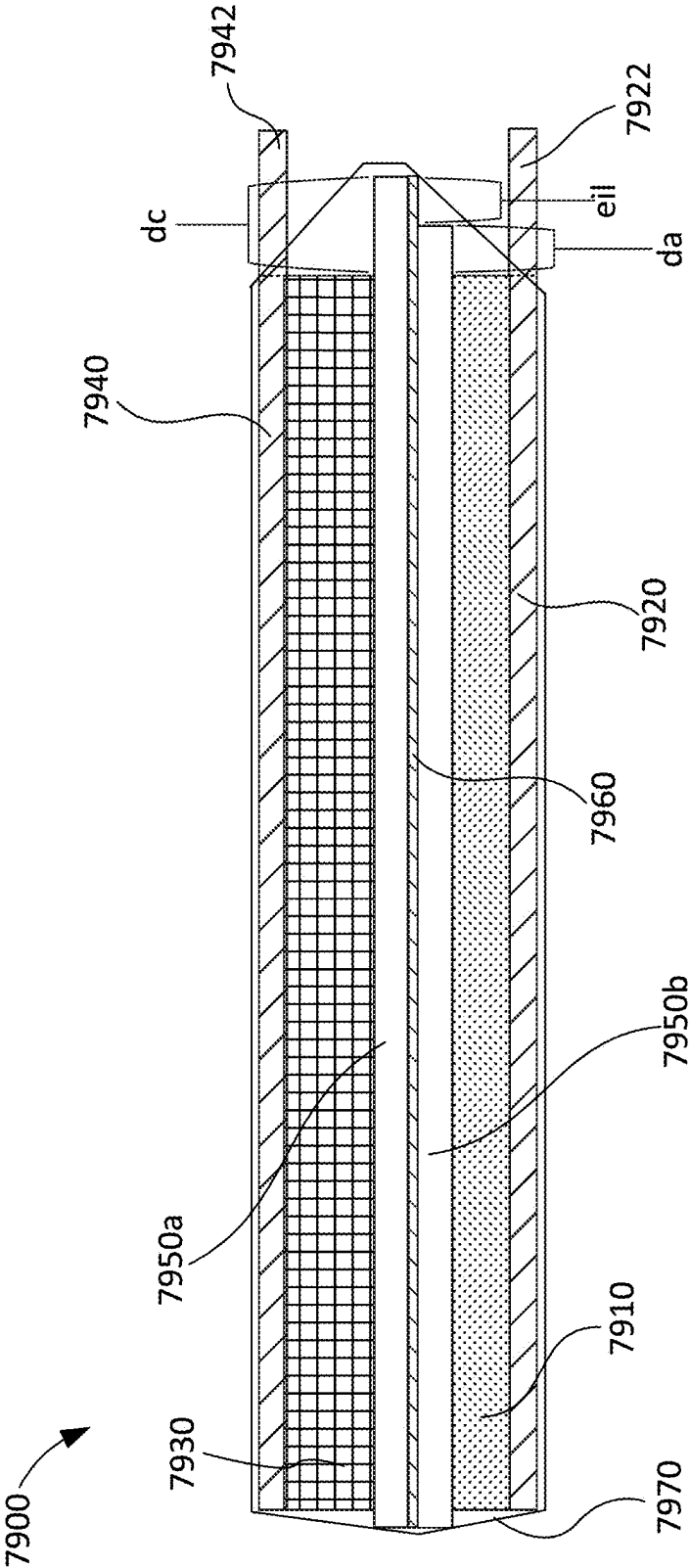


FIG. 79A



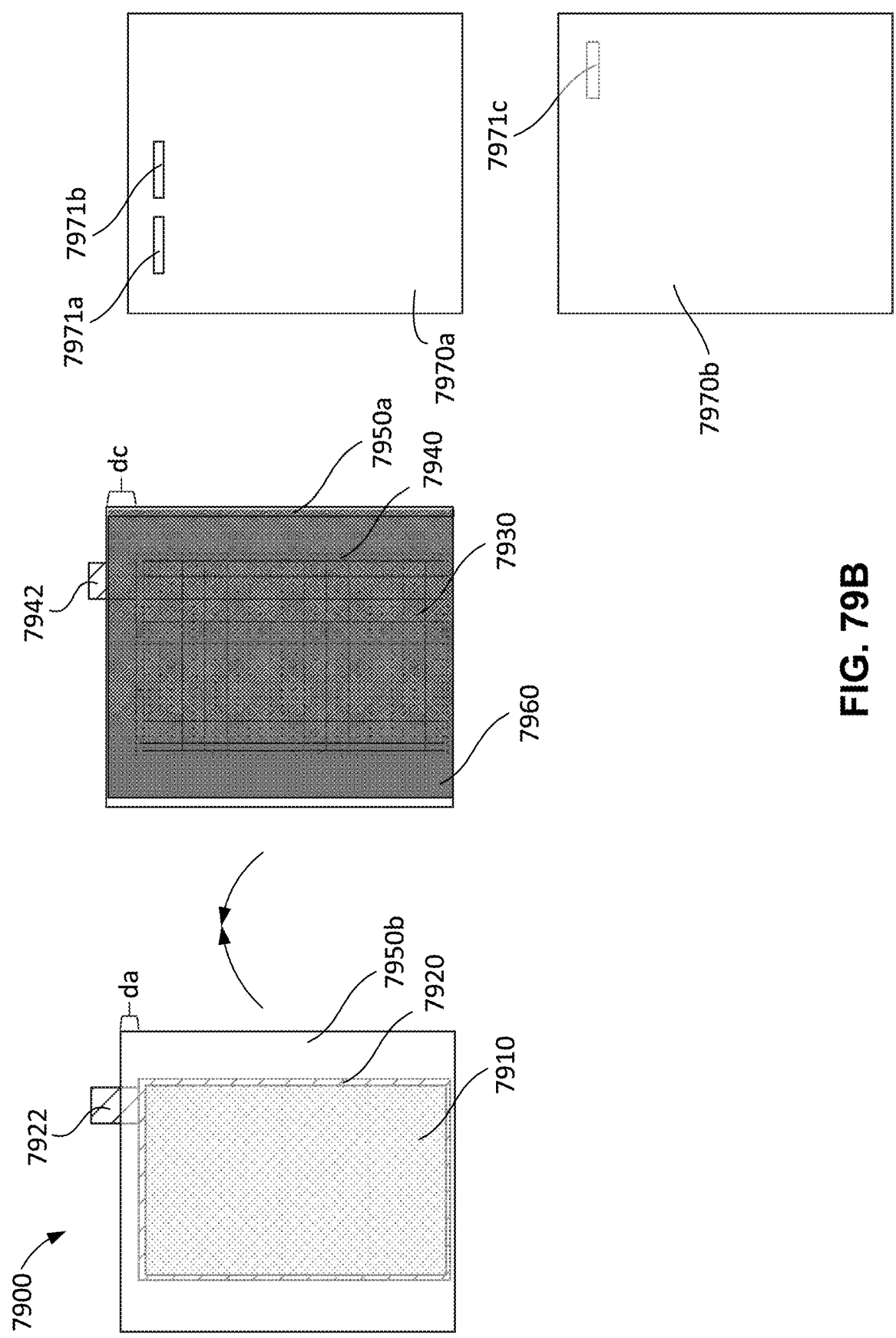


FIG. 79B

FIG. 80

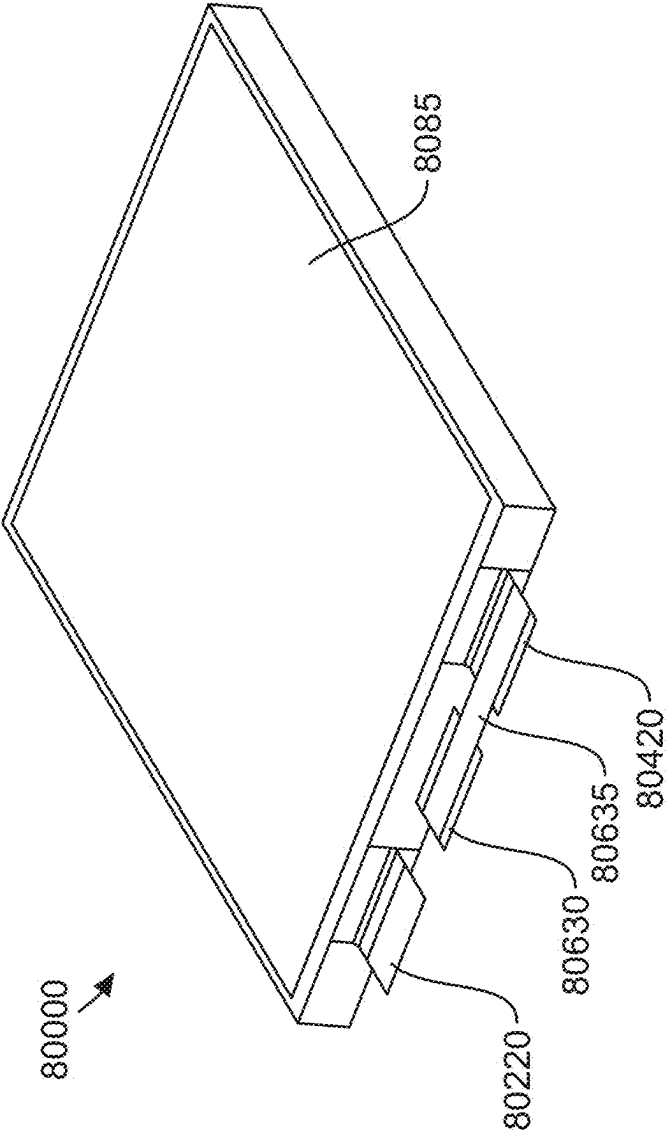


FIG. 81A

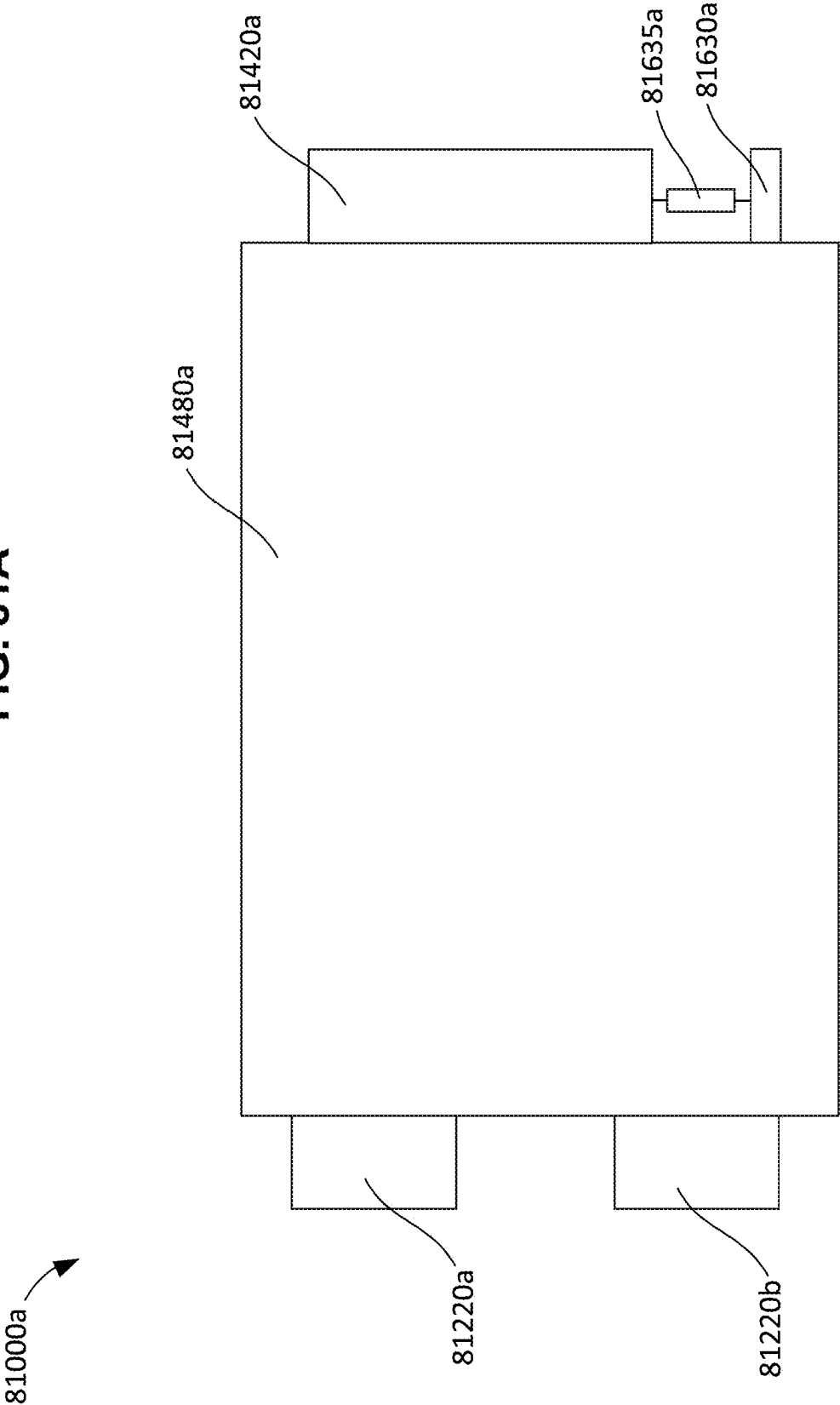


FIG. 81B

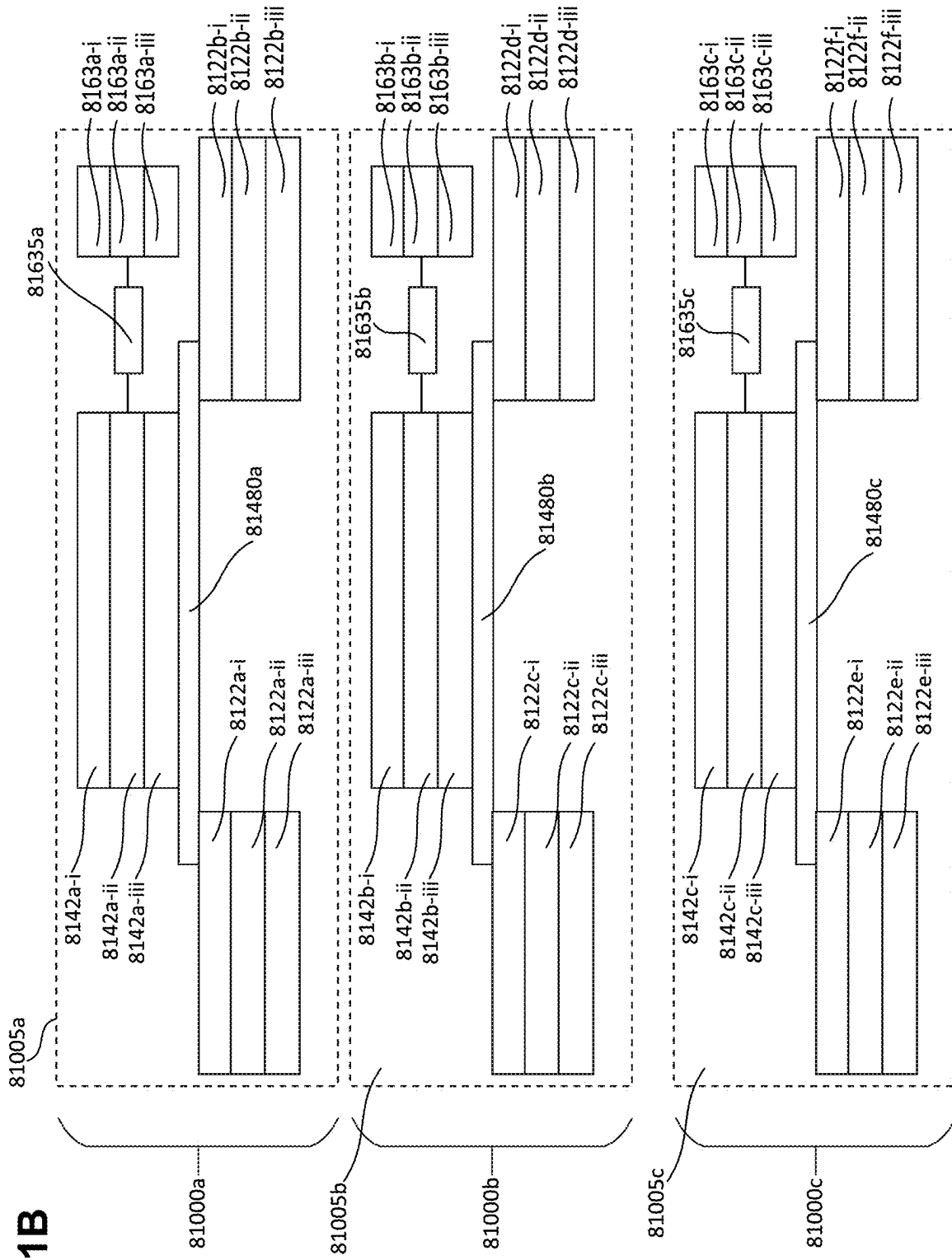
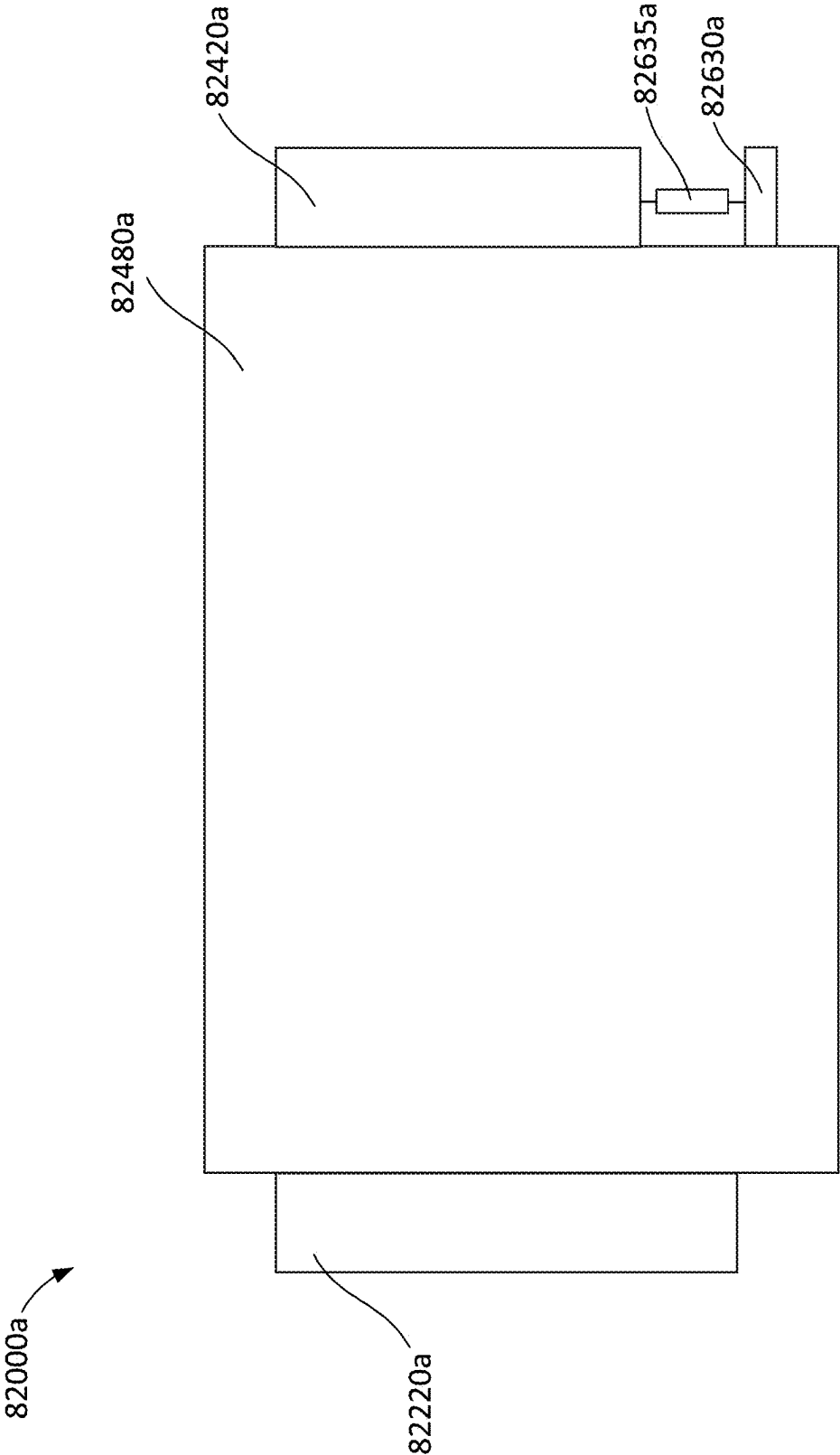


FIG. 82A



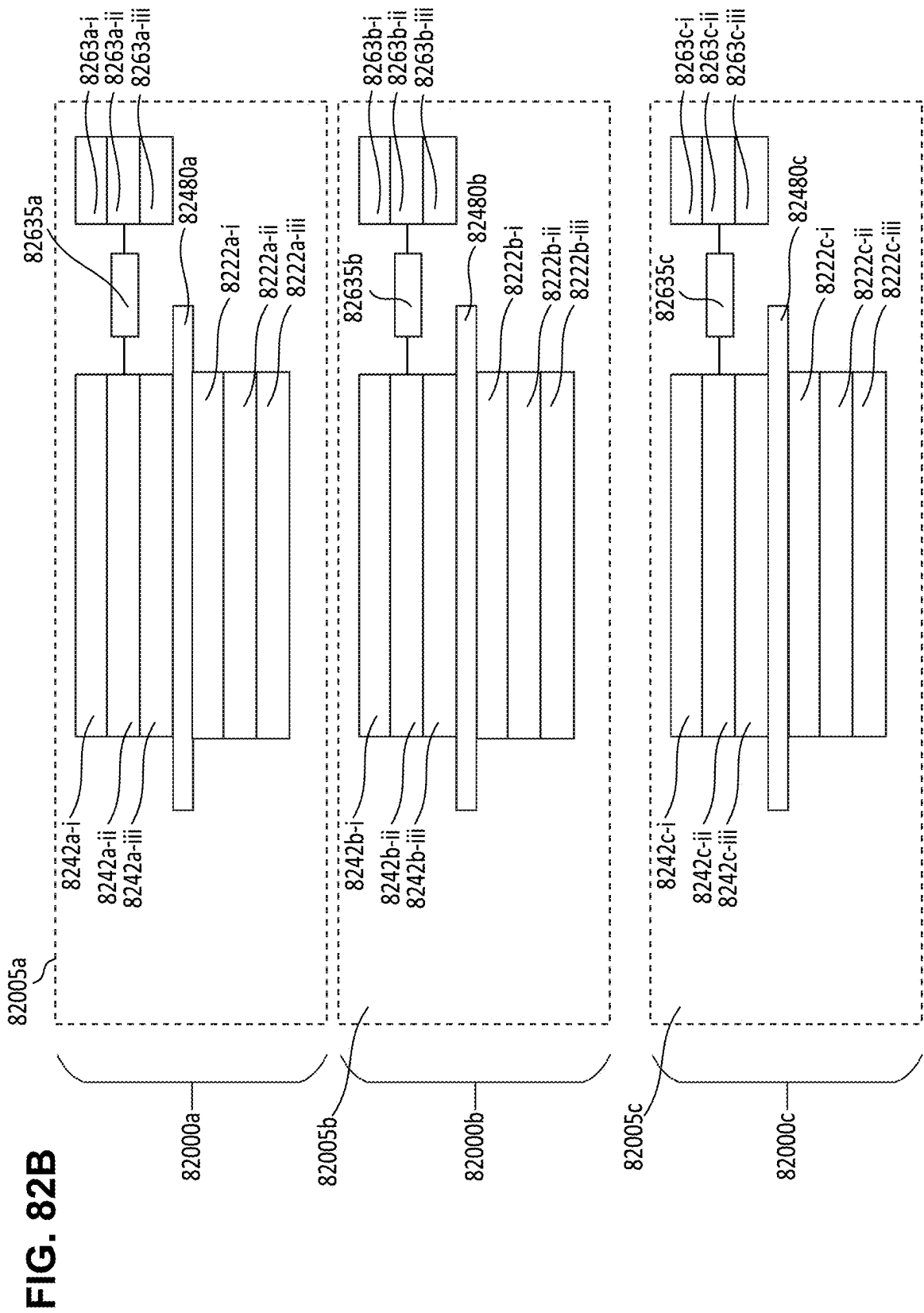


FIG. 83A

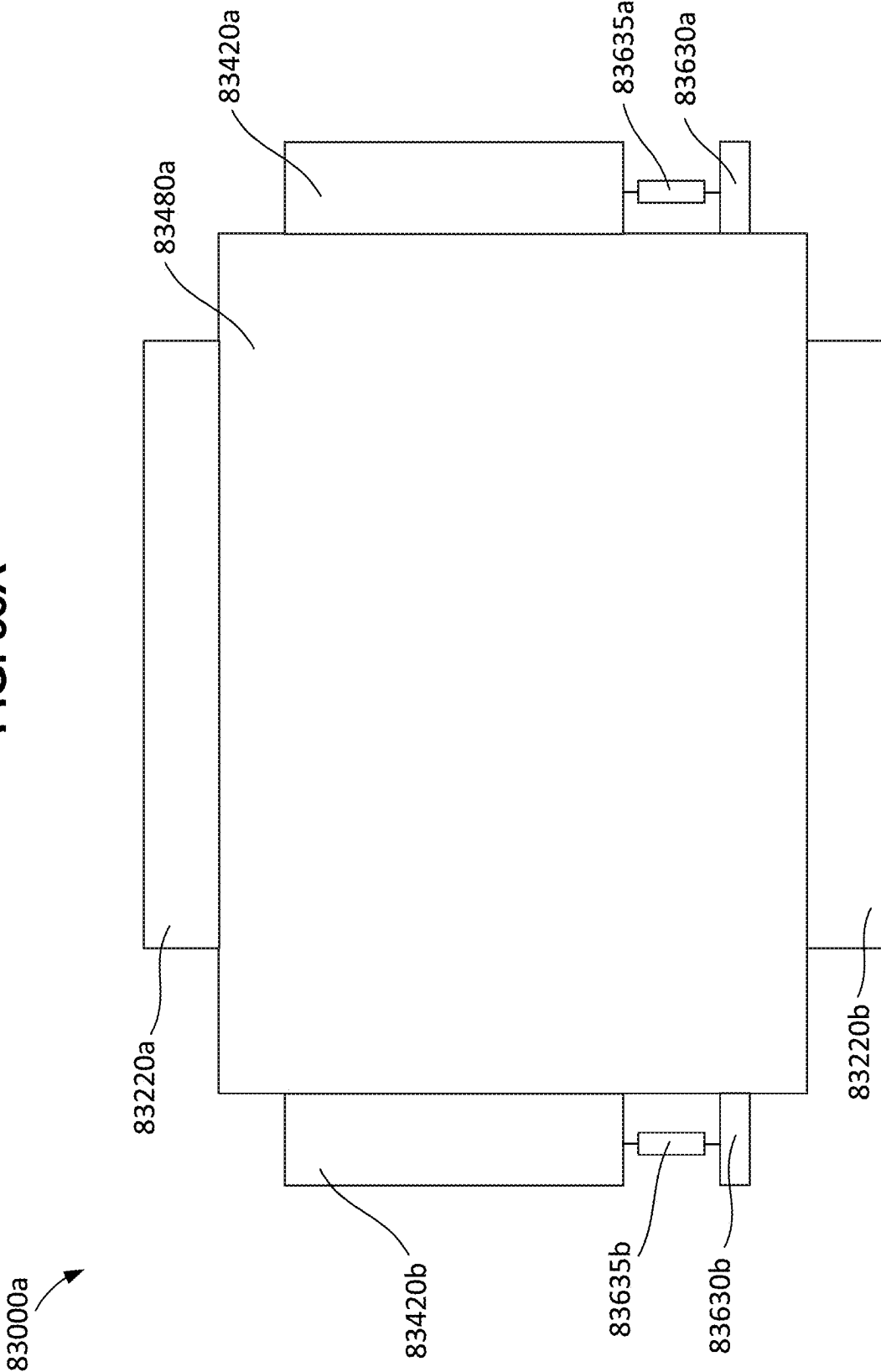


FIG. 83B

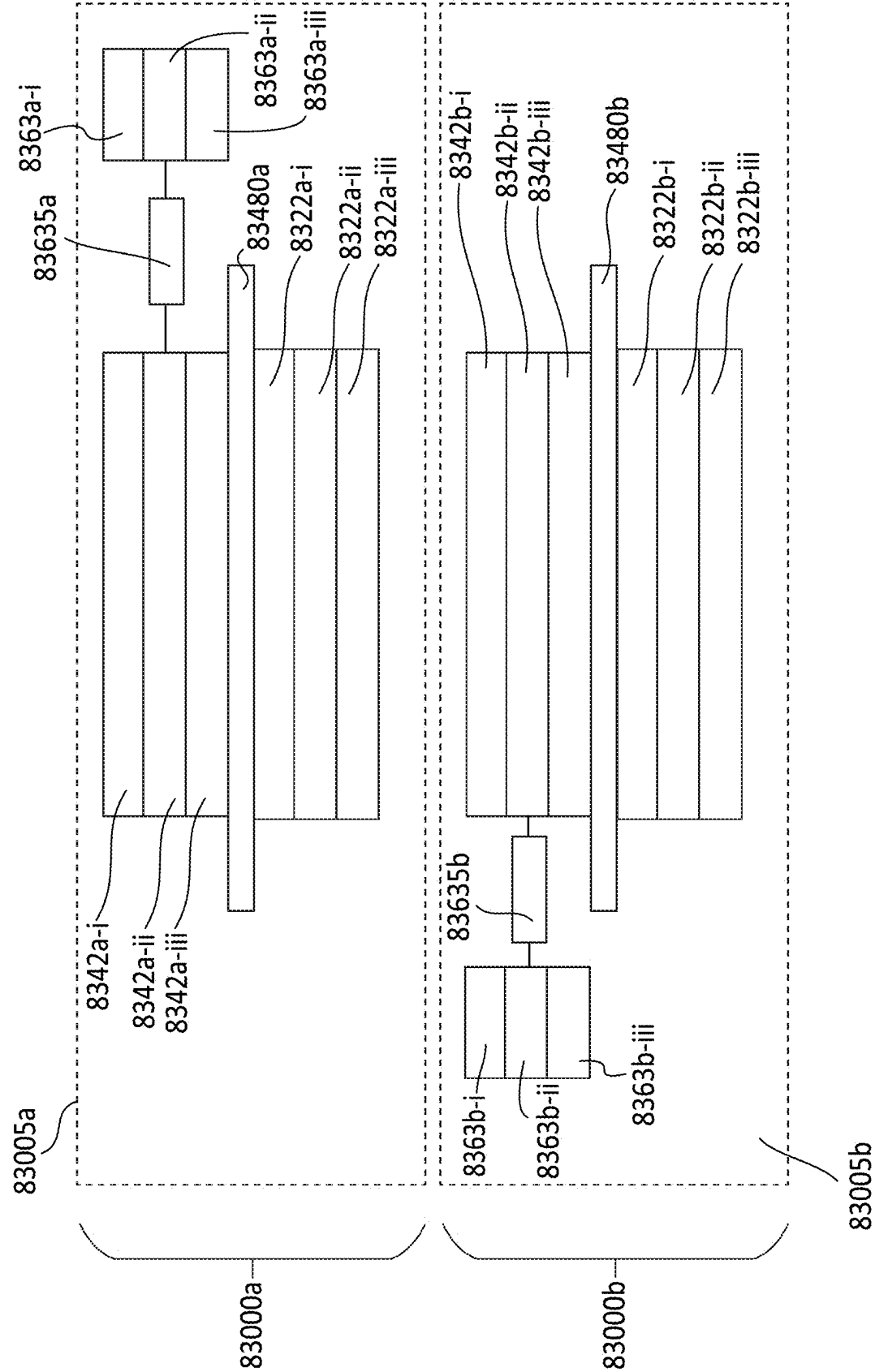


FIG. 84A

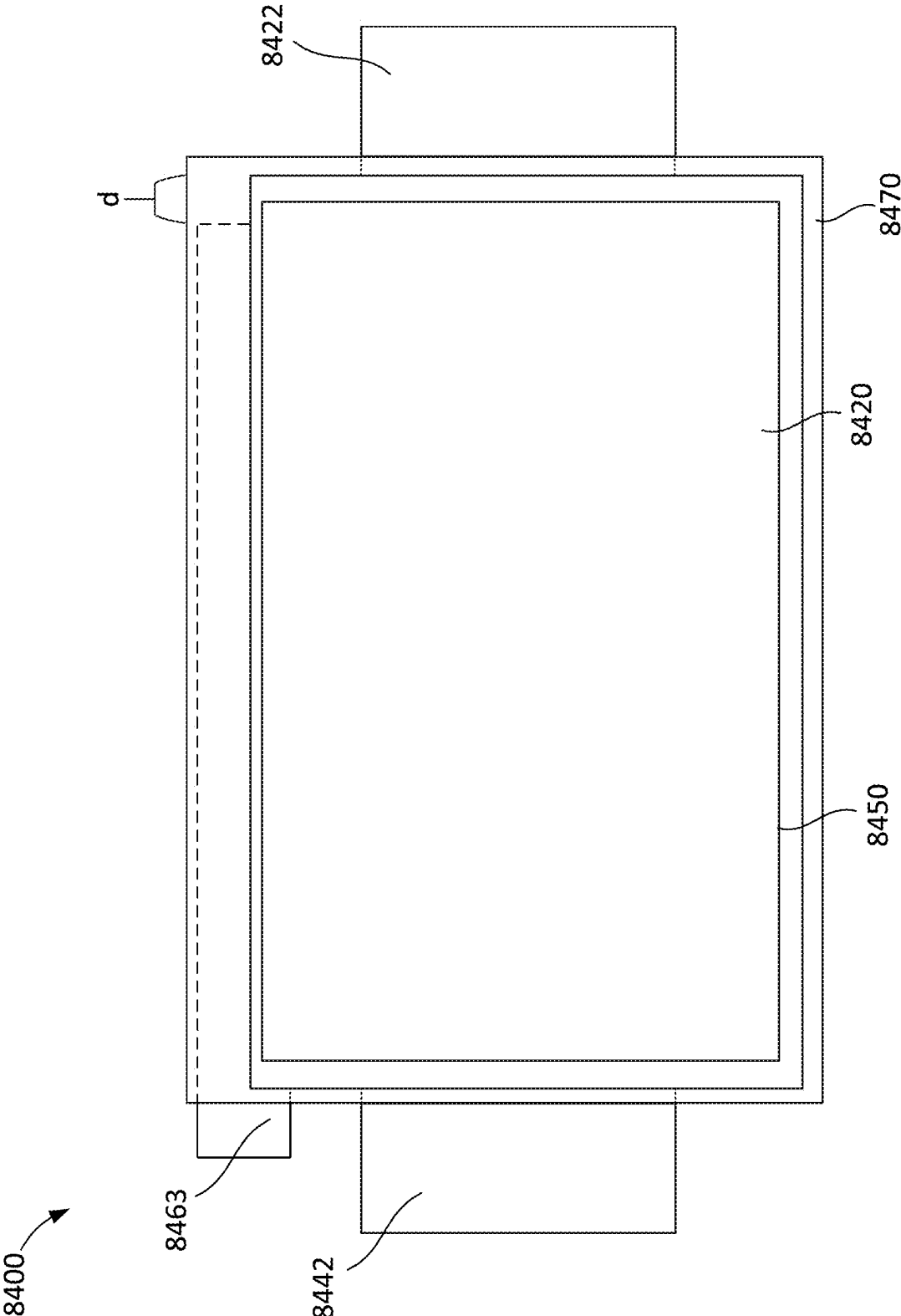


FIG. 84B

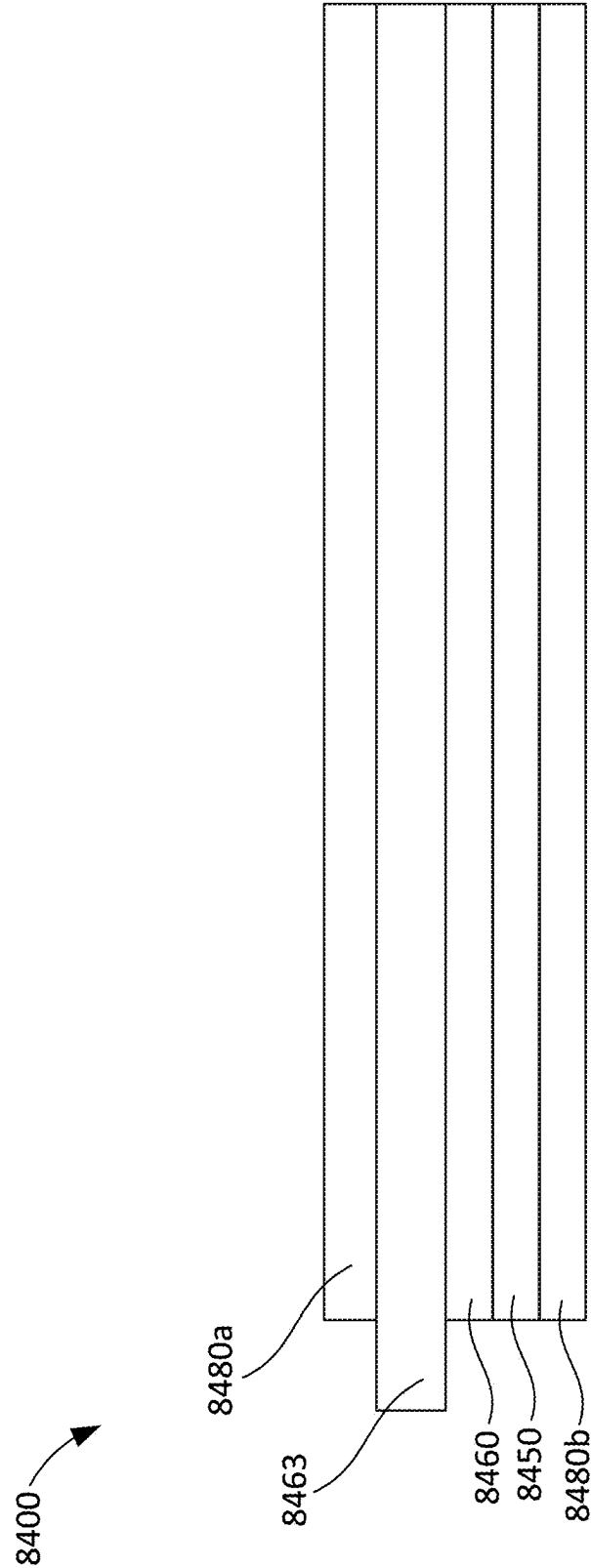


FIG. 85

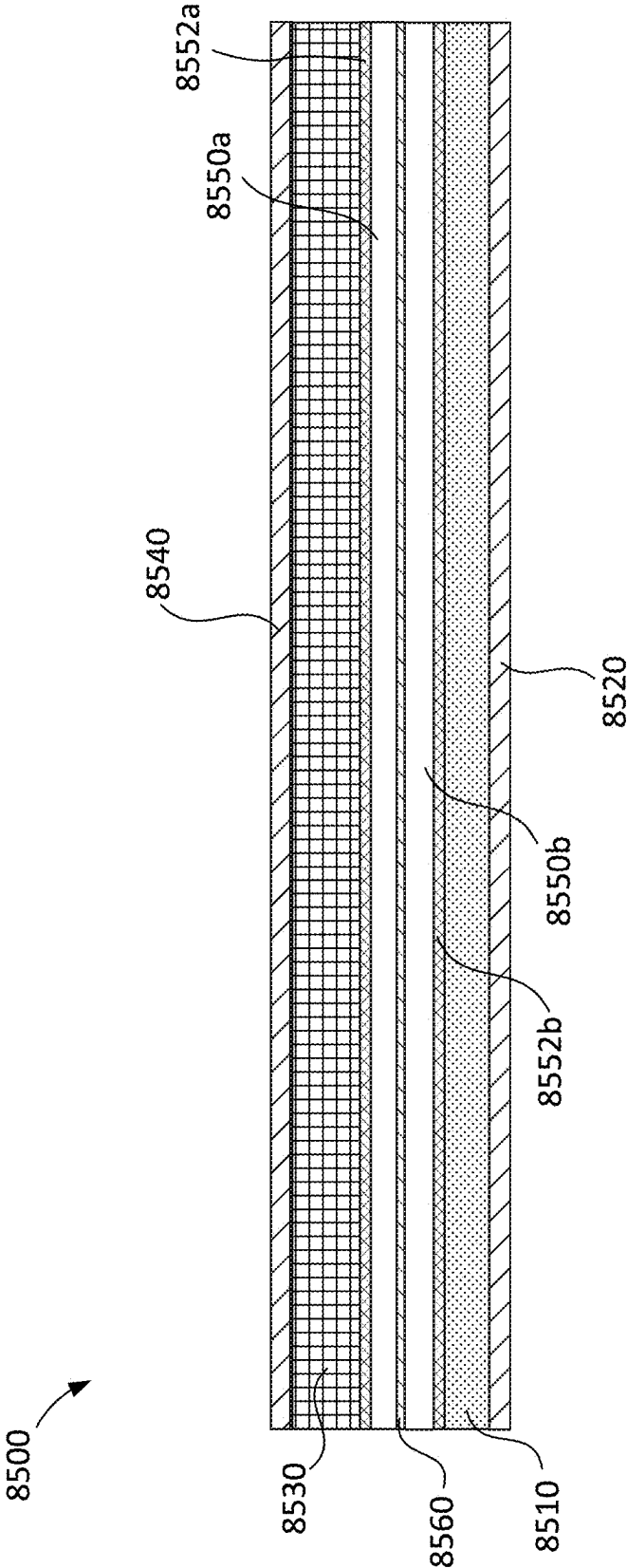


FIG. 86

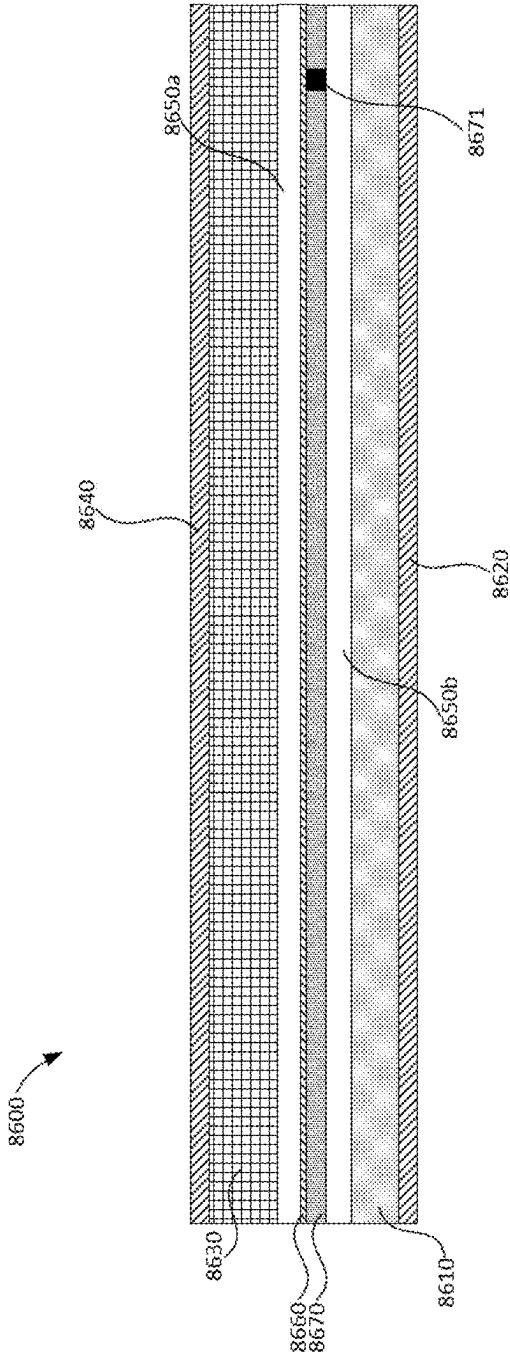
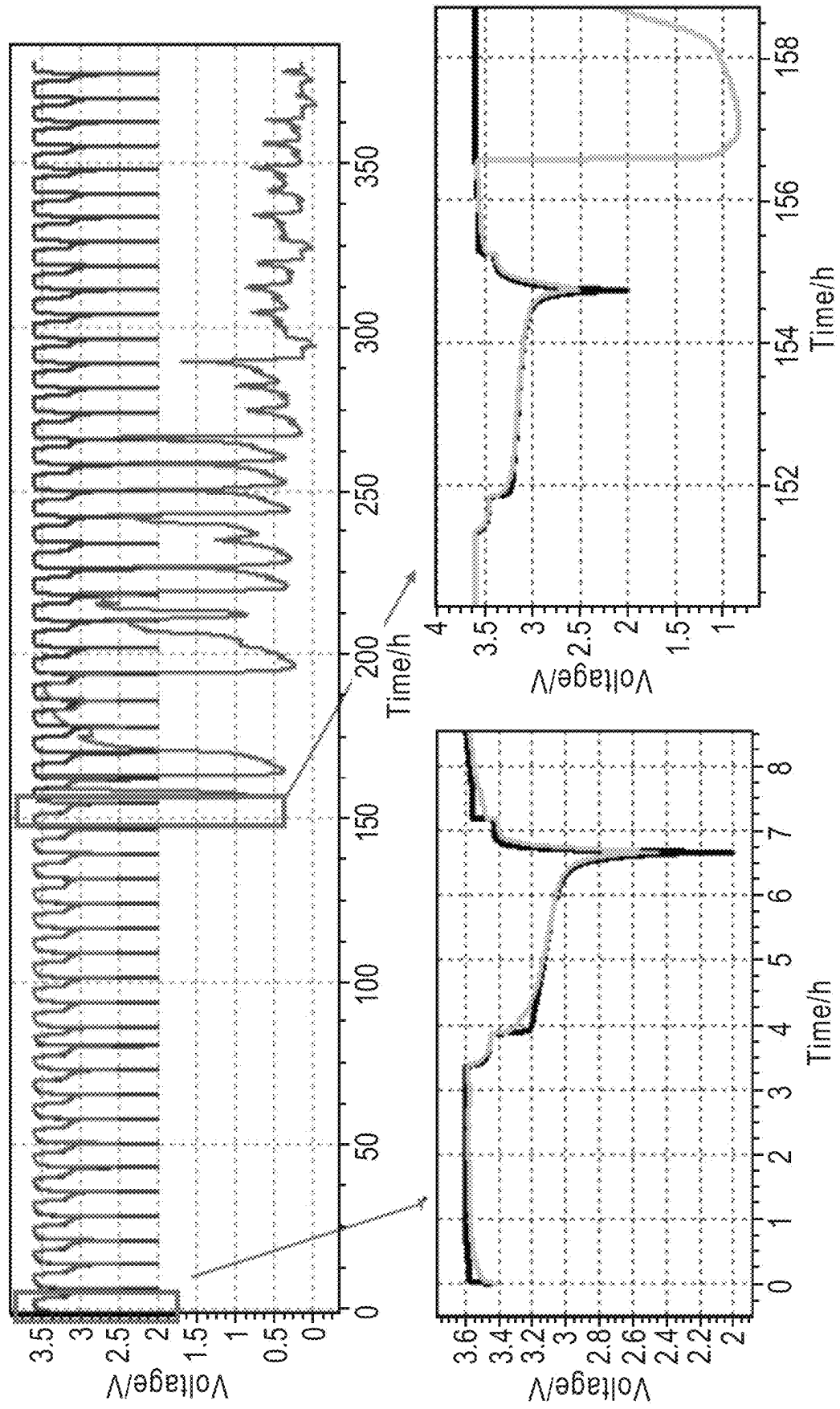


FIG. 87



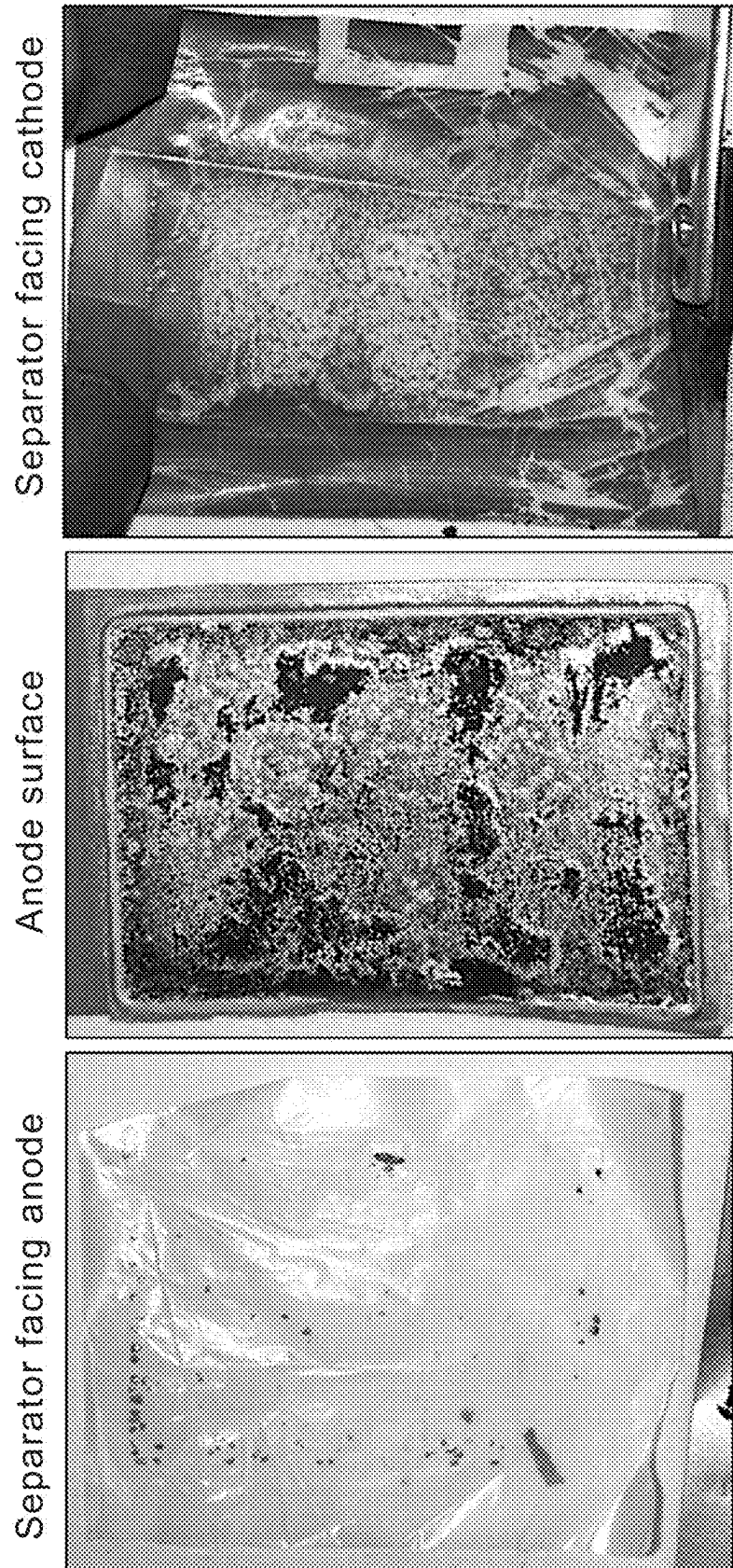


FIG. 88

FIG. 89

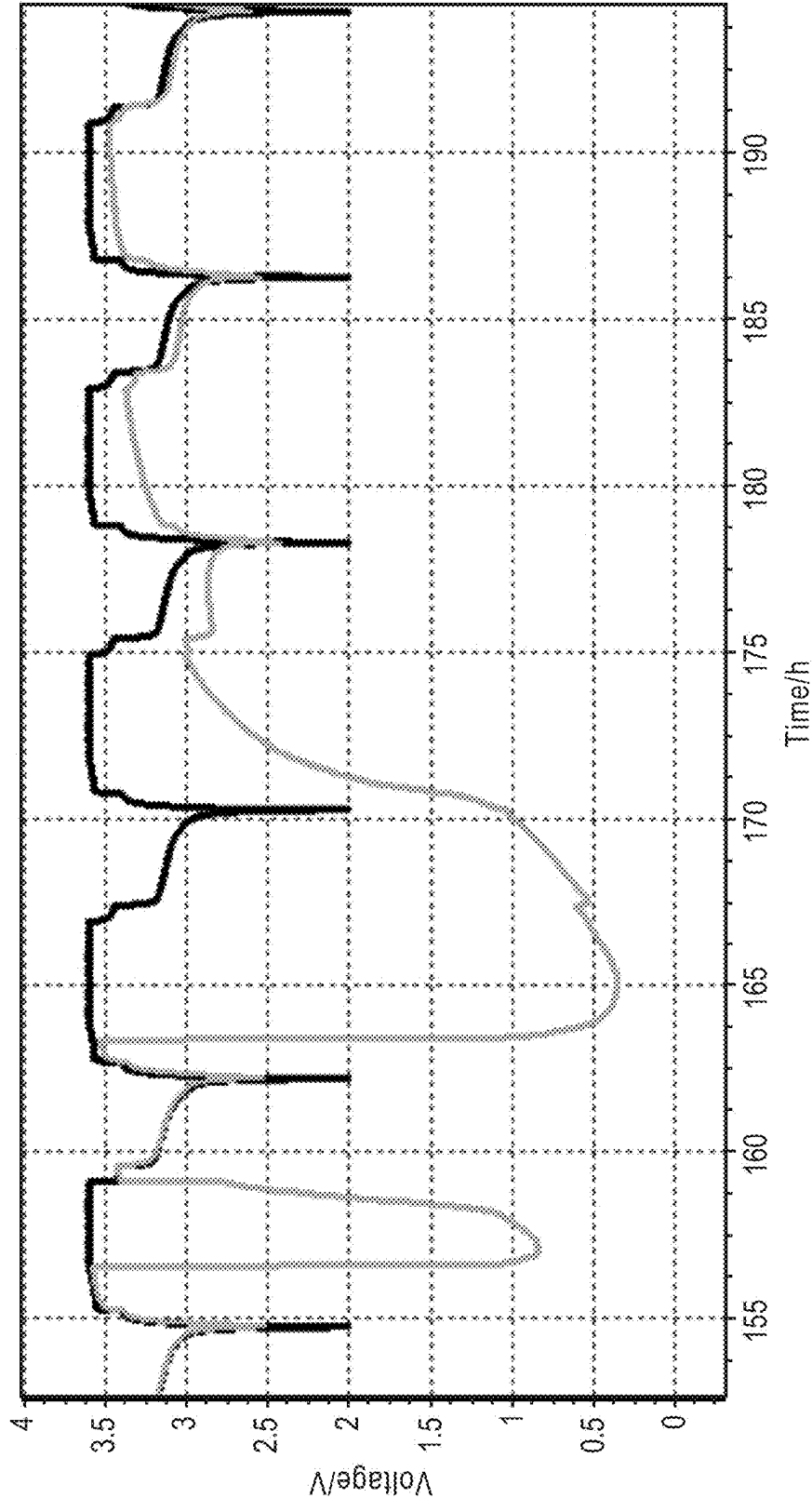


FIG. 90

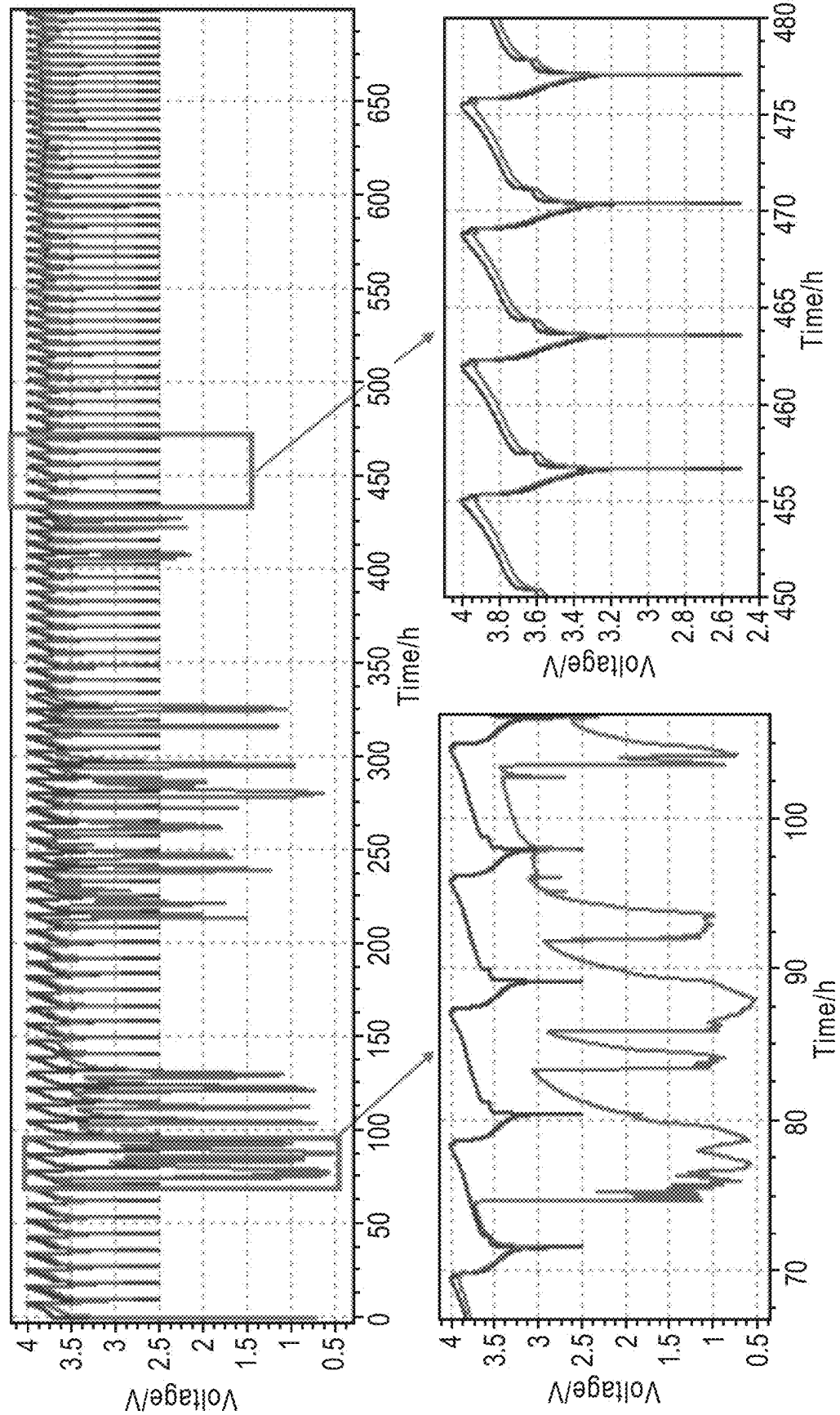


FIG. 91

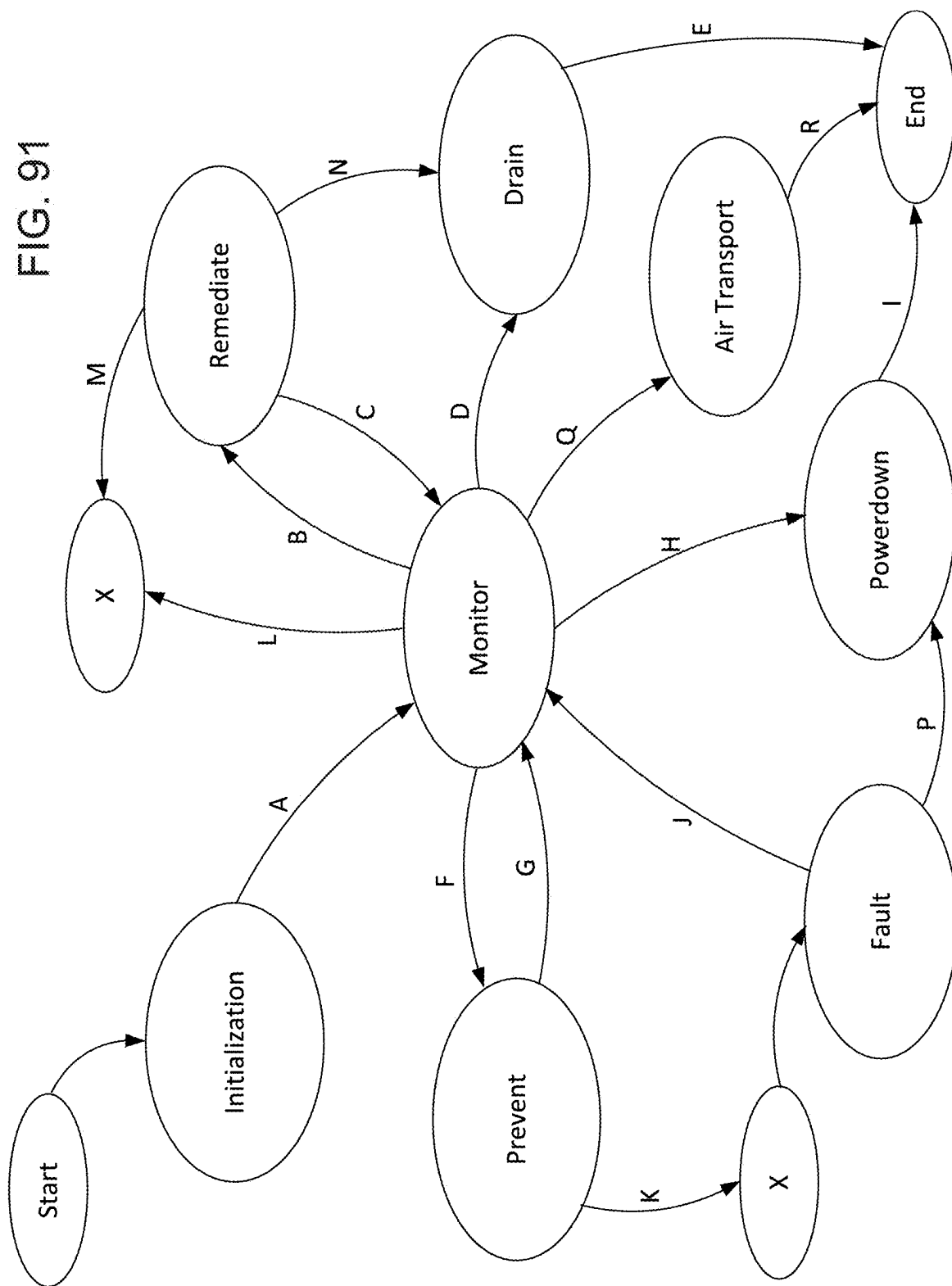


FIG. 92

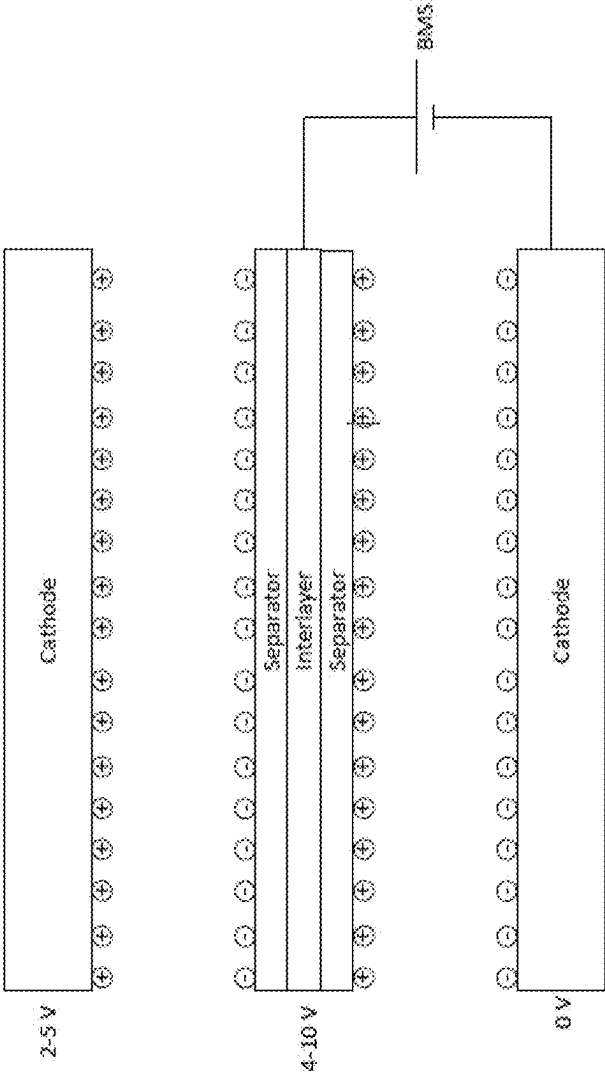
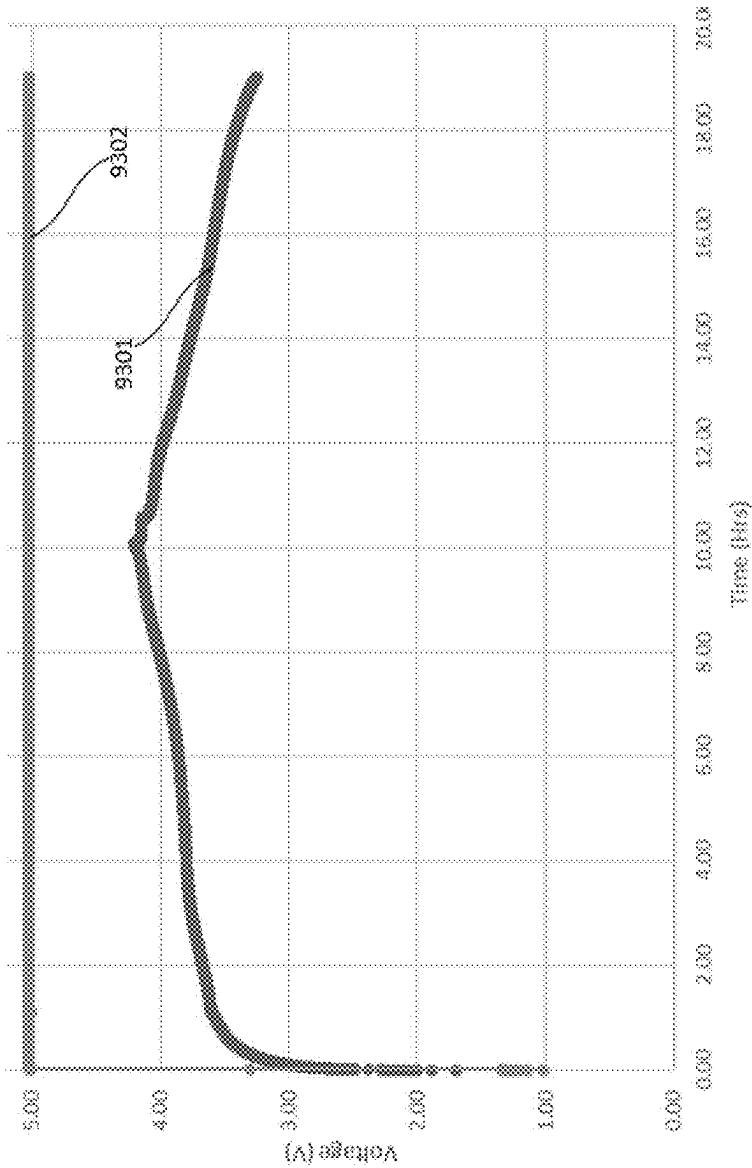


FIG. 93



SYSTEMS AND METHODS FOR MINIMIZING AND PREVENTING DENDRITE FORMATION IN ELECTROCHEMICAL CELLS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/741,181, titled “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” and filed on Jun. 12, 2024, which is a continuation of U.S. patent application Ser. No. 18/650,789, titled “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” and filed on Apr. 30, 2024, now U.S. Pat. No. 12,119,458, which is a continuation of U.S. patent application Ser. No. 18/543,515, titled “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” and filed on Dec. 18, 2023, now U.S. Pat. No. 11,984,564, which claims priority to and the benefit of U.S. Provisional Patent Application No. 63/433,269, filed Dec. 16, 2022 and titled, “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” U.S. Provisional Patent Application No. 63/450,208, filed Mar. 6, 2023 and titled, “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” U.S. Provisional Patent Application No. 63/461,506, filed Apr. 24, 2023 and titled, “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” U.S. Provisional Patent Application No. 63/470,679, filed Jun. 2, 2023 and titled, “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” U.S. Provisional Patent Application No. 63/528,213, filed Jul. 21, 2023 and titled, “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” and U.S. Provisional Patent Application No. 63/546,980, filed Nov. 2, 2023 and titled, “Systems and Methods for Minimizing and Preventing Dendrite Formation in Electrochemical Cells,” the disclosures of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] Embodiments described herein relate to electrochemical cells formulated to minimize damage from dendrite formation.

BACKGROUND

[0003] Dendrite formation in electrochemical cells can lead to short circuiting and heat generation. Heat generation in electrochemical cells is a safety issue that can have dangerous results. Thermal runaway can lead to fires and thermal decomposition of the electrochemical cell materials. By minimizing the size to which dendrites can grow, significant safety issues can be avoided.

SUMMARY

[0004] Embodiments described herein relate to electrochemical cells with dendrite prevention mechanisms, and methods of producing and operating the same. In some aspects, an electrochemical cell can include an anode disposed on an anode current collector, a cathode disposed on a cathode current collector, the cathode having a first thick-

ness at a proximal end of the cathode and a second thickness at a distal end of the cathode, the second thickness greater than the first thickness, a first separator disposed on the anode, a second separator disposed on the cathode, an interlayer disposed between the first separator and the second separator, the interlayer including electroactive material and having a proximal end and a distal end, and a power source electrically connected to the proximal end of the cathode and the proximal end of the interlayer, the power source configured to maintain a voltage difference between the cathode and the interlayer below a threshold value. In some embodiments, the threshold value can be about 0.01 V. In some embodiments, the interlayer can include $\text{Li}_{(1-x)}\text{NMC}$ wherein x is between 0 and 1. In some embodiments, the anode can include graphite. In some embodiments, the second thickness can be greater than the first thickness by about 500 nm to about 5 μm .

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a block diagram of an electrochemical cell with an interlayer, according to an embodiment.

[0006] FIGS. 2A-2B are illustrations of an electrochemical cell with an interlayer, according to an embodiment.

[0007] FIGS. 3A-3B are illustrations of an electrochemical cell with an interlayer, according to an embodiment.

[0008] FIGS. 4A-4B are illustrations of an electrochemical cell with multiple interlayers, according to an embodiment.

[0009] FIG. 5 is an illustration of an electrochemical cell with an interlayer, according to an embodiment.

[0010] FIG. 6 is an illustration of an electrochemical cell with an interlayer, according to an embodiment.

[0011] FIGS. 7A-7B are illustrations of separators with an interlayer disposed therebetween, according to an embodiment.

[0012] FIGS. 8A-8B are illustrations of separators with an interlayer disposed therebetween, according to an embodiment.

[0013] FIGS. 9A-9B are illustrations of separators with an interlayer disposed therebetween, according to an embodiment.

[0014] FIGS. 10A-10C are illustrations of separators with an interlayer disposed therebetween, according to an embodiment.

[0015] FIGS. 11A-11B are illustrations of separators with an interlayer disposed therebetween, according to an embodiment.

[0016] FIG. 12 is a flow diagram of a method of operating an electrochemical cell, according to an embodiment.

[0017] FIG. 13 is an illustration of an electrochemical cell with an interlayer, according to an embodiment.

[0018] FIG. 14 is an illustration of an electrochemical cell with an interlayer, according to an embodiment.

[0019] FIGS. 15A-15B are illustrations of an electrochemical cell with an interlayer, according to an embodiment.

[0020] FIGS. 16A-16B are illustrations of an electrochemical cell stack with an interlayer, according to an embodiment.

[0021] FIGS. 17A-17B are illustrations of an electrochemical cell with an interlayer, according to an embodiment.

[0022] FIG. 18 is an illustration of an electrochemical cell with an interlayer, according to an embodiment.

[0083] FIGS. 79A-79B are illustrations of an electrochemical cell with an interlayer, according to an embodiment.

[0084] FIG. 80 is an illustration of an electrochemical cell stack, according to an embodiment.

[0085] FIGS. 81A-81B are illustrations of a collection of electrochemical cell stacks, according to an embodiment.

[0086] FIGS. 82A-82B are illustrations of a collection of electrochemical cell stacks, according to an embodiment.

[0087] FIGS. 83A-83B are illustrations of a collection of electrochemical cell stacks, according to an embodiment.

[0088] FIGS. 84A-84B are illustrations of an electrochemical cell, according to an embodiment.

[0089] FIG. 85 is an illustration of an electrochemical cell, according to an embodiment.

[0090] FIG. 86 is an illustration of an electrochemical cell, according to an embodiment.

[0091] FIG. 87 is a graphic representation of a point in time when a dendrite forms and interfaces with the interlayer.

[0092] FIG. 88 is a photograph of dendritic formation at interfaces with an interlayer.

[0093] FIG. 89 is a graphic representation of an electrochemical cell with an interlayer maintained with a passive resistor.

[0094] FIG. 90 is a graphical representation of a prevention of a dendrite from penetrating a separator and creating a localized short circuit.

[0095] FIG. 91 is a sample control operating mode state transition diagram.

[0096] FIG. 92 is an illustration of the formation of a dual ion system.

[0097] FIG. 93 is a graphical representation of a cell voltage and an interlayer voltage.

DETAILED DESCRIPTION

[0098] Embodiments described herein relate to cells with interlayers, and methods of operating the same. An interlayer can include a layer of electroactive material placed between an anode and a cathode of an electrochemical cell. The interlayer can be disposed between a first separator and a second separator. Interlayers can be used to detect dendrites before they grow too large, such that the dendrites would cause safety hazards. A battery management system (BMS) can be connected to the electrochemical cell to detect when a dendrite enters the interlayer and safely discharge the remaining energy in the electrochemical cell. In some embodiments, the discharge energy can be used to power other devices, such as heaters, removing cell energy to create a safe condition.

[0099] In some embodiments, the BMS can be used to draw energy through the interlayer, causing the dendrite to dissolve. This effectively removes the dendrite from the electrochemical cell. In some embodiments, energy for dissolution of the dendrite can be produced via a power supply in the BMS. In some embodiments, energy for dissolution of the dendrite can be produced by drawing energy from the cathode to increase the interlayer voltage relative to the anode.

[0100] In some embodiments, the BMS can be used to detect the voltage of the interlayer with respect to the anode, in order to detect the formation of the dendrite. The detection can include the estimation of the relative voltage of the interlayer to both the anode and the cathode. If the voltage

of the interlayer decreases with respect to the cathode (e.g., if the voltage difference between the interlayer and the cathode is greater than about 0.1 V, greater than about 0.2 V, greater than about 0.3 V, greater than about 0.4 V, greater than about 0.5 V, greater than about 0.6 V, greater than about 0.7 V, greater than about 0.8 V, greater than about 0.9 V, greater than about 1 V, greater than about 1.5 V, greater than about 2 V, greater than about 2.5 V, or greater than about 3 V, inclusive of all values and ranges therebetween), a signal can be provided to a vehicle housing the electrochemical cell to set a warning that the vehicle needs service. The threshold voltage can be a function of the design of the electrochemical cell and the interlayer. In some embodiments, the voltage difference between the interlayer and the anode can be used to trigger the service warning. In some embodiments, a combination of voltages between the anode, cathode, and/or the interlayer can be used to trigger the service warning. In some embodiments, a rate of change of the voltage of the interlayer can be used to evaluate warnings and faults in the electrochemical cell and/or in the vehicle. In some embodiments, the rate of change of the interlayer voltage can be used to perform control functions to eliminate the dendrite.

[0101] In some embodiments, a significant voltage difference between the interlayer and the cathode (e.g., at least about 0.5 V, at least about 1 V, at least about 1.5 V, at least about 2 V, at least about 2.5 V, at least about 3 V, at least about 3.5 V, at least about 4 V, at least about 4.5 V, at least about 5 V, at least about 5.5 V, or at least about 6 V, inclusive of all values and ranges therebetween) can trigger a warning signal that electrochemical cell failure and/or vehicle failure is imminent. In some embodiments, the BMS can limit discharge current of the electrochemical cell to create a reduction of power to the vehicle. This can be by directly limiting power and/or by communication of limits to a vehicle controller or another controller in the vehicle, depending on the vehicle's design.

[0102] In some embodiments, voltage can be measured between the anode and the interlayer. In some embodiments, the voltage can be measured between the cathode and the interlayer. In some embodiments, the voltages can be measured via a proportional-integral (PI) loop. The voltage between the anode and the interlayer and the voltage between the cathode and the interlayer preferably remain consistent throughout a charging process. In some embodiments, an external component can be used to maintain the interlayer near the cathode voltage. In some embodiments, the external component can include a diode, a resistor, a fuse, a transistor (Bi junction, field-effect transistor (FET), etc.), or any combination thereof.

[0103] In some embodiments, the interlayer can be chemically configured to remove the dendrite as the dendrite protrudes into the interlayer. For example, a high potential applied to the interlayer can oxidize and dissolve the dendrite. In some embodiments, the interlayer can include one or more solid layers that physically block dendrites from penetrating the interlayer. In some embodiments, the solid layer can include a solid-state electrolyte.

[0104] In some embodiments, a resistance can be applied to the interlayer. The resistance can provide a continuous excitation of the interlayer such that a dendrite would not be able to form across the dendrite and both separator layers. Such a prevention method can be used as part of an overall

control strategy where the voltage potential, current, resistance to interlayer could be changed based on a control algorithm.

[0105] A control system can act in an active prevention mode, where the potential of the interlayer is modulated or changed to apply different voltage potentials. The voltage potentials can be increased (i.e., changed to be more similar to cathode side) or decreased (changed to be more similar to anode side) to maintain the cell function. When the dendrite forms and interfaces with the interlayer, the voltage of the interlayer is increased to be more similar to the cathode potential, with respect to the anode. The dendrite is dissolved or remediated, and the voltage potential of the interlayer returns to near the voltage potential of the cathode with respect to the anode.

[0106] Dendrite growth in lithium cells is often detected via a thermal event (i.e., a sudden spike in temperature). In many cases, cell damage has already occurred once the thermal event is detected. Embodiments described herein relate to measurement of voltage potential of a separator layer (i.e., including an interlayer) relative to the anode and/or the cathode. Voltage potential is used to detect dendrite growth into the separator layer. Dendrite growth causes a voltage change in the separator layer relative to the anode and/or the cathode. Detection of the voltage change allows direct sensing of the dendrite growth before a safety event occurs. In some embodiments, the voltage potential of the interlayer can be altered or modulated to stop the growth of the dendrite or make the dendrite shrink. The voltage can be actively changed by a control system to remediate the dendrite formation at a separator layer. Voltage increases relative to an anode can prevent dendritic growth through a separator. Voltage decreases relative to an anode can dissolve dendritic growth in the active area. Modulation of the voltage on the interlayer separator can affect the ability of the electrodes to flow current into the cell. For example, inputting a voltage to the separator layer that is greater than the anode can reduce, or prevent charge current from flowing into the cell. This will provide a means to control current flow within the cell without utilizing an traditional external switching device, or as a primary or secondary means of current flow control.

[0107] Further descriptions of electrochemical cells with multiple separators and interlayers can be found in U.S. Patent Publication No. 2022/0352597 (“the ‘597 publication”), filed Apr. 29, 2022 and titled “Electrochemical Cells with Multiple Separators and Methods of Producing the Same,” the disclosure of which is hereby incorporated by reference in its entirety.

[0108] In some embodiments, electrodes described herein can include conventional solid electrodes. In some embodiments, the solid electrodes can include binders. In some embodiments, electrodes described herein can include semi-solid electrodes. Semi-solid electrodes described herein can be made: (i) thicker (e.g., greater than 100 μm -up to 2,000 μm or even greater) due to the reduced tortuosity and higher electronic conductivity of the semi-solid electrode, (ii) with higher loadings of active materials, and (iii) with a simplified manufacturing process utilizing less equipment. These relatively thick semi-solid electrodes decrease the volume, mass and cost contributions of inactive components with respect to active components, thereby enhancing the commercial appeal of batteries made with the semi-solid electrodes. In some embodiments, the semi-solid electrodes

described herein are binderless and/or do not use binders that are used in conventional battery manufacturing. Instead, the volume of the electrode normally occupied by binders in conventional electrodes, is now occupied by: 1) electrolyte, which has the effect of decreasing tortuosity and increasing the total salt available for ion diffusion, thereby countering the salt depletion effects typical of thick conventional electrodes when used at high rate, 2) active material, which has the effect of increasing the charge capacity of the battery, or 3) conductive additive, which has the effect of increasing the electronic conductivity of the electrode, thereby countering the high internal impedance of thick conventional electrodes. The reduced tortuosity and a higher electronic conductivity of the semi-solid electrodes described herein, results in superior rate capability and charge capacity of electrochemical cells formed from the semi-solid electrodes. Since the semi-solid electrodes described herein, can be made substantially thicker than conventional electrodes, the ratio of active materials (i.e., the semi-solid cathode and/or anode) to inactive materials (i.e., the current collector and separator) can be much higher in a battery formed from electrochemical cell stacks that include semi-solid electrodes relative to a similar battery formed from electrochemical cell stacks that include conventional electrodes. This substantially increases the overall charge capacity and energy density of a battery that includes the semi-solid electrodes described herein.

[0109] In some embodiments, the electrode materials described herein can be a flowable semi-solid or condensed liquid composition. In some embodiments, the electrode materials described herein can be binderless or substantially free of binder. A flowable semi-solid electrode can include a suspension of an electrochemically active material (anodic or cathodic particles or particulates), and optionally an electronically conductive material (e.g., carbon) in a non-aqueous liquid electrolyte. Said another way, the active electrode particles and conductive particles are co-suspended in an electrolyte to produce a semi-solid electrode. Examples of battery architectures utilizing semi-solid suspensions are described in International Patent Publication No. WO 2012/024499, entitled “Stationary, Fluid Redox Electrode,” and International Patent Publication No. WO 2012/088442, entitled “Semi-Solid Filled Battery and Method of Manufacture,” the entire disclosures of which are hereby incorporated by reference.

[0110] As used in this specification, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, a term “a member” is intended to mean a single member or a combination of members, “a material” is intended to mean one or more materials, or a combination thereof.

[0111] The term “substantially” when used in connection with “cylindrical,” “linear,” and/or other geometric relationships is intended to convey that the structure so defined is nominally cylindrical, linear or the like. As one example, a portion of a support member that is described as being “substantially linear” is intended to convey that, although linearity of the portion is desirable, some non-linearity can occur in a “substantially linear” portion. Such non-linearity can result from manufacturing tolerances, or other practical considerations (such as, for example, the pressure or force applied to the support member). Thus, a geometric construction modified by the term “substantially” includes such geometric properties within a tolerance of plus or minus 5%

of the stated geometric construction. For example, a “substantially linear” portion is a portion that defines an axis or center line that is within plus or minus 5% of being linear.

[0112] As used herein, the term “set” and “plurality” can refer to multiple features or a singular feature with multiple parts. For example, when referring to a set of electrodes, the set of electrodes can be considered as one electrode with multiple portions, or the set of electrodes can be considered as multiple, distinct electrodes. Additionally, for example, when referring to a plurality of electrochemical cells, the plurality of electrochemical cells can be considered as multiple, distinct electrochemical cells or as one electrochemical cell with multiple portions. Thus, a set of portions or a plurality of portions may include multiple portions that are either continuous or discontinuous from each other. A plurality of particles or a plurality of materials can also be fabricated from multiple items that are produced separately and are later joined together (e.g., via mixing, an adhesive, or any suitable method).

[0113] As used herein, the term “semi-solid” refers to a material that is a mixture of liquid and solid phases, for example, such as a particle suspension, a slurry, a colloidal suspension, an emulsion, a gel, or a micelle.

[0114] FIG. 1 is a block diagram of an electrochemical cell 100 with an interlayer 160, according to an embodiment. As shown, the electrochemical cell 100 includes an anode 110 disposed on an anode current collector 120, a cathode 130 disposed on a cathode current collector 140, a first separator 150a, and a second separator 150b disposed between the anode 110 and the cathode 130, with the interlayer 160 disposed between the first separator 150a and the second separator 150b.

[0115] In some embodiments, the anode 110 and/or the cathode 130 can include at least about 0.1%, at least about 0.2%, at least about 0.3%, at least about 0.4%, at least about 0.5%, at least about 0.6%, at least about 0.7%, at least about 0.8%, at least about 0.9%, at least about 1%, at least about 2%, at least about 3%, at least about 4%, at least about 5%, at least about 6%, at least about 7%, at least about 8%, at least about 9%, at least about 10%, at least about 11%, at least about 12%, at least about 13%, at least about 14%, at least about 15%, at least about 16%, at least about 17%, at least about 18%, at least about 19%, at least about 20%, at least about 21%, at least about 22%, at least about 23%, or at least about 24% by volume of liquid electrolyte solution. In some embodiments, the anode 110 and/or the cathode 130 can include no more than about 25%, no more than about 24%, no more than about 23%, no more than about 22%, no more than about 21%, no more than about 20%, no more than about 19%, no more than about 18%, no more than about 17%, no more than about 16%, no more than about 15%, no more than about 14%, no more than about 13%, no more than about 12%, no more than about 11%, no more than about 10%, no more than about 9%, no more than about 8%, no more than about 7%, no more than about 6%, no more than about 5%, no more than about 4%, no more than about 3%, no more than about 2%, no more than about 1%, no more than about 0.9%, no more than about 0.8%, no more than about 0.7%, no more than about 0.6%, no more than about 0.5%, no more than about 0.4%, no more than about 0.3%, or no more than about 0.2% by volume of liquid electrolyte solution.

[0116] Combinations of the above-referenced volumetric percentages of liquid electrolyte solution in the anode 110

and/or the cathode 130 are also possible (e.g., at least about 0.1% and no more than about 25% or at least about 5% and no more than about 10%), inclusive of all values and ranges therebetween. In some embodiments, the anode 110 and/or the cathode 130 can include about 0.1%, about 0.2%, about 0.3%, about 0.4%, about 0.5%, about 0.6%, about 0.7%, about 0.8%, about 0.9%, about 1%, about 2%, about 3%, about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, about 10%, about 11%, about 12%, about 13%, about 14%, about 15%, about 16%, about 17%, about 18%, about 19%, about 20%, about 21%, about 22%, about 23%, about 24%, or about 25% by volume of liquid electrolyte solution.

[0117] In some embodiments, the anode current collector 120 and/or the cathode current collector 140 can be composed of copper, aluminum, titanium, or other metals that do not form alloys or intermetallic compounds with lithium, carbon, and/or coatings comprising such materials disposed on another conductor. In some embodiments, the anode current collector 120 and/or the cathode current collector 140 can have a thickness of at least about 1 μm , at least about 5 μm , at least about 10 μm , at least about 15 μm , at least about 20 μm , at least about 25 μm , at least about 30 μm , at least about 35 μm , at least about 40 μm , or at least about 45 μm . In some embodiments, the anode current collector 120 and/or the cathode current collector 140 can have a thickness of no more than about 50 μm , no more than about 45 μm , no more than about 40 μm , no more than about 35 μm , no more than about 30 μm , no more than about 25 μm , no more than about 20 μm , no more than about 15 μm , no more than about 10 μm , or no more than about 5 μm . Combinations of the above-referenced thicknesses of the anode current collector 120 and/or the cathode current collector 140 are also possible (e.g., at least about 1 μm and no more than about 50 μm or at least about 10 μm and no more than about 30 μm), inclusive of all values and ranges therebetween. In some embodiments, the anode current collector 120 and/or the cathode current collector 140 can have a thickness of about 1 μm , about 5 μm , about 10 μm , about 15 μm , about 20 μm , about 25 μm , about 30 μm , about 35 μm , about 40 μm , about 45 μm , or about 50 μm .

[0118] In some embodiments, the anode 110 can include a first electrolyte and the cathode 130 can include a second electrolyte. In other words, and the anode 110 can include an anolyte and the cathode 130 can include a catholyte. In some embodiments, the electrochemical cell 100 can include an anolyte disposed on the anode side of the separators 150. In some embodiments, the electrochemical cell 100 can include a catholyte disposed on the cathode side of the separators 150. In some embodiments, the electrochemical cell 100 can include a selectively permeable membrane. In some embodiments, the selectively permeable membrane can be disposed between the first separator 150a and the second separator 150b. Electrochemical cells with anolytes, catholytes, and/or selectively permeable membranes are described in U.S. Pat. No. 10,734,672 (“the ‘672 patent”), filed Jan. 8, 2019, and titled, “Electrochemical Cells Including Selectively Permeable Membranes, Systems and Methods of Manufacturing the Same,” the disclosure of which is hereby incorporated by reference in its entirety.

[0119] As shown, the first separator 150a is disposed on the anode 110 while the second separator 150b is disposed on the cathode 130. In some embodiments, the separators 150 can be disposed on their respective electrodes during production of the electrochemical cell 100. In some embodi-

ments, the first separator **150a** and/or the second separator **150b** can be composed of polyethylene, polypropylene, high density polyethylene, polyethylene terephthalate, polystyrene, a thermosetting polymer, hard carbon, a thermosetting resin, a polyimide, a ceramic coated separator, an inorganic separator, cellulose, glass fiber, a polyethylenoxide (PEO) polymer in which a lithium salt is complexed to provide lithium conductivity, Nation™ membranes which are proton conductors, or any other suitable separator material, or combinations thereof. In some embodiments, the first separator **150a** can be composed of the same material as the second separator **150b**. In some embodiments, the first separator **150a** can be composed of a different material from the second separator **150b**.

[0120] In some embodiments, the first separator **150a** and/or the second separator **150b** can have a porosity of at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, or at least about 90%. In some embodiments, the first separator **150a** and/or the second separator **150b** can have a porosity of no more than about 95%, no more than about 90%, no more than about 85%, no more than about 80%, no more than about 75%, no more than about 70%, no more than about 65%, no more than about 60%, no more than about 55%, no more than about 50%, no more than about 45%, no more than about 40%, no more than about 35%, no more than about 30%, no more than about 25%, no more than about 20%, or no more than about 15%.

[0121] Combinations of the above-referenced porosity percentages of the first separator **150a** and/or the second separator **150b** are also possible (e.g., at least about 10% and no more than about 95% or at least about 20% and no more than about 40%), inclusive of all values and ranges therebetween. In some embodiments, the first separator **150a** and/or the second separator **150b** can have a porosity of about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, or about 95%.

[0122] In some embodiments, the first separator **150a** can have a different porosity from the second separator **150b**. In some embodiments, the porosities of the first separator **150a** and the second separator **150b** can be selected based on the difference between the anolyte and the catholyte. For example, if the catholyte has a higher vapor pressure and faster evaporation properties than the anolyte, then the second separator **150b** can have a lower porosity than the first separator **150a**. The lower porosity of the second separator **150b** can at least partially prevent the catholyte from evaporating during production.

[0123] In some embodiments, the first separator **150a** can be composed of a different material from the second separator **150b**. In some embodiments, the materials of the first separator **150a** and the second separator **150b** can be selected to facilitate wettability of the first separator **150a** with the anolyte and the second separator **150b** with the catholyte **150**. For example, an ethylene carbonate/propylene carbonate-based catholyte can wet a polyethylene separator better than a polyimide separator, based on the molecular properties of the materials. An ethylene carbonate/

dimethyl carbonate-based anolyte can wet a polyimide separator better than a polyethylene separator. A full wetting of the first separator **150a** and the second separator **150b** can give way to better transport of electroactive species via the separators **150**. This transport can be facilitated particularly well when the first separator **150a** physically contacts the second separator **150b**.

[0124] As shown, the electrochemical cell **100** includes two separators **150**. In some embodiments, the electrochemical cell **100** can include 3, 4, 5, 6, 7, 8, 9, 10, or more than about 10 separators **150**. In some embodiments, a layer of liquid electrolyte (not shown) can be disposed between the first separator **150a** and the second separator **150b**. A layer of liquid electrolyte can promote better adhesion between the separators **150**.

[0125] In some embodiments, the first separator **150a** and/or the second separator **150b** can have a thickness of at least about 0.5 μm , at least about 1 μm , at least about 2 μm , at least about 3 μm , at least about 4 μm , at least about 5 μm , at least about 6 μm , at least about 7 μm , at least about 8 μm , at least about 9 μm , at least about 10 μm , at least about 15 μm , at least about 20 μm , or at least about 25 μm . In some embodiments, the first separator **150a** and/or the second separator **150b** can have a thickness of no more than about 30 μm , no more than about 25 μm , no more than about 20 μm , no more than about 15 μm , no more than about 10 μm , no more than about 9 μm , no more than about 8 μm , no more than about 7 μm , no more than about 6 μm , no more than about 5 μm , no more than about 4 μm , no more than about 3 μm , no more than about 2 μm , or no more than about 1 μm . Combinations of the above-referenced thicknesses of the first separator **150a** and/or the second separator **150b** are also possible (e.g., at least about 0.5 μm and no more than about 30 μm or at least about 5 μm and no more than about 20 μm), inclusive of all values and ranges therebetween. In some embodiments, the first separator **150a** and/or the second separator **150b** can have a thickness of about 0.5 μm , about 1 μm , about 2 μm , about 3 μm , about 4 μm , about 5 μm , about 6 μm , about 7 μm , about 8 μm , about 9 μm , about 10 μm , about 15 μm , about 20 μm , about 25 μm , or about 30 μm . In some embodiments, the first separator **150a** can have a thickness the same or substantially similar to the thickness of the second separator **150b**. In some embodiments, the first separator **150a** can have a thickness greater or less than a thickness of the second separator **150b**.

[0126] In some embodiments, the first separator **150a**, the second separator **150b**, and the interlayer **160** can form a film. In some embodiments, the film can have a total thickness of at least about 5 μm , at least about 6 μm , at least about 7 μm , at least about 8 μm , at least about 9 μm , at least about 10 μm , at least about 15 μm , at least about 20 μm , at least about 25 μm , at least about 30 μm , at least about 35 μm , at least about 40 μm , or at least about 45 μm . In some embodiments, the film can have a total thickness of no more than about 50 μm , no more than about 45 μm , no more than about 40 μm , no more than about 35 μm , no more than about 30 μm , no more than about 25 μm , no more than about 20 μm , no more than about 15 μm , no more than about 10 μm , no more than about 9 μm , no more than about 8 μm , no more than about 7 μm , or no more than about 6 μm . Combinations of the above-referenced thicknesses are also possible (e.g., at least about 5 μm and no more than about 50 μm or at least about 10 μm and no more than about 40 μm), inclusive of all values and ranges therebetween. In some embodiments, the

film can have a total thickness of about 5 μm , about 6 μm , about 7 μm , about 8 μm , about 9 μm , about 10 μm , about 15 μm , about 20 μm , about 25 μm , about 30 μm , about 35 μm , about 40 μm , about 45 μm , or about 50 μm .

[0127] In some embodiments, the first separator **150a** and/or the second separator **150b** can include a solid-state electrolyte sheet. In some embodiments, the solid-state electrolyte sheet can replace the first separator **150a** and/or the second separator **150b**. In some embodiments, the first separator **150a** and/or the second separator **150b** can be made with a separator film. In some embodiments, the first separator **150a** and/or the second separator **150b** can include a coating polymer, a spray polymer, and/or a print polymer. In some embodiments, the first separator **150a** and/or the second separator **150b** can include a ceramic powder. In some embodiments, the first separator **150a** and/or the second separator **150b** can be absent of a ceramic powder. In some embodiments, the first separator **150a** and/or the second separator **150b** can include a ceramic with a liquid electrolyte and/or a solid-state electrolyte.

[0128] The interlayer **160** can dissolve dendrites via voltage manipulation. In other words, current can be supplied to the interlayer **160**, the anode **110**, and/or the cathode **130** to create a potential difference between the interlayer **160** and the anode **110** or the interlayer **160** and the cathode **130** that dissolves dendrites that have formed in the interlayer **160**. In some embodiments, the interlayer **160** can include a conductive layer. In some embodiments, the interlayer **160** can include a liquid electrolyte. In some embodiments, the interlayer **160** can include a solid-state electrolyte. In some embodiments, the interlayer **160** can include Ketjen Black, AA-stacked graphene, AB-stacked graphene, carbon, hard carbon, soft carbon, graphite, lithium iron phosphate (LFP), lithium manganese iron phosphate (LMFP), lithium manganese oxide (LMO), LiNiO_2 (LNO), nickel manganese cobalt (NMC), lithium nickel manganese oxide (LNMO), lithium cobalt oxide (LCO), Iron (III) fluoride (FeF_3), sulfur, vanadium (V) oxide (V_2O_5), bismuth trifluoride (BiF_3), iron (IV) sulfate (FeS_2), or any combination thereof. In some embodiments, the interlayer **160** can create a physical block that prevents vertical growth of the dendrite, such that the dendrite is forced to grow horizontally.

[0129] In some embodiments, the interlayer **160** can include an intercalate cathode (e.g., LMOP, LNO, NMC, LFP, LNMO, LCO, and/or LMFP). In some embodiments, the interlayer **160** can include a convertible cathode (e.g., FeF_3 , sulfur, V_2O_5 , BiF_3 , FeS_2). In some embodiments, the interlayer **160** can include a high voltage bearable anode. In some embodiments, the interlayer **160** can include a traditional anode (e.g., hard carbon, graphite, and/or silicon). In some embodiments, the interlayer **160** can include a metal. In some embodiments, the interlayer **160** can include a metal alloy. In some embodiments, the metal alloy can include lithium, tin, aluminum, silver, and/or copper. In some embodiments, the interlayer **160** can include a metal oxide. In some embodiments, the metal oxide can include silicon oxide (SiO_2), zinc oxide (ZnO), copper oxide (Cu_2O), lithium titanate (LTO), and/or titanium (IV) oxide (TiO_2). In some embodiments, the interlayer **160** can include a semi-solid electrode. In some embodiments, the interlayer **160** can include a coating, a spray, and/or a print polymer. In some embodiments, the interlayer **160** can include a ceramic powder. In some embodiments, the interlayer **160** can include a premade film with a solid-state electrolyte.

[0130] In some embodiments, the interlayer **160** can include conductive materials. In some embodiments, the interlayer **160** can include allotropes of carbon including activated carbon, hard carbon, soft carbon, Ketjen, carbon black, graphitic carbon, carbon fibers, carbon microfibers, vapor-grown carbon fibers (VGCF), fullerenic carbons including “buckyballs”, carbon nanotubes (CNTs), multi-wall carbon nanotubes (MWNTs), single wall carbon nanotubes (SWNTs), graphene, graphene sheets or aggregates of graphene sheets, and materials comprising fullerenic fragments, or any combination thereof. In some embodiments, the interlayer **160** can include a solid-state electrode material. In some embodiments, the solid-state electrolyte can include an oxide-based electrolyte. In some embodiments, the solid-state electrolyte material can include lithium lanthanum zirconium oxide (LLZO), $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (LATP), lithium phosphorus oxynitride (LiPON), li-ion conducting solid-state electrolyte ceramics (LLTO), and/or $\text{Li}_3\text{BO}_3\text{—Li}_2\text{SO}_4\text{—Li}_2\text{CO}_3$ (LiBSCO). In some embodiments, the solid-state electrolyte material can include one or more oxide-based solid electrolyte materials including a garnet structure, a perovskite structure, a phosphate-based Lithium Super Ionic Conductor (LISICON) structure, a glass structure such as $\text{La}_{0.51}\text{Li}_{0.34}\text{TiO}_{2.94}$, $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$, $\text{Li}_{1.4}\text{Al}_{0.4}\text{Ti}_{1.6}(\text{PO}_4)_3$, $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, $\text{Li}_{6.66}\text{La}_3\text{Zr}_1.6\text{Ta}_{0.4}\text{O}_{12.9}$ (LLZO), $50\text{Li}_4\text{SiO}_4\text{—}50\text{Li}_3\text{BO}_3$, $\text{Li}_{2.9}\text{PO}_{3.3}\text{N}_{0.46}$ (lithium phosphorous oxynitride, LiPON), $\text{Li}_{3.6}\text{Si}_{0.6}\text{P}_{0.4}\text{O}_4$, Li_3BN_2 , $\text{Li}_3\text{BO}_3\text{—Li}_2\text{SO}_4$, and/or sulfide containing solid electrolyte materials including a thio-LISICON structure, a glassy structure and a glass-ceramic structure such as $\text{Li}_{1.07}\text{Al}_{0.69}\text{Ti}_{1.46}(\text{PO}_4)_3$, $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$, $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS), $30\text{Li}_2\text{S}\text{—}26\text{B}_2\text{S}_3\text{—}44\text{LiI}$, $63\text{Li}_2\text{S}\text{—}36\text{SiS}_{2.1}\text{Li}_3\text{PO}_4$, $57\text{Li}_2\text{S}\text{—}38\text{SiS}_2\text{—}5\text{Li}_4\text{SiO}_4$, $70\text{Li}_2\text{S}\text{—}30\text{P}_2\text{S}_5$, $50\text{Li}_2\text{S}\text{—}50\text{GeS}_2$, $\text{Li}_2\text{P}_3\text{S}_{11}$, $\text{Li}_{3.25}\text{Po}\text{—}95\text{S}_4$, and $\text{Li}_{9.54}\text{S}_{11.74}\text{P}_{1.44}\text{S}_{11.7}\text{Cl}_{0.3}$, and/or closo-type complex hydride solid electrolyte, $\text{LiBH}_4\text{—LiI}$, $\text{LiBH}_4\text{—LiNH}_2$, $\text{LiBH}_4\text{—P}_2\text{S}_5$, $\text{Li}(\text{CBxH}_{x+1})\text{—LiI}$, $\text{Li}(\text{CB}_9\text{H}_{10})\text{—and/or LiI}$. In some embodiments, the solid-state electrolyte material can be sulfide-based. In some embodiments, the solid-state electrolyte can include lithium phosphorus sulfide (LPS), $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS), lithium tin phosphorus sulfide (LSPS), and/or $\text{Li}_{5.5}\text{PS}_{4.5}\text{Cl}_{1.5}$ (LPSCI). In some embodiments, the solid-state electrolyte material can include a complex hydride solid electrolyte. In some embodiments, the solid-state electrolyte material can include $\text{LiBH}_4\text{—LiI}$ and/or $\text{LiBH}_4\text{—P}_2\text{S}_5$.

[0131] In some embodiments, when the interlayer **160** includes a solid-state electrolyte, the interlayer **160** can have a porosity of at least about 0%, at least about 5%, at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, or at least about 90%. In some embodiments, when the interlayer **160** includes a solid-state electrolyte, the interlayer **160** can have a porosity of no more than about 95%, no more than about 90%, no more than about 85%, no more than about 80%, no more than about 75%, no more than about 70%, no more than about 65%, no more than about 60%, no more than about 55%, no more than about 50%, no more than about 45%, no more than about 40%, no more than about 35%, no more than about 30%, no more than about 25%, no more than about 20%, no more than about 15%, no more than about 10%, or no more than about

5%. Combinations of the above-referenced porosities are also possible (e.g., at least about 0% and no more than about 95% or at least about 10% and no more than about 50%), inclusive of all values and ranges therebetween. In some embodiments, when the interlayer includes a solid-state electrolyte, the interlayer **160** can have a porosity of about 0%, about 5%, about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, or about 95%.

[0132] In some embodiments, when the interlayer **160** includes a liquid electrolyte, the interlayer **160** can have a porosity of at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, or at least about 90%. In some embodiments, when the interlayer **160** includes a liquid electrolyte, the interlayer **160** can have a porosity of no more than about 95%, no more than about 90%, no more than about 85%, no more than about 80%, no more than about 75%, no more than about 70%, no more than about 65%, no more than about 60%, no more than about 55%, no more than about 50%, no more than about 45%, no more than about 40%, no more than about 35%, no more than about 30%, no more than about 25%, no more than about 20%, or no more than about 15%. Combinations of the above-referenced porosities are also possible (e.g., at least about 0% and no more than about 95% or at least about 10% and no more than about 50%), inclusive of all values and ranges therebetween. In some embodiments, when the interlayer includes a liquid electrolyte, the interlayer **160** can have a porosity of about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, or about 95%.

[0133] In some embodiments, the interlayer **160** can be pre-coated onto the first separator **150a** and/or the second separator **150b**. In some embodiments, the interlayer **160** can aid in identifying a contamination amount of lithium or another metal via a BMS. The BMS can then add more voltage and current to the interlayer **160** to dissolve the contamination. The BMS can keep the state of charge (SOC) of the interlayer **160** between a lower bound (e.g., about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, or about 40%, inclusive of all values and ranges therebetween) and an upper bound (e.g., about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or about 90%, inclusive of all values and ranges therebetween). Keeping the interlayer **160** between a lower bound and an upper bound of voltage can aid in diminishing dendrite formation while the electrochemical cell **100** is not in use (i.e., via addition of voltage and/or current). In some embodiments, the interlayer **160** can include a tab (not shown) that can be used to monitor the voltage of the interlayer **160** while the electrochemical cell is hot pressed (e.g., via a two-sided hot press or a four-sided hot press with a jelly roll design to fit into a prismatic can).

[0134] In some embodiments, the voltage between the anode **110** and the interlayer **160** (and/or the voltage between the anode **110** and the cathode **130**) can be kept above a threshold (e.g., about 2 V, about 2.1 V, about 2.2 V, about 2.3 V, about 2.4 V, about 2.5 V, about 2.6 V, about 2.7

V, about 2.8 V, about 2.9 V, about 3 V, about 3.1 V, about 3.2 V, about 3.3 V, about 3.4 V, about 3.5 V, about 3.6 V, about 3.7 V, about 3.8 V, about 3.9 V, about 4 V, about 4.1 V, about 4.2 V, about 4.3 V, about 4.4 V, about 4.5 V, about 4.6 V, about 4.7 V, about 4.8 V, about 4.9 V, about 5 V, about 5.1 V, about 5.2 V, about 5.3 V, about 5.4 V, about 5.5 V, about 5.6 V, about 5.7 V, about 5.8 V, about 5.9 V, or about 6 V, inclusive of all values and ranges therebetween). Keeping the cell voltage above a threshold can prevent dendrite formation. The inclusion of a high-stability salt in the electrolyte can facilitate the application of a high voltage (i.e., at least about 5 V) between the anode **110** and the interlayer **160**.

[0135] In some embodiments, a high salt concentration (e.g., at least about 2 M, at least about 2.1 M, at least about 2.2 M, at least about 2.3 M, at least about 2.4 M, at least about 2.5 M, at least about 2.6 M, at least about 2.7 M, at least about 2.8 M, at least about 2.9 M, at least about 3 M, at least about 3.1 M, at least about 3.2 M, at least about 3.3 M, at least about 3.4 M, at least about 3.5 M, at least about 3.6 M, at least about 3.7 M, at least about 3.8 M, at least about 3.9 M, at least about 4 M, at least about 4.1 M, at least about 4.2 M, at least about 4.3 M, at least about 4.4 M, at least about 4.5 M, at least about 4.6 M, at least about 4.7 M, at least about 4.8 M, at least about 4.9 M, or at least about 5 M) in the electrolyte can cause side reactions at higher voltages. Maintaining the voltage difference between the anode **110** and the interlayer **160** can prevent these side reactions. Higher stability carbons in the interlayer **160** (e.g., ketjen, graphite, graphene, CNT, and/or hard carbon). In some embodiments, a high-voltage stable metal (e.g., aluminum, gold, platinum) can be used to promote stability in the interlayer **160**.

[0136] At a low state of charge (SOC), the interlayer **160** can have a voltage lower than a voltage necessary to dissolve dendrites. In some embodiments, a BMS can be activated to provide extra voltage to the interlayer **160** to ensure the voltage is high enough for metal dissolution. If no extra power is available to deliver to the interlayer **160**, cycling conditions can be narrowed (e.g., to about 5%, about 10%, about 15%, or about 20% SOC, inclusive of all values and ranges therebetween) in order to limit the formation of dendrites. In some embodiments, the voltage between the anode **110** and the interlayer **160** can be kept above the threshold value by monitoring the potential of the anode **110** and the interlayer **160** and controlling the SOC (i.e., by monitoring and adjusting the cycling). In some embodiments, the voltage between the anode **110** and the interlayer **160** can be maintained by keeping a fixed SOC (e.g., at least about 15%, at least about 20%, at least about 25%, inclusive of all values and ranges therebetween).

[0137] FIGS. 2A-2B are illustrations of an electrochemical cell **200** with an interlayer **260**, according to an embodiment. As shown, the electrochemical cell **200** includes an anode **210** disposed on an anode current collector **220**, a cathode **230** disposed on a cathode current collector **240**, with a first separator **250a** and a second separator **250b** disposed between the anode **210** and the cathode **230**. The interlayer **260** is disposed between the first separator **250a** and the second separator **250b**. In some embodiments, the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260** can be the same or substantially similar to the anode **110**, the anode

current collector **120**, the cathode **130**, the cathode current collector **140**, the first separator **150a**, the second separator **150b**, and the interlayer **160**, as described above with reference to FIG. 1. Thus, certain aspects of the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260** are not described in greater detail herein.

[0138] As shown, a voltage V_1 is measured between the anode current collector **220** and the interlayer **260**, a voltage V_2 is measured between the anode current collector **220** and the cathode current collector **240**. As shown, a diode **D** is coupled to the interlayer **260** and the cathode current collector **240** to direct the flow of current in a single direction from the interlayer **260** to the cathode current collector **240**. In some embodiments, the diode **D** can be replaced by a resistor, a fuse, and/or a transistor (e.g., a bi-junction transistor, FET). The anode current collector **220** is coupled to the cathode current collector **240** via a resistor R_1 and a switch **S**.

[0139] In FIG. 2A, the switch **S** is open and the electrochemical cell **200** is functioning normally. In FIG. 2B, a dendrite **225** has formed. The dendrite **225** creates a short circuit between the anode **210** and the interlayer **260**, such that the voltage V_1 is reduced below a threshold value. Upon detecting that V_1 is below the threshold value, the switch **S** closes, and the electrochemical cell **200** discharges through the resistor R_1 . In some embodiments, the switch **S** can be closed during charge of the electrochemical cell **200**. In some embodiments, the switch **S** can be closed during discharge of the electrochemical cell **200**. In some embodiments, the threshold value of the voltage V_1 can be about 0.001 V, about 0.002 V, about 0.003 V, about 0.004 V, about 0.005 V, about 0.006 V, about 0.007 V, about 0.008 V, about 0.009 V, about 0.01 V, about 0.02 V, about 0.03 V, about 0.04 V, about 0.05 V, about 0.06 V, about 0.07 V, about 0.08 V, about 0.09 V, about 0.1 V, about 0.2 V, about 0.3 V, about 0.4 V, about 0.5 V, about 0.6 V, about 0.7 V, about 0.8 V, about 0.9 V, or about 1 V, inclusive of all values and ranges therebetween.

[0140] In some embodiments, multiple electrochemical cells can be connected in series and/or in parallel. As shown, when the switch **S** closes, a hardware triggered discharge can easily be used to automatically apply an external load to quickly discharge the electrochemical cell **200**. This discharge can benefit the functional safety of the electrochemical cell **200**. The Failure-in-time (FIT) number of the electrochemical cell **200** or multiple electrochemical cells can be detected for the reliability of detection of dendrites (e.g., based on a contaminant concentration (ppm) value corresponding to a particular voltage). Each electrochemical cell connected in parallel can require a separate monitor. This would increase the controls cost of such a system without individual hardware detection. In some embodiments all of the interlayers from parallel cells can be connected in parallel. Using a switching device can allow duplication of single components (i.e., single transistors are cheaper than BMS monitoring chips). When electrochemical cells are connected fully in series, a battery measurement chip can be used, connecting V_1 to a first channel and connecting V_2 to a next higher channel. In some embodiments, the BMS can measure/calculate a voltage V_f equal to $V_2 - V_1$ to allow a control decision to be made to protect the system. The BMS can react to a short circuit detection via

standard balance circuitry or through additional controls. Low integration level monitors can be appropriate for such measurements, as they have a lower controls overhead and cost than more automated BMS devices. Using multiple chip sets can also create redundant safety measurements for voltage and temperature sensing giving additional flexibility to the safety design.

[0141] The resistor R_1 eases or reduces the flow of current through a direct path between the anode **210** and the cathode **230**. In some embodiments, the current flowing along the closed switch **S** can be used to power an external device. In some embodiments, the current flowing along the closed switch **S** can be used to provide heat to an external device or heater. As shown, the diode **D** connects the interlayer **260** to the cathode current collector **240**. In some embodiments, the diode **D** can connect the interlayer **260** to the anode current collector **220** while the voltage is measured between the interlayer **260** and the cathode current collector **240**. In other words, the electrochemical energy can flow in the opposite direction to what is depicted in FIGS. 2A-2B.

[0142] In some embodiments, the anode **210** can include a semi-solid anode. In some embodiments, the anode **210** can include a conventional solid anode (i.e., with a binder). In some embodiments, the cathode **230** can include a semi-solid cathode. In some embodiments, the cathode **230** can include a conventional solid cathode (i.e., with a binder).

[0143] In some embodiments, the electrochemical cell **200** can be subject to high-potential (hipot) testing. In some embodiments, the cathode **230** can be subject to hipot testing. In some embodiments, the anode **210** can be subject to hipot testing. In some embodiments, a full hipot of the electrochemical cell **200** can be conducted with the interlayer **260** acting as a reference layer. In some embodiments, the hipot testing can be conducted with conventional electrode layers.

[0144] In some embodiments, the first separator **250a** and/or the second separator **250b** can be composed of a material that has minimal heat deformation. For example, the first separator **250a** and/or the second separator **250b** can be composed of cellulose, a cellulosic compound, a thermos-cured resin, a high melting temperature polymer, polyimide, or any combination thereof. In some embodiments, the first separator **250a** and/or the second separator **250b** can have a melting temperature of at least about 100° C., at least about 110° C., at least about 120° C., at least about 130° C., at least about 140° C., at least about 150° C., at least about 160° C., at least about 170° C., at least about 180° C., at least about 190° C., at least about 200° C., at least about 210° C., at least about 220° C., at least about 230° C., at least about 240° C., at least about 250° C., at least about 260° C., at least about 270° C., at least about 280° C., at least about 290° C., or at least about 300° C. This level of heat resistance prevents shrinkage by the first separator **250a** and/or the second separator **250b**. In some embodiments, the electrochemical cell **200** can be formed such that the first separator **250a** and the second separator **250b** do not contact pouch film enclosing the electrochemical cell **200**. Since dendrite formation is prevented, the first separator **250a** and the second separator **250b** need not extend substantially outward from the anode current collector **220** or the cathode current collector **240**.

[0145] In some embodiments, the interlayer **260** can have a thickness of at least about 500 nm, at least about 1 μ m, at least about 2 μ m, at least about 3 μ m, at least about 4 μ m, at least about 5 μ m, at least about 6 μ m, at least about 7 μ m,

at least about 8 μm , at least about 9 μm , at least about 10 μm , at least about 20 μm , at least about 30 μm , at least about 40 μm , at least about 50 μm , at least about 60 μm , at least about 70 μm , at least about 80 μm , at least about 90 μm , at least about 100 μm , at least about 150 μm , at least about 200 μm , at least about 250 μm , at least about 300 μm , at least about 350 μm , at least about 400 μm , at least about 450 μm , at least about 500 μm , at least about 550 μm , at least about 600 μm , at least about 650 μm , at least about 700 μm , at least about 750 μm , at least about 800 μm , at least about 850 μm , at least about 900 μm , or at least about 950 μm . In some embodiments, the interlayer 260 can have a thickness of no more than about 1,000 μm , no more than about 950 μm , no more than about 900 μm , no more than about 850 μm , no more than about 800 μm , no more than about 750 μm , no more than about 700 μm , no more than about 650 μm , no more than about 600 μm , no more than about 550 μm , no more than about 500 μm , no more than about 450 μm , no more than about 400 μm , no more than about 350 μm , no more than about 300 μm , no more than about 250 μm , no more than about 200 μm , no more than about 150 μm , no more than about 100 μm , no more than about 90 μm , no more than about 80 μm , no more than about 70 μm , no more than about 60 μm , no more than about 50 μm , no more than about 40 μm , no more than about 30 μm , no more than about 20 μm , no more than about 10 μm , no more than about 9 μm , no more than about 8 μm , no more than about 7 μm , no more than about 6 μm , no more than about 5 μm , no more than about 4 μm , no more than about 3 μm , no more than about 2 μm , or no more than about 1 μm . Combinations of the above-referenced thickness values are also possible (e.g., at least about 500 nm and no more than about 1,000 μm or at least about 200 μm and no more than about 700 μm), inclusive of all values and ranges therebetween. In some embodiments, the interlayer 260 can have a thickness of about 10 μm , about 20 μm , about 30 μm , about 40 μm , about 50 μm , about 60 μm , about 70 μm , about 80 μm , about 90 μm , about 100 μm , about 150 μm , about 200 μm , about 250 μm , about 300 μm , about 350 μm , about 400 μm , about 450 μm , about 500 μm , about 550 μm , about 600 μm , about 650 μm , about 700 μm , about 750 μm , about 800 μm , about 850 μm , about 900 μm , about 950 μm , or about 1,000 μm .

[0146] In some embodiments, the interlayer 260 can have a conductivity of at least about 1×10^{-3} S/m, at least about 2×10^{-3} S/m, at least about 3×10^{-3} S/m, at least about 4×10^{-3} S/m, at least about 5×10^{-3} S/m, at least about 6×10^{-3} S/m, at least about 7×10^{-3} S/m, at least about 8×10^{-3} S/m, at least about 9×10^{-3} S/m, at least about 1×10^{-2} S/m, at least about 2×10^{-2} S/m, at least about 3×10^{-2} S/m, at least about 4×10^{-2} S/m, at least about 5×10^{-2} S/m, at least about 6×10^{-2} S/m, at least about 7×10^{-2} S/m, at least about 8×10^{-2} S/m, at least about 9×10^{-2} S/m, at least about 0.1 S/m, at least about 0.2 S/m, at least about 0.3 S/m, at least about 0.4 S/m, at least about 0.5 S/m, at least about 0.6 S/m, at least about 0.7 S/m, at least about 0.8 S/m, at least about 0.9 S/m. In some embodiments, the interlayer 260 can have a conductivity of no more than about 1 S/m, no more than about 0.9 S/m, no more than about 0.8 S/m, no more than about 0.7 S/m, no more than about 0.6 S/m, no more than about 0.5 S/m, no more than about 0.4 S/m, no more than about 0.3 S/m, no more than about 0.2 S/m, no more than about 0.1 S/m, no more than about 9×10^{-2} S/m, no more than about 8×10^{-2} S/m, no more than about 7×10^{-2} S/m, no more than about 6×10^{-2} S/m, no more than about 5×10^{-2} S/m, no more than

about 4×10^{-2} S/m, no more than about 3×10^{-2} S/m, no more than about 2×10^{-2} S/m, no more than about 1×10^{-2} S/m, no more than about 9×10^{-3} S/m, no more than about 8×10^{-3} S/m, no more than about 7×10^{-3} S/m, no more than about 6×10^{-3} S/m, no more than about 5×10^{-3} S/m, no more than about 4×10^{-3} S/m, no more than about 3×10^{-3} S/m, or no more than about 2×10^{-3} S/m. Combinations of the above-referenced conductivity values are also possible (e.g., at least about 1×10^{-3} S/m and no more than about 1 S/m or at least about at least about 2×10^{-3} S/m and no more than about 0.1 S/cm), inclusive of all values and ranges therebetween. In some embodiments, the interlayer 260 can have a conductivity of about 1×10^{-3} S/m, about 2×10^{-3} S/m, about 3×10^{-3} S/m, about 4×10^{-3} S/m, about 5×10^{-3} S/m, about 6×10^{-3} S/m, about 7×10^{-3} S/m, about 8×10^{-3} S/m, about 9×10^{-3} S/m, about 1×10^{-2} S/m, about 2×10^{-2} S/m, about 3×10^{-2} S/m, about 4×10^{-2} S/m, about 5×10^{-2} S/m, about 6×10^{-2} S/m, about 7×10^{-2} S/m, about 8×10^{-2} S/m, about 9×10^{-2} S/m, about 0.1 S/m, about 0.2 S/m, about 0.3 S/m, about 0.4 S/m, about 0.5 S/m, about 0.6 S/m, about 0.7 S/m, about 0.8 S/m, about 0.9 S/m, or about 1 S/m.

[0147] In some embodiments, the interlayer 260 can be electrically coupled to a cathode external to the electrochemical cell 200. In some embodiments, the external cathode can have a higher resistance than the interlayer 260. In some embodiments, the interlayer 260 can be electrically coupled to an anode external to the electrochemical cell 200. In some embodiments, the external anode can have a higher resistance than the interlayer 260. In some embodiments, the interlayer 260 can be electrically connected to an external electrochemical cell. In some embodiments, the interlayer 260 can be electrically connected to an external capacitor. In some embodiments, the interlayer 260 can be electrically connected to an external resistor.

[0148] In some embodiments, the potential of the interlayer 260 can be manipulated to dissolve or oxidize the dendrite 225 as the dendrite 225 penetrates the interlayer 260. This is particularly valid when the interlayer 260 includes cathode materials therein. In some embodiments, the voltage difference between the anode 210 and the interlayer 260 can be tuned to at least about 1 V, at least about 1.5 V, at least about 2 V, at least about 2.5 V, at least about 3 V, at least about 3.5 V, at least about 4 V, or at least about 4.5 V. In some embodiments, the voltage difference between the anode 210 and the interlayer 260 can be tuned to no more than about 5 V, no more than about 4.5 V, no more than about 4 V, no more than about 3.5 V, no more than about 3 V, no more than about 2.5 V, no more than about 2 V, or no more than about 1.5 V. Combinations of the above-referenced voltage ranges are also possible (e.g., at least about 1 V and no more than about 5 V or at least about 2 V and no more than about 4 V), inclusive of all values and ranges therebetween.

[0149] In some embodiments, the voltage difference between the anode 210 and the interlayer 260 can be tuned to about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V.

[0150] In some embodiments, the voltage difference between the cathode 230 and the interlayer 260 can be tuned to at least about 1 V, at least about 1.5 V, at least about 2 V, at least about 2.5 V, at least about 3 V, at least about 3.5 V, at least about 4 V, at least about 4.5 V, at least about 5 V, or at least about 5.5 V. In some embodiments, the voltage

difference between the cathode **230** and the interlayer **260** can be tuned to no more than about 6 V, no more than about 5.5 V, no more than about 5 V, no more than about 4.5 V, no more than about 4 V, no more than about 3.5 V, no more than about 3 V, no more than about 2.5 V, no more than about 2 V, or no more than about 1.5 V. Combinations of the above-referenced voltage ranges are also possible (e.g., at least about 1 V and no more than about 5 V or at least about 2 V and no more than about 4 V), inclusive of all values and ranges therebetween. In some embodiments, the voltage difference between the cathode **230** and the interlayer **260** can be tuned to about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V.

[0151] In some embodiments, the impedance through the structural components of the electrochemical cell **200** (i.e., the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, and the separator **250**) can be less than the impedance through the resistor R_1 . In some embodiments, the impedance through the structural components of the electrochemical cell **200** can be less than the impedance through the diode **D**. In some embodiments, the ratio of the impedance through the structural components of the electrochemical cell **200** to the impedance through the resistor R_1 can be less than about 1:2, less than about 1:3, less than about 1:4, less than about 1:5, less than about 1:6, less than about 1:7, less than about 1:8, less than about 1:9, less than about 1:10, less than about 1:20, less than about 1:30, less than about 1:40, less than about 1:50, less than about 1:60, less than about 1:70, less than about 1:80, less than about 1:90, or less than about 1:100. In some embodiments, the ratio of the impedance through the structural components of the electrochemical cell **200** to the impedance through the diode **D** can be less than about 1:2, less than about 1:3, less than about 1:4, less than about 1:5, less than about 1:6, less than about 1:7, less than about 1:8, less than about 1:9, less than about 1:10, less than about 1:20, less than about 1:30, less than about 1:40, less than about 1:50, less than about 1:60, less than about 1:70, less than about 1:80, less than about 1:90, or less than about 1:100.

[0152] In some embodiments, the first separator **250a** can have a first length and the second separator **250b** can have a second length, the second length greater than the first length. In some embodiments, the first separator **250a** can have a first width and the second separator **250b** can have a second width, the second width greater than the first width. In some embodiments, the interlayer **260** can have a length and/or width greater than the first width and/or the second width, such that a portion of the interlayer **260** is adjacent to the second separator **250b** but not the first separator **250a**. In some embodiments, a tab (not shown) can be attached (e.g., via welding) to the interlayer **260**. In some embodiments, the tab can be attached to a portion of the interlayer **260** extending beyond the first separator **250a**.

[0153] In some embodiments, a portion of the first separator **250a** and/or the second separator **250b** can be removed (e.g., the first separator **250a** and/or the second separator **250b** can be punched), such that the tab can be attached to the interlayer **260** through the removed section of the first separator **250a** and/or the second separator **250b**. In some embodiments, the electrochemical cell **200** can be contained inside a pouch (not shown). In some embodiments, the pouch can include a first film attached to the anode current collector **220** and a second film attached to the cathode

current collector **240**. In some embodiments, portion of the pouch can be removed (e.g., punched), such that the tab can be attached to the interlayer **260** through the removed section of the pouch.

[0154] FIGS. 3A-3B are illustrations of an electrochemical cell **300** with an interlayer **360**, according to an embodiment. As shown, the electrochemical cell **300** includes an anode **310** disposed on an anode current collector **320**, a cathode **330** disposed on a cathode current collector **340**, with a first separator **350a** and a second separator **350b** disposed between the anode **310** and the cathode **330**. The interlayer **360** is disposed between the first separator **350a** and the second separator **350b**. In some embodiments, the anode **310**, the anode current collector **320**, the cathode **330**, the cathode current collector **340**, the first separator **350a**, the second separator **350b**, and the interlayer **360** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **310**, the anode current collector **320**, the cathode **330**, the cathode current collector **340**, the first separator **350a**, the second separator **350b**, and the interlayer **360** are not described in greater detail herein.

[0155] As shown, a voltage V_1 is measured between the anode current collector **320** and the interlayer **360**, a voltage V_2 is measured between the anode current collector **320** and the cathode current collector **340**. As shown, the anode current collector **320** is coupled to the cathode current collector **340** via a resistor R_1 and a switch **S**. As shown, a resistor R_2 is coupled to the interlayer **360** and the cathode current collector **340** to resist current flow in a direction from the interlayer **360** to the cathode current collector **340**.

[0156] In FIG. 3A, the switch **S** is open and the electrochemical cell **300** is functioning normally. In FIG. 3B, a dendrite **325** has formed. The dendrite **325** creates a short circuit between the anode **320** and the interlayer **360**, such that the voltage V_1 is reduced below a threshold value. Upon detecting that V_1 is below the threshold value, the switch **S** closes (see FIG. 3B), and the electrochemical cell **300** discharges through the resistor R_1 .

[0157] As shown, the resistor R_1 connects the interlayer **360** to the cathode current collector **340**. In some embodiments, the resistor R_1 can connect the interlayer **360** to the anode current collector **320** while the voltage is measured between the interlayer **360** and the cathode current collector **340**. In other words, the electrochemical energy can flow in the opposite direction to what is depicted in FIGS. 3A-3B.

[0158] FIGS. 4A-4B are illustrations of an electrochemical cell **400** with multiple interlayers, according to an embodiment. As shown, the electrochemical cell **400** includes an anode **410** disposed on an anode current collector **420**, a cathode **430** disposed on a cathode current collector **440**, with a first separator **450a**, a second separator **450b**, and a third separator **450c** disposed between the anode **410** and the cathode **430**. A first interlayer **460a** is disposed between the first separator **450a** and the second separator **450b**. A second interlayer **460b** is disposed between the second separator **450b** and the third separator **450c**. In some embodiments, the anode **410**, the anode current collector **420**, the cathode **430**, the cathode current collector **440**, the first separator **450a**, the second separator **450b**, and the first interlayer **460a** can be the same or substantially similar to

the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **410**, the anode current collector **420**, the cathode **430**, the cathode current collector **440**, the first separator **450a**, the second separator **450b**, and the first interlayer **460a** are not described in greater detail herein.

[0159] As shown, a voltage V_1 is measured between the anode current collector **420** and the second interlayer **460b**. A voltage V_2 is measured between the second interlayer **460b** and the first interlayer **460a**. As shown, a first diode D_1 is coupled to the second interlayer **460b** and the cathode current collector **440** to direct the flow of current in a single direction from the second interlayer **460** to the cathode current collector **440**. As shown, a second diode D_2 is coupled to the first interlayer **460a** and the cathode **440**. The anode current collector **420** is coupled to the cathode current collector **440** via a resistor R_1 and a switch S .

[0160] In FIG. 4A, the switch S is open and the electrochemical cell **400** is functioning normally. In FIG. 4B, a dendrite **425** has formed. The dendrite **425** creates a short circuit between the anode **420** and the first interlayer **460a** and/or the second interlayer **460b**, such that the voltage V_1 and/or the voltage V_2 are reduced below a threshold value. In some embodiments, a BMS can redirect the flow of current based on V_1 decreasing to less than a threshold value. In some embodiments, the BMS can redirect the flow of current based on V_2 decreasing to less than a threshold value. In some embodiments, the BMS can redirect the flow of current based on (V_1+V_2) decreasing to less than a threshold value. In some embodiments, the threshold value can be about 0.001 V, about 0.002 V, about 0.003 V, about 0.004 V, about 0.005 V, about 0.006 V, about 0.007 V, about 0.008 V, about 0.009 V, about 0.01 V, about 0.02 V, about 0.03 V, about 0.04 V, about 0.05 V, about 0.06 V, about 0.07 V, about 0.08 V, about 0.09 V, about 0.1 V, about 0.2 V, about 0.3 V, about 0.4 V, about 0.5 V, about 0.6 V, about 0.7 V, about 0.8 V, about 0.9 V, or about 1 V, inclusive of all values and ranges therebetween.

[0161] As shown, the electrochemical cell **400** includes 3 separators **450a**, **450b**, **450c** (collectively referred to as separators **450**). In some embodiments, the electrochemical cell **400** can include at least about 3, at least about 4, at least about 5, at least about 6, at least about 7, at least about 8, at least about 9, at least about 10, at least about 15, at least about 20, at least about 25, at least about 30, at least about 35, at least about 40, at least about 45, at least about 50, at least about 55, at least about 60, at least about 65, at least about 70, at least about 75, at least about 80, at least about 85, at least about 90, or at least about 95 separators **450**. In some embodiments, the electrochemical cell **400** can include no more than about 100, no more than about 95, no more than about 90, no more than about 85, no more than about 80, no more than about 75, no more than about 70, no more than about 65, no more than about 60, no more than about 55, no more than about 50, no more than about 45, no more than about 40, no more than about 35, no more than about 30, no more than about 25, no more than about 20, no more than about 15, no more than about 10, no more than about 9, no more than about 8, no more than about 7, no more than about 6, no more than about 5, or no more than about 4 separators **450**. Combinations of the above-referenced numbers of separators **450** are also possible (e.g., at least about

3 and no more than about 100 or at least about 10 and no more than about 40), inclusive of all values and ranges therebetween. In some embodiments, the electrochemical cell **400** can include about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 15, about 20, about 25, about 30, about 35, about 40, about 45, about 50, about 55, about 60, about 65, about 70, about 75, about 80, about 85, about 90, about 95, or about 100 separators **450**.

[0162] As shown, the electrochemical cell **400** includes 2 interlayers **460a**, **460b** (collectively referred to as interlayers **460**). Multiple interlayers **460** allow for multiple points along the electrochemical cell, where voltages and voltage differences can be measured. In some embodiments, the electrochemical cell **400** can include at least about 2, at least about 3, at least about 4, at least about 5, at least about 6, at least about 7, at least about 8, at least about 9, at least about 10, at least about 15, at least about 20, at least about 25, at least about 30, at least about 35, at least about 40, at least about 45, at least about 50, at least about 55, at least about 60, at least about 65, at least about 70, at least about 75, at least about 80, at least about 85, at least about 90, or at least about 95 interlayers **460**. In some embodiments, the electrochemical cell **400** can include no more than about 100, no more than about 95, no more than about 90, no more than about 85, no more than about 80, no more than about 75, no more than about 70, no more than about 65, no more than about 60, no more than about 55, no more than about 50, no more than about 45, no more than about 40, no more than about 35, no more than about 30, no more than about 25, no more than about 20, no more than about 15, no more than about 10, no more than about 9, no more than about 8, no more than about 7, no more than about 6, no more than about 5, no more than about 4, or no more than about 3 interlayers **460**. Combinations of the above-referenced numbers of interlayers **460** are also possible (e.g., at least about 3 and no more than about 100 or at least about 10 and no more than about 40), inclusive of all values and ranges therebetween. In some embodiments, the electrochemical cell **400** can include about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 15, about 20, about 25, about 30, about 35, about 40, about 45, about 50, about 55, about 60, about 65, about 70, about 75, about 80, about 85, about 90, about 95, or about 100 interlayers **460**.

[0163] As shown, the diode D_1 connects the second interlayer **460b** to the cathode current collector **440** and the diode D_2 connects the first interlayer **460a** to the cathode current collector **440**. In some embodiments, the diode D_1 can connect the first interlayer **460a** to the anode current collector **420** and the diode D_2 can connect the second interlayer **460b** to the anode current collector **420** while voltage is measured between the first interlayer **460a** and the second interlayer **460b** and between the first interlayer **460a** and the cathode **430**. In other words, the electrochemical energy can flow in the opposite direction to what is depicted in FIGS. 4A-4B.

[0164] In some embodiments, the separator **450a**, the separator **450b**, and/or the separator **450c** can include a solid-state electrolyte. In some embodiments, the solid-state electrolyte layer can be incorporated into the electrochemical cell **400** instead of the separator **450a**, the separator **450b**, and/or the separator **450c**. In some embodiments, the separator **450b** can have a lower melting temperature than the separator **450a**. In some embodiments, the separator **450c** can have a lower melting temperature than the separator

rator **450a**. In some embodiments, the separator **450c** can have a lower melting temperature than the separator **450a**. In some embodiments, the separator **450a**, the separator **450b**, and/or the separator **450c** can include a gel electrolyte. In some embodiments, the gel electrolyte can be included in the electrochemical cell **400** instead of either of the separators **450**. In some embodiments, either of the separators **450** can include non-ionic conductive powders.

[0165] In some embodiments, the resistance across the diode D_1 can be greater than the resistance across the diode D_2 . This allows more current to flow through the interlayer **460a** than through the interlayer **460b**, which can cause a dendrite that has penetrated the interlayer **460b** to dissipate more rapidly. In some embodiments, a V_1 measurement of zero can cause a 'caution' alert to be sent to a user (e.g., via a BMS). For example, the caution can be a yellow light. In some embodiments, a V_2 measurement of zero can cause a 'warning' or 'danger' alert to be sent to the user. For example, a red light can be switched on for the user to see. In some embodiments, a V_2 measurement of zero can trigger safety actions. In some embodiments, the safety actions can include an external short circuit through a BMS. In some embodiments, the safety actions can include a rerouting of current through a heater (e.g., via a BMS).

[0166] In some embodiments, the separator **450b** and/or the separator **450c** can have a thickness of at least about 0.2 μm , at least about 0.3 μm , at least about 0.4 μm , at least about 0.5 μm , at least about 0.6 μm , at least about 0.7 μm , at least about 0.8 μm , at least about 0.9 μm , at least about 1 μm , at least about 1.5 μm , at least about 2 μm , or at least about 2.5 μm . In some embodiments, the separator **450b** and/or the separator **450c** can have a thickness of no more than about 3 μm , no more than about 2.5 μm , no more than about 2 μm , no more than about 1.5 μm , no more than about 1 μm , no more than about 0.9 μm , no more than about 0.8 μm , no more than about 0.7 μm , no more than about 0.6 μm , no more than about 0.5 μm , no more than about 0.4 μm , or no more than about 0.3 μm . Combinations of the above-referenced thickness values are also possible (e.g., at least about 0.2 μm and no more than about 3 μm or at least about 0.5 μm and no more than about 1 μm), inclusive of all values and ranges therebetween. In some embodiments, the separator **450b** and/or the separator **450c** can have a thickness of about 0.2 μm , about 0.3 μm , about 0.4 μm , about 0.5 μm , about 0.6 μm , about 0.7 μm , about 0.8 μm , about 0.9 μm , about 1 μm , about 1.5 μm , about 2 μm , about 2.5 μm , or about 3 μm .

[0167] In some embodiments, the separator **450b** and/or the separator **450c** can be excluded from the electrochemical cell **400**. For example, the electrochemical cell **400** can include lithium metal anodes with or without mesh type current collectors. Lithium metal can be filled into the mesh current collector. In such a case, the V_1 monitoring capability is lost, but V_2 can still be monitored for safety purposes.

[0168] FIG. 5 is an illustration of an electrochemical cell **500** with an interlayer **560**, according to an embodiment. As shown, the electrochemical cell **500** includes an anode **510** disposed on an anode current collector **520**, a cathode **530** disposed on a cathode current collector **540**, with a first separator **550a** and a second separator **550b** disposed between the anode **510** and the cathode **530**. The interlayer **560** is disposed between the first separator **550a** and the second separator **550b**. In some embodiments, the anode **510**, the anode current collector **520**, the cathode **530**, the

cathode current collector **540**, the first separator **550a**, the second separator **550b**, and the interlayer **560** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **510**, the anode current collector **520**, the cathode **530**, the cathode current collector **540**, the first separator **550a**, the second separator **550b**, and the interlayer **560** are not described in greater detail herein.

[0169] As shown, the cathode **530** has a non-uniform thickness. Without the interlayer **560**, the SOC of the anode **510** and/or the cathode **530** can become varied due to coating thicknesses or impedance variations. Lithium ions are drawn to move inside the electrode to equalize the same potential. However, cathode materials (e.g., LNO or NMC) can have a greater potential slope than anode materials (e.g., graphite), depending on SOC, and lithium ions can move faster in the cathode material than in the anode material. Adding a lithium-reduced NMC (e.g., $\text{Li}_{(1-x)}\text{NMC}$) in the interlayer **560** can reduce the SOC difference inside the cathode **530**. In charging, electrons can go through the interlayer **560**, but in discharging, the material does not move. The interlayer **560** can become a buffer capacity to equalize the SOC difference inside the cathode **530**. In some embodiments, the thickness of the cathode **530** can be made non-uniform to balance distribution of lithium ions. In other words, the lithium ions can have a uniform or near uniform (e.g., within about 5%, within about 4%, within about 3%, within about 2%, or within about 1%) volumetric density of lithium ions (i.e., lithium ions per unit volume of electrode material) throughout the length of the cathode **530**.

[0170] As shown, a first side of the cathode **530** has a first thickness t_1 and a second side of the cathode **530** has a second thickness t_2 . The variable thickness of the cathode **530** (t_1 and t_2) can facilitate cathode and anode alignment and avoid misalignment when the electrochemical cell **500** when the electrochemical cell **500** is rolled (i.e., in a jelly roll format and placed into a can). In some embodiments, the first side of the cathode **530** can be adjacent to a tab where voltage is measured. In some embodiments, the second side of the cathode **530** can be adjacent to a tab where voltage is measured. In other words, the cathode **530** can be thicker on the side proximal to where voltage is measured, or the cathode **530** can be thicker on the side distal to where the voltage is measured.

[0171] In some embodiments, t_1 can be at least about 100 μm , at least about 150 μm , at least about 200 μm , at least about 250 μm , at least about 300 μm , at least about 350 μm , at least about 400 μm , at least about 450 μm , at least about 500 μm , at least about 550 μm , at least about 600 μm , at least about 650 μm , at least about 700 μm , at least about 750 μm , at least about 800 μm , at least about 850 μm , at least about 900 μm , at least about 950 μm , at least about 1,000 μm , at least about 1,100 μm , at least about 1,200 μm , at least about 1,300 μm , at least about 1,400 μm , at least about 1,500 μm , at least about 1,600 μm , at least about 1,700 μm , at least about 1,800 μm , or at least about 1,900 μm . In some embodiments, t_1 can be no more than about 2,000 μm , no more than about 2,000 μm , no more than about 2,000 μm , no more than about 1,900 μm , no more than about 1,800 μm , no more than about 1,700 μm , no more than about 1,600 μm , no more than about 1,500 μm , no more than about 1,400 μm , no

more than about 1,300 μm , no more than about 1,200 μm , no more than about 1,100 μm , no more than about 1,000 μm , no more than about 950 μm , no more than about 900 μm , no more than about 850 μm , no more than about 800 μm , no more than about 750 μm , no more than about 700 μm , no more than about 650 μm , no more than about 600 μm , no more than about 550 μm , no more than about 500 μm , no more than about 450 μm , no more than about 400 μm , no more than about 350 μm , no more than about 300 μm , no more than about 250 μm , no more than about 200 μm , or no more than about 150 μm . Combinations of the above-referenced thicknesses are also possible (e.g., at least about 100 μm and no more than about 2,000 μm or at least about 150 μm and no more than about 500 μm), inclusive of all values and ranges therebetween. In some embodiments, t1 can be about 100 μm , about 150 μm , about 200 μm , about 250 μm , about 300 μm , about 350 μm , about 400 μm , about 450 μm , about 500 μm , about 550 μm , about 600 μm , about 650 μm , about 700 μm , about 750 μm , about 800 μm , about 850 μm , about 900 μm , about 950 μm , about 1,000 μm , about 1,100 μm , about 1,200 μm , about 1,300 μm , about 1,400 μm , about 1,500 μm , about 1,600 μm , about 1,700 μm , about 1,800 μm , about 1,900 μm , or about 2,000 μm .

[0172] In some embodiments, t2 can be at least about 100 μm , at least about 150 μm , at least about 200 μm , at least about 250 μm , at least about 300 μm , at least about 350 μm , at least about 400 μm , at least about 450 μm , at least about 500 μm , at least about 550 μm , at least about 600 μm , at least about 650 μm , at least about 700 μm , at least about 750 μm , at least about 800 μm , at least about 850 μm , at least about 900 μm , at least about 950 μm , at least about 1,000 μm , at least about 1,100 μm , at least about 1,200 μm , at least about 1,300 μm , at least about 1,400 μm , at least about 1,500 μm , at least about 1,600 μm , at least about 1,700 μm , at least about 1,800 μm , or at least about 1,900 μm . In some embodiments, t2 can be no more than about 2,000 μm , no more than about 2,000 μm , no more than about 1,900 μm , no more than about 1,800 μm , no more than about 1,700 μm , no more than about 1,600 μm , no more than about 1,500 μm , no more than about 1,400 μm , no more than about 1,300 μm , no more than about 1,200 μm , no more than about 1,100 μm , no more than about 1,000 μm , no more than about 950 μm , no more than about 900 μm , no more than about 850 μm , no more than about 800 μm , no more than about 750 μm , no more than about 700 μm , no more than about 650 μm , no more than about 600 μm , no more than about 550 μm , no more than about 500 μm , no more than about 450 μm , no more than about 400 μm , no more than about 350 μm , no more than about 300 μm , no more than about 250 μm , no more than about 200 μm , or no more than about 150 μm . Combinations of the above-referenced thicknesses are also possible (e.g., at least about 100 μm and no more than about 2,000 μm or at least about 150 μm and no more than about 500 μm), inclusive of all values and ranges therebetween. In some embodiments, t2 can be about 100 μm , about 150 μm , about 200 μm , about 250 μm , about 300 μm , about 350 μm , about 400 μm , about 450 μm , about 500 μm , about 550 μm , about 600 μm , about 650 μm , about 700 μm , about 750 μm , about 800 μm , about 850 μm , about 900 μm , about 950 μm , about 1,000 μm , about 1,100 μm , about 1,200 μm , about 1,300 μm , about 1,400 μm , about 1,500 μm , about 1,600 μm , about 1,700 μm , about 1,800 μm , about 1,900 μm , or about 2,000 μm .

[0173] In some embodiments, t2 can be larger than t1 by at least about 50 nm, at least about 100 nm, at least about 200 nm, at least about 300 nm, at least about 400 nm, at least about 500 nm, at least about 600 nm, at least about 700 nm, at least about 800 nm, at least about 900 nm, at least about 1 μm , at least about 2 μm , at least about 3 μm , at least about 4 μm , at least about 5 μm , at least about 6 μm , at least about 7 μm , at least about 8 μm , at least about 9 μm , at least about 10 μm , at least about 15 μm , at least about 20 μm , at least about 25 μm , at least about 30 μm , at least about 35 μm , at least about 40 μm , or at least about 45 μm . In some embodiments, t2 can be larger than t1 by no more than about 50 μm , no more than about 45 μm , no more than about 40 μm , no more than about 35 μm , no more than about 30 μm , no more than about 25 μm , no more than about 20 μm , no more than about 15 μm , no more than about 10 μm , no more than about 9 μm , no more than about 8 μm , no more than about 7 μm , no more than about 6 μm , no more than about 5 μm , no more than about 4 μm , no more than about 3 μm , no more than about 2 μm , no more than about 1 μm , no more than about 900 nm, no more than about 800 nm, no more than about 700 nm, no more than about 600 nm, no more than about 500 nm, no more than about 400 nm, no more than about 300 nm, no more than about 200 nm, or no more than about 100 nm. Combinations of the above-referenced thickness differences are also possible (e.g., at least about 50 nm and no more than about 50 μm or at least about 500 nm and no more than about 10 μm), inclusive of all values and ranges therebetween.

[0174] In some embodiments, t2 can be greater than t1 by at least about 0.1%, at least about 0.2%, at least about 0.3%, at least about 0.4%, at least about 0.5%, at least about 0.6%, at least about 0.7%, at least about 0.8%, at least about 0.9%, at least about 1%, at least about 2%, at least about 3%, at least about 4%, at least about 5%, at least about 6%, at least about 7%, at least about 8%, or at least about 9%. In some embodiments, t2 can be greater than t1 by no more than about 10%, no more than about 9%, no more than about 8%, no more than about 7%, no more than about 6%, no more than about 5%, no more than about 4%, no more than about 3%, no more than about 2%, no more than about 1%, no more than about 0.9%, no more than about 0.8%, no more than about 0.7%, no more than about 0.6%, no more than about 0.5%, no more than about 0.4%, no more than about 0.3%, or no more than about 0.2%. Combinations of the above-referenced thickness differences are also possible (e.g., at least about 0.1% and no more than about 10% or at least about 0.5% and no more than about 5%), inclusive of all values and ranges therebetween. In some embodiments, t2 can be greater than t1 by about 0.1%, about 0.2%, about 0.3%, about 0.4%, about 0.5%, about 0.6%, about 0.7%, about 0.8%, about 0.9%, about 1%, about 2%, about 3%, about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, or about 10%.

[0175] As shown, the cathode **530** has a non-uniform thickness. In some embodiments, the anode **510** can have a non-uniform thickness. In some embodiments, the first separator **550a** and/or the second separator **550b** can have a non-uniform thickness to conform to the non-uniform thickness of the cathode **530** and/or the anode **510**. In other words, the first separator **550a** and/or the second separator **550b** can have a smaller thickness on the side of the electrochemical cell **200** including t1 and a larger thickness on the side of the electrochemical cell **200** including t2. In

some embodiments, the first separator **550a** and/or the second separator **550b** can include a malleable structure, such that the non-uniform thickness of the cathode **530** and/or the anode **510** are absorbed and do not affect the uniformity of the thickness of the electrochemical cell **500**.

[0176] FIG. 6 is an illustration of an electrochemical cell **600** with an interlayer **660**, according to an embodiment. As shown, the electrochemical cell **600** includes an anode **610** disposed on an anode current collector **620** with a lithium metal layer **670** interposed therebetween, a cathode **630** disposed on a cathode current collector **640**, with a first separator **650a** and a second separator **650b** disposed between the anode **610** and the cathode **630**. The interlayer **660** is disposed between the first separator **650a** and the second separator **650b**. In some embodiments, the anode **610**, the anode current collector **620**, the cathode **630**, the cathode current collector **640**, the first separator **650a**, the second separator **650b**, and the interlayer **660** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **610**, the anode current collector **620**, the cathode **630**, the cathode current collector **640**, the first separator **650a**, the second separator **650b**, and the interlayer **660** are not described in greater detail herein.

[0177] The lithium metal layer **670** is positioned adjacent to the anode **610**. In some embodiments, the lithium metal layer **670** can aid in even electron distribution in the anode **610**. In some embodiments, the anode **610** can include hard carbon coated onto the lithium metal layer **670**. In some embodiments, the anode **610** can include silicon coated onto the lithium metal layer **670**. In some embodiments, the anode **610** can include graphite coated onto the lithium metal layer **670**. In some embodiments, the anode **610** can include indium coated onto the lithium metal layer **670**.

[0178] In some embodiments, the lithium metal layer **670** can have a thickness of at least about 50 μm , at least about 60 μm , at least about 70 μm , at least about 80 μm , at least about 90 μm , at least about 100 μm , at least about 150 μm , at least about 200 μm , at least about 250 μm , at least about 300 μm , at least about 350 μm , at least about 400 μm , or at least about 450 μm . In some embodiments, the lithium metal layer **670** can have a thickness of no more than about 500 μm , no more than about 450 μm , no more than about 400 μm , no more than about 350 μm , no more than about 300 μm , no more than about 250 μm , no more than about 200 μm , no more than about 150 μm , no more than about 100 μm , no more than about 90 μm , no more than about 80 μm , no more than about 70 μm , or no more than about 60 μm . Combinations of the above-referenced thicknesses are also possible (e.g., at least about 50 μm and no more than about 500 μm or at least about 100 μm and no more than about 400 μm), inclusive of all values and ranges therebetween. In some embodiments, the lithium metal layer **670** can have a thickness of about 50 μm , about 60 μm , about 70 μm , about 80 μm , about 90 μm , about 100 μm , about 150 μm , about 200 μm , about 250 μm , about 300 μm , about 350 μm , about 400 μm , about 450 μm , or about 500 μm .

[0179] FIGS. 7A-7B are illustrations of separators **750a**, **750b** (collectively referred to as separators **750**) with an interlayer **760** disposed therebetween, according to an embodiment. As shown, the interlayer **760** is disposed on the

separator **750a**. A frame **762** is disposed on the interlayer **760** and around an outside edge of the interlayer **760**. A tab **763** is coupled to the frame **762** and extends beyond the separators **750**, such that the tab **763** can couple to a voltage source or a voltage measurement lead. The second separator **750b** is removed in FIG. 7A to show details of the components between the separators **750**. FIG. 7B shows the second separator **750b** in place, such that the components between the separators **750** are not visible. In some embodiments, the first separator **750a**, the second separator **750b**, and the interlayer **760** can be the same or substantially similar to the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the first separator **750a**, the second separator **750b**, and the interlayer **760** are not described in greater detail herein.

[0180] The frame **762** is disposed around an outside edge of the interlayer **760**. In some embodiments, the frame **762** is composed of a conductive material. In some embodiments, the frame **762** can be composed of aluminum, copper, conductive ceramic, a conductive polymer, carbon fiber paper, nickel, titanium, or any combination thereof. In some embodiments, the frame **762** can be coupled to the first separator **750a** and/or the second separator **750b** via an adhesive. In some embodiments, the adhesive can be applied during the electrochemical cell manufacturing process. In some embodiments, the adhesive can be electrically conductive. In some embodiments, the tab **763** can be welded to the frame **762**. In some embodiments, the tab **763** can be ultrasonically welded to the frame **762**. In some embodiments, the tab **763** can be coupled to a pouch tab (not shown) via an adhesive. In some embodiments, the interlayer **760** can have multiple layers. In some embodiments, the interlayer **760** can include gold, carbon, indium, tin, or any combination thereof. In some embodiments, the interlayer **760** can be coated to the tab **763**, the first separator **750a**, and/or the second separator **750b**. In some embodiments, the tab **763** can be directly welded to the interlayer **760**, the first separator **750a**, and/or the second separator **750b**.

[0181] As shown, the frame **762** overlaps the interlayer **760** by an overlapping width w . The width w can vary to reduce the resistance of the frame **762** and can be tuned based on the material of the frame **762**, the thickness of the frame **762**, and/or the capacity of the electrochemical cell. In some embodiments, w can be at least about 250 μm , at least about 500 μm , at least about 750 μm , at least about 1 mm, at least about 1.5 mm, at least about 2 mm, at least about 2.5 mm, at least about 3 mm, at least about 3.5 mm, at least about 4 mm, at least about 4.5 mm, at least about 5 mm, at least about 5.5 mm, at least about 6 mm, at least about 6.5 mm, at least about 7 mm, at least about 7.5 mm, at least about 8 mm, at least about 8.5 mm, at least about 9 mm, or at least about 9.5 mm. In some embodiments, w can be no more than about 10 mm, no more than about 9.5 mm, no more than about 9 mm, no more than about 8.5 mm, no more than about 8 mm, no more than about 7.5 mm, no more than about 7 mm, no more than about 6.5 mm, no more than about 6 mm, no more than about 5.5 mm, no more than about 5 mm, no more than about 4.5 mm, no more than about 4 mm, no more than about 3.5 mm, no more than about 3 mm, no more than about 2.5 mm, no more than about 2 mm, no more than about 1.5 mm, no more than about 1 mm, no more than about 750 μm , or no more than about 500 μm . Combinations of the above-referenced values of w are also possible (e.g.,

at least about 250 μm and no more than about 10 mm or at least about 1 mm and no more than about 5 mm), inclusive of all values and ranges therebetween. In some embodiments, the width w can be about 250 μm , about 500 μm , about 750 μm , about 1 mm, about 1.5 mm, about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, about 5 mm, about 5.5 mm, about 6 mm, about 6.5 mm, about 7 mm, about 7.5 mm, about 8 mm, about 8.5 mm, about 9 mm, about 9.5 mm, or about 10 mm.

[0182] As shown, the interlayer 760 includes an active area 761. A distance from the inside edge of the frame to the outside edge of the active area is represented as d_a . In some embodiments, the distance d_a can be at least about 1 mm, at least about 1.5 mm, at least about 2 mm, at least about 2.5 mm, at least about 3 mm, at least about 3.5 mm, at least about 4 mm, at least about 4.5 mm, at least about 5 mm, at least about 5.5 mm, at least about 6 mm, at least about 6.5 mm, at least about 7 mm, at least about 7.5 mm, at least about 8 mm, at least about 8.5 mm, at least about 9 mm, or at least about 9.5 mm. In some embodiments, the distance d_a can be no more than about 10 mm, no more than about 9.5 mm, no more than about 9 mm, no more than about 8.5 mm, no more than about 8 mm, no more than about 7.5 mm, no more than about 7 mm, no more than about 6.5 mm, no more than about 6 mm, no more than about 5.5 mm, no more than about 5 mm, no more than about 4.5 mm, no more than about 4 mm, no more than about 3.5 mm, no more than about 3 mm, no more than about 2.5 mm, no more than about 2 mm, or no more than about 1.5 mm.

[0183] Combinations of the above-referenced distances for d_a are also possible (e.g., at least about 1 mm and no more than about 10 mm or at least about 2 mm and no more than about 8 mm), inclusive of all values and ranges therebetween. In some embodiments, the distance d_a can be about 1 mm, about 1.5 mm, about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, about 5 mm, about 5.5 mm, about 6 mm, about 6.5 mm, about 7 mm, about 7.5 mm, about 8 mm, about 8.5 mm, about 9 mm, about 9.5 mm, or about 10 mm.

[0184] FIGS. 8A-8B are illustrations of separators 850a, 850b (collectively referred to as separators 850) with an interlayer 860 disposed therebetween, according to an embodiment. As shown, the interlayer 860 is disposed on the separator 850a. A wire 862 is disposed around an outside perimeter of the interlayer 860. A terminal end of the wire 862 acts as a tab 863 extending beyond the separators 850, such that the tab 863 can couple to a voltage source or a voltage measurement lead. The second separator 850b is removed in FIG. 8A to show details of the components between the separators 850. FIG. 8B shows the second separator 850b in place, such that the components between the separators 850 are not visible. In some embodiments, the first separator 850a, the second separator 850b, and the interlayer 860 can be the same or substantially similar to the first separator 750a, the second separator 750b, and the interlayer 760, as described above with reference to FIGS. 7A-7B. Thus, certain aspects of the first separator 850a, the second separator 850b, and the interlayer 860 are not described in greater detail herein.

[0185] In some embodiments, the wire 862 can be composed of aluminum. In some embodiments, the wire 862 can be composed of copper. In some embodiments, the gauge of the wire 862 can be selected based on the capacity of the electrochemical cell (i.e., a thicker wire can be incorporated

into an electrochemical cell with a higher capacity). In some embodiments, the wire 862 can have 32 American Wire Gauge (AWG), 31 AWG, 30 AWG, 29 AWG, 28 AWG, 27 AWG, 26 AWG, 25 AWG, 24 AWG, 23 AWG, 22 AWG, 21 AWG, 20 AWG, 19 AWG, or 18 AWG, inclusive of all sizes therebetween. In some embodiments, the wire 862 can be attached to the first separator 850a and/or the second separator 850b via an adhesive. In some embodiments, the adhesive can be applied to the wire 862 during the electrochemical cell manufacturing process. In some embodiments, the adhesive can be electrically conductive.

[0186] In some embodiments, a coating can be molded around the wire 862. In some embodiments, the coating can include a polymer coating. In some embodiments, the coating can include polyethylene. In some embodiments, the wire 862 can be connected directly to a diode (not shown). In some embodiments, the wire 862 can be connected to a diode via soldering, butt splicing, or any other suitable connection. In some embodiments, an electrochemical cell that includes the wire 862 can have a lower mass than an electrochemical cell that includes a frame (e.g., the frame 762, as described above with reference to FIG. 7).

[0187] As shown, the wire 862 is offset from the edge of the interlayer 860 by an offset distance d . In some embodiments, d can be the same or substantially similar on all sides of the interlayer 860. In some embodiments, d can vary from one side of the interlayer 860 to another. In some embodiments, the wire 862 can be positioned on the edge of the interlayer 860, such that d is about 0 mm. In some embodiments, d can be at least about 0 mm, at least about 0.5 mm, at least about 1 mm, at least about 1.5 mm, at least about 2 mm, at least about 2.5 mm, at least about 3 mm, at least about 3.5 mm, at least about 4 mm, at least about 4.5 mm, at least about 5 mm, at least about 5.5 mm, at least about 6 mm, at least about 6.5 mm, at least about 7 mm, at least about 7.5 mm, at least about 8 mm, at least about 8.5 mm, at least about 9 mm, or at least about 9.5 mm. In some embodiments, d can be no more than about 10 mm, no more than about 9.5 mm, no more than about 9 mm, no more than about 8.5 mm, no more than about 8 mm, no more than about 7.5 mm, no more than about 7 mm, no more than about 6.5 mm, no more than about 6 mm, no more than about 5.5 mm, no more than about 5 mm, no more than about 4.5 mm, no more than about 4 mm, no more than about 3.5 mm, no more than about 3 mm, no more than about 2.5 mm, no more than about 2 mm, no more than about 1.5 mm, no more than about 1 mm, or no more than about 0.5 mm. Combinations of the above-referenced values of d are also possible (e.g., at least about 0 mm and no more than about 10 mm or at least about 1 mm and no more than about 5 mm), inclusive of all values and ranges therebetween. In some embodiments, d can be about 0 mm, about 0.5 mm, about 1 mm, about 1.5 mm, about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, about 5 mm, about 5.5 mm, about 6 mm, about 6.5 mm, about 7 mm, about 7.5 mm, about 8 mm, about 8.5 mm, about 9 mm, about 9.5 mm, or about 10 mm.

[0188] FIGS. 9A-9B are illustrations of separators 950a, 950b (collectively referred to as separators 950) with an interlayer 960 disposed therebetween, according to an embodiment. As shown, the interlayer 960 is disposed on the separator 950a. A wire 962 is disposed around portions of an outside perimeter of the interlayer 960 and the wire 962 is patterned across the interlayer 960. A terminal end of the

wire 962 acts as a tab 963 extending beyond the separators 950, such that the tab 963 can couple to a voltage source or a voltage measurement lead. The second separator 950b is removed in FIG. 9A to show details of the components between the separators 950. FIG. 9B shows the second separator 950b in place, such that the components between the separators 950 are not visible. In some embodiments, the first separator 950a, the second separator 950b, the interlayer 960, the wire 962, and the tab 963 can be the same or substantially similar to the first separator 850a, the second separator 850b, the interlayer 860, the wire 862, and the tab 863, as described above with reference to FIGS. 8A-8B. Thus, certain aspects of the first separator 950a, the second separator 950b, the interlayer 960, the wire 962, and the tab 963 are not described in greater detail herein.

[0189] In some embodiments, the pattern of the wire 962 can be optimized to decrease resistance of the electrochemical cell. For example, the pattern the wire 962 follows along the interlayer 960 can minimize the distance from a dendrite to the wire 962 to a diode, a resistor, a resistor, a fuse, a transistor, or any combination thereof. This can minimize the amount of growth a dendrite can experience in the interlayer 960, before redirecting the current and discharging the electrochemical cell. As shown, the wire 962 is formed in serpentine pattern with rounded edges. In some embodiments, the wire 962 can be formed in a serpentine pattern with sharp edges. In some embodiments, the wire 962 can be formed in a spiral pattern.

[0190] FIGS. 10A-10C are illustrations of separators 1050a, 1050b (collectively referred to as separators 1050) with an interlayer 1060 disposed therebetween, according to an embodiment. As shown, the interlayer 1060 is disposed on the separator 1050a. A conductive coating 1065 is disposed on the of the interlayer 1060 with a tab 1063 coupled to the interlayer 1060, the conductive coating 1065, the first separator 1050a, and/or the second separator 1050b. The second separator 1050b and the conductive coating 1065 are removed in FIG. 10A. The second separator 1050b is removed from FIG. 10B to show details of the components between the separators 1050. FIG. 10C shows the second separator 1050b in place, such that the components between the separators 1050 are not visible. In some embodiments, the first separator 1050a, the second separator 1050b, and the interlayer 1060 can be the same or substantially similar to the first separator 750a, the second separator 750b, and the interlayer 760, as described above with reference to FIGS. 7A-7B. Thus, certain aspects of the first separator 1050a, the second separator 1050b, and the interlayer 1060 are not described in greater detail herein.

[0191] The conductive coating 1065 is disposed on the interlayer 1060. In some embodiments, the conductive coating 1065 can be composed of a conductive polymer. In some embodiments, the conductive coating 1065 can be composed of poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT: PSS), polyaniline (PANI), polypyrrole (PPy), or any combination thereof. In some embodiments, the tab 1063 can be in the form of a wire (e.g., the same or substantially similar to the tab 963, as described above with reference to FIGS. 9A-9B). In some embodiments, the tab 1063 can be composed of aluminum, copper, a conductive ceramic, a conductive polymer, a carbon fiber paper, or any combination thereof.

[0192] In some embodiments, the conductive coating 1065 can be coated onto the second separator 1050b. In some

embodiments, the tab 1063 can be adhered to the first separator 1050a and/or the second separator 1050b and in conduct with the conductive coating 1065. In some embodiments, the tab 1063 can be coupled to a diode (not shown) via soldering, welding, butt splicing, or any other suitable coupling means. In some embodiments, the tab 1063 can be welded to a pouch tab (not shown) and connected to the diode. In some embodiments, the conductive coating 1065 can cover the entire active area of the interlayer 1065. In some embodiments, the conductive coating 1065 can be installed without a frame (e.g., the frame 762, as described above with reference to FIGS. 7A-7B).

[0193] FIGS. 11A-11B are illustrations of separators 950a, 950b (collectively referred to as separators 1150) with an interlayer 1160 disposed therebetween, according to an embodiment. As shown, the interlayer 1160 is disposed on the separator 1150a. A tab 1163 is coupled directly to the interlayer 1160. The second separator 1150b is removed in FIG. 11A to show details of the components between the separators 1150. FIG. 11B shows the second separator 1150b in place, such that the components between the separators 1150 are not visible. In some embodiments, the first separator 1150a, the second separator 1150b, the interlayer 1160, and the tab 1163 can be the same or substantially similar to the first separator 750a, the second separator 750b, the interlayer 760, and the tab 763, as described above with reference to FIGS. 7A-7B. Thus, certain aspects of the first separator 1150a, the second separator 1150b, the interlayer 1160, and the tab 1163 are not described in greater detail herein.

[0194] In some embodiments, the tab 1163 can be composed of aluminum, copper, conductive ceramic, conductive polymer, carbon fiber paper, or any combination thereof. In some embodiments, the tab 1163 can be pressed into the interlayer 1160. For example, the interlayer 1160 can include a carbon/binder slurry and the tab 1163 can be pressed into the carbon/binder slurry with such a force that the tab 1163 and/or the interlayer 1160 yields. In some embodiments, the tab 1163 can be attached to the either of the separators 1150 via an adhesive, a staple, ultrasonic welding, adhesive tape, adhesive glue, or any combination thereof. In some embodiments, the tab 1163 can be in contact with the interlayer 1160 via pressure applied to the electrochemical cell. In some embodiments, the configuration shown in FIGS. 11A-11B can reduce the mass of an electrochemical cell and improve energy density, in comparison with other embodiments.

[0195] In some embodiments, the distance from the sealing/adhesive area of the tab 1163 to the edge of the interlayer 1160 can be at least about 1 mm, at least about 1.5 mm, at least about 2 mm, at least about 2.5 mm, at least about 3 mm, at least about 3.5 mm, at least about 4 mm, at least about 4.5 mm, at least about 5 mm, at least about 5.5 mm, at least about 6 mm, at least about 6.5 mm, at least about 7 mm, at least about 7.5 mm, at least about 8 mm, at least about 8.5 mm, at least about 9 mm, or at least about 9.5 mm. In some embodiments, the distance from the sealing/adhesive area of the tab 1163 to the edge of the interlayer 1160 can be no more than about 10 mm, no more than about 9.5 mm, no more than about 9 mm, no more than about 8.5 mm, no more than about 8 mm, no more than about 7.5 mm, no more than about 7 mm, no more than about 6.5 mm, no more than about 6 mm, no more than about 5.5 mm, no more than about 5 mm, no more than about 4.5 mm, no more than about 4 mm, no more than about 3.5 mm, no more than about 3 mm, no

more than about 2.5 mm, no more than about 2 mm, or no more than about 1.5 mm. Combinations of the above-referenced distances are also possible (e.g., at least about 1 mm and no more than about 10 mm or at least about 2 mm and no more than about 8 mm), inclusive of all values and ranges therebetween. In some embodiments, the distance from the sealing/adhesive area of the tab **1163** to the edge of the interlayer **1160** can be about 1 mm, about 1.5 mm, about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, about 5 mm, about 5.5 mm, about 6 mm, about 6.5 mm, about 7 mm, about 7.5 mm, about 8 mm, about 8.5 mm, about 9 mm, about 9.5 mm, or about 10 mm.

[0196] In some embodiments, the width of the sealing/adhesive area of the tab **1163** (i.e., the distance along the length of the tab **1163**, or in the direction up and down the page in FIG. **11A**) can be at least about 1 mm, at least about 1.5 mm, at least about 2 mm, at least about 2.5 mm, at least about 3 mm, at least about 3.5 mm, at least about 4 mm, at least about 4.5 mm, at least about 5 mm, at least about 5.5 mm, at least about 6 mm, at least about 6.5 mm, at least about 7 mm, at least about 7.5 mm, at least about 8 mm, at least about 8.5 mm, at least about 9 mm, or at least about 9.5 mm. In some embodiments, the width of the sealing/adhesive area of the tab **1163** can be no more than about 10 mm, no more than about 9.5 mm, no more than about 9 mm, no more than about 8.5 mm, no more than about 8 mm, no more than about 7.5 mm, no more than about 7 mm, no more than about 6.5 mm, no more than about 6 mm, no more than about 5.5 mm, no more than about 5 mm, no more than about 4.5 mm, no more than about 4 mm, no more than about 3.5 mm, no more than about 3 mm, no more than about 2.5 mm, no more than about 2 mm, or no more than about 1.5 mm. Combinations of the above-referenced widths are also possible (e.g., at least about 1 mm and no more than about 10 mm or at least about 2 mm and no more than about 8 mm), inclusive of all values and ranges therebetween. In some embodiments, the width of the sealing/adhesive area of the tab **1163** can be about 1 mm, about 1.5 mm, about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, about 5 mm, about 5.5 mm, about 6 mm, about 6.5 mm, about 7 mm, about 7.5 mm, about 8 mm, about 8.5 mm, about 9 mm, about 9.5 mm, or about 10 mm.

[0197] As shown, the interlayer **1160** includes an active area **1161**. In some embodiments, the distance from the edge of the tab **1163** to the edge of the active area **1161** can be at least about 1 mm, at least about 1.5 mm, at least about 2 mm, at least about 2.5 mm, at least about 3 mm, at least about 3.5 mm, at least about 4 mm, at least about 4.5 mm, at least about 5 mm, at least about 5.5 mm, at least about 6 mm, at least about 6.5 mm, at least about 7 mm, at least about 7.5 mm, at least about 8 mm, at least about 8.5 mm, at least about 9 mm, or at least about 9.5 mm. In some embodiments, the distance from the edge of the tab **1163** to the edge of the active area **1161** can be no more than about 10 mm, no more than about 9.5 mm, no more than about 9 mm, no more than about 8.5 mm, no more than about 8 mm, no more than about 7.5 mm, no more than about 7 mm, no more than about 6.5 mm, no more than about 6 mm, no more than about 5.5 mm, no more than about 5 mm, no more than about 4.5 mm, no more than about 4 mm, no more than about 3.5 mm, no more than about 3 mm, no more than about 2.5 mm, no more than about 2 mm, or no more than about 1.5 mm. Combinations of the above-referenced distances are also possible (e.g., at least about 1 mm and no more than about 10 mm or at least

about 2 mm and no more than about 8 mm), inclusive of all values and ranges therebetween. In some embodiments, the distance from the edge of the tab **1163** to the edge of the active area **1161** can be about 1 mm, about 1.5 mm, about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, about 5 mm, about 5.5 mm, about 6 mm, about 6.5 mm, about 7 mm, about 7.5 mm, about 8 mm, about 8.5 mm, about 9 mm, about 9.5 mm, or about 10 mm.

[0198] FIG. **12** is a flow diagram of a method **10** of operating an electrochemical cell, according to an embodiment. As shown, the method **10** includes measuring a voltage between a first electrode and a second electrode at step **11** and measuring a voltage between the first electrode and a first interlayer at step **12**. The method **10** optionally includes measuring voltage between the first interlayer and a second electrode at step **13** and measuring a voltage between the first electrode and the second interlayer at step **14**. The method **10** further includes closing a circuit between the first electrode and the second electrode at step **15**.

[0199] Measuring the voltage between the first electrode and the second electrode at step **11** determines the voltage, at which the electrochemical cell is operating. Measuring the voltage between the first electrode and the first interlayer at step **12** provides a basis of comparison for detection of dendrites. If the voltage between the first electrode and the first interlayer decreases to less than a threshold value, this can be an indication of a short circuit between the first electrode and the first interlayer. In response, a circuit is closed between the first electrode and the second electrode at step **15**. In some embodiments, the threshold voltage can be a voltage difference between the first electrode and the first interlayer. In some embodiments the threshold voltage can be about 0.001 V, about 0.002 V, about 0.003 V, about 0.004 V, about 0.005 V, about 0.006 V, about 0.007 V, about 0.008 V, about 0.009 V, about 0.01 V, about 0.02 V, about 0.03 V, about 0.04 V, about 0.05 V, about 0.06 V, about 0.07 V, about 0.08 V, about 0.09 V, about 0.1 V, about 0.2 V, about 0.3 V, about 0.4 V, about 0.5 V, about 0.6 V, about 0.7 V, about 0.8 V, about 0.9 V, or about 1 V, inclusive of all values and ranges therebetween. In some embodiments, the threshold voltage can be a threshold voltage fraction of the voltage measured between the first electrode and the interlayer to the voltage measured between the first electrode and the second electrode. In some embodiments, the threshold voltage fraction can be about 0.001, about 0.002, about 0.003, about 0.004, about 0.005, about 0.006, about 0.007, about 0.008, about 0.009, about 0.01, about 0.02, about 0.03, about 0.04, about 0.05, about 0.06, about 0.07, about 0.08, about 0.09, about 0.1, about 0.2, about 0.3, about 0.4, or about 0.5, inclusive of all values and ranges therebetween.

[0200] The method **10** optionally includes measuring the voltage between the first interlayer and the second electrode at step **13**. In some embodiments, the circuit is closed at step **15** in response to the voltage measured between the first interlayer and the second electrode at step **13**. In some embodiments, the closing of the circuit at step **15** can be in response to the voltage measured between the first interlayer and the second electrode decreasing to less than a threshold value. In some embodiments, the threshold voltage can be a voltage difference between the first electrode and the second interlayer. In some embodiments, the threshold voltage can be a voltage difference between the first interlayer and the second electrode. In some embodiments the threshold voltage can be about 0.001 V, about 0.002 V, about 0.003 V,

about 0.004 V, about 0.005 V, about 0.006 V, about 0.007 V, about 0.008 V, about 0.009 V, about 0.01 V, about 0.02 V, about 0.03 V, about 0.04 V, about 0.05 V, about 0.06 V, about 0.07 V, about 0.08 V, about 0.09 V, about 0.1 V, about 0.2 V, about 0.3 V, about 0.4 V, about 0.5 V, about 0.6 V, about 0.7 V, about 0.8 V, about 0.9 V, or about 1 V, inclusive of all values and ranges therebetween. In some embodiments, the threshold voltage can be a threshold voltage fraction of the voltage measured between the first electrode and the second interlayer and the voltage measured between the first electrode and the second electrode. In some embodiments, the threshold voltage can be a threshold voltage fraction of the voltage measured between the first interlayer and the second electrode and the voltage measured between the first electrode and the second electrode. In some embodiments, the threshold voltage fraction can be about 0.001, about 0.002, about 0.003, about 0.004, about 0.005, about 0.006, about 0.007, about 0.008, about 0.009, about 0.01, about 0.02, about 0.03, about 0.04, about 0.05, about 0.06, about 0.07, about 0.08, about 0.09, about 0.1, about 0.2, about 0.3, about 0.4, or about 0.5, inclusive of all values and ranges therebetween.

[0201] Step 15 includes closing a circuit between the first electrode and the second electrode. In some embodiments, closing the circuit can include redirecting current through a resistor. In some embodiments, closing the circuit can include redirecting current through a separate device (e.g., a lightbulb, an additional electrochemical cell, etc.).

[0202] FIG. 13 is an illustration of an electrochemical cell 1300 with an interlayer 1360, according to an embodiment. As shown, the electrochemical cell 1300 includes an anode 1310 disposed on an anode current collector 1320, a cathode 1330 disposed on a cathode current collector 1340, with a first separator 1350a and a second separator 1350b (collectively referred to as separators 1350) disposed between the anode 1310 and the cathode 1330. The interlayer 1360 is disposed between the first separator 1350a and the second separator 1350b. A first film 1380a is coupled to the anode current collector 1320 and a second film 1380b is coupled to the cathode current collector 1340. The first film 1380a and the second film 1380b combine to form a pouch and are collectively referred to herein as a pouch 1380. A tab 1363 extends from the interlayer 1360. A staple 1367 is wrapped around the separators 1350 and the interlayer 1360. In some embodiments, the anode 1310, the anode current collector 1320, the cathode 1330, the cathode current collector 1340, the first separator 1350a, the second separator 1350b, and the interlayer 1360 can be the same or substantially similar to the anode 210, the anode current collector 220, the cathode 230, the cathode current collector 240, the first separator 250a, the second separator 250b, and the interlayer 260, as described above with reference to FIGS. 2A-2B. In some embodiments, the tab 1363 can be the same or substantially similar to the tab 763, as described above with reference to FIGS. 7A-7B. Thus, certain aspects of the anode 1310, the anode current collector 1320, the cathode 1330, the cathode current collector 1340, the first separator 1350a, the second separator 1350b, and the interlayer 1360 are not described in greater detail herein.

[0203] As shown, the tab 1363 protrudes from the pouch 1380, such that the tab 1363 can connect to a voltage source, a voltage measurement point, a diode, a resistor, a transistor, a fuse, or any combination thereof. The staple 1367 encircles the separators 1350 and the tab 1363, such that the staple

1367 presses the separators 1350 and the tab 1363 together. In some embodiments, the staple 1367 partially encircles the separators 1350 and the tab 1363. In some embodiments, the staple 1367 fully encircles the separators 1350 and the tab 1363. In some embodiments, the staple 1367 can add structural stability to the tab 1363 and the separators 1350, such that the 1367 aids in preventing the tab 1363 from being removed from the electrochemical cell 1300. As shown, the staple 1367 is positioned inside the first film 1380a and the second film 1380b. In other words, the staple 1367 is inside the pouch 1380. In some embodiments, the staple 1367 can be positioned outside the pouch 1380. In some embodiments, the staple 1367 can be composed of metal. In some embodiments, the staple 1367 can be composed of plastic. In some embodiments, the staple 1367 can create a sealing area around the separators 1350 and the tab 1363. In some embodiments, the sealing area can be formed via heat sealing. In some embodiments, the sealing area can be formed via pressing the staple 1367 onto the separators 1350 at a high pressure. In some embodiments, the sealing area can be formed via an adhesive between the staple 1367 and the separators 1350. In some embodiments, the sealing area can be formed via an adhesive between the staple 1367 and the pouch 1380.

[0204] As shown, the tab 1363 is positioned between the interlayer 1360 and the first separator 1350a. In some embodiments, the tab 1363 can be positioned between the interlayer 1360 and the second separator 1350b. In some embodiments, the tab 1363 can be sealed to the interlayer 1360, the first separator 1350a, and/or the second separator 1350b (e.g., via an adhesive, heat sealing, etc.). In some embodiments, the tab 1363 can be coupled to the interlayer 1360 before assembly of the electrochemical cell 1300. In some embodiments, the tab 1363 can be coupled to the interlayer 1360 after assembly of the electrochemical cell 1300. In some embodiments, the tab 1363 can protrude through a hole in the pouch 1380. In some embodiments, the hole can be cut into the pouch 1380 before installing the tab 1363. In some embodiments, the first film 1380a and/or the second film 1380b can be cut into a desired shape prior to coupling the first film 1380a and/or the second film 1380b to the anode current collector 1320 and/or the cathode current collector 1340. The anode current collector 1320 and the cathode current collector 1340 can also have tabs connected thereto. In some embodiments, the tabs connected to the anode current collector 1320 and the cathode current collector 1340 can penetrate precut holes in the first film 1380a and/or the second film 1380b.

[0205] FIG. 14 is an illustration of an electrochemical cell 1400 with an interlayer 1460, according to an embodiment. As shown, the electrochemical cell 1400 includes an anode 1410 disposed on an anode current collector 1420, a cathode 1430 disposed on a cathode current collector 1440, with a first separator 1450a and a second separator 1450b (collectively referred to as separators 1450) disposed between the anode 1410 and the cathode 1430. The interlayer 1460 is disposed between the first separator 1450a and the second separator 1450b. A first film 1480a is coupled to the anode current collector 1420 and a second film 1480b is coupled to the cathode current collector 1440. The first film 1480a and the second film 1480b combine to form a pouch and are collectively referred to herein as a pouch 1480. A tab 1463 extends from the interlayer 1460 and is bonded to the first film 1480a and the second film 1480b via a first sealing

region **1468a** and a second sealing region **1468b**, respectively. In some embodiments, the anode **1410**, the anode current collector **1420**, the cathode **1430**, the cathode current collector **1440**, the first separator **1450a**, the second separator **1450b**, the interlayer **1460**, the tab **1463**, and the pouch **1480** can be the same or substantially similar to the anode **1310**, the anode current collector **1320**, the cathode **1330**, the cathode current collector **1340**, the first separator **1350a**, the second separator **1350b**, the interlayer **1360**, the tab **1363**, and the pouch **1380**, as described above with reference to FIG. 13. Thus, certain aspects of the anode **1410**, the anode current collector **1420**, the cathode **1430**, the cathode current collector **1440**, the first separator **1450a**, the second separator **1450b**, the interlayer **1460**, the tab **1463**, and the pouch **1480** are not described in greater detail herein.

[0206] In some embodiments, the first sealing region **1468a** and/or the second sealing region **1468b** (collectively referred to as heat sealing regions **1468**) can include a heat seal. In some embodiments, the sealing regions **1468** can include an adhesive layer bonding the tab **1463** to the pouch **1480**. In some embodiments, the sealing regions **1468** can include tape. In some embodiments, the tape can be wrapped around the outside of the first film **1480a** and/or the second film **1480b**. In some embodiments, the separators **1450** can include ceramic separators. The stiffer ceramic materials can improve the stability of the sealing regions **1468** and the tab **1463** in general.

[0207] FIGS. 15A-15B show an electrochemical cell **1500** arranged in a jelly roll assembly. The electrochemical cell **1500** includes a first separator **1550a** and a second separator **1550b** (collectively referred to as separators **1550**) with an interlayer **1560** disposed therebetween. An anode (not shown) and a cathode (not shown) are disposed on either side of the separators **1550**. As shown, the second separator **1550b** has the interlayer **1560** coated thereon, and the second separator **1550b** is longer than the first separator **1550a**, such that a portion of the interlayer **1560** is exposed. An interlayer tab **1563** is coupled to the exposed portion of the interlayer **1560**. FIG. 15A shows components of the electrochemical cell **1500** without a casing **1590**, and in FIG. 15B, the components of the electrochemical cell **1500** have been placed inside the casing **1590**. An anode tab **1522** and a cathode tab **1542** extend from inside to outside the casing **1590**. The interlayer tab **1563** extends from inside to outside the casing **1590**. In some embodiments, multiple pairs of separators **1550** can be used to form the jelly roll. In some embodiments, multiple sections of the interlayer **1560** material can be exposed, such that multiple tabs **1563** can be connected to the interlayer. Any of the tabs **1563** can be connected outside of the casing **1590** to diodes, transistors, fuses, voltage measurements, voltage sources, etc. In some embodiments, one or more of the separators **1550** can be created by porous coating with or without containing ceramic particles. In some embodiments, one or more of the separators **1550** can be composed of polyvinylidene fluoride (PVDF), styrene-butadiene (SBR), carboxymethyl cellulose (CMC), polyethylene oxides (PEO), polytetrafluoroethylene (PTFE), (perfluoroalkoxy alkanes) PFA, polyacrylonitrile (PAN), poly(acrylic acid) PAA, Poly Olefin, polysulfone (PES), polyethylene terephthalate (PET), polyphenylene sulfide (PPS), polybenzimidazole (PBI), polyamide-imides (PAI), polyimide (PI), polyether ether ketone (PEEK), ultraviolet (UV) curable resin, urethane/epoxy acrylate, or any

combination thereof. In some embodiments, the ceramic can include Al_2O_3 , boehmite, MgO , $\text{Al}(\text{HO})_3$, ZrO_2 .

[0208] FIGS. 16A-16B show a stack of electrochemical cells with interlayers. As shown, the stack includes anodes **1610a**, **1610b**, **1610c** (collectively referred to as anodes **1610**) and cathodes **1630a**, **1630b**, **1630c**, **1630d** (collectively referred to as cathodes **1630**). A first separator **1650a** and a second separator **1650b** (collectively referred to as separators **1650**) are woven between the anodes **1610** and the cathodes **1630** in a serpentine or zig-zag pattern. An interlayer **1660** is disposed between the separators **1650**. FIG. 16A shows a cross-sectional view of the separators **1650** with the interlayer **1660** therebetween as they are woven through the electrodes. As shown, the second separator **1650b** is longer than the first separator **1650a**, such that a portion of the interlayer **1660** disposed on the second separator **1650b** is uncovered. FIG. 16B shows a bottom view of the stack, with the portion of the interlayer **1660** exposed. An interlayer tab **1663** is coupled to the interlayer **1660** and extends outward. Anode tabs **1622** and cathode tabs **1624** extend from the anodes **1610** and the cathodes **1630**.

[0209] FIGS. 17A-17B are illustrations of an electrochemical cell **1700** with an interlayer **1760**, according to an embodiment. As shown, the electrochemical cell **1700** includes an anode **1710** disposed on an anode current collector **1720**, a cathode **1730** disposed on a cathode current collector **1740**, with a first separator **1750a** and a second separator **1750b** disposed between the anode **1710** and the cathode **1730**. The interlayer **1760** is disposed between the first separator **1750a** and the second separator **1750b**. In some embodiments, the anode **1710**, the anode current collector **1720**, the cathode **1730**, the cathode current collector **1740**, the first separator **1750a**, the second separator **1750b**, and the interlayer **1760** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **1710**, the anode current collector **1720**, the cathode **1730**, the cathode current collector **1740**, the first separator **1750a**, the second separator **1750b**, and the interlayer **1760** are not described in greater detail herein.

[0210] As shown, the interlayer **1760** includes a solid-state electrolyte layer **1762**. FIG. 17B shows the formation of a dendrite **1725**. The solid-state electrolyte layer **1762** blocks the dendrite **1725** and prevents the dendrite **1725** from penetrating the interlayer **1762**. As shown, the solid-state electrolyte layer **1762** is appended to the second separator **1750b**. In some embodiments, the solid-state electrolyte layer **1762** can be appended to the first separator **1750a**. In some embodiments, the solid-state electrolyte layer **1762** can be in a relatively central position in the interlayer **1760**, such that the solid-state electrolyte layer **1762** is appended to neither the first separator **1750a** nor the second separator **1750b**. In some embodiments, the interlayer **1760** can include multiple solid-state electrolyte layers **1762**.

[0211] FIG. 18 is an illustration of an electrochemical cell **1800** with an interlayer **1860**, according to an embodiment. As shown, the electrochemical cell **1800** includes an anode **1810** disposed on an anode current collector **1820**, a cathode **1830** disposed on a cathode current collector **1840**, with a first separator **1850a** and a second separator **1850b** disposed

between the anode **1810** and the cathode **1830**. The interlayer **1860** is disposed between the first separator **1850a** and the second separator **1850b**. In some embodiments, the anode **1810**, the anode current collector **1820**, the cathode **1830**, the cathode current collector **1840**, the first separator **1850a**, the second separator **1850b**, and the interlayer **1860** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **1810**, the anode current collector **1820**, the cathode **1830**, the cathode current collector **1840**, the first separator **1850a**, the second separator **1850b**, and the interlayer **1860** are not described in greater detail herein.

[0212] As shown, the electrochemical cell **1800** includes a circuit between the cathode **1830** and the interlayer **1860**, enabling the transfer of electrical energy between the cathode **1830** and the interlayer **1860**. By connecting the cathode **1830** and the interlayer **1860** electrically, the voltage in the interlayer **1860** can be increased to such a level that dissolves and/or oxidizes dendrites that have penetrated into the interlayer **1860**. As shown, a voltage V_1 is measured between the anode current collector **1820** and the interlayer **1860** and a voltage V_2 is measured between the anode current collector **1820** and the cathode current collector **1840**. Current flows from the interlayer **1860** to the cathode current collector **1840** via an electrical pathway that includes a resistor R_1 . The current can follow a path that includes a switch S . The current can also follow a path that includes a diode D and a resistor R_2 . In some embodiments the switch can be closed during current flow allowing normal charge and discharge to happen in the cell. In some embodiments, the switch S can be opened during a controlled charge. During charge, when the switch S is open, the voltage potential of the interlayer can increase by the same value of the voltage drop of the diode D and the resistor R_2 . In some embodiments, the diode D can be selected and placed for a specific forward or reverse breakdown voltage. In some embodiments, the breakdown voltage of the diode D can be about 0.4 V, about 0.5 V, about 0.6 V, about 0.7 V, about 0.8 V, about 0.9 V, about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V, inclusive of all values and ranges therebetween.

[0213] In some embodiments, the resistance of the resistor R_1 can be the same or substantially similar to the resistance of the resistor R_2 . In some embodiments, the resistance of the resistor R_1 can be different from the resistance of the resistor R_2 . In some embodiments, the resistor R_1 and the resistor R_2 can have resistances that represent other impedances inherent to the electrochemical cell **1800**. In some embodiments, the switch S can be replaced with a diode, a metal oxide silicon field effect transistor (MOSFET), a bipolar junction transistor (BJT), or any other suitable component. In some embodiments, the switch S can bypass the function of the diode D , creating a selective dendrite treatment mode or normal operation.

[0214] FIG. 19 is an illustration of an electrochemical cell **1900** with an interlayer **1960**, according to an embodiment. As shown, the electrochemical cell **1900** includes an anode **1910** disposed on an anode current collector **1920**, a cathode **1930** disposed on a cathode current collector **1940**, with a first separator **1950a** and a second separator **1950b** disposed

between the anode **1910** and the cathode **1930**. The interlayer **1960** is disposed between the first separator **1950a** and the second separator **1950b**. In some embodiments, the anode **1910**, the anode current collector **1920**, the cathode **1930**, the cathode current collector **1940**, the first separator **1950a**, the second separator **1950b**, and the interlayer **1960** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **1910**, the anode current collector **1920**, the cathode **1930**, the cathode current collector **1940**, the first separator **1950a**, the second separator **1950b**, and the interlayer **1960** are not described in greater detail herein.

[0215] As shown, flow of current from the interlayer **1960** to the cathode current collector **1940** can be controlled via a transistor Q . As shown, a voltage V_1 is measured between the anode current collector **1920** and the cathode current collector **1940** and a voltage V_2 is measured between the anode current collector **1920** and the interlayer **1960**. Current can flow from the interlayer **1960** to the cathode current collector **1940** via the transistor Q (and an optional resistor R). The transistor Q can act as a switching device. In some embodiments, the transistor Q can be controlled by a BMS. In some embodiments, the transistor Q can be controlled by a local hardware circuit or other system control apparatus. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **1900**. In some embodiments, the transistor Q can include a BJT. In some embodiments, the transistor Q can be replaced with a diode. In some embodiments, the transistor Q can include a junction field effect transistor (JFET). In some embodiments, the transistor Q can include a MOSFET. In some embodiments, the transistor Q can be replaced with a switch.

[0216] FIG. 20 is an illustration of an electrochemical cell **2000** with an interlayer **2060**, according to an embodiment. As shown, the electrochemical cell **2000** includes an anode **2010** disposed on an anode current collector **2020**, a cathode **2030** disposed on a cathode current collector **2040**, with a first separator **2050a** and a second separator **2050b** disposed between the anode **2010** and the cathode **2030**. The interlayer **2060** is disposed between the first separator **2050a** and the second separator **2050b**. In some embodiments, the anode **2010**, the anode current collector **2020**, the cathode **2030**, the cathode current collector **2040**, the first separator **2050a**, the second separator **2050b**, and the interlayer **2060** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **2010**, the anode current collector **2020**, the cathode **2030**, the cathode current collector **2040**, the first separator **2050a**, the second separator **2050b**, and the interlayer **2060** are not described in greater detail herein.

[0217] As shown, flow of current between the interlayer **1960** and the cathode current collector **1940** can be controlled via a diode D . In the event the current through the diode D falls below a threshold value, the diode D can 'turn off,' preventing continuous current flow to the interlayer **1960**. As shown, a voltage V_1 is measured between the anode current collector **2020** and the cathode current collector **2040**.

lector **2040** and a voltage V_2 is measured between the anode current collector **2020** and the interlayer **2060**. If the voltage of the interlayer **2060** decreases to a value below the diode forward voltage, current can flow through the diode **D** (and an optional resistor **R**), increasing the potential of the interlayer **2060**. In some embodiments, the resistance value can represent other impedances inherent to the electrochemical cell **2000**.

[0218] FIG. 21 is an illustration of an electrochemical cell **2100** with an interlayer **2160**, according to an embodiment. As shown, the electrochemical cell **2100** includes an anode **2110** disposed on an anode current collector **2120**, a cathode **2130** disposed on a cathode current collector **2140**, with a first separator **2150a** and a second separator **2150b** disposed between the anode **2110** and the cathode **2130**. The interlayer **2160** is disposed between the first separator **2150a** and the second separator **2150b**. In some embodiments, the anode **2110**, the anode current collector **2120**, the cathode **2130**, the cathode current collector **2140**, the first separator **2150a**, the second separator **2150b**, and the interlayer **2160** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **2110**, the anode current collector **2120**, the cathode **2130**, the cathode current collector **2140**, the first separator **2150a**, the second separator **2150b**, and the interlayer **2160** are not described in greater detail herein.

[0219] As shown, the electrochemical cell **2100** is configured as a voltage doubling mechanism, commonly called a 'charge pump'. Common components such as SEMTECH SC632A can be used to accomplish this or similar functions. As shown, a voltage V_1 is measured between the anode current collector **2120** and the cathode current collector **2140**. The electrochemical cell **2100** includes switches S_1 and S_2 , resistor **R**, and capacitors C_1 and C_2 . A voltage V_2 is measured between the anode current collector **2120** and the interlayer **2160**. As shown, switch S_1 and switch S_2 control the flow of current through the electrochemical cell **2100**. Through control of the switching sequence of the switch S_1 and the switch S_2 , a voltage equal to or double the voltage of the cathode **2130** relative to the anode **2110** can be applied to the interlayer **2160**. For example, the switch S_1 can be switched to the up position while the switch S_2 is switched to the down position to charge the capacitor C_1 . The capacitor C_2 can charge continuously via the resistor **R**. In order to apply a doubling voltage, the switch S_1 can be moved to the down position, connecting to the interlayer **2160**, and the switch S_2 is moved to the up position, electrically connecting the capacitor C_1 to the top side of the capacitor C_2 . In such a case, the total voltage applied to the interlayer **2160** is equal to $C_1 \times V_1 + C_2 \times V_1$, where V_1 is the voltage applied to the interlayer **2160**. In cases where C_1 is equal to C_2 , this equates to $2 \times V_1$. For example, if only cell voltage V_1 is applied, then the switch S_2 can remain in the down position and the switch S_1 can alternate between the up and the down position. In some embodiments, the input voltage can be taken from the electrochemical cell **2100** or from another source. The capacitor C_2 can also be replaced with a secondary voltage source, either galvanically isolated from the cell or on a common reference.

[0220] The energy transferred to the interlayer **2160** can be controlled via the resistance of the components of the

electrochemical cell **2100** (including the resistor **R**) and/or via other resistors (not shown). In some embodiments, the energy transferred to the interlayer **2160** can be controlled via selection of the capacitor C_1 or use of other types of active components. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **2100**. In some embodiments, the switch S_1 and/or the switch S_2 can be replaced by a diode, a MOSFET, a BJT, or any other suitable device.

[0221] FIG. 22 is an illustration of an electrochemical cell **2200** with an interlayer **2260**, according to an embodiment. As shown, the electrochemical cell **2200** includes an anode **2210** disposed on an anode current collector **2220**, a cathode **2230** disposed on a cathode current collector **2240**, with a first separator **2250a** and a second separator **2250b** disposed between the anode **2210** and the cathode **2230**. The interlayer **2260** is disposed between the first separator **2250a** and the second separator **2250b**. In some embodiments, the anode **2210**, the anode current collector **2220**, the cathode **2230**, the cathode current collector **2240**, the first separator **2250a**, the second separator **2250b**, and the interlayer **2260** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **2210**, the anode current collector **2220**, the cathode **2230**, the cathode current collector **2240**, the first separator **2250a**, the second separator **2250b**, and the interlayer **2260** are not described in greater detail herein.

[0222] As shown, the circuitry in the electrochemical cell **2200** can have a bias circuit. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **2200**. As shown, a voltage V_1 is measured between the anode current collector **2220** and the cathode current collector **2240** and a voltage V_2 is measured between the anode current collector **2220** and the cathode current collector **2240**. When the voltage of the interlayer **2260** decreases to a value of less than the voltage of the cathode current collector **2240**, current can be routed to flow from the cathode current collector **2240** to the interlayer **2260**, thereby increasing the potential of the interlayer **2260**. As shown, a resistor **R** is placed between the interlayer **2260** and the cathode current collector **2240**. In some embodiments, the resistance of the resistor **R** can be large enough to prevent self-discharge of the electrochemical cell **2200**.

[0223] FIG. 23 is an illustration of an electrochemical cell **2300** with an interlayer **2360**, according to an embodiment. As shown, the electrochemical cell **2300** includes an anode **2310** disposed on an anode current collector **2320**, a cathode **2330** disposed on a cathode current collector **2340**, with a first separator **2350a** and a second separator **2350b** disposed between the anode **2310** and the cathode **2330**. The interlayer **2360** is disposed between the first separator **2350a** and the second separator **2350b**. In some embodiments, the

anode **2310**, the anode current collector **2320**, the cathode **2330**, the cathode current collector **2340**, the first separator **2350a**, the second separator **2350b**, and the interlayer **2360** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **2310**, the anode current collector **2320**, the cathode **2330**, the cathode current collector **2340**, the first separator **2350a**, the second separator **2350b**, and the interlayer **2360** are not described in greater detail herein.

[0224] As shown, the circuitry in the electrochemical cell **2300** can have a bias circuit. A first voltage V_1 is measured between the anode current collector **2320** and the cathode current collector **2340**. A second voltage V_2 is measured between the interlayer **2360** and the cathode current collector **2340**. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance can represent other impedances inherent to the electrochemical cell **2300**. As shown a resistor R and a diode D are placed between the interlayer **2360** and the cathode current collector **2340**. When the voltage of the interlayer **2360** decreases below the voltage of the current collector **2340**, leakage current can flow through the diode D, allowing the potential of the interlayer **2360** to increase. In some embodiments, the diode D can include a rectification diode (a diode not intended to function in reverse breakdown voltage). In embodiments where the diode D includes a rectification diode, current can be restricted in two ways. Reverse bias leakage current of the diode D (e.g., as defined by the manufacturer of the diode D) can be limited by specification of the diode D. Use of a standard diode can ensure that only reverse recovery current (as specified by the manufacturer of the diode) can flow through the circuit. When the recovery current is reached, the diode D can open, allowing only leakage current (as specified by the manufacturer). Also, the reverse recovery current of the diode D (e.g., as defined by the manufacturer of the diode D) can be limited by resistor R. In the event that the voltage of the interlayer **2360** exceeds the voltage of the cathode **2340**, the diode D can allow a full rated current to flow, clamping the voltage potential to the voltage of the cathode **2340** and the forward voltage of the diode D.

[0225] FIG. 24 is an illustration of an electrochemical cell **2400** with an interlayer **2460**, according to an embodiment. As shown, the electrochemical cell **2400** includes an anode **2410** disposed on an anode current collector **2420**, a cathode **2430** disposed on a cathode current collector **2440**, with a first separator **2450a** and a second separator **2450b** disposed between the anode **2410** and the cathode **2430**. The interlayer **2460** is disposed between the first separator **2450a** and the second separator **2450b**. In some embodiments, the anode **2410**, the anode current collector **2420**, the cathode **2430**, the cathode current collector **2440**, the first separator **2450a**, the second separator **2450b**, and the interlayer **2460** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **2410**, the anode current collector **2420**, the cathode

2430, the cathode current collector **2440**, the first separator **2450a**, the second separator **2450b**, and the interlayer **2460** are not described in greater detail herein.

[0226] As shown, the electrochemical cell **2400** can act as an isolated DC-DC cell. A first voltage V_1 is measured between the anode current collector **2420** and the cathode current collector **2440**. A second voltage V_2 is measured between the interlayer **2460** and the cathode current collector **2440**. As shown, the electrochemical cell **2400** includes a resistor R and a converter BB powered by a module M. The converter can be configured based on the needs of the system as a buck type converter, a boost type converter or a buck-boost style converter. As used in reference to FIG. 24, the term buck-boost can be understood to include all types of switching regulators. The regulator can include a transformer or be a transformer-less topology. The buck-boost converter BB is a DC-DC converter that has an output voltage magnitude that is either greater than or less than the input voltage magnitude as defined by the needs of the system, including pre-selected and in real time control. In some embodiments, the buck-boost converter BB can be powered via an independent DC source. In some embodiments, the resistance can represent other impedances inherent to the electrochemical cell **2400**. When the voltage of the interlayer **2460** decreases to below a defined level (e.g., about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V, inclusive of all values and ranges therebetween), the current can be applied to the buck-boost converter BB (e.g., via the module M) to cause a defined voltage potential (e.g., about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V inclusive of all values and ranges there between) to be applied to the interlayer **2460**. In some embodiments, the buck-boost converter BB can be replaced by a buck converter. In some embodiments, the buck-boost converter BB can be replaced by a boost converter. In some embodiments, the buck-boost converter BB can be switched on or off via a BMS, a local hardware circuit, or any other suitable system control method. The buck, boost, and buck-boost converter topologies can be implemented in many ways by those skilled in the art.

[0227] FIG. 25 is an illustration of an electrochemical cell **2500** with an interlayer **2560**, according to an embodiment. As shown, the electrochemical cell **2500** includes an anode **2510** disposed on an anode current collector **2520**, a cathode **2530** disposed on a cathode current collector **2540**, with a first separator **2550a** and a second separator **2550b** disposed between the anode **2510** and the cathode **2530**. The interlayer **2560** is disposed between the first separator **2550a** and the second separator **2550b**. In some embodiments, the anode **2510**, the anode current collector **2520**, the cathode **2530**, the cathode current collector **2540**, the first separator **2550a**, the second separator **2550b**, and the interlayer **2560** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **2510**, the anode current collector **2520**, the cathode **2530**, the cathode current collector **2540**, the first separator **2550a**, the second separator **2550b**, and the interlayer **2560** are not described in greater detail herein.

[0228] As shown, the electrochemical cell **2500** can have an isolated AC-DC converter attached to the cell. A first voltage V_1 is measured between the anode current collector **2520** and the cathode current collector **2540**. A second voltage V_2 is measured between the interlayer **2560** and the cathode current collector **2540**. As shown, the electrochemical cell **2500** includes a resistor R and a converter BB powered by an AC device. The converter can be configured based on the needs of the system as a buck type converter, a boost type converter or a buck-boost style converter. As used in reference to FIG. **24**, the term buck-boost can be understood to include all types of switching regulators. The regulator can include a transformer or be a transformer-less topology. In some embodiments, the AC device can include a rectifier. In some embodiments, the AC device can include a transformer with a rectifier. When the voltage of the interlayer **2560** decreases to below a defined level (e.g., about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V, inclusive of all values and ranges therebetween), the current can be applied to the buck-boost converter BB (e.g., via the AC source) to cause a voltage potential as defined by the needs of the system, including pre-selected and in real time control. to be applied to the interlayer **2460**. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **2500**.

[0229] FIG. **26** is an illustration of an electrochemical cell **2600** with an interlayer **2660**, according to an embodiment. As shown, the electrochemical cell **2600** includes an anode **2610** disposed on an anode current collector **2620**, a cathode **2630** disposed on a cathode current collector **2640**, with a first separator **2650a** and a second separator **2650b** disposed between the anode **2610** and the cathode **2630**. The interlayer **2660** is disposed between the first separator **2650a** and the second separator **2650b**. In some embodiments, the anode **2610**, the anode current collector **2620**, the cathode **2630**, the cathode current collector **2640**, the first separator **2650a**, the second separator **2650b**, and the interlayer **2660** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. Thus, certain aspects of the anode **2610**, the anode current collector **2620**, the cathode **2630**, the cathode current collector **2640**, the first separator **2650a**, the second separator **2650b**, and the interlayer **2660** are not described in greater detail herein.

[0230] As shown, the electrochemical cell **2600** can have a voltage regulator. A first voltage V_1 is measured between the anode current collector **2620** and the cathode current collector **2640**. A second voltage V_2 is measured between the interlayer **2660** and the cathode current collector **2640**. Current can flow from the interlayer **2660** to the cathode current collector **2640** via a regulator RG (with optional resistors R_1 and R_2 on either side of the regulator RG). In some embodiments, the regulator RG can include a linear regulator. In some embodiments, the regulator RG can include a switching or active regulator. When the voltage of the interlayer **2660** decreases below a defined level (e.g., about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V, inclusive of all values and ranges therebetween), the regulator RG can cause current to flow to the

interlayer **2660** to increase or decrease the voltage of the interlayer **2660**. In some embodiments, the regulator RG can be controlled by a BMS, a local hardware circuit, or any other suitable system control mechanism. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **2600**.

[0231] In some embodiments, the resistances R_1 and R_2 can be representative of other impedances inherent to the system. When the voltage of the interlayer **2660** decreases to below a threshold value, current from the regulator RG can flow (e.g., via a BMS), causing a voltage potential to be applied to the interlayer **2660**. In some embodiments, the regulator RG can be controlled via BMS or by local hardware circuit or another system control method.

[0232] FIG. **27** is an illustration of an electrochemical cell **2700** with an interlayer **2760**, according to an embodiment. As shown, the electrochemical cell **2700** includes an anode **2710** disposed on an anode current collector **2720**, a cathode **2730** disposed on a cathode current collector **2740**, with a first separator **2750a** and a second separator **2750b** disposed between the anode **2710** and the cathode **2730**. The interlayer **2760** is disposed between the first separator **2750a** and the second separator **2750b**. In some embodiments, the anode **2710**, the anode current collector **2720**, the cathode **2730**, the cathode current collector **2740**, the first separator **2750a**, the second separator **2750b**, and the interlayer **2760** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. Thus, certain aspects of the anode **2710**, the anode current collector **2720**, the cathode **2730**, the cathode current collector **2740**, the first separator **2750a**, the second separator **2750b**, and the interlayer **2760** are not described in greater detail herein.

[0233] As shown, the electrochemical cell **2700** includes a voltage amplification system, including switches S_1 , S_2 , S_3 , S_4 . This circuit represents devices referred to as “charge pumps” and boost converters used in switching power supply applications such as the LT **3482** and the MAX**633**. These converters are used to create a boosted voltage for use in electronic devices. For representation, a basic functionality of a simplified charge pump is described herein. A capacitor C_1 is placed between the switch S_1 and the switch S_2 . A capacitor C_2 is placed between the switch S_3 and the switch S_4 . The electrochemical cell **2700** further includes a capacitor C_3 and a resistor R on a current flow pathway. A first voltage V_1 is measured between the anode current collector **2720** and the cathode current collector **2740**. A second voltage V_2 is measured between the interlayer **2660** and the anode current collector **2740**. Depending on how the switches S_1 , S_2 , S_3 , S_4 are configured, a voltage of at least about 2, or at least about 3 times the voltage difference between the anode current collector **2720** and the cathode current collector **2740** can be applied to the interlayer **2760**. For example, with switch S_1 and switch S_3 in the up position while switch S_2 and switch S_4 are in the down position, the capacitors C_1 and C_2 charge. Then, switching the switch S_1 and the switch S_3 to the down position while switching the switch S_2 and the switch S_4 to the up position, triple the

voltage V_1 is applied to the interlayer **2760**. This can be referred to as a “charge pump” procedure. In some embodiments, any of the switches S_1, S_2, S_3, S_4 can be replaced with a diode, a MOSFET, a BJT, or any other suitable device. In some embodiments, any of the switches S_1, S_2, S_3, S_4 can be controlled by a BMS, a hardware device, a control chip, an oscillator, or any other suitable controller device.

[0234] In some embodiments, the energy transferred to the interlayer **2760** can be controlled by the resistance of the system, or other resistors. In some embodiments, the energy transferred to the interlayer **2760** can be controlled by the selection of the capacitance or use of other types of active components. In some embodiments, additional stages of voltage amplification can be added to increase the total voltage applied to the interlayer **2760**. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **2700**.

[0235] FIG. 28 is an illustration of an electrochemical cell **2800** with an interlayer **2860**, according to an embodiment. As shown, the electrochemical cell **2800** includes an anode **2810** disposed on an anode current collector **2820**, a cathode **2830** disposed on a cathode current collector **2840**, with a first separator **2850a** and a second separator **2850b** disposed between the anode **2810** and the cathode **2830**. The interlayer **2860** is disposed between the first separator **2850a** and the second separator **2850b**. In some embodiments, the anode **2810**, the anode current collector **2820**, the cathode **2830**, the cathode current collector **2840**, the first separator **2850a**, the second separator **2850b**, and the interlayer **2860** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode **2810**, the anode current collector **2820**, the cathode **2830**, the cathode current collector **2840**, the first separator **2850a**, the second separator **2850b**, and the interlayer **2860** are not described in greater detail herein.

[0236] As shown, current flow and voltage potential between the cathode current collector **2840** and the interlayer **2860** can be controlled via a transistor Q_1 (and optional resistor R_1). As shown, flow of current between the interlayer **2860** and the cathode current collector **2840** can be controlled via a transistor Q_2 (and optional resistor R_2). As shown, a voltage V_1 is measured between the anode current collector **2820** and the cathode current collector **2840** and a voltage V_2 is measured between the anode current collector **2820** and the interlayer **2860**. In some embodiments, the transistor Q_1 and/or the transistor Q_2 can act as a switching device. In some embodiments, the transistor Q_1 and/or the transistor Q_2 can be controlled by a BMS. In some embodiments, the transistor Q_1 and/or the transistor Q_2 can be controlled by a local hardware circuit or other system control apparatus. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **2800**. In some embodi-

ments, the transistor Q_1 and/or the transistor Q_2 can include a BJT in a PNP or NPN configuration. In some embodiments, the transistor Q_1 and/or the transistor Q_2 can be replaced with a diode. In some embodiments, the transistor Q_1 and/or the transistor Q_2 can include a JFET. In some embodiments, the transistor Q_1 and/or the transistor Q_2 can include a MOSFET. In some embodiments, the transistor Q_1 and/or the transistor Q_2 can be replaced with a switch. In some embodiments, the transistor Q_1 can be the same type of transistor or circuit device as the transistor Q_2 . In some embodiments, the transistor Q_1 can be a different type of transistor or circuit device from the transistor Q_2 . In some embodiments, the transistor Q_1 and the transistor Q_2 can be operated independently. In some embodiments, the transistor Q_1 and the transistor Q_2 can be operated in concert. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **2800**.

[0237] FIG. 29 is an illustration of an electrochemical cell **2900** with an interlayer **2960**, according to an embodiment. As shown, the electrochemical cell **2900** includes an anode **2910** disposed on an anode current collector **2920**, a cathode **2930** disposed on a cathode current collector **2940**, with a first separator **2950a** and a second separator **2950b** disposed between the anode **2910** and the cathode **2930**. The interlayer **2960** is disposed between the first separator **2950a** and the second separator **2950b**. In some embodiments, the anode **2910**, the anode current collector **2920**, the cathode **2930**, the cathode current collector **2940**, the first separator **2950a**, the second separator **2950b**, and the interlayer **2960** can be the same or substantially similar to the anode **2810**, the anode current collector **2820**, the cathode **2830**, the cathode current collector **2840**, the first separator **2850a**, the second separator **2850b**, and the interlayer **2860**, as described above with reference to FIG. 28. Thus, certain aspects of the anode **2910**, the anode current collector **2920**, the cathode **2930**, the cathode current collector **2940**, the first separator **2950a**, the second separator **2950b**, and the interlayer **2960** are not described in greater detail herein.

[0238] As shown, a voltage V_1 is measured between the anode current collector **2920** and the cathode current collector **2940** and a voltage V_2 is measured between the anode current collector **2920** and the interlayer **2960**. As shown, the electrochemical cell **2900** includes transistors Q_1, Q_2 and resistors R_1, R_2 . As shown, the resistor R_2 is positioned upstream of the transistor Q_2 while the resistor R_1 is located downstream of the transistor Q_1 . This differs from FIG. 28, in which the resistor R_2 is located downstream of the transistor Q_2 . Otherwise, each of the components of the electrochemical cell **2900** can be the same or substantially similar to the components of the electrochemical cell **2800**, as described above with reference to FIG. 28. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **2900**.

[0239] FIG. 30 is an illustration of an electrochemical cell 3000 with an interlayer 3060, according to an embodiment. As shown, the electrochemical cell 3000 includes an anode 3010 disposed on an anode current collector 3020, a cathode 3030 disposed on a cathode current collector 3040, with a first separator 3050a and a second separator 3050b disposed between the anode 3010 and the cathode 3030. The interlayer 3060 is disposed between the first separator 3050a and the second separator 3050b. In some embodiments, the anode 3010, the anode current collector 3020, the cathode 3030, the cathode current collector 3040, the first separator 3050a, the second separator 3050b, and the interlayer 3060 can be the same or substantially similar to the anode 2810, the anode current collector 2820, the cathode 2830, the cathode current collector 2840, the first separator 2850a, the second separator 2850b, and the interlayer 2860, as described above with reference to FIG. 28. Thus, certain aspects of the anode 3010, the anode current collector 3020, the cathode 3030, the cathode current collector 3040, the first separator 3050a, the second separator 3050b, and the interlayer 3060 are not described in greater detail herein.

[0240] As shown, the electrochemical cell includes transistors Q_1 , Q_2 and resistors R_1 , R_2 , R_3 . This circuit is commonly called a ‘push-pull’ amplifier. V_1 is measured between the anode current collector 3020 and the cathode current collector 3040 and a voltage V_2 is measured between the anode current collector 3020 and the interlayer 3060. As shown, the resistor R_1 is located upstream of the transistor Q_1 and the resistor R_2 is located downstream of the transistor Q_2 . The resistor R_3 limits the flow of current through a base of the transistor Q_1 and the transistor Q_2 . The emitters connect to the interlayer 3060 in order to flow current into the interlayer 3060 relative to the cathode current collector 3040 and the anode current collector 3020. The circuit can be used to set any voltage potential desired between the cathode current collector 3040 and the anode current collector 3020 by selection of the voltage potential of and the current applied through the resistor R_3 . In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3000.

[0241] FIG. 31 is an illustration of an electrochemical cell 3100 with an interlayer 3160, according to an embodiment. As shown, the electrochemical cell 3100 includes an anode 3110 disposed on an anode current collector 3120, a cathode 3130 disposed on a cathode current collector 3140, with a first separator 3150a and a second separator 3150b disposed between the anode 3110 and the cathode 3130. The interlayer 3160 is disposed between the first separator 3150a and the second separator 3150b. In some embodiments, the anode 3110, the anode current collector 3120, the cathode 3130, the cathode current collector 3140, the first separator 3150a, the second separator 3150b, and the interlayer 3160 can be the same or substantially similar to the anode 2810, the anode current collector 2820, the cathode 2830, the cathode current collector 2840, the first separator 2850a, the second separator 2850b, and the interlayer 2860, as described above with reference to FIG. 28. Thus, certain aspects of the anode 3110, the anode current collector 3120, the cathode 3130, the cathode current collector 3140, the

first separator 3150a, the second separator 3150b, and the interlayer 3160 are not described in greater detail herein.

[0242] As shown, a diode D is positioned in a circuit between the cathode current collector 3140 and the interlayer 3160 to provide a continuous bias potential and a transistor Q is positioned in a circuit between the interlayer 3160 and the anode current collector 3120 (with optional resistors R_1 , R_2) to provide a variable potential. As shown, a voltage V_1 is measured between the anode current collector 3120 and the cathode current collector 3140 and a voltage V_2 is measured between the anode current collector 3120 and the interlayer 3160. The circuit can be used to shift the potential of the interlayer 3160 closer to the current collector 3120. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3100.

[0243] FIG. 32 is an illustration of an electrochemical cell 3200 with an interlayer 3260, according to an embodiment. As shown, the electrochemical cell 3200 includes an anode 3210 disposed on an anode current collector 3220, a cathode 3230 disposed on a cathode current collector 3240, with a first separator 3250a and a second separator 3250b disposed between the anode 3210 and the cathode 3230. The interlayer 3260 is disposed between the first separator 3250a and the second separator 3250b. In some embodiments, the anode 3210, the anode current collector 3220, the cathode 3230, the cathode current collector 3240, the first separator 3250a, the second separator 3250b, and the interlayer 3260 can be the same or substantially similar to the anode 2810, the anode current collector 2820, the cathode 2830, the cathode current collector 2840, the first separator 2850a, the second separator 2850b, and the interlayer 2860, as described above with reference to FIG. 28. Thus, certain aspects of the anode 3210, the anode current collector 3220, the cathode 3230, the cathode current collector 3240, the first separator 3250a, the second separator 3250b, and the interlayer 3260 are not described in greater detail herein.

[0244] As shown, a transistor Q is positioned in a circuit between the interlayer 3160 and the anode current collector 3120 (with optional resistors R_1 , R_2). As shown, a voltage V_1 is measured between the anode current collector 3220 and the cathode current collector 3240 and a voltage V_2 is measured between the anode current collector 3220 and the interlayer 3260. The electrochemical cell 3200 does not include a diode. Otherwise, the components of the electrochemical cell 3200 can be the same or substantially similar to the electrochemical cell 3100, as described above with reference to FIG. 31. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3200.

[0245] FIG. 33 is an illustration of an electrochemical cell 3300 with an interlayer 3360, according to an embodiment. As shown, the electrochemical cell 3300 includes an anode 3310 disposed on an anode current collector 3320, a cathode 3330 disposed on a cathode current collector 3340, with a first separator 3350a and a second separator 3350b disposed

between the anode 3310 and the cathode 3330. The interlayer 3360 is disposed between the first separator 3350a and the second separator 3350b. In some embodiments, the anode 3310, the anode current collector 3320, the cathode 3330, the cathode current collector 3340, the first separator 3350a, the second separator 3350b, and the interlayer 3360 can be the same or substantially similar to the anode 2810, the anode current collector 2820, the cathode 2830, the cathode current collector 2840, the first separator 2850a, the second separator 2850b, and the interlayer 2860, as described above with reference to FIG. 28. Thus, certain aspects of the anode 3310, the anode current collector 3320, the cathode 3330, the cathode current collector 3340, the first separator 3350a, the second separator 3350b, and the interlayer 3360 are not described in greater detail herein.

[0246] As shown, the electrochemical cell 3300 includes diodes D_1 , D_2 . An optional switch S can bypass the diode D_2 , creating a selective dendrite treatment mode or normal operation. In some embodiments, the electrochemical cell can be absent of the switch S. In some embodiments, the switch S can be replaced with a transistor, a MOSFET, a JFET, or any other device that can bypass the diode D_2 . As shown, a voltage V_1 is measured between the anode current collector 3320 and the cathode current collector 3340 and a voltage V_2 is measured between the anode current collector 3320 and the interlayer 3360. As shown, a voltage V_1 is measured between the anode current collector 3320 and the cathode current collector 3340 and a voltage V_2 is measured between the anode current collector 3320 and the interlayer 3360. As shown, the circuit design of the electrochemical cell 3300 can act as a charge diode with a bypass. In some embodiments, the switch S can be opened during a controlled charge method. The diode D_2 can cause a voltage drop of the interlayer 3360 relative to the cathode 3330 equal to the forward voltage drop of the diode D_2 . In some embodiments, the voltage drop can be about 0.1 V, about 0.2 V, about 0.3 V, about 0.4 V, about 0.5 V, about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V, inclusive of all values and ranges therebetween. The diode D_1 can cause a voltage drop of the interlayer 3360 relative to the cathode 3330 equal to the forward voltage drop of the diode D_2 plus the voltage of the cathode current collector 3340 making the voltage of the interlayer 3360 higher than the voltage of the cathode current collector 3340. In some embodiments, the voltage drop can be about 0.1 V, about 0.2 V, about 0.3 V, about 0.4 V, about 0.5 V, about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V, inclusive of all values and ranges there between. The switch S can be closed to allow normal flow of current to the electrochemical cell 3300 without imparting voltage changes to the interlayer 3360. The switch S can be replaced by a diode, a MOSFET, a JFET, or any other suitable device or combinations thereof. The switch S can also be eliminated from the circuit using only the Diodes or other devices to apply the voltage potential on current flow. In some embodiments, the diodes D_1 and D_2 can also be replaced with switches diode, a MOSFET, a JFET, or any other suitable device or combinations thereof. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3300.

rate components, or they can represent other impedances inherent to the electrochemical cell 3300.

[0247] FIG. 34 is an illustration of an electrochemical cell 3400 with multiple interlayers 3460a, 3460b (collectively referred to as interlayers 3460), according to an embodiment. As shown, the electrochemical cell 3400 includes an anode 3410 disposed on an anode current collector 3420, a cathode 3430 disposed on a cathode current collector 3440, with a first separator 3450a, a second separator 3450b, and a third separator 3450c disposed between the anode 3410 and the cathode 3430. The interlayer 3460a is disposed between the first separator 3450a and the second separator 3450b and the second interlayer 3460b is disposed between the second separator 3450b and the third separator 3450c. In some embodiments, the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3460 can be the same or substantially similar to the anode 2810, the anode current collector 2820, the cathode 2830, the cathode current collector 2840, the separators 2850a, 2850b, and the interlayer 2860, as described above with reference to FIG. 28. Thus, certain aspects of the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c and the interlayers 3460 are not described in greater detail herein.

[0248] As shown, a voltage V_1 is measured between the anode current collector 3420 and the cathode current collector 3440, a voltage V_2 is measured between the anode current collector 3420 and the interlayer 3460a, and a voltage V_3 is measured between the anode current collector 3440 and the interlayer 3460b. As shown, the electrochemical cell 3400 includes switches S_{1a} , S_{1b} , S_{2a} , S_{2b} and optional resistors R_1 , R_2 , R_3 , R_4 . As shown, the switch S_{1a} controls the flow of current between the anode current collector 3440 and the interlayer 3460a. As shown, the switch S_{1b} controls the flow of current between the anode current collector 3440 and the interlayer 3460b. As shown, the switch S_{2a} controls the flow of current between the interlayer 3460a and the cathode current collector 3420. As shown, the switch S_{2b} controls the flow of current between the interlayer 3460b and the cathode current collector 3420.

[0249] In some embodiments, either of the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} can be replaced with a diode, a MOSFET, a JFET, or any other suitable device or combinations thereof. In some embodiments, the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} can each be replaced with the same device. In some embodiments, either of the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} can be replaced with different devices. In some embodiments, either of the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} can be controlled by a BMS. In some embodiments, either of the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} by a local hardware circuit or other system control method. In some embodiments, the interlayers 3460 can be controlled in concert, electrically connecting any of the anode 3410, the cathode 3440, the interlayer 3460a, and/or the interlayer 3460b. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3400.

[0250] FIG. 35 is an illustration of an electrochemical cell 3500 with multiple interlayers 3560a, 3560b (collectively referred to as interlayers 3560), according to an embodiment. As shown, the electrochemical cell 3500 includes an anode 3510 disposed on an anode current collector 3520, a cathode 3530 disposed on a cathode current collector 3540, with a first separator 3550a, a second separator 3550b, and a third separator 3550c disposed between the anode 3510 and the cathode 3530. The interlayer 3560a is disposed between the first separator 3550a and the second separator 3550b and the second interlayer 3560b is disposed between the second separator 3550b and the third separator 3550c. In some embodiments, the anode 3510, the anode current collector 3520, the cathode 3530, the cathode current collector 3540, the first separator 3550a, the second separator 3550b, the third separator 3550c, and the interlayers 3560 can be the same or substantially similar to the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3460, as described above with reference to FIG. 34. Thus, certain aspects of the anode 3510, the anode current collector 3520, the cathode 3530, the cathode current collector 3540, the first separator 3550a, the second separator 3550b, the third separator 3550c and the interlayers 3560 are not described in greater detail herein.

[0251] As shown, a voltage V_1 is measured between the anode current collector 3520 and the cathode current collector 3540, a voltage V_2 is measured between the anode current collector 3520 and the interlayer 3560a, and a voltage V_3 is measured between the anode current collector 3540 and the interlayer 3560b. As shown, the electrochemical cell 3500 includes transistors Q_1 , Q_2 , Q_3 , Q_4 and optional resistors R_1 , R_2 , R_3 , R_4 , R_5 , R_6 . The emitters connect to the interlayer 3560a in order to flow current into the interlayer 3560a relative to the anode current collector 3520 and the cathode current collector 3540. The circuit can be used to set any voltage potential desired between the cathode current collector 3540 and the anode current collector 3520 by selection of the voltage potential of and the current applied through R_3 . The emitters connect to the interlayer 3560b in order to flow current into the interlayer 3560b relative to the cathode current collector 3540 and the anode current collector 3520. The circuit can be used to set any voltage potential desired between the cathode current collector 3540 and the anode current collector 3520 by selection of the voltage potential of and the current applied through R_3 . In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3600. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell 3000, as described above with reference to FIG. 30.

[0252] FIG. 36 is an illustration of an electrochemical cell 3600 with multiple interlayers 3660a, 3660b (collectively referred to as interlayers 3660), according to an embodiment. As shown, the electrochemical cell 3600 includes an anode 3610 disposed on an anode current collector 3620, a cathode 3630 disposed on a cathode current collector 3640, with a first separator 3650a, a second separator 3650b, and

a third separator 3650c disposed between the anode 3610 and the cathode 3630. The interlayer 3660a is disposed between the first separator 3650a and the second separator 3650b and the second interlayer 3660b is disposed between the second separator 3650b and the third separator 3650c. In some embodiments, the anode 3610, the anode current collector 3620, the cathode 3630, the cathode current collector 3640, the first separator 3650a, the second separator 3650b, the third separator 3650c, and the interlayers 3660 can be the same or substantially similar to the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3460, as described above with reference to FIG. 34. Thus, certain aspects of the anode 3610, the anode current collector 3620, the cathode 3630, the cathode current collector 3640, the first separator 3650a, the second separator 3650b, the third separator 3650c and the interlayers 3660 are not described in greater detail herein.

[0253] As shown, a voltage V_1 is measured between the anode current collector 3620 and the cathode current collector 3640, a voltage V_2 is measured between the anode current collector 3620 and the interlayer 3660a, and a voltage V_3 is measured between the anode current collector 3640 and the interlayer 3660b. As shown, the electrochemical cell 3500 includes transistors Q_1 , Q_2 , Q_3 , Q_4 and optional resistors R_1 , R_2 , R_3 , R_4 , R_5 , R_6 . The resistor R_2 controls the flow of current between the transistor Q_1 and the transistor Q_2 . The resistor R_3 controls the flow of current between the transistor Q_3 and the transistor Q_4 . In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3600.

[0254] FIG. 37 is an illustration of an electrochemical cell 3700 with multiple interlayers 3760a, 3760b (collectively referred to as interlayers 3760), according to an embodiment. As shown, the electrochemical cell 3700 includes an anode 3710 disposed on an anode current collector 3720, a cathode 3730 disposed on a cathode current collector 3740, with a first separator 3750a, a second separator 3750b, and a third separator 3750c disposed between the anode 3710 and the cathode 3730. The interlayer 3760a is disposed between the first separator 3750a and the second separator 3750b and the second interlayer 3760b is disposed between the second separator 3750b and the third separator 3750c. In some embodiments, the anode 3710, the anode current collector 3720, the cathode 3730, the cathode current collector 3740, the first separator 3750a, the second separator 3750b, the third separator 3750c, and the interlayers 3760 can be the same or substantially similar to the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3460, as described above with reference to FIG. 34. Thus, certain aspects of the anode 3710, the anode current collector 3720, the cathode 3730, the cathode current collector 3740, the first separator 3750a, the second separator 3750b, the third separator 3750c and the interlayers 3760 are not described in greater detail herein.

[0255] As shown, a voltage V_1 is measured between the anode current collector 3720 and the cathode current collector 3740.

lector 3740, a voltage V_2 is measured between the anode current collector 3720 and the interlayer 3760a, and a voltage V_3 is measured between the anode current collector 3740 and the interlayer 3760b. As shown, the electrochemical cell 3700 includes transistors Q_1 , Q_2 , and optional resistors R_1 , R_2 , R_3 , R_4 . As shown, the transistor Q_1 and the resistor R_3 are placed on a circuit between the interlayer 3760a and the anode current collector 3720. As shown, the transistor Q_2 and the resistor R_4 are placed on a circuit between the interlayer 3760b and the anode current collector 3720. As shown, the resistor R_1 is placed on a circuit between the cathode current collector 3740 and the interlayer 3760a. As shown, the resistor R_2 is placed on a circuit between the cathode current collector 3740 and the interlayer 3760b. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3700. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell 3200, as described above with reference to FIG. 32.

[0256] FIG. 38 is an illustration of an electrochemical cell 3800 with multiple interlayers 3860a, 3860b (collectively referred to as interlayers 3860), according to an embodiment. As shown, the electrochemical cell 3800 includes an anode 3810 disposed on an anode current collector 3820, a cathode 3830 disposed on a cathode current collector 3840, with a first separator 3850a, a second separator 3850b, and a third separator 3850c disposed between the anode 3810 and the cathode 3830. The interlayer 3860a is disposed between the first separator 3850a and the second separator 3850b and the second interlayer 3860b is disposed between the second separator 3850b and the third separator 3850c. In some embodiments, the anode 3810, the anode current collector 3820, the cathode 3830, the cathode current collector 3840, the first separator 3850a, the second separator 3850b, the third separator 3850c, and the interlayers 3860 can be the same or substantially similar to the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3560, as described above with reference to FIG. 34. Thus, certain aspects of the anode 3810, the anode current collector 3820, the cathode 3830, the cathode current collector 3840, the first separator 3850a, the second separator 3850b, the third separator 3850 and the interlayers 3860 are not described in greater detail herein.

[0257] As shown, a voltage V_1 is measured between the anode current collector 3820 and the cathode current collector 3840, a voltage V_2 is measured between the anode current collector 3820 and the interlayer 3860a, and a voltage V_3 is measured between the anode current collector 3840 and the interlayer 3860b. As shown, the electrochemical cell 3800 includes transistors Q_1 , Q_2 , and optional resistors R_1 , R_2 , R_3 , R_4 . As shown, the transistor Q_1 and the resistor R_1 are placed on a circuit between the interlayer 3860a and the cathode current collector 3840. As shown, the transistor Q_2 and the resistor R_2 are placed on a circuit between the interlayer 3860b and the cathode current collector 3840. As shown, the resistor R_3 is placed on a circuit between the anode current collector 3820 and the interlayer

3860a. As shown, the resistor R_4 is placed on a circuit between the anode current collector 3820 and the interlayer 3860b. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3800.

[0258] FIG. 39 is an illustration of an electrochemical cell 3900 with multiple interlayers 3960a, 3960b (collectively referred to as interlayers 3960), according to an embodiment. As shown, the electrochemical cell 3900 includes an anode 3910 disposed on an anode current collector 3920, a cathode 3930 disposed on a cathode current collector 3940, with a first separator 3950a, a second separator 3950b, and a third separator 3950c disposed between the anode 3910 and the cathode 3930. The interlayer 3960a is disposed between the first separator 3950a and the second separator 3950b and the second interlayer 3960b is disposed between the second separator 3950b and the third separator 3950c. In some embodiments, the anode 3910, the anode current collector 3920, the cathode 3930, the cathode current collector 3940, the first separator 3950a, the second separator 3950b, the third separator 3950c, and the interlayers 3960 can be the same or substantially similar to the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3560, as described above with reference to FIG. 34. Thus, certain aspects of the anode 3910, the anode current collector 3920, the cathode 3930, the cathode current collector 3940, the first separator 3950a, the second separator 3950b, the third separator 3950 and the interlayers 3960 are not described in greater detail herein.

[0259] As shown, a voltage V_1 is measured between the anode current collector 3920 and the cathode current collector 3940, a voltage V_2 is measured between the anode current collector 3920 and the interlayer 3960a, and a voltage V_3 is measured between the anode current collector 3940 and the interlayer 3960b. As shown, the electrochemical cell 3900 includes transistors Q_1 , Q_2 , diodes D_1 , D_2 and optional resistors R_1 , R_2 , R_3 , R_4 . As shown, the diode D_1 and the resistor R_1 are placed on a circuit between the interlayer 3960a and the cathode current collector 3940. As shown, the diode D_2 and the resistor R_2 are placed on a circuit between the interlayer 3960b and the cathode current collector 3940. As shown, the resistor R_3 and the transistor Q_1 are placed on a circuit between the anode current collector 3920 and the interlayer 3960a. As shown, the transistor Q_2 and the resistor R_4 are placed on a circuit between the anode current collector 3920 and the interlayer 3960b. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 3900. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell 3100, as described above with reference to FIG. 31.

[0260] FIG. 40 is an illustration of an electrochemical cell 4000 with multiple interlayers 4060a, 4060b (collectively referred to as interlayers 4060), according to an embodi-

ment. As shown, the electrochemical cell 4000 includes an anode 4010 disposed on an anode current collector 4020, a cathode 4030 disposed on a cathode current collector 4040, with a first separator 4050a, a second separator 4050b, and a third separator 4050c disposed between the anode 4010 and the cathode 4030. The interlayer 4060a is disposed between the first separator 4050a and the second separator 4050b and the second interlayer 4060b is disposed between the second separator 4050b and the third separator 4050c. In some embodiments, the anode 4010, the anode current collector 4020, the cathode 4030, the cathode current collector 4040, the first separator 4050a, the second separator 4050b, the third separator 4050c, and the interlayers 4060 can be the same or substantially similar to the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3560, as described above with reference to FIG. 34. Thus, certain aspects of the anode 4010, the anode current collector 4020, the cathode 4030, the cathode current collector 4040, the first separator 4050a, the second separator 4050b, the third separator 4050c and the interlayers 4060 are not described in greater detail herein.

[0261] As shown, a voltage V_1 is measured between the anode current collector 4020 and the cathode current collector 4040, a voltage V_2 is measured between the anode current collector 4020 and the interlayer 4060a, and a voltage V_3 is measured between the anode current collector 4040 and the interlayer 4060b. As shown, the electrochemical cell 4000 includes transistors Q_1 , Q_2 , diodes D_1 , D_2 and optional resistors R_1 , R_2 , R_3 , R_4 . As shown, the transistor Q_1 and the resistor R_1 are placed on a circuit between the interlayer 4060a and the cathode current collector 4040. As shown, the transistor Q_2 and the resistor R_2 are placed on a circuit between the interlayer 4060b and the cathode current collector 4040. As shown, the resistor R_3 and the diode D_1 are placed on a circuit between the anode current collector 4020 and the interlayer 4060a. As shown, the diode D_2 and the resistor R_4 are placed on a circuit between the anode current collector 4020 and the interlayer 4060b. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 4000.

[0262] FIG. 41 is an illustration of an electrochemical cell 4100 with multiple interlayers 4160a, 4160b (collectively referred to as interlayers 4160), according to an embodiment. As shown, the electrochemical cell 4100 includes an anode 4110 disposed on an anode current collector 4120, a cathode 4130 disposed on a cathode current collector 4140, with a first separator 4150a, a second separator 4150b, and a third separator 4150c disposed between the anode 4110 and the cathode 4130. The interlayer 4160a is disposed between the first separator 4150a and the second separator 4150b and the second interlayer 4160b is disposed between the second separator 4150b and the third separator 4150c. In some embodiments, the anode 4110, the anode current collector 4120, the cathode 4130, the cathode current collector 4140, the first separator 4150a, the second separator 4150b, the third separator 4150c, and the interlayers 4160 can be the same or substantially similar to the anode 3410,

the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3560, as described above with reference to FIG. 34. Thus, certain aspects of the anode 4110, the anode current collector 4120, the cathode 4130, the cathode current collector 4140, the first separator 4150a, the second separator 4150b, the third separator 4150c and the interlayers 4160 are not described in greater detail herein.

[0263] As shown, a voltage V_1 is measured between the anode current collector 4120 and the cathode current collector 4140, a voltage V_2 is measured between the anode current collector 4120 and the interlayer 4160a, and a voltage V_3 is measured between the anode current collector 4140 and the interlayer 4160b. As shown, the electrochemical cell 4100 includes resistors R_1 , R_2 . As shown, the resistor R_1 is placed on a circuit between the interlayer 4160a and the cathode current collector 4140. As shown, the transistor Q_2 and the resistor R_2 are placed on a circuit between the interlayer 4160b and the cathode current collector 4140. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 4100. When the voltage of either of the interlayers 4160 decreases to a voltage less than the voltage of the cathode 4130, current can flow from the cathode 4130 to either of the interlayers 4160, allowing the potential of the interlayers 4160 to increase.

[0264] FIG. 42 is an illustration of an electrochemical cell 4200 with multiple interlayers 4260a, 4260b (collectively referred to as interlayers 4260), according to an embodiment. As shown, the electrochemical cell 4200 includes an anode 4210 disposed on an anode current collector 4220, a cathode 4230 disposed on a cathode current collector 4240, with a first separator 4250a, a second separator 4250b, and a third separator 4250c disposed between the anode 4210 and the cathode 4230. The interlayer 4260a is disposed between the first separator 4250a and the second separator 4250b and the second interlayer 4260b is disposed between the second separator 4250b and the third separator 4250c. In some embodiments, the anode 4210, the anode current collector 4220, the cathode 4230, the cathode current collector 4240, the first separator 4250a, the second separator 4250b, the third separator 4250c, and the interlayers 4260 can be the same or substantially similar to the anode 3410, the anode current collector 3420, the cathode 3430, the cathode current collector 3440, the first separator 3450a, the second separator 3450b, the third separator 3450c, and the interlayers 3560, as described above with reference to FIG. 34. Thus, certain aspects of the anode 4210, the anode current collector 4220, the cathode 4230, the cathode current collector 4240, the first separator 4250a, the second separator 4250b, the third separator 4250c and the interlayers 4260 are not described in greater detail herein.

[0265] As shown, a voltage V_1 is measured between the anode current collector 4220 and the cathode current collector 4240, a voltage V_2 is measured between the anode current collector 4220 and the interlayer 4260a, and a voltage V_3 is measured between the anode current collector 4240 and the interlayer 4260b. As shown, the electrochemical cell 4200 includes regulators RG_1 , RG_2 and optional

resistors R_1 , R_2 , R_3 . In some embodiments, the regulator RG_1 and/or the regulator RG_2 can include a buck converter, a buck-boost converter, and/or a boost converter. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **4200**. When the voltage of either of the interlayers **4260** decreases to below a defined level, the regulator RG_1 and/or the regulator RG_2 can cause a current to flow, causing a voltage potential to be applied to one or more of the interlayers **4260**. In some embodiments, the switching of the regulators RG_1 , RG_2 can be controlled by a BMS or by a local hardware circuit or other system control method. In some embodiments, the regulators RG_1 , RG_2 can be operated in concert. In some embodiments, the regulators RG_1 , RG_2 can be operated independently. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **2500**, as described above with reference to FIG. 25.

[0266] In some embodiments, the resistances R_1 , R_2 , R_3 can be representative of actual resistors, or can represent impedances inherent to the system (i.e., the electrochemical cell **4200**). When the voltage of the interlayer **4260** decreases to less than a threshold value, current can flow from the regulators RG_1 , RG_2 , causing a potential to be applied to the interlayer **4260**. The regulator RG_1 and/or the regulator RG_2 can be controlled via one or more switching devices (not shown). In some embodiments, the switching devices can be controlled by a BMS or by a local hardware circuit or other system control method. In some embodiments, each of the components of the electrochemical cell **4200** can be operated in concert, or independently. In some embodiments, the voltage V_1 , the voltage V_2 , and/or the voltage V_3 can be monitored in real time.

[0267] FIG. 43 is an illustration of an electrochemical cell **4300** with multiple interlayers **4360a**, **4360b** (collectively referred to as interlayers **4360**), according to an embodiment. As shown, the electrochemical cell **4300** includes an anode **4310** disposed on an anode current collector **4320**, a cathode **4330** disposed on a cathode current collector **4340**, with a first separator **4350a**, a second separator **4350b**, and a third separator **4350c** disposed between the anode **4310** and the cathode **4330**. The interlayer **4360a** is disposed between the first separator **4350a** and the second separator **4350b** and the second interlayer **4360b** is disposed between the second separator **4350b** and the third separator **4350c**. In some embodiments, the anode **4310**, the anode current collector **4320**, the cathode **4330**, the cathode current collector **4340**, the first separator **4350a**, the second separator **4350b**, the third separator **4350c**, and the interlayers **4360** can be the same or substantially similar to the anode **3410**, the anode current collector **3420**, the cathode **3430**, the cathode current collector **3440**, the first separator **3450a**, the second separator **3450b**, the third separator **3450c**, and the interlayers **3560**, as described above with reference to FIG. 34. Thus, certain aspects of the anode **4310**, the anode current collector **4320**, the cathode **4330**, the cathode current collector **4340**, the first separator **4350a**, the second separator **4350b**, the third separator **4350c** and the interlayers **4360** are not described in greater detail herein.

[0268] As shown, a voltage V_1 is measured between the anode current collector **4320** and the cathode current collector **4340**, a voltage V_2 is measured between the anode current collector **4320** and the interlayer **4360a**, and a voltage V_3 is measured between the anode current collector **4320** and the interlayer **4360b**. The electrochemical cell **4300** includes isolated DC circuits. As shown, the electrochemical cell **4300** includes buck-boosts BB_1 , BB_2 and optional resistors R_1 , R_2 . As shown, the buck-boosts BB_1 , BB_2 are coupled to modules M_1 , M_2 . In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **4300**. When the voltage of either of the interlayers **4360** decreases to below a defined level, the buck-boost BB_1 and/or the buck-boost BB_2 can cause a current to flow, causing a voltage potential to be applied to one or more of the interlayers **4360**. In some embodiments, either of the buck-boosts BB_1 , BB_2 can be replaced with a buck, a boost, a charge pump, a voltage multiplier and/or a diode ladder. In some embodiments, the switching of the buck-boosts BB_1 , BB_2 can be controlled by a BMS or by a local hardware circuit or other system control method. In some embodiments, the buck-boosts BB_1 , BB_2 can be operated in concert. In some embodiments, the buck-boosts BB_1 , BB_2 can be operated independently. In some embodiments, either of the modules M_1 , M_2 can include a battery or a battery pack, or any other source of DC energy. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **2400**, as described above with reference to FIG. 24.

[0269] FIG. 44 is an illustration of an electrochemical cell **4400** with multiple interlayers **4460a**, **4460b** (collectively referred to as interlayers **4460**), according to an embodiment. As shown, the electrochemical cell **4400** includes an anode **4410** disposed on an anode current collector **4420**, a cathode **4430** disposed on a cathode current collector **4440**, with a first separator **4450a**, a second separator **4450b**, and a third separator **4450c** disposed between the anode **4410** and the cathode **4430**. The interlayer **4460a** is disposed between the first separator **4450a** and the second separator **4450b** and the second interlayer **4460b** is disposed between the second separator **4450b** and the third separator **4450c**. In some embodiments, the anode **4410**, the anode current collector **4420**, the cathode **4430**, the cathode current collector **4440**, the first separator **4450a**, the second separator **4450b**, the third separator **4450c**, and the interlayers **4460** can be the same or substantially similar to the anode **3410**, the anode current collector **3420**, the cathode **3430**, the cathode current collector **3440**, the first separator **3450a**, the second separator **3450b**, the third separator **3450c**, and the interlayers **3560**, as described above with reference to FIG. 34. Thus, certain aspects of the anode **4410**, the anode current collector **4420**, the cathode **4430**, the cathode current collector **4440**, the first separator **4450a**, the second separator **4450b**, the third separator **4450c** and the interlayers **4460** are not described in greater detail herein.

[0270] As shown, a voltage V_1 is measured between the anode current collector **4420** and the cathode current collector **4440**, a voltage V_2 is measured between the anode current collector **4420** and the interlayer **4460a**, and a

voltage V_3 is measured between the anode current collector **4440** and the interlayer **4460b**. The electrochemical cell **4400** includes isolated DC circuits. As shown, the electrochemical cell **4400** includes buck-boosts BB_1 , BB_2 and optional resistors R_1 , R_2 . As shown, the buck-boosts BB_1 , BB_2 are coupled to AC power sources. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **4400**. When the voltage of either of the interlayers **4460** decreases to below a defined level, the buck-boost BB_1 and/or the buck-boost BB_2 can cause a current to flow, causing a voltage potential to be applied to one or more of the interlayers **4460**. In some embodiments, the switching of the AC power sources can be controlled by a BMS or by a local hardware circuit or other system control method. In some embodiments, the AC power sources can be operated in concert. In some embodiments, the AC power sources can be operated independently. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **2500**, as described above with reference to FIG. 25.

[0271] FIG. 45 is an illustration of an electrochemical cell **4500** with multiple interlayers **4560a**, **4560b** (collectively referred to as interlayers **4560**), according to an embodiment. As shown, the electrochemical cell **4500** includes an anode **4510** disposed on an anode current collector **4520**, a cathode **4530** disposed on a cathode current collector **4540**, with a first separator **4550a**, a second separator **4550b**, and a third separator **4550c** disposed between the anode **4510** and the cathode **4530**. The interlayer **4560a** is disposed between the first separator **4550a** and the second separator **4550b** and the second interlayer **4560b** is disposed between the second separator **4550b** and the third separator **4550c**. In some embodiments, the anode **4510**, the anode current collector **4520**, the cathode **4530**, the cathode current collector **4540**, the first separator **4550a**, the second separator **4550b**, the third separator **4550c**, and the interlayers **4560** can be the same or substantially similar to the anode **3410**, the anode current collector **3420**, the cathode **3430**, the cathode current collector **3440**, the first separator **3450a**, the second separator **3450b**, the third separator **3450c**, and the interlayers **3560**, as described above with reference to FIG. 34. Thus, certain aspects of the anode **4510**, the anode current collector **4520**, the cathode **4530**, the cathode current collector **4540**, the first separator **4550a**, the second separator **4550b**, the third separator **4550c** and the interlayers **4560** are not described in greater detail herein.

[0272] As shown, a voltage V_1 is measured between the anode current collector **4520** and the cathode current collector **4540**, a voltage V_2 is measured between the anode current collector **4520** and the interlayer **4560a**, and a voltage V_3 is measured between the anode current collector **4540** and the interlayer **4560b**. As shown, the electrochemical cell **4500** includes switches S_{1a} , S_{1b} , S_{2a} , S_{2b} , capacitors C_1 , C_2 , C_3 , and optional resistor R . As shown, the electrochemical cell **4500** is configured as a voltage doubling mechanism. Through control of the switching sequence of the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} , a voltage equal to or double the voltage of the cathode **4530** relative to the anode **4510** can be applied to the interlayer **4560a** and/or the interlayer

4560b. For example, the switch S_{1a} can be switched to the up position while the switch S_{1b} is switched to the down position to charge the capacitor C_1 . The capacitor C_3 can charge continuously via the resistor R . In order to apply a doubling voltage, the switch S_{1a} can be moved to the down position, connecting to the interlayer **4560a**, and the switch S_{1b} is moved to the up position, electrically connecting the capacitor C_1 to the top side of the capacitor C_3 . In such a case, the total voltage applied to the interlayer **4560a** is equal to $C_1 \times V_1 + C_2 \times V_1$, where V_1 is the voltage applied to the interlayer **4560a**. In cases where C_1 is equal to C_3 , this equates to $2 \times V_1$. For example, if only cell voltage V_1 is applied, then the switch S_{1b} can remain in the down position and the switch S_{1a} can alternate between the up and the down position.

[0273] The energy transferred to the interlayer **4560a** and/or the interlayer **4560b** can be controlled via the resistance of the components of the electrochemical cell **4500** (including the resistor R) and/or via other resistors (not shown). In some embodiments, the energy transferred to the interlayers **4560** can be controlled via selection of the capacitor C_1 , C_2 , or use of other types of active components. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **4500**. In some embodiments, any of the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} , can be replaced by a diode, a MOSFET, a BJT, or any other suitable device. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **2100**, as described above with reference to FIG. 21.

[0274] FIG. 46 is an illustration of an electrochemical cell **4600** with multiple interlayers **4660a**, **4660b** (collectively referred to as interlayers **4660**), according to an embodiment. As shown, the electrochemical cell **4600** includes an anode **4610** disposed on an anode current collector **4620**, a cathode **4630** disposed on a cathode current collector **4640**, with a first separator **4650a**, a second separator **4650b**, and a third separator **4650c** disposed between the anode **4610** and the cathode **4630**. The interlayer **4660a** is disposed between the first separator **4650a** and the second separator **4650b** and the second interlayer **4660b** is disposed between the second separator **4650b** and the third separator **4650c**. In some embodiments, the anode **4610**, the anode current collector **4620**, the cathode **4630**, the cathode current collector **4640**, the first separator **4650a**, the second separator **4650b**, the third separator **4650c**, and the interlayers **4660** can be the same or substantially similar to the anode **3410**, the anode current collector **3420**, the cathode **3430**, the cathode current collector **3440**, the first separator **3450a**, the second separator **3450b**, the third separator **3450c**, and the interlayers **3560**, as described above with reference to FIG. 34. Thus, certain aspects of the anode **4610**, the anode current collector **4620**, the cathode **4630**, the cathode current collector **4640**, the first separator **4650a**, the second separator **4650b**, the third separator **4650c** and the interlayers **4660** are not described in greater detail herein.

[0275] As shown, a voltage V_1 is measured between the anode current collector **4620** and the cathode current collector **4640**, a voltage V_2 is measured between the anode current collector **4620** and the interlayer **4660a**, and a

voltage V_3 is measured between the anode current collector **4640** and the interlayer **4660b**. As shown, the electrochemical cell **4600** includes switches S_{1a} , S_{1b} , S_{2a} , S_{2b} , S_{3a} , S_{3b} , S_{4a} , S_{4b} , capacitors C_1 , C_2 , C_3 , C_4 , C_5 , and optional resistor R . The capacitor C_5 can charge continuously via the resistor R .

[0276] Depending on how the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} , S_{3a} , S_{3b} , S_{4a} , S_{4b} , are configured, a voltage of at least about 2, or at least about 3 times the voltage difference between the anode current collector **4620** and the cathode current collector **4660** can be applied to the interlayer **4660a** and/or the interlayer **4660b**. For example, with switch S_{1a} and switch S_{2a} in the up position while switch S_{1b} and switch S_{2b} are in the down position, the capacitors C_1 and C_2 charge. Then, switching the switch S_{1a} and the switch S_{2a} to the down position while switching the switch S_{1b} and the switch S_{2b} to the up position, triple the voltage V_1 is applied to the interlayer **4660**. This is a charge pump procedure. In some embodiments, any of the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} , S_{3a} , S_{3b} , S_{4a} , S_{4b} , can be replaced with a diode, a MOSFET, a BJT, or any other suitable device. In some embodiments, any of the switches S_{1a} , S_{1b} , S_{2a} , S_{2b} , S_{3a} , S_{3b} , S_{4a} , S_{4b} , can be controlled by a BMS, a hardware device, a control chip, an oscillator, or any other suitable controller device. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **4600**. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **2700**, as described above with reference to FIG. 27.

[0277] FIG. 47 is an illustration of an electrochemical cell **4700** with multiple interlayers **4760a**, **4760b** (collectively referred to as interlayers **4760**), according to an embodiment. As shown, the electrochemical cell **4700** includes an anode **4710** disposed on an anode current collector **4720**, a cathode **4730** disposed on a cathode current collector **4740**, with a first separator **4750a**, a second separator **4750b**, and a third separator **4750c** disposed between the anode **4710** and the cathode **4730**. The interlayer **4760a** is disposed between the first separator **4750a** and the second separator **4750b** and the second interlayer **4760b** is disposed between the second separator **4750b** and the third separator **4750c**. In some embodiments, the anode **4710**, the anode current collector **4720**, the cathode **4730**, the cathode current collector **4740**, the first separator **4750a**, the second separator **4750b**, the third separator **4750c**, and the interlayers **4760** can be the same or substantially similar to the anode **3410**, the anode current collector **3420**, the cathode **3430**, the cathode current collector **3440**, the first separator **3450a**, the second separator **3450b**, the third separator **3450c**, and the interlayers **3560**, as described above with reference to FIG. 34. Thus, certain aspects of the anode **4710**, the anode current collector **4720**, the cathode **4730**, the cathode current collector **4740**, the first separator **4750a**, the second separator **4750b**, the third separator **4750c** and the interlayers **4760** are not described in greater detail herein.

[0278] As shown, a voltage V_1 is measured between the anode current collector **4720** and the cathode current collector **4740**, a voltage V_2 is measured between the anode current collector **4720** and the interlayer **4760a**, and a

voltage V_3 is measured between the anode current collector **4740** and the interlayer **4760b**. As shown, the electrochemical cell **4700** includes diodes D_1 , D_2 and optional resistors R_1 , R_2 . As shown, the diodes D_1 , D_2 direct current from the cathode current collector **4740** to the interlayer **4760a** and from the cathode current collector **4740** to the interlayer **4760b**, respectively. The electrochemical cell **4700** acts as a diode conductance cell. When the voltage of the interlayer **4760a** and/or the interlayer **4760b** decreases to a value of less than the forward voltage of the diodes D_1 , D_2 , current can flow through either of the diodes D_1 , D_2 , increasing the potential of the interlayer **4760a** and/or the interlayer **4760b**. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **4700**. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **200**, as described above with reference to FIGS. 2A-2B.

[0279] FIG. 48 is an illustration of an electrochemical cell **4800** with multiple interlayers **4860a**, **4860b** (collectively referred to as interlayers **4860**), according to an embodiment. As shown, the electrochemical cell **4800** includes an anode **4810** disposed on an anode current collector **4820**, a cathode **4830** disposed on a cathode current collector **4840**, with a first separator **4850a**, a second separator **4850b**, and a third separator **4850c** disposed between the anode **4810** and the cathode **4830**. The interlayer **4860a** is disposed between the first separator **4850a** and the second separator **4850b** and the second interlayer **4860b** is disposed between the second separator **4850b** and the third separator **4850c**. In some embodiments, the anode **4810**, the anode current collector **4820**, the cathode **4830**, the cathode current collector **4840**, the first separator **4850a**, the second separator **4850b**, the third separator **4850c**, and the interlayers **4860** can be the same or substantially similar to the anode **3410**, the anode current collector **3420**, the cathode **3430**, the cathode current collector **3440**, the first separator **3450a**, the second separator **3450b**, the third separator **3450c**, and the interlayers **3560**, as described above with reference to FIG. 34. Thus, certain aspects of the anode **4810**, the anode current collector **4820**, the cathode **4830**, the cathode current collector **4840**, the first separator **4850a**, the second separator **4850b**, the third separator **4850c** and the interlayers **4860** are not described in greater detail herein. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **2000**, as described above with reference to FIG. 20.

[0280] As shown, a voltage V_1 is measured between the anode current collector **4820** and the cathode current collector **4840**, a voltage V_2 is measured between the anode current collector **4820** and the interlayer **4860a**, and a voltage V_3 is measured between the anode current collector **4840** and the interlayer **4860b**. As shown, the electrochemical cell **4800** includes diodes D_1 , D_2 and optional resistors R_1 , R_2 . As shown, the diodes D_1 , D_2 direct current from the interlayer **4860a** to the cathode current collector **4840** and from the interlayer **4860b** to the cathode current collector **4840**, respectively. The electrochemical cell **4800** includes a backflow circuit bias. When the voltage of the interlayer **4860a** and/or the interlayer **4860b** decreases to a value of

less than the forward voltage of the diodes D_1 , D_2 , current can flow through either of the diodes D_1 , D_2 , increasing the potential of the interlayer **4860a** and/or the interlayer **4860b**. In some embodiments, if either of the diodes D_1 , D_2 includes a rectification diode (i.e., a diode not intended to function in a reverse breakdown voltage), the current can be restricted in two ways. First the reverse bias leakage current can be defined by the manufacturer of the component. Second the reverse recovery current of the diode D_1 and/or the diode D_2 can be defined by the manufacturer of the component. In the event the current exceeds a threshold value, the diode D_1 and/or the diode D_2 can switch off, preventing excess current flow to the interlayer **4860a** and/or the interlayer **4860b**. In some embodiments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **4800**. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **200**, as described above with reference to FIGS. 2A-2B.

[0281] FIG. 49 is an illustration of an electrochemical cell **4900** with multiple interlayers **4960a**, **4960b** (collectively referred to as interlayers **4960**), according to an embodiment. As shown, the electrochemical cell **4900** includes an anode **4910** disposed on an anode current collector **4920**, a cathode **4930** disposed on a cathode current collector **4940**, with a first separator **4950a**, a second separator **4950b**, and a third separator **4950c** disposed between the anode **4910** and the cathode **4930**. The interlayer **4960a** is disposed between the first separator **4950a** and the second separator **4950b** and the second interlayer **4960b** is disposed between the second separator **4950b** and the third separator **4950c**. In some embodiments, the anode **4910**, the anode current collector **4920**, the cathode **4930**, the cathode current collector **4940**, the first separator **4950a**, the second separator **4950b**, the third separator **4950c**, and the interlayers **4960** can be the same or substantially similar to the anode **3410**, the anode current collector **3420**, the cathode **3430**, the cathode current collector **3440**, the first separator **3450a**, the second separator **3450b**, the third separator **3450c**, and the interlayers **3560**, as described above with reference to FIG. 34. Thus, certain aspects of the anode **4910**, the anode current collector **4920**, the cathode **4930**, the cathode current collector **4940**, the first separator **4950a**, the second separator **4950b**, the third separator **4950c** and the interlayers **4960** are not described in greater detail herein.

[0282] As shown, the electrochemical cell **4900** includes diodes D_1 and D_2 and optional resistors R_1 , R_2 , R_3 . The electrochemical cell **4900** includes charge diode capability. The diode D_2 can cause a voltage drop of the interlayer **4960a** relative to the cathode **4930** equal to the forward voltage drop of the diode D_2 . This voltage drop can be about 0.1, about 0.2, about 0.3, about 0.4, about 0.5, about 0.6, about 0.7, about 0.8, about 0.9, or about 1 times the full voltage difference between the cathode **4930** and the anode **4910**, inclusive of all values and ranges therebetween. The diode D_2 can operate in forward breakdown, reverse breakdown, and/or any other topology. The diode D_1 can be selected for a specific forward or reverse voltage and can be placed for forward or reverse voltage and can be placed for forward or reverse breakdown voltage. In some embodi-

ments, the resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell **4900**. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell **3300**, as described above with reference to FIG. 33. In some embodiments, the interlayers **4960** can be connected together as shown, or the interlayers **4960** can be connected to different circuit types as described above in other embodiments or other types of circuits used to set the voltage potential of the interlayers **4960**.

[0283] FIG. 50 is an illustration of an electrochemical cell **5000** with multiple interlayers **5060a**, **5060b** (collectively referred to as interlayers **5060**), according to an embodiment. As shown, the electrochemical cell **5000** includes an anode **5010** disposed on an anode current collector **5020**, a cathode **5030** disposed on a cathode current collector **5040**, with a first separator **5050a**, a second separator **5050b**, and a third separator **5050c** disposed between the anode **5010** and the cathode **5030**. The interlayer **5060a** is disposed between the first separator **5050a** and the second separator **5050b** and the second interlayer **5060b** is disposed between the second separator **5050b** and the third separator **5050c**. In some embodiments, the anode **5010**, the anode current collector **5020**, the cathode **5030**, the cathode current collector **5040**, the first separator **5050a**, the second separator **5050b**, the third separator **5050c**, and the interlayers **5060** can be the same or substantially similar to the anode **3410**, the anode current collector **3420**, the cathode **3430**, the cathode current collector **3440**, the first separator **3450a**, the second separator **3450b**, the third separator **3450c**, and the interlayers **3560**, as described above with reference to FIG. 34. Thus, certain aspects of the anode **5010**, the anode current collector **5020**, the cathode **5030**, the cathode current collector **5040**, the first separator **5050a**, the second separator **5050b**, the third separator **5050c** and the interlayers **5060** are not described in greater detail herein.

[0284] As shown, the electrochemical cell **5000** includes diodes D_1 and D_2 , switch S and optional resistors R_1 , R_2 , R_3 . The electrochemical cell **5000** includes charge diode capability. The diode D_2 can cause a voltage drop of the interlayer **5060a** relative to the cathode **5030** equal to the forward voltage drop of the diode D_2 . This voltage drop can be about 0.1, about 0.2, about 0.3, about 0.4, about 0.5, about 0.6, about 0.7, about 0.8, about 0.9, or about 1 times the full voltage difference between the cathode **5030** and the anode **5010**, inclusive of all values and ranges therebetween. The diode D_2 can operate in forward breakdown, reverse breakdown, and/or any other topology. The diode D_1 can be selected for a specific forward or reverse voltage and can be placed for forward or reverse voltage and can be placed for forward or reverse breakdown voltage. During charge, when the switch S is open, the voltage of the interlayer **5060a** and/or the interlayer **5060b** can increase by the same value as the voltage drop across any of the diodes D_1 , D_2 or the resistors R_1 , R_2 , R_3 . In some embodiments, the switch S can bypass the function of the diodes D_1 , D_2 , creating a selective dendrite treatment mode or normal operation. In some embodiments, the switch S_2 can be replaced with a transistor, a MOSFET, a JFET, or any other devices used to bypass the function of the diodes D_1 , D_2 . In some embodiments, the

resistances through each current flow path can be the same or substantially similar. In some embodiments, the resistances through each current flow path can be different from each other. In some embodiments, the resistance values can be from separate components, or they can represent other impedances inherent to the electrochemical cell 5000. In some embodiments, the functionality of the circuit can be the same or substantially similar to that of the electrochemical cell 3000, as described above with reference to FIG. 30. In some embodiments, the interlayers 5060 can be connected together as shown, or the interlayers 4960 can be connected to different circuit types as described above in other embodiments or other types of circuits used to set the voltage potential of the interlayers 4960.

[0285] FIGS. 51A-51C are illustrations of a method of producing a prelithiated electrochemical cell 5100, according to an embodiment. As shown in FIG. 51A, the electrochemical cell 5100 includes an anode 5110 disposed on an anode current collector 5120, a cathode 5130 disposed on a cathode current collector 5140, with a first separator 5150a and a second separator 5150b disposed between the anode 5110 and the cathode 5130. The interlayer 5160 is disposed between the first separator 5150a and the second separator 5150b. In some embodiments, the anode 5110, the anode current collector 5120, the cathode 5130, the cathode current collector 5140, the first separator 5150a, the second separator 5150b, and the interlayer 5160 can be the same or substantially similar to the anode 210, the anode current collector 220, the cathode 230, the cathode current collector 240, the first separator 250a, the second separator 250b, and the interlayer 260, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode 5110, the anode current collector 5120, the cathode 5130, the cathode current collector 5140, the first separator 5150a, the second separator 5150b, and the interlayer 5160 are not described in greater detail herein.

[0286] As shown in FIG. 51A, a voltage V_1 is measured between the anode current collector 5120 and the cathode current collector 5140 and a voltage V_2 is measured between the anode current collector 5120 and the interlayer 5160. As shown in FIG. 51A, a switch S is placed on a circuit and joins the interlayer 5160 to the cathode current collector 5140 and to the anode current collector 5120 (with optional resistors R_1 , R_2). The cathode 5130 includes a lithium-rich cathode material (i.e., a cathode material that includes a greater stoichiometric amount of lithium than the anode). The cathode material can produce a lithium-rich salt (i.e., if the cathode material has multiple lithium atoms per ionic compound). In some embodiments, the cathode material can include Li_3N , Li_2O , Li_3PO_4 , Li_2S , or any combination thereof.

[0287] During initial cycling and gas formation depicted in FIG. 51A, current is passed from the cathode current collector 5140 to the interlayer 5160. In other words, a short circuit is formed between the cathode 5130 and the interlayer 5160. This forms lithium plating 5161 on the interlayer 5160, due to migration of lithium ions from the cathode current collector 5140 to the interlayer 5160 (as shown in FIG. 51B). In some embodiments, the current passing from the cathode current collector 5140 to the interlayer 5160 can increase the voltage of the interlayer 5160a relative to the cathode current collector 5140 to about 0.5 V, about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 3.5 V, about 4 V, about 4.5 V, about 5 V, about 5.5 V, or about 6 V,

inclusive of all values and ranges therebetween. The short circuit between the cathode current collector 5140 and the interlayer 5160 aids in formation of a solid-electrolyte interphase (SEI) layer on particles in the interlayer 5160. This stabilizes the interlayer 5160, and makes the interlayer 5160 more effective in limiting the growth of dendrites through the interlayer 5160. This prevents short circuiting during the operation of the electrochemical cell.

[0288] FIG. 51C shows the switch S configured such that current can pass between the interlayer 5160 and the anode current collector 5120. This can aid in balancing the SOC between the interlayer 5160 and the anode 5110. This balancing can improve stability during operation or dendrite detection. This also helps prevent the interlayer 5160 from becoming lithium reactive. In some embodiments, the interlayer 5160 can include a mesh. In some embodiments, a tab (e.g., a weld tab) can be connected to the mesh of the interlayer 5160.

[0289] FIGS. 52A-52C are illustrations of a method of producing a prelithiated electrochemical cell 5200, according to an embodiment. As shown in FIG. 52A, the electrochemical cell 5200 includes an anode 5210 disposed on an anode current collector 5220, a cathode 5230 disposed on a cathode current collector 5240, with a first separator 5250a and a second separator 5250b disposed between the anode 5210 and the cathode 5230. The interlayer 5260 is disposed between the first separator 5250a and the second separator 5250b. In some embodiments, the anode 5210, the anode current collector 5220, the cathode 5230, the cathode current collector 5240, the first separator 5250a, the second separator 5250b, and the interlayer 5260 can be the same or substantially similar to the anode 5110, the anode current collector 5120, the cathode 5130, the cathode current collector 5140, the first separator 5150a, the second separator 5150b, and the interlayer 5160, as described above with reference to FIGS. 51A-51C. Thus, certain aspects of the anode 5210, the anode current collector 5220, the cathode 5230, the cathode current collector 5240, the first separator 5250a, the second separator 5250b, and the interlayer 5260 are not described in greater detail herein.

[0290] As shown in FIG. 52A, a voltage V_1 is measured between the anode current collector 5220 and the cathode current collector 5240 and a voltage V_2 is measured between the anode current collector 5220 and the interlayer 5260. As shown in FIG. 52A, a switch S is placed on a circuit and joins the interlayer 5260 to the anode current collector 5220 and to the anode current collector 5120 (with optional resistors R_1 , R_2). In other words, a short circuit is formed between the anode 5210 and the interlayer 5260. The anode 5210 includes lithium. In some embodiments, the anode 5210 can include lithium metal. In some embodiments, the anode 5210 can include lithium intercalated into graphite.

[0291] While the anode 5210 is electrically connected to the interlayer 5260, lithium ions and electrons pass from the anode 5210 to the interlayer 5260. As shown, in FIG. 52B, the anode 5210 has reduced in size. This is due to the migration of lithium ions to the interlayer. In some embodiments, the migration of lithium ions from the anode 5210 to the interlayer 5260 can reduce the mass of the anode 5210 by about 50%, about 60%, about 70%, about 80%, about 90%, or about 100%, inclusive of all values and ranges therebetween. In some embodiments, all or substantially all of the mass of the anode 5210 can be depleted by the migration of the lithium ions.

[0292] In FIG. 52B, the switch S is switched to form a short circuit between the interlayer 5260 and the cathode 5230. This aids in SEI growth and stabilization on the particles in the interlayer 5260 and gas generation. In FIG. 52C, the switch S is switched back to a position to form a short circuit between the interlayer 5260 and the anode 5210. This can aid in balancing the state of charge between the interlayer 5260 and the anode 5210.

[0293] In some embodiments, the sequence depicted in FIGS. 51A-51C and the sequence depicted in FIGS. 52A-52C can be executed in tandem. In other words, lithium from both the anode and the cathode can be used to plate the interlayer. In some embodiments, either of the electrochemical cells described herein can be subject to hipot testing. In some embodiments, the hipot testing can include driving a thin conductive rod through an anode tab, a cathode tab, and a tab electrically connected to the interlayer. A high voltage is then applied across the rod. A stably formed SEI layer in the interlayer can aid in preventing any component failures in the electrochemical cell. In some embodiments, the voltage can be about 5V, about 6V, about 7V, about 8V, about 9V, or about 10V, inclusive of all values and ranges therebetween.

[0294] FIG. 53 is an illustration of a tab welding scheme to a separator 5350, according to an embodiment. As shown, the separator 5350 is a carbon-coated separator. In some embodiments, the carbon coating on the separator can include CNT, CNF, carbon black, and/or graphene. The carbon-coating surface of the separator 5350 can contact an interlayer (not shown). The interlayer can be the same or substantially similar to any of the aforementioned interlayers. A metal section 5364 is coupled to the separator 5350 and contacts the carbon coating. In some embodiments, the metal section 5364 can include a folded metal foil. The folded metal foil can support the welding process. In some embodiments, the folded metal foil can be composed of aluminum, copper, and/or any other suitable material. In some embodiments, the metal section 5364 can include a sputtered metal. The sputtered metal can act as a substrate for welding of the tab 5363.

[0295] A tab 5363 is welded to the metal section 5364 via a weld area W. In some embodiments, the welding can be via ultrasonic welding. In some embodiments, the tab 5363 can be a mesh tab. The mesh tab can be light to avoid the tear of welding on the separator 5350. In some embodiments, the tab 5363 can include PE or a conductive polymer to be heat treated and then laminated to an interlayer. In some embodiments, the tab 5363 can be laminated to the separator 5350. In some embodiments, the tab 5363 can be stitched to the interlayer via a wire or thread, including metal wire, polyamide thread, polyester thread, or other polymer based material. In some embodiments, a male/female pinch can be used and a bolt can be tightened to fix the tab 5363 to the separator 5350.

[0296] FIG. 54 is an illustration of a tab welding scheme to a separator 5450, according to an embodiment. As shown, a tab 5463 is appended to the separator 5450 via a sealing area S. In some embodiments, the separator 5450 and the tab 5463 can be the same or substantially similar to the separator 5350 and the tab 5363, as described above with reference to FIG. 53. Thus, certain aspects of the separator 5450 and the tab 5463 are not described in greater detail herein. The separator 5450 can be carbon coated. In some embodiments, the tab 5463 can be coupled to the separator via an adhesive.

In some embodiments, the adhesive can include a conductive adhesive. If the adhesive dissolves over time, an externally applied pressure can keep the tab 5463 in place long term. In other words, in some embodiments, the adhesive can be used to keep the tab in place only during assembly. In some embodiments, the tab 5463 can be soldered to the separator 5450.

[0297] FIG. 55 is an illustration of a tab welding scheme to a separator 5550, according to an embodiment. As shown, a tab 5563 is appended to the separator 5550 via a sealing area S. In some embodiments, the separator 5550 and the tab 5563 can be the same or substantially similar to the separator 5350 and the tab 5363, as described above with reference to FIG. 53. Thus, certain aspects of the separator 5550 and the tab 5563 are not described in greater detail herein. As shown, the separator 5550 includes an extension 5551, which provides additional surface for sealing or welding. The extension 5551 is also carbon coated.

[0298] FIG. 56 is an illustration of a tab welding scheme to a separator 5650, according to an embodiment. As shown, a tab 5663 is appended to the separator 5650 via a double-sided tape 5666 and a heat seal. In some embodiments, the separator 5650 and the tab 5663 can be the same or substantially similar to the separator 5350 and the tab 5363, as described above with reference to FIG. 53. Thus, certain aspects of the separator 5650 and the tab 5663 are not described in greater detail herein. During production, the heat seal is applied to the double-sided tape 5666 and around the perimeter of the separator 5650. A sealing region 5652 forms, around the perimeter of the separator 5650. The double-sided tape 5666 melts during the sealing process. The tab is aligned so the double-sided tape 5666 is in the sealing region 5652.

[0299] FIG. 57 is an illustration of a tab welding scheme to a separator 5650, according to an embodiment. As shown, a tab 5763 is appended to a metal foil strip 5764, which is coupled to the separator 5750. The carbon coating coats the separator 5650 up to a line L. In other words, the carbon coating overlaps with the metal foil strip 5764, but does not cover the metal foil strip 5764. In some embodiments, the separator 5750 and the tab 5763 can be the same or substantially similar to the separator 5350 and the tab 5363, as described above with reference to FIG. 53. Thus, certain aspects of the separator 5750 and the tab 5763 are not described in greater detail herein. The tab 5763 is welded to the metal foil strip 5764.

[0300] FIG. 58 is an illustration of an electrochemical cell 5800 with a tab sealing scheme, according to an embodiment. Not shown in FIG. 58 are the anode, anode current collector, cathode, cathode current collector, and separator of the electrochemical cell. These components can have any of the properties of the cells described in the aforementioned embodiments. An anode tab 5822, a cathode tab 5842, and an interlayer tab 5863 each extend outward from their respective electrodes and/or current collectors and are held in place via a unit cell sealing region 5852. An anode tab extender 5824 is welded to the anode tab 5822 via welding region W1. A cathode tab extender 5844 is welded to the cathode tab 5842 via welding region W2. An interlayer tab extender 5864 is welded to the interlayer tab 5862 via welding region W3. A pouch heat sealing region 5871 is depicted as sealing the anode tab extender 5824, the cathode tab extender 5844, and the interlayer tab extender 5864 in

place. The pouch heat sealing region **5871** is part of a pouch (e.g., an aluminized pouch that houses multiple unit cells).

[0301] In some embodiments, the interlayer tab **5863** can extend from a coupling to a carbon coated separator, as depicted in FIGS. **53-56**. In some embodiments, the interlayer tab **5863** can include a thin foil tab. In some embodiments, the interlayer tab **5863**, the anode tab **5822**, and the cathode tab **5842** can include thin metal tabs and the interlayer tab extender **5864**, the anode tab extender **5824**, and the cathode tab extender **5844** can be thicker and more robust tabs that protrude through the pouch. This scheme can be employed in cells with a single interlayer as well as cells with multiple interlayers. In some embodiments, the separator can include a carbon-coated area that extends outside of the heat-sealed unit cell area. This can ease the attachment of the interlayer tab **5863** and/or the interlayer tab extender **5864** to the separator during manufacture.

[0302] FIGS. **59A-59C** are illustrations of an electrochemical cell stack **5900** with multiple interlayers **5960a**, **5960b** (collectively referred to as interlayers **5960**), according to an embodiment. As shown, the electrochemical cell stack **5900** includes an anodes **5910a**, **5910b** (collectively referred to as anodes **5910**) disposed on an anode current collectors **5920**, cathodes **5930a**, **5930b**, **5930c**, **5950d** (collectively referred to as cathodes **5930**) disposed on cathode current collectors **5940a**, **5940b** (collectively referred to as cathode current collectors **5940**), with a first separator **5950a** and a second separator **5950b** disposed between the anode **5910a** and the cathode **5930b** and a third separator **5950c** and a fourth separator **5950d** disposed between the anode **5910b** and the cathode **5930c**. The interlayer **5960a** is disposed between the first separator **5950a** and the second separator **5950b** and the interlayer **5960b** is disposed between the third separator **5950c** and the fourth separator **5950d**. In some embodiments, the anodes **5910**, the anode current collector **5920**, the cathodes **5930**, the cathode current collectors **5940**, the first separator **5950a**, the second separator **5950b**, the third separator **5950c**, the fourth separator **5950d** and the interlayer **5960** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. Thus, certain aspects of the anodes **5910**, the anode current collector **5920**, the cathodes **5930**, the cathode current collectors **5940**, the first separator **5950a**, the second separator **5950b**, the third separator **5950c**, the fourth separator **5950d** and the interlayer **5960** are not described in greater detail herein.

[0303] As shown in FIG. **59A**, the interlayer **5960a** and the second separator **5950b** extend beyond the separator **5950a** by a distance d . This offset aids in detecting misalignment between the anodes **5910** and the cathodes **5930**. As shown, in FIG. **59B**, the anode **5910a** has become misaligned with the cathode **5930b**. In FIG. **59C**, the interlayer **5960a** and the separator **5950b** bend toward the cathode **5930b**, such that the interlayer **5960a** contacts the cathode **5930b**. This creates a short circuit between the cathode **5930b** and the interlayer **5960a**. This misalignment would be detected by way of a zero voltage or a significantly reduced voltage difference between the interlayer **5960a** and the cathode **5930b**.

[0304] In some embodiments, the distance d can be at least about 50 μm , at least about 100 μm , at least about 200 μm ,

at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 1.1 mm, at least about 1.2 mm, at least about 1.3 mm, at least about 1.4 mm, at least about 1.5 mm, at least about 1.6 mm, at least about 1.7 mm, at least about 1.8 mm, or at least about 1.9 mm. In some embodiments, the distance d can be no more than about 2 mm, no more than about 1.9 mm, no more than about 1.8 mm, no more than about 1.7 mm, no more than about 1.6 mm, no more than about 1.5 mm, no more than about 1.4 mm, no more than about 1.3 mm, no more than about 1.2 mm, no more than about 1.1 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , no more than about 200 μm , or no more than about 100 μm . Combinations of the above-referenced distances d are also possible (e.g., at least about 50 μm and no more than about 2 mm or at least about 100 μm and no more than about 1 mm), inclusive of all values and ranges therebetween. In some embodiments, the distance d can be about 50 μm , about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.1 mm, about 1.2 mm, about 1.3 mm, about 1.4 mm, about 1.5 mm, about 1.6 mm, about 1.7 mm, about 1.8 mm, about 1.9 mm, or about 2 mm.

[0305] In some embodiments, the anodes **5910**, the anode current collector **5920**, the cathodes **5930**, the cathode current collectors **5940**, the separators **5950a**, **5950b**, **5950c**, **5950d** (collectively referred to as separators **5950**), and the interlayers **5960** can be oriented in a single sheet stacking orientation (as shown), a z-stacking configuration, a cylindrical winding configuration, or a prismatic winding configuration. Each of these configurations can allow for the contact to occur between the interlayers **5960** and an adjacent electrode. As shown, the interlayer **5960a** contacts one of the cathodes **5930**. In some embodiments, the short circuit can be caused by contact between one of the interlayers **5960** and one of the anodes **5910**.

[0306] FIG. **60** is an illustration of an electrochemical cell **6000** with multiple interlayers **6060a**, **6060b** (collectively referred to as interlayers **6060**), according to an embodiment. As shown, the electrochemical cell **6000** includes an anode **6010** disposed on an anode current collector **6020**, a cathode **6030** disposed on a cathode current collector **6040**, with a first separator **6050a**, a second separator **6050b**, and a third separator **6050c** disposed between the anode **6010** and the cathode **6030**. The interlayer **6060a** is disposed between the first separator **6050a** and the second separator **6050b** and the second interlayer **6060b** is disposed between the second separator **6050b** and the third separator **6050c**. In some embodiments, the anode **6010**, the anode current collector **6020**, the cathode **6030**, the cathode current collector **6040**, the first separator **6050a**, the second separator **6050b**, the third separator **6050c**, and the interlayers **6060** can be the same or substantially similar to the anodes **5910**, the anode current collector **5920**, the cathodes **5930**, the cathode current collectors **5940**, the first separator **5950a**, the second separator **5950b**, the third separator **5950c**, and the interlayers **5960**, as described above with reference to FIGS. **59A-59C**. Thus, certain aspects of the anode **6010**, the anode current collector **6020**, the cathode **6030**, the cathode current collector **6040**, the first separator **6050a**, the second separator

6050b, the third separator **6050c** and the interlayers **6060** are not described in greater detail herein.

[0307] As shown, the interlayer **6060a** extends beyond the separator **6050a** by a distance *d*, while the interlayer **6060b** is not offset from its adjacent separator **6050c**. Such a configuration allows for the detection of electrode misalignments adjacent only to the interlayer **6060a** and not to the interlayer **6060b**. In some embodiments, the distance *d* can be any of the distances *d* described above with reference to FIGS. 59A-59C.

[0308] FIG. 61 is an illustration of an electrochemical cell **6100** with multiple interlayers **6160a**, **6160b** (collectively referred to as interlayers **6160**), according to an embodiment. As shown, the electrochemical cell **6100** includes an anode **6110** disposed on an anode current collector **6120**, a cathode **6130** disposed on a cathode current collector **6140**, with a first separator **6150a**, a second separator **6150b**, and a third separator **6150c** disposed between the anode **6110** and the cathode **6130**. The interlayer **6160a** is disposed between the first separator **6150a** and the second separator **6150b** and the second interlayer **6160b** is disposed between the second separator **6150b** and the third separator **6150c**. In some embodiments, the anode **6110**, the anode current collector **6120**, the cathode **6130**, the cathode current collector **6140**, the first separator **6150a**, the second separator **6150b**, the third separator **6150c**, and the interlayers **6160** can be the same or substantially similar to the anodes **5910**, the anode current collector **5920**, the cathodes **5930**, the cathode current collectors **5940**, the first separator **5950a**, the second separator **5950b**, the third separator **5950c**, and the interlayers **5960**, as described above with reference to FIGS. 59A-59C. Thus, certain aspects of the anode **6110**, the anode current collector **6120**, the cathode **6130**, the cathode current collector **6140**, the first separator **6150a**, the second separator **6150b**, the third separator **6150c** and the interlayers **6160** are not described in greater detail herein.

[0309] As shown, the interlayer **6160a** and the separator **6150b** are offset from the separator **6150a** by a distance *d1*. As shown, the interlayer **6160b** and the separator **6150c** are offset from the separator **6150b** by a distance *d2*. This double offset allows for both the interlayer **6160a** and the interlayer **6160b** to contact the cathode **6130** in the case of a misalignment. In some embodiments, the interlayer **6160a** and/or the interlayer **6160b** can extend by offset distances on the side adjacent to the anode **6110**, such that a misalignment between the anode **6110** and either of the interlayers **6160** is detectable.

[0310] In some embodiments, the distance *d1* and/or the distance *d2* can be at least about 50 μm , at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 1.1 mm, at least about 1.2 mm, at least about 1.3 mm, at least about 1.4 mm, at least about 1.5 mm, at least about 1.6 mm, at least about 1.7 mm, at least about 1.8 mm, or at least about 1.9 mm. In some embodiments, the distance *d1* and/or the distance *d2* can be no more than about 2 mm, no more than about 1.9 mm, no more than about 1.8 mm, no more than about 1.7 mm, no more than about 1.6 mm, no more than about 1.5 mm, no more than about 1.4 mm, no more than about 1.3 mm, no more than about 1.2 mm, no more than about 1.1 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no

more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , no more than about 200 μm , or no more than about 100 μm . Combinations of the above-referenced distances *d* are also possible (e.g., at least about 50 μm and no more than about 2 mm or at least about 100 μm and no more than about 1 mm), inclusive of all values and ranges therebetween. In some embodiments, the distance *d1* and/or the distance *d2* can be about 50 μm , about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.1 mm, about 1.2 mm, about 1.3 mm, about 1.4 mm, about 1.5 mm, about 1.6 mm, about 1.7 mm, about 1.8 mm, about 1.9 mm, or about 2 mm.

[0311] FIGS. 62A-62B are illustrations of a tabbing arrangement, according to an embodiment. FIG. 62A shows a separator **6250** and an interlayer **6260**. The interlayer **6260** includes a carbon layer **6261** and a coating **6262**. A weld tab **6263** with a tab coating **6264** is also shown in FIG. 62A.

[0312] In some embodiments, the carbon layer **6261** can include conductive carbon. In some embodiments, the carbon layer **6261** can include binder. In some embodiments, the carbon layer **6261** can include about 5 wt %, about 10 wt %, about 15 wt %, about 20 wt %, about 25 wt %, about 30 wt %, about 35 wt %, about 40 wt %, about 45 wt %, about 50 wt %, about 55 wt %, about 60 wt %, about 65 wt %, about 70 wt %, about 75 wt %, about 80 wt %, about 85 wt %, about 90 wt %, about 95 wt %, or about 100 wt % conductive carbon, inclusive of all values and ranges therebetween. In some embodiments, the carbon layer **6261** can include about 5 wt %, about 10 wt %, about 15 wt %, about 20 wt %, about 25 wt %, about 30 wt %, about 35 wt %, about 40 wt %, about 45 wt %, about 50 wt %, about 55 wt %, about 60 wt %, about 65 wt %, about 70 wt %, about 75 wt %, about 80 wt %, about 85 wt %, about 90 wt %, about 95 wt %, or about 100 wt % binder, inclusive of all values and ranges therebetween.

[0313] In some embodiments, the binder can include polyvinylidene chloride (PVDC). In some embodiments, the binder can include PVDF. In some embodiments, the binder and the conductive carbon can be bound together via melting and/or polymerizing (e.g., crosslinking) the binder and physically connecting the binder to the conductive carbon. In some embodiments, the binder and the conductive carbon can be melt-mixed together. In some embodiments, the binder can be melted via heating, ultrasonic application, and/or ultraviolet application. In some embodiments, the binder can be melted under pressure. In some embodiments, the mixing and/or melting of the binder with the conductive carbon can occur inside an electrochemical cell (i.e., after the electrochemical cell has been built). In some embodiments, the mixing and/or melting of the binder with the conductive carbon can occur outside of an electrochemical cell (e.g., in the construction of individual unit cells).

[0314] In some embodiments, the carbon layer **6261** can be mixed with a solvent before disposing the carbon layer **6261** onto the separator **6250**. In some embodiments, the binder in the coating **6262** can be mixed with a solvent. In some embodiments, the carbon layer **6261** can include a first solvent and the coating **6262** can include a second solvent, the second solvent different from the first solvent. In some embodiments, the second solvent can be the same as the first solvent. In some embodiments, the carbon layer **6261** can be coated onto the separator **6250** with water, carboxymethyl

cellulose (CMC), or any combination thereof. In some embodiments, the coating **6262** can include PVDF, n-methyl pyrrolidone (NMP), or any combination thereof.

[0315] As shown in FIG. **62A**, a section of the carbon layer **6261** is not coated by the coating **6262**. This allows for tabbing via the tab **6263**. In FIG. **62B**, the tab **6263** and the tab coating **6264** have been merged with the interlayer **6260**. In some embodiments, the tab **6263** can have any of the properties of the tab **763**, as described above with reference to FIGS. **7A-7B**. In some embodiments, the tab coating **6264** can be the same or substantially similar to the coating **6262**. In some embodiments, the tab **6263** and the tab coating **6264** can be pressed to the coating **6262**. In some embodiments, the tab **6263** and the tab coating **6264** can be heat-pressed to the coating **6262**. In some embodiments, the tab **6263** and the tab coating **6264** can be laminated to the coating **6262**. In some embodiments, the tab **6263** and the tab coating **6264** can adhere to the coating **6262** via an adhesive. In some embodiments, the tab coating **6264** and the coating **6262** can be melted together. In some embodiments, the melting can be via heat, ultrasonic, UV, and/or with pressure. In some embodiments, the tab **6263** and the tab coating **6264** can be merged with the coating **6262** inside the electrochemical cell (i.e., after the electrochemical cell has been formed. In some embodiments, the tab **6263** and the tab coating **6264** can be merged with the coating **6262** outside the electrochemical cell.

[0316] FIG. **63** is a block diagram of an electrochemical cell system **63000**, according to an embodiment. As shown, the electrochemical cell system **63000** includes electrochemical cells **6300a**, **6300b**, **6300c**, **6300d**, **6300e** (collectively referred to as electrochemical cells **6300**) inside a casing **6390**. As shown, the electrochemical cells **6300** include interlayers that extend beyond the breadth of the other components of their respective electrochemical cells, marked with 'd.' This can be the same or substantially similar to the interlayers **6090**, **6190**, as described above with respect to FIGS. **60** and **61**, respectively. The electrochemical cell system **63000** further includes a system controller **6301**, a battery management system (BMS) **6302**, and a dendrite control **6303**. As shown, the system controller **6301** controls the action of the dendrite control **6303** and the BMS **6302**. The dendrite control **6303** controls the movement of current among the electrochemical cells **6300** and their various interlayers to prevent dendrite growth or dissipate dendrites. The dendrite control **6303** specifically interfaces to the interlayers in the electrochemical cells **6300** of the electrochemical cell system **63000**. In some embodiments, the electrochemical cells **6300** and the casing **6390** can be the same or substantially similar to the electrochemical cell **1500** and the casing **1590**, as described above with reference to FIGS. **15A-15B**. Thus, certain aspects of the electrochemical cells **6300** and the casing **6390** are not described in greater detail herein. The electrochemical cells **6300** each include one or more interlayers.

[0317] Series voltage monitors can be used to detect the interlayer voltage relative to the anode or the cathode. Differential voltage monitors can also be used based on system implementation. The electrochemical cell system **63000** can compare the interlayer voltage to the voltage of an anode or a cathode. If the voltage, relative to the cathode, is within a defined range (e.g., about -0.2 V to about -1 V, about -0.1 V, about -0.2 V, about -0.3 V, about -0.4 V, about -0.5 V, about -0.6 V, about -0.7 V, about -0.8 V, about -0.9

V, about -1 V, about -1.1 V, about -1.2 V, about -1.3 V, about -1.4 V, or about -1.5 V, inclusive of all values and ranges therebetween), then no dendritic growth has been detected. If the voltage, relative to the cathode is below a defined level (e.g., less than about -0.5 V, less than about -0.6 V, less than about -0.7 V, less than about -0.8 V, less than about -0.9 V, less than about -1 V, less than about -1.1 V, less than about -1.2 V, less than about -1.3 V, less than about -1.4 V, or less than about -1.5 V, inclusive of all values and ranges therebetween), then dendritic growth has been detected.

[0318] In testing, dendritic growth has created perturbations of the interlayer voltage. The system controller **6301** can sense the voltage of the electrodes and also use advanced filtering and waveform evolution to detect dendritic growth. This filtering can be in the form of active hardware filtering or digital filters of various types. Hardware for filtering can include high pass, low pass, band pass, proportional, integration, differential, amplitude, frequency conversion, or other types of active or passive designs. A hardware filtering can be via an external circuit (e.g., a BMS), that can detect the voltage and current change and quantify the variations. The digital filtering can be done in a standard processor, a field-programmable gate array (FPGA), or a digital system processor (DSP) using compact, low speed, or high-speed filtering methods. The model used for digital filter can include finite impulse response (FIR), infinite impulse response (IIR), S model, Fast Fourier Transform (FFT), advanced Fast Fourier Transform (AFFT), or any other appropriate filtering or detection model method or combinations thereof in order to isolate the characteristic event.

[0319] In some embodiments, a device can be added to one or more of the electrochemical cells **6300** or to the system controller **6301** that can apply a constant potential, resistance, or current to the interlayer of the electrochemical cells **6300**. In some embodiments, the device can provide a continuous excitation of the interlayer, such that a dendrite would not be able to form across the separator layer. In some embodiments, the device can be similar to devices described in the '597 publication. In some embodiments, the device can incorporate a single passive device, or any number of components in an electrical control methodology. In some embodiments, a prevention method can be used as part of an overall control strategy, in which the voltage potential, current, and resistance passing through the interlayer can be changed based on a control algorithm. In some embodiments, dendrites can be detected via actively modulating, pulsing, and/or alternating a controlled potential of the interlayer to remediate or prevent a dendrite from forming.

[0320] In some embodiments, the potential of the interlayers can be changed, depending on how the dendritic growth is dissolved or blocked. A high potential, close to or exceeding the cathode voltage can be effective to dissolve a metallic dendrite including lithium, iron, nickel, chromium, and/or copper. A low potential, close to or below the anode voltage can be effective to store the lithium ion in the hard carbon layer for pre-lithiation purposes. The system controller **6301** can act in an active prevention mode, where the potential of the interlayer can be modulated or changed to apply different voltage potentials and either increasing the voltage potentials (cathode side) or decreasing the voltage potentials (anode side) to maintain a cell function or health.

[0321] FIG. **64** is a block diagram of an electrochemical cell system **64000**, according to an embodiment. As shown,

the electrochemical cell system **64000** includes electrochemical cells **6400a**, **6400b**, **6400c**, **6400d**, **6400e** (collectively referred to as electrochemical cells **6400**) inside a casing **6490**. The electrochemical cell system **64000** further includes a system controller **6401**, which includes a battery control **6404**. The battery control **6404** can control movement of current among the electrochemical cells **6400**. The battery control **6404** includes each of the functions needed to control the electrochemical cell system **64000**. The battery control **6404** can be carried out via a BMS. In some embodiments, as shown in FIG. **64**, the functions required to control the battery can be contained in the system controller, eliminating the separation of the BMS. In some embodiments, the electrochemical cells **6400** and the casing **6490** can be the same or substantially similar to the electrochemical cells **6300** and the casing **6390**, as described above with reference to FIG. **63**. Thus, certain aspects of the electrochemical cells **6400** and the casing **6490** are not described in greater detail herein.

[0322] FIG. **65** is a block diagram of an electrochemical cell system **65000**, according to an embodiment. As shown, the electrochemical cell system **65000** includes electrochemical cells **6500a**, **6500b**, **6500c**, **6500d**, **6500e** (collectively referred to as electrochemical cells **6500**) inside a casing **6590**. The electrochemical cell system **65000** further includes a system controller **6501**, which includes a battery control **6504** and a dendrite control **6503**. The dendrite control **6503** can operate independently of the system controller **6501**. In some embodiments, the electrochemical cells **6500**, the casing **6590**, and the dendrite control **6503** can be the same or substantially similar to the electrochemical cells **6300**, the casing **6390**, and the dendrite control **6303**, as described above with reference to FIG. **63**. In some embodiments, the battery control **6504** can be the same or substantially similar to the battery control **6404**, as described above with reference to FIG. **64**. Thus, certain aspects of the electrochemical cells **6500**, the casing **6590**, the dendrite control **6503**, and the battery control **6504** are not described in greater detail herein.

[0323] FIG. **66** is a block diagram of an electrochemical cell system **66000**, according to an embodiment. As shown, the electrochemical cell system **66000** includes electrochemical cells **6600a**, **6600b**, **6600c**, **6600d**, **6600e** (collectively referred to as electrochemical cells **6600**) inside a casing **6690**. A system controller **6601** includes a dendrite control **6603** and a battery control **6604**. The dendrite control **6603** and the battery control **6604** can each independently control the movement of current in the electrochemical cells **6600**. In some embodiments, as shown in FIG. **66**, the functions required to control the electrochemical cell system **66000** and dendrites can be contained in the system controller, eliminating the separation of the BMS or dendrite control **6603**. In some embodiments, the electrochemical cells **6600**, the casing **6690**, the system controller **6601**, the dendrite control **6603**, and the battery control **6604** can be the same or substantially similar to the electrochemical cells **6500**, the casing **6590**, the system controller **6501**, the dendrite control **6503**, and the battery control **6504**, as described above with reference to FIG. **65**. Thus, certain aspects of the electrochemical cells **6600**, the casing **6690**, the system controller **6601**, the dendrite control **6603**, and the battery control **6604** are not described in greater detail herein.

[0324] FIG. **67** is a block diagram of an electrochemical cell system **67000**, according to an embodiment. As shown, the electrochemical cell system **67000** includes system level functions, battery functions, cell interfaces, and electrochemical cells. The system level functions include system thermal interfaces, system operation, and system/cell current management subsystems. The system thermal interface monitors the temperature of the electrochemical cells to avoid temperature gradients and associated capacity loss. The system operation includes operating protocols for the electrochemical cells. The system/cell current subsystem manages current movement throughout the electrochemical cells. FIG. **67** as shown is a representative structure that can be included in a larger system or used as a standalone architecture. Additional functions and feature can be used or the functional blocks can be moved to other controllers or systems.

[0325] The battery functions can include, but are not limited to: a current/power limit algorithm, a cell voltage algorithm, a cell balance control, a state of charge (SOC)/state of health (SOH)/state of function (SOF) monitor, thermal control, dendrite detection, and a dendrite control algorithm. The current/power limit algorithm limits the amount of current that can flow through the electrochemical cells by onboard control or through the system interface. The cell voltage algorithm limits the maximum voltage of either of the electrodes with respect to the other electrodes in the electrochemical cells. The cell balance control prevents significant variations in cell voltage. The SOC/SOH/SOF monitor screens the general health of the electrochemical cells. The thermal control subsystem adjusts the movement of current based on the temperatures of the electrochemical cells and interfaces to heaters/chillers (not shown) as needed to heat and cool the electrochemical cell system **67000**. The dendrite detection and dendrite control algorithms monitor voltage in the interlayers of the electrochemical cells and adjust current or potential to prevent or remove dendrites.

[0326] The cell interfaces can include, but are not limited to: current measurement, cell balance, cell voltage measurement, cell temperature measurement, interlayer measurement, and interlayer control algorithms. The current measurement algorithm measures the movement of current between electrochemical cells. The cell balance algorithm shifts the movement of current based on imbalances in the cells. The cell voltage measurement and cell temperature measurement algorithms measure the voltage and temperature of the cells. The interlayer measurement algorithm measures the voltage of the interlayers in the electrochemical cells. The interlayer control algorithm controls movement of current or potential into and out of the interlayers.

[0327] FIG. **68** is an illustration of an electrochemical cell **6800** with an interlayer **6860**, according to an embodiment. As shown, the electrochemical cell **6800** includes an anode **6810** disposed on an anode current collector **6820**, a cathode **6830** disposed on a cathode current collector **6840**, with a first separator **6850a** and a second separator **6850b** disposed between the anode **6810** and the cathode **6830**. The interlayer **6860** is disposed between the first separator **6850a** and the second separator **6850b**. A pre-lithiation layer **6855** is disposed between the cathode **6830** and the separator **6850a**. In some embodiments, the anode **6810**, the anode current collector **6820**, the cathode **6830**, the cathode current collector **6840**, the first separator **6850a**, the second separator **6850b**, and the interlayer **6860** can be the same or substan-

tially similar to the anode **2810**, the anode current collector **2820**, the cathode **2830**, the cathode current collector **2840**, the first separator **2850a**, the second separator **2850b**, and the interlayer **2860**, as described above with reference to FIG. **28**. Thus, certain aspects of the anode **6810**, the anode current collector **6820**, the cathode **6830**, the cathode current collector **6840**, the first separator **6850a**, the second separator **6850b**, and the interlayer **6860** are not described in greater detail herein.

[0328] A first voltage V_1 is measured between the anode current collector **6820** and the interlayer **6860**. A second voltage V_2 is measured between the anode current collector **6820** and the anode current collector **6820** and the cathode current collector **6840**. The pre-lithiation layer **6855** can be a functional layer. In some embodiments, the pre-lithiation layer **6855** can be coated on the separator **6850a**. In some embodiments, the pre-lithiation layer **6855** can include functional lithium-containing materials with a high specific capacity and a low decomposition potential. During initial charge, the pre-lithiation layer **6855** can decompose to provide extra lithium ions for pre-lithiation of the anode **6810**. In some embodiments, the pre-lithiation layer **6855** can include LiFePO_4 , Li_2FeO_4 , or any combination thereof. During operation, a continuous charge and discharge with an interlayer potential greater than about 1V can diminish a lithium dendrite that has penetrated the interlayer **6860** and revert charge back to the pre-lithiation layer **6855**.

[0329] FIG. **69** is an illustration of an electrochemical cell **6900** with an interlayer **6960**, according to an embodiment. As shown, the electrochemical cell **6900** includes an anode **6910** disposed on an anode current collector **6920**, a cathode **6930** disposed on a cathode current collector **6940**, with a first separator **6950a** and a second separator **6950b** disposed between the anode **6910** and the cathode **6930**. The interlayer **6960** is disposed between the first separator **6950a** and the second separator **6950b**. A positive temperature coefficient (PTC) layer **6965** is disposed between the interlayer **6960** and the separator **6950a**. In some embodiments, the anode **6910**, the anode current collector **6920**, the cathode **6930**, the cathode current collector **6940**, the first separator **6950a**, the second separator **6950b**, and the interlayer **6960** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. Thus, certain aspects of the anode **6910**, the anode current collector **6920**, the cathode **6930**, the cathode current collector **6940**, the first separator **6950a**, the second separator **6950b**, and the interlayer **6960** are not described in greater detail herein.

[0330] The PTC layer **6965** can resist the flow of current therethrough when the PTC layer **6965** takes on heat. In some embodiments, the PTC layer **6965** can have a temperature-dependent electrical resistance. In some embodiments, the PTC layer **6965** can be composed of polycrystalline materials, conductive polymers, barium carbonate, titanium oxide, tantalum, silica, manganese, activated carbon, hard carbon, graphite, carbon grains, or any combination thereof. In some embodiments, the PTC layer **6965** can include a polymer with carbon grains embedded therein. In some embodiments, the PTC layer **6965** can include any of the properties of the PTC materials described in U.S. Patent Publication No. 2023/0022329 (“the ‘329 publication”), filed Jul. 20, 2022 and titled “Electrodes and

Electrochemical Cells with Positive Temperature Coefficient Materials and Methods of Producing the Same,” the disclosure of which is hereby incorporated by reference in its entirety.

[0331] As shown, the PTC layer **6965** is disposed between the interlayer **6960** and the separator **6950a**. In some embodiments, the PTC layer **6965** can be disposed between the interlayer **6960** and the separator **6950b**. In some embodiments, the PTC layer **6965** can be disposed between the separator **6950a** and the cathode **6930**. In some embodiments, the PTC layer **6965** can be disposed between the separator **6950b** and the anode **6910**.

[0332] In use, a short circuit is detected (i.e., via the interlayer **6960**). Energy is then directed away from the positive electrode to reduce the SOC of the electrochemical cell **6900**. The voltage is then reduced. In other words, an external short circuit is initiated. Activation of the PTC layer **6965** is slow during an internal short circuit, and the PTC layer **6965** does not react fast enough to stop the flow of current through the electrochemical cell **6900**. The external short circuit provides enough heat through the cell to induce the PTC layer **6965** to substantially increase its electrical resistance.

[0333] FIG. **70A** is an illustration of an electrochemical cell **7000** with an interlayer **7060**, according to an embodiment. As shown, the electrochemical cell **7000** includes a controller **7005**, an anode **7010** disposed on an anode current collector **7020**, a cathode **7030** disposed on a cathode current collector **7040**, with a first separator **7050a** and a second separator **7050b** disposed between the anode **7010** and the cathode **7030**. The interlayer **7060** is disposed between the first separator **7050a** and the second separator **7050b**. In some embodiments, the anode **7010**, the anode current collector **7020**, the cathode **7030**, the cathode current collector **7040**, the first separator **7050a**, the second separator **7050b**, and the interlayer **7060** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. Thus, certain aspects of the anode **7010**, the anode current collector **7020**, the cathode **7030**, the cathode current collector **7040**, the first separator **7050a**, the second separator **7050b**, and the interlayer **7060** are not described in greater detail herein.

[0334] As shown, a voltage V_1 is measured between the anode current collector **7020** and the interlayer **7060**, a voltage V_2 is measured between the anode current collector **7020** and the cathode current collector **7040**. As shown, a switch S_1 and a resistor R_1 control a cell drain control module. The drain control module is effectively a cell shutdown module. By engaging the switch S_1 , the electrochemical cell **7000** is shut down and the voltage is drained to zero or near zero. The drain control module can be used if other efforts to prevent or remediate a detected dendrite are unsuccessful.

[0335] A switch S_2 and resistors R_2 and R_3 control a dendrite prevention and remediation control module. The switch S_2 is closed to create a circuit with the resistor R_3 to activate the prevention control module. The switch S_2 is closed to create a circuit with the resistor R_2 to activate the remediation control module. The prevention control module pulls up the voltage of the interlayer **7060** to be more similar to the voltage of the cathode **7030**. This prevents dendrites

from forming or growing through the interlayer 7060. The remediation control module also pulls up the voltage of the interlayer 7060 to be closer to that of the cathode 7030. This shrinks and aids in eliminating dendrites that have already formed in the interlayer 7060. The resistance of the resistor R_2 is less than the resistance R_3 , so that more energy from the dendrites can flow through the resistor R_2 , as compared to prevention. In some embodiments, the resistor R_2 can have a resistance less than that of the resistor R_3 by a factor of about 1.1, about 1.2, about 1.3, about 1.4, about 1.5, about 1.6, about 1.7, about 1.8, about 1.9, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 20, about 30, about 40, about 50, about 60, about 70, about 80, about 90, or about 100, inclusive of all values and ranges therebetween.

[0336] A switch S_3 and resistor R_4 control a cell remediation pulldown control module. The controller 7005 receives information about voltage V_1 and V_2 and controls switches S_1 , S_2 , S_3 as well as an air transport switch.

[0337] The controller 7005 can operate via a BMS to manage the growth of dendrites. In some embodiments, the controller 7005 can include main functions and function decompositions. The main functions can include (1) providing communications, (2) providing current flow control, (3) providing safe operation, (4) providing dendrite control module, and/or (5) providing cell protection. The (1) providing communications function can include (a) providing calibration and/or (b) providing operating status of the electrochemical cell 7000. The (2) providing current flow control module can include (a) providing current flow between the cathode 7030 and the interlayer 7060, (b) providing current flow between the cathode 7030 and the anode 7010, and/or (c) providing no current flow in the electrochemical cell 7000. In some embodiments, the (3) providing safe operation module can include (a) providing dendrite remediation, (b) preventing a thermal event, and/or (c) providing a safe state. In some embodiments, the (4) providing dendrite control module can include (a) providing dendrite detection, (b) providing dendrite prevention, (c) providing dendrite remediation, and/or (d) providing voltage sense. In some embodiments, the (5) providing cell protection module can include (a) providing fault protection, and/or (b) providing data logging.

[0338] In use, the electrochemical cell can practice a protocol that includes detection, prevention, and remediation. Detection can include using a PI loop to measure the voltage difference between the total cell voltage (i.e., the voltage between the anode current collector 7020 and the cathode current collector 7040) and the interlayer voltage (i.e., the voltage between the interlayer 7060 and the anode current collector 7020). If the interlayer voltage decreases to less than a set threshold value, the potential of the interlayer 7060 is increased by electrically connecting the interlayer 7060 to the cathode current collector 7040 via the switch S_2 through the resistor R_3 . This takes advantage of the pullup control module.

[0339] Dendrite prevention can be exercised by applying a resistance to the interlayer 7060. This resistance can provide continuous excitation of the interlayer 7060, such that a dendrite cannot form across either of the separators 7050. This prevention method can be used as part of an

overall control strategy where the voltage potential, current, and resistance to the interlayer 7060 can be changed based on a control algorithm (e.g., a control algorithm used by the controller 7005).

[0340] The control system can act in an active prevention mode where the potential of the interlayer 7060 is modulated or changed to apply different potentials and increasing them (i.e., to be closer to the potential of the cathode 7030) or decreasing them (i.e., to be closer to the potential of the anode 7010) via the switch S_3 and the resistor R_4 to maintain the functioning of the electrochemical cell 7000. When the dendrite forms and interfaces with the interlayer 7060, the voltage of the interlayer 7060 can be pulled up toward the potential of the cathode 7030. The dendrite is dissolved or remediated, and the voltage potential of the interlayer 7060 returns to near the voltage potential of the cathode 7030 with respect to the anode 7010.

[0341] As shown, the cell drain control module closes the switch S_1 and runs current through the resistor R_1 to drain the electrochemical energy from the electrochemical cell 7000. This module can be used after the aforementioned dendrite remediation mechanisms fail. This can be enabled in extreme situations where the controller 7005 deems that a thermal event can be pending. The dendrite prevention pullup control module can be enabled to slowly pass a current potential to the interlayer 7060 to prevent dendrites from forming. The dendrite remediation pullup control mechanism can be enabled to more rapidly pass a current potential to the interlayer 7060 to raise the voltage of the interlayer 7060 to be closer to that of the cathode 7030. This can be used to remediate any dendrites the controller 7005 identifies during a detection phase. The cell remediation pulldown control module can be used during early development testing to lower the voltage of the interlayer 7060 to be more similar to that of the anode 7010.

[0342] In some embodiments, multiple versions or iterations of the electrochemical cell 7000 can be connected in series and/or in parallel. In some embodiments, the control of the electrochemical cell 7000 or multiple electrochemical cells connected in series and/or in parallel can be via a BMS. The types of switches and resistors employed in the electrochemical cell 7000 can be implemented in many ways in the detailed circuit design. The air transport switch can be used to drain the electrochemical cell 7000 to a desired SOC (e.g., about 5%, about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, or about 50% SOC, inclusive of all values and ranges therebetween) for air transport or any other shipping or movement of the electrochemical cell 7000. This can reduce the probability of dangerous electrical discharges

[0343] During operation, the controller 7005 can send various signals to the components of the electrochemical cell 7000. Table 1 shows example signals that can be sent from the controller 7005 to the components of the electrochemical cell 7000.

TABLE 1

Sample signal functions of controller 7005			
Signal	Function	Default State	Additional Function
Cell +/-	Cell positive and negative connections continuing to main BMS power circuits.	N/A	
SW1-DRAIN	Discrete signal from controller 7005 to enable/disable the cell drain function.	OFF	
SW2-PREVENT	Discrete signal from the controller 7005 to enable/disable the dendrite remediate function.	OFF	Pulled to ground in case control is lost.
SW2-REMEDiate	Discrete Signal from Controller to enable/disable dendrite remediate function.	OFF	Pulled to ground in case control is lost.
SW3-PU-REMEDiate	Discrete signal from controller to enable/disable the dendrite remediate function via pullup.	OFF	Pulled to ground in case control is lost.
SW3-PD-REMEDiate	Discrete signal from controller to enable/disable the dendrite remediate function via pulldown.	OFF	Pulled to ground in case control is lost.
CELL-V-SENSE	Analog signal to the controller 7005 to sense positive or negative cell stack voltage.	N/A	Used to determine differential voltage to the INTERLAYER(1, 2)-V-SENSE voltage used in detection function.
INTERLAYER(1, 2)-V-SENSE	Analog signal to the controller 7005 to sense the interlayer to negative voltage.	N/A	To be used to determine differential voltage to the CELL_V_SENSE voltage used in the detection function.
SW4-AIR-TRANSPORT	Discrete input from the user button to drain the electrochemical cell to about 15% SOC for air transport.	N/A	

[0344] In some embodiments, the controller 7005 can implement one or more software control methods. In some embodiments, the controller 7005 can implement a dendrite detection method. Using a software monitoring loop, a voltage difference is measured between the total cell voltage (i.e., the voltage between the anode 7010 and the cathode 7030) and the interlayer voltage (i.e., the voltage between the anode 7010 and the interlayer 7060). If the interlayer voltage decreases to less than a threshold value (e.g., about 1 V, about 1.5 V, about 2 V, about 2.5 V, about 3 V, about 4 V, etc.), the potential of the interlayer 7060 is increased by connecting the interlayer 7060 to the cathode 7030 (i.e., through the resistor R_2 or the resistor R_3).

[0345] In some embodiments, the controller 7005 can implement a prevention control method via software. In implementing the prevention control method, a resistance can be applied to the interlayer 7060. Specifically, the switch S_2 can close to switch on the resistor R_3 to provide current flow from the cathode 7030 to the interlayer 7060 to help prevent dendrite growth. The resistance provides a continuous excitation of the interlayer 7060, such that a dendrite would not be able to form across either of the separators 7050. The prevention method can be used as part of an overall control strategy, wherein the voltage potential, current, and resistance to the interlayer 7060 can be changed based on a software control algorithm.

[0346] In some embodiments, the controller 7005 can implement a remediation control method via software. In implementing the remediation control method, the controller 7005 can act in an active prevention mode, where the

potential of the interlayer 7060 is modulated or changed to apply different potentials. The potential of the interlayer 7060 can be increased (cathode side) or decreased (anode side) to maintain the cell function. When the dendrite forms and interfaces with the interlayer 7060, the voltage of the interlayer 7060 can be pulled up to the potential of the cathode 7030, with respect to the anode 7010. The dendrite is dissolved or remediated, and the voltage potential of the interlayer 7060 can return to be more similar to the voltage potential of the cathode 7030 with respect to the anode 7010.

[0347] The software implemented by the controller 7005 can include many operating modes. In some embodiments, the controller 7005 can implement an initialization mode, which executes once the controller 7005 has powered up. The controller 7005 can implement a monitoring mode.

[0348] The monitoring mode can act as a main loop, in which voltages are read and a dendrite detection algorithm is repeatedly executed. In some embodiments, the controller 7005 can implement a remediation mode, in which voltages are read and a remediation algorithm executes. In some embodiments, the controller 7005 can implement a draining mode. This discharges the cell voltage (i.e., the voltage between the anode 7010 and the cathode 7030) down to about 0 V to prevent a thermal event. This mode should be implemented to protect the safety and property of the user. In some embodiments, the controller 7005 can include a power down mode, implemented via a BMS master controller when the BMS is powering down. In some embodiments, the controller 7005 can implement a prevention mode, in which voltages are read and a prevention algorithm

executes. This mode can be executed at various intervals to allow a small potential of current to flow from the cathode 7030 to the interlayer 7060 to prevent formation of dendrites.

[0349] In some embodiments, the controller 7005 can implement a fault mode. The fault mode is a state entered when a fault is detected by the controller 7005. The electrochemical cell 7000 can operate in a safe state when fault mode is entered. The safe state can be defined by the control system functional design as determined by the system architectural design of the given device. Faults can be implemented in software control systems to protect the hardware of the electrochemical cell 7000 and the safety of the user. Examples of faults can include dendrite circuit overtemperature fault, overtemperature of the electrochemical cell 7000 fault, undertemperature of the electrochemical cell 7000 fault, drain cell active fault (in which the aforementioned drain functionality has been activated), dendrite remediate failed fault (in which the aforementioned remediation functionality has failed), dendrite prevention failed fault (in which the aforementioned prevention functionality has failed), interlayer voltage differential warning fault (in which the differential between the voltage of the interlayer 7060 and the anode 7010 has decreased to less than a threshold value), interlayer voltage differential critical fault (in which the differential between the voltage of the interlayer 7060 and the anode 7010 has decreased to less than a more critical threshold value), overall cell voltage overvoltage fault, and/or overall cell voltage undervoltage fault.

[0350] In some embodiments, signals from throughout the electrochemical cell 7000 can be transferred (e.g., via a controller area network (CAN) bus) to the controller 7005. In some embodiments, such signals can include overall cell voltage between the anode 7010 and the cathode 7030, maximum overall cell voltage during the current power cycle, minimum overall cell voltage during the current power cycle, dendrite detection counter, fault status, voltage difference between the anode 7010 and the interlayer 7060, maximum voltage difference between the anode 7010 and the interlayer 7060, minimum voltage difference between the anode 7010 and the interlayer 7060, drain switch status and override, prevent switch status and override, remediate switch status and override, and power down command.

[0351] In some embodiments, the software implemented by the controller 7005 can include a detection algorithm. Via the detection algorithm, the controller 7005 can use software to detect the difference between dendrite growth, a cell short circuit, and cell gas generation. Once detected, the controller 7005 can take appropriate reactionary steps to attempt to remediate and report the system status via CAN messaging.

[0352] In some embodiments, the software implemented by the controller 7005 can execute signal filtering. For voltage signal inputs, several potential software filter algorithm methods can be implemented. These can include a median filter, a moving average filter, and a low-pass filter. Such filter algorithms can also be used with other types of sensors. The proper filter can depend on factors such as signal dynamics, accuracy, and loss limit.

[0353] In some embodiments, the software implemented by the controller 7005 can include a dendrite control algorithm. For controlling the dendrite growth remediation, a closed loop control system can be used, implementing a proportional-integral-derivative (PID) algorithm, as diagrammed in FIG. 70B. The setpoint in a PID algorithm can

be a desired voltage differential calculated by subtracting the interlayer voltage from the cell voltage, for example, and dividing the difference by 2 (although other detection levels may be used). The process variable can be the actual real time voltage differential calculated by subtracting the software filtered interlayer voltage from the software filtered overall cell voltage and dividing the difference by 2. The output from the PID algorithm can be the control output that initiates the dendrite remediation action. In some embodiments, the control output can include a switch that applies a known resistance between the cathode 7030 and the interlayer 7060. The derivative time (T_d) can be determined during testing to tune the PID algorithm but can be tuned to be as small as possible. The proportional, integral, and derivative gains applied in the algorithm can also be determined during testing to tune the algorithm.

[0354] FIG. 71 is an illustration of an electrochemical cell 7100 with interlayers 7160a, 7160b (collectively referred to as interlayers 7160), according to an embodiment. As shown, the electrochemical cell 7100 includes a controller 7105, an anode 7110 disposed on an anode current collector 7120, a cathode 7130 disposed on a cathode current collector 7140, with a first separator 7150a, a second separator 7150b, and a third separator 7150c disposed between the anode 7110 and the cathode 7130. The interlayer 7160a is disposed between the first separator 7150a and the second separator 7150b and the interlayer 7160b is disposed between the second separator 7150b and the third separator 7150c. In some embodiments, the anode 7110, the anode current collector 7120, the cathode 7130, the cathode current collector 7140, the first separator 7150a, the second separator 7150b, the third separator 7150c, and the interlayers 7160 can be the same or substantially similar to the anode 410, the anode current collector 420, the cathode 430, the cathode current collector 440, the first separator 450a, the second separator 450b, the third separator 450c, and the interlayers 460, as described above with reference to FIG. 4. In some embodiments, the controller 7105 can be the same or substantially similar to the controller 7005, as described above with reference to FIG. 70. Thus, certain aspects of the controller 7105, the anode 7110, the anode current collector 7120, the cathode 7130, the cathode current collector 7140, the first separator 7150a, the second separator 7150b, the third separator 7150c, and the interlayers 7160 are not described in greater detail herein.

[0355] As shown, a voltage V_1 is measured between the anode current collector 7120 and the interlayer 7160a, a voltage V_2 is measured between the anode current collector 7120 and the interlayer 7160b, and a voltage V_3 is measured between the anode current collector 7120 and the cathode current collector 7140. As shown, a switch S_1 and a resistor R_1 control a cell drain control module. A switch S_2 and resistors R_2 and R_3 control a dendrite prevention and remediation control module. A switch S_3 and resistor R_4 control a cell remediation pulldown control module. The controller 7105 receives information about voltage V_1 and V_2 and controls switches S_1 , S_2 , S_3 as well as an air transport switch.

[0356] In some embodiments, the electrochemical cell 7100 and its controller 7105 can have any of the functionalities of the electrochemical cell 7000, and the controller 7005, as described above with reference to FIG. 70. A difference between these two cells is the presence of multiple interlayers 7160 in the electrochemical cell 7100. The controller 7105 can include functionalities to pullup or

pulldown the voltages of the interlayer **7160b**. As shown, the interlayer **7160a** is used for voltage and dendrite sensing and is not operably coupled to any switches. In some embodiments, the interlayer **7160a** can be operably coupled to one or more switches, such that the interlayer **7160a** can be subject to pullup and/or pulldown of voltage.

[0357] FIG. 72 shows interrelationships between a controller **7205** and related devices, according to an embodiment. As shown, the controller **7205** is in communication with a BMS **7201**, an electrochemical cell **7200**, and a surrounding environment. In some embodiments, the electrochemical cell **7200** can be the same or substantially similar to the electrochemical cell **7100**, as described above with reference to FIG. 71. The controller **7205** is in communication with the electrochemical cell **7200** via cell tabs (e.g., an anode tab and a cathode tab). Voltage sense information and dendrite information are sent from the electrochemical cell **7200** to the controller **7205**. The controller **7205** processes the received information and sends commands to various switches in the electrochemical cell **7200** via the BMS, based on the detection of dendrites. Additionally, the controller **7205** receives signals from the BMS **7201**. These signals can include voltage measurement, current measurement, or any other information pertinent to the cells. The controller **7205** receives information from the surrounding environment, including temperature information, humidity information, and/or incident force information. The controller **7205** can simultaneously transmit electromagnetic compatibility (EMC) information to the surrounding environment.

[0358] FIG. 73 is an illustration of an electrochemical cell **7300** with interlayers **7360a**, **7360b** (collectively referred to as interlayers **7360**), according to an embodiment. As shown, the electrochemical cell **7300** includes an anode **7310** disposed on an anode current collector **7320**, a cathode **7330** disposed on a cathode current collector **7340**, with a first separator **7350a** and a second separator **7350b** disposed between the anode **7310** and the cathode **7330**. An interlayer **7360a** is disposed between the first separator **7350a** and the second separator **7350b** and the interlayer **7360b** is disposed between the second separator **7350b** and the anode **7310**. In some embodiments, the anode **7310**, the anode current collector **7320**, the cathode **7330**, the cathode current collector **7340**, the first separator **7350a**, the second separator **7350b**, the third separator **7350c**, and the interlayers **7360** can be the same or substantially similar to the anode **410**, the anode current collector **420**, the cathode **430**, the cathode current collector **440**, the first separator **450a**, the second separator **450b**, the third separator **450c**, and the interlayers **460**, as described above with reference to FIG. 4.

[0359] As shown, a voltage V_1 is measured between the anode current collector **7320** and the interlayer **7360a**, a voltage V_2 is measured between the anode current collector **7320** and the interlayer **7360b**, and a voltage V_3 is measured between the anode current collector **7320** and the cathode current collector **7340**. A diode D_1 connects the interlayer **7360b** and the cathode current collector **7340**. A diode D_2 connects the interlayer **7360a** to the cathode current collector **7340**. Current can flow from the anode current collector **7320** to the cathode current collector **7340** via a switch **S** and a resistor **R**.

[0360] As shown, the interlayer **7360b** is in direct contact with the anode **7310**. In some embodiments, the interlayer **7360b** can be directly coupled to the anode **7310**. With this

direct coupling, the interlayer **7360b** is shorted from the beginning of the operation of the electrochemical cell **7300**. In some embodiments, the interlayer **7360b** can be welded and/or brazed to the anode **7310**. In some embodiments, the anode **7310** can include a lithium metal. In some embodiments, the anode **7310** can include a zero-lithium anode. In use, closing the switch **S** can create an external short circuit in response to signals from measuring the voltage V_1 . The interlayer **7360b** can be dedicated to its function as a coating but can be attached to the anode **7310**. In some embodiments, the interlayer **7360a** and/or the interlayer **7360b** can include carbon. In some embodiments, the interlayer **7360a** and/or the interlayer **7360b** can include allotropes of carbon including activated carbon, hard carbon, soft carbon, Ketjen, carbon black, graphitic carbon, carbon fibers, carbon micro-fibers, vapor-grown carbon fibers (VGCF), fullerenic carbons including “buckyballs”, carbon nanotubes (CNTs), multiwall carbon nanotubes (MWNTs), single wall carbon nanotubes (SWNTs), graphene, graphene sheets or aggregates of graphene sheets, and materials comprising fullerenic fragments, or any combination thereof.

[0361] In some embodiments, the interlayer **7360b** can include scarified additives to form an insulative and/or stabilized SEI layer. In some embodiments, the anode **7310** can be connected to an external circuit. The external circuit can have a high resistance sufficient to prevent a short circuit between the anode **7310** and the interlayer **7360b**. In some embodiments, the resistance of the external circuit can be variable. For example, the external circuit can have a first voltage during formation and/or initial charging of the electrochemical cell **7300** and a second voltage during discharge of the electrochemical cell **7300**. During formation, the resistance can be lower than during cycling in order to facilitate SEI growth. The resistance can then be increased during cycling to limit self-discharge. This can facilitate controlled deposition between the interlayer **7360b** and the anode **7310**, such that the deposited material does not penetrate the separator **7350b** or the separator **7350a**. In some embodiments, the resistance of the electrochemical cell **7300** and the components thereof can be tuned to control SEI formation with a desired speed and morphology. This control can be exercised by changing the external resistance to the interlayer **7360b** and effecting direct contact between the interlayer **7360b** and different hosts (e.g., carbon, metals) that alloy with lithium or polymers. In some embodiments, the electrochemical cell **7300** can include a SEI forming additive. In some embodiments, the SEI forming additives can be included in the electrolyte. In some embodiments, the SEI forming additive can include decomposition additives, such as a nitrate based salt (LiNO_3 , $\text{Mg}(\text{NO}_3)_2$, KNO_3), and/or a sulfite based salt. In some embodiments, the SEI forming additives can include a fluorine-rich solvent or additive, such as fluorinated ether. In some embodiments, the SEI forming additive can include a carbonate, and/or an acetate. In some embodiments, the SEI forming additive can include a morphology-controlled additive such as cesium bistriflimide, and/or LiASF_6 . In some embodiments, the SEI forming additive can include fluoroethylene carbonate (FEC) and 2-fluoropyridine (2-FP). In some embodiments, the SEI forming additives can include vinylene carbonate (VC), dimethyl sulfate (DMS), poly(sulfur-random-1,3-diisopropenylbenzene) (PSD), N,N -dimethylethanolamine (DMEA), trimethylsilyl(fluorosulfonyl) (n-nonafluorobutanesulfonyl) imide (TMS-FNFSI), tripropargyl phosphite

(TPP), and/or tris(2, 2,2-trifluoroethyl) borate (TTFEB). In some embodiments, the SEI forming additive can include organic salts (e.g., lithium bis(oxalato) borate (LiBOB), lithium difluoro (oxalate) borate (LiDFOB)).

[0362] FIG. 74 is an illustration of an electrochemical cell 7400 with an interlayer 7460, according to an embodiment. As shown, the electrochemical cell 7400 includes an anode 7410 disposed on an anode current collector 7420, a cathode 7430 disposed on a cathode current collector 7440, with a separator 7450 and a solid-state electrolyte (SSE) layer 7452 disposed between the anode 7410 and the cathode 7430. An interlayer 7460 is disposed between the separator 7350 and the SSE layer 7452. In some embodiments, the anode 7410, the anode current collector 7420, the cathode 7430, the cathode current collector 7440, the separator 7450, and the interlayer 7460 can be the same or substantially similar to the anode 210, the anode current collector 220, the cathode 230, the cathode current collector 240, the separator 250, and the interlayer 260, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode 7410, the anode current collector 7420, the cathode 7430, the cathode current collector 7440, the separator 7450, and the interlayer 7460 are not described in greater detail herein.

[0363] In some embodiments, the anode 7410 can include lithium metal. The interface between the SSE layer 7452 and a lithium metal anode delivers high coulombic efficiency without liquid electrolyte. Further, a dry interface (i.e., between a metal and a solid electrolyte layer) leads to good cycling and capacity retention. However, the dry interface alone does not necessarily stop lithium dendrite formation. Thus, the cathode 7430 includes a liquid electrolyte (i.e., a catholyte). In some embodiments, the cathode 7430 can include a semi-solid electrode. In some embodiments, the cathode 7430 can include a conventional solid electrode (i.e., with binder) with a liquid electrolyte. In some embodiments, the interlayer 7460 can include carbon and a solid-state electrolyte to act as a conductive layer. In some embodiments, the SSE layer 7452 can include sulfide. The combination of the SSE layer 7452 and the interlayer 7460 can prevent dendrites from forming. At the same time, liquid electrolyte can be prevented from entering the anode side of the electrochemical cell 7400 so the interface between the anode 7410 and the SSE layer 7452 remains dry. Also, this arrangement can prevent the liquid electrolyte from dissolving the SSE layer 7452 and the solid-state electrolyte in the interlayer 7460, particularly for sulfide-type solid-state electrolyte.

[0364] In production, carbon (e.g., activated carbon, hard carbon, soft carbon, Ketjen, carbon black, graphitic carbon, carbon fibers, carbon microfibers, VGCFs, fullerenic carbons, CNTs, MWNTs, SWNTs, graphene, graphene sheets or aggregates of graphene shetes, materials comprising fullerenic fragments, or any combination thereof), binder, are combined with a solid-state electrolyte to form the interlayer 7460. This creates a barrier with little or no porosity for preventing catholyte from migrating from the cathode 7430 to the anode 7410. This prevents dissolution of the SSE layer 7452. Alternatively, the separator 7450 can be coated with binder to produce the same effect. In some embodiments, the interlayer 7460 can have a porosity of less than about 5%, less than about 4%, less than about 3%, less than about 2%, less than about 1%, less than about 0.9%, less than about 0.8%, less than about 0.7%, less than about 0.6%, less than about 0.5%, less than about 0.4%, less than about 0.3%, less

than about 0.2%, less than about 0.1%, or about 0%. In some embodiments, the separator 7450 can include a ceramic coating with a polymer.

[0365] FIGS. 75A-75C are illustrations of an electrochemical cell 7500 with an interlayer 7560, according to an embodiment. As shown, the electrochemical cell 7500 includes an anode 7510 disposed on an anode current collector 7520, a cathode 7530 disposed on a cathode current collector 7540, with a first separator 7550a and a second separator 7550b disposed between the anode 7510 and the cathode 7530. The interlayer 7560 is disposed between the first separator 7550a and the second separator 7550b. In some embodiments, the anode 7510, the anode current collector 7520, the cathode 7530, the cathode current collector 7540, the first separator 7550a, the second separator 7550b, and the interlayer 7560 can be the same or substantially similar to the anode 210, the anode current collector 220, the cathode 230, the cathode current collector 240, the first separator 250a, the second separator 250b, and the interlayer 260, as described above with reference to FIGS. 2A-2B. Thus, certain aspects of the anode 7510, the anode current collector 7520, the cathode 7530, the cathode current collector 7540, the first separator 7550a, the second separator 7550b, and the interlayer 7560 are not described in greater detail herein.

[0366] As shown, the first separator 7550a and the interlayer 7560 extend beyond the outside edges of the second separator 7550b by an extension distance d. This allows for the interlayer 7560 to potentially contact the cathode 7530 if the cathode 7530 becomes misaligned from the anode 7510. This effectively blocks the cathode 7530 from contacting the anode 7510 and short circuiting. Rather, the cathode 7530 contacts the interlayer 7560 and short circuits. The extension of the interlayer 7560 acts as an edge detection point, and measuring the voltage between the cathode 7530 and the interlayer 7560 can detect such a short circuit (i.e., when the voltage difference is decreased to less than a threshold value). The electrochemical cell 7500 can subsequently be disabled in order to prevent a more catastrophic short circuit event.

[0367] In some embodiments, the extension distance d can be at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 1.1 mm, at least about 1.2 mm, at least about 1.3 mm, at least about 1.4 mm, at least about 1.5 mm, at least about 1.6 mm, at least about 1.7 mm, at least about 1.8 mm, or at least about 1.9 mm. In some embodiments, the extension distance d can be no more than about 2 mm, no more than about 1.9 mm, no more than about 1.8 mm, no more than about 1.7 mm, no more than about 1.6 mm, no more than about 1.5 mm, no more than about 1.4 mm, no more than about 1.3 mm, no more than about 1.2 mm, no more than about 1.1 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , or no more than about 200 μm . Combinations of the above-referenced extension distances d are also possible (e.g., at least about 100 μm and no more than about 2 mm or at least about 300 μm and no more than about 1 mm), inclusive of all values and ranges therebetween. In some embodiments, the extension distance d can be about 100 μm , about 200 μm , about

300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.1 mm, about 1.2 mm, about 1.3 mm, about 1.4 mm, about 1.5 mm, about 1.6 mm, about 1.7 mm, about 1.8 mm, about 1.9 mm, or about 2 mm.

[0368] FIG. 75A shows the electrochemical cell 7500 in a condition where the anode 7510 is aligned properly with the cathode 7530. In FIG. 75B, the cathode 7530 has become misaligned with the anode 7510. In FIG. 75C, the first separator 7550a and the interlayer 7560 contact the cathode 7530 to prevent short-circuiting between the anode 7510 and the cathode 7530. As shown, the first separator 7550a and the interlayer 7560 are bent to contact the cathode 7530. In some embodiments, the cathode 7530 can bend to contact the interlayer 7560. In some embodiments, the cathode 7530, the first separator 7550a, and the interlayer 7560 can each bend to contact each other. As shown, the first separator 7550a and the interlayer 7560 are wider than the second separator 7550b, such that the interlayer 7560 can contact the cathode 7530. In some embodiments, the second separator 7550b and the interlayer 7560 can be longer than the first separator 7550a, such that the interlayer 7560 can contact the anode 7510. As shown, the first separator 7550a and the interlayer 7560 extend beyond the second separator 7550b on all sides. In some embodiments, the first separator 7550a and the interlayer 7560 can extend beyond the second separator 7550b on one side, two side, three sides, or any number of sides. As shown, the edge of the interlayer 7560 is approximately flush with the edge of the first separator 7550a. In some embodiments, the edge of the first separator 7550a can extend beyond the edge of the interlayer 7560 (e.g., by a distance of about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.1 mm, about 1.2 mm, about 1.3 mm, about 1.4 mm, about 1.5 mm, about 1.6 mm, about 1.7 mm, about 1.8 mm, about 1.9 mm, or about 2 mm, inclusive of all values and ranges therebetween).

[0369] FIGS. 76A-76C are illustrations of an electrochemical cell 7600 with interlayers 7660a, 7660b (collectively referred to as interlayers 7660). As shown, the electrochemical cell 7600 operates as a bicell with anodes 7610a, 7610b (collectively referred to as anodes 7610) disposed on an anode current collector 7610, cathodes 7630a, 7630b (collectively referred to as cathodes 7630) disposed on cathode current collectors 7640a, 7640b (collectively referred to as cathode current collectors 7640), separators 7650a, 7650b disposed between the anode 7610a and the cathode 7630a, the interlayer 7660a disposed between the separators 7650a, 7650b, and the interlayer 7660b disposed between separators 7650c, 7650d (separators 7650a, 7650b, 7650c, 7650d collectively referred to as separators 7650). In some embodiments, the anodes 7610, the anode current collector 7620, the cathodes 7630, the cathode current collectors 7640, the separators 7650, and the interlayers 7660 can be the same or substantially similar to anodes 7510, the anode current collector 7520, the cathodes 7530, the cathode current collectors 7540, the separators 7550, and the interlayers 7560, as described above with reference to FIGS. 75A-75C. Thus, certain aspects of the anodes 7610, the anode current collector 7620, the cathodes 7630, the cathode current collectors 7640, the separators 7650, and the interlayers 7660 are not described in greater detail herein.

[0370] As shown, the interlayer 7660a and the separator 7650b extend beyond the separator 7650a by an extension distance d1. As shown, the interlayer 7660b and the separator 7650c extend beyond the separator 7650d by an extension distance d2. In some embodiments, the extension distances d1 and/or d2 can be at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 1.1 mm, at least about 1.2 mm, at least about 1.3 mm, at least about 1.4 mm, at least about 1.5 mm, at least about 1.6 mm, at least about 1.7 mm, at least about 1.8 mm, or at least about 1.9 mm. In some embodiments, the extension distances d1 and/or d2 can be no more than about 2 mm, no more than about 1.9 mm, no more than about 1.8 mm, no more than about 1.7 mm, no more than about 1.6 mm, no more than about 1.5 mm, no more than about 1.4 mm, no more than about 1.3 mm, no more than about 1.2 mm, no more than about 1.1 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , or no more than about 200 μm . Combinations of the above-referenced extension distances d1 and/or d2 are also possible (e.g., at least about 100 μm and no more than about 2 mm or at least about 300 μm and no more than about 1 mm), inclusive of all values and ranges therebetween. In some embodiments, the extension distances d1 and/or d2 can be about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.1 mm, about 1.2 mm, about 1.3 mm, about 1.4 mm, about 1.5 mm, about 1.6 mm, about 1.7 mm, about 1.8 mm, about 1.9 mm, or about 2 mm.

[0371] FIG. 76A shows the cathodes 7630 properly aligned with the anodes 7610. FIG. 76B show the cathodes 7630 misaligned with the anodes 7610. In FIG. 76C, the interlayer 7660a and the separator 7650b are bent, such that the cathode 7630a contacts the interlayer 7660a, while the interlayer 7660b and the separator 7650c are bent, such that the cathode 7630b contacts the interlayer 7660b. In some embodiments, the anodes 7610 and the cathodes 7630 can have the opposite orientation to how they are depicted in FIGS. 76A-76C. In other words, the electrochemical cell 7600 can be built out from a central cathode current collector 7640 with anodes 7610 on the outside.

[0372] FIGS. 77A-77C are illustrations of an electrochemical cell 7700 with multiple interlayers, according to an embodiment. As shown, the electrochemical cell 7700 includes an anode 7710 disposed on an anode current collector 7720, a cathode 7730 disposed on a cathode current collector 7740, with a first separator 7750a, a second separator 7750b, and a third separator 7750c (collectively referred to as separators 7750) disposed between the anode 7710 and the cathode 7730. A first interlayer 7760a is disposed between the first separator 7750a and the second separator 7750b. A second interlayer 7760b is disposed between the second separator 7750b and the third separator 7750c. In some embodiments, the anode 7710, the anode current collector 7720, the cathode 7730, the cathode current collector 7740, the first separator 7750a, the second separator 7750b, the third separator 7750c, the first interlayer 7760a, and the second interlayer 7760b can be the same or substantially similar to the anode 410, the anode current

collector **420**, the cathode **430**, the cathode current collector **440**, the first separator **450a**, the second separator **450b**, the third separator **450c**, the first interlayer **460a**, and the second interlayer **460b**. Thus, certain aspects of the anode **7710**, the anode current collector **7720**, the cathode **7730**, the cathode current collector **7740**, the first separator **7750a**, the second separator **7750b**, the third separator **7750c**, the first interlayer **7760a**, and the second interlayer **7760b** are not described in greater detail herein.

[0373] As shown, the interlayer **7760b** extends beyond the separator **7750c** by an extension distance *d*. In some embodiments, the extension distance *d* can be at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 1.1 mm, at least about 1.2 mm, at least about 1.3 mm, at least about 1.4 mm, at least about 1.5 mm, at least about 1.6 mm, at least about 1.7 mm, at least about 1.8 mm, or at least about 1.9 mm. In some embodiments, the extension distance *d* can be no more than about 2 mm, no more than about 1.9 mm, no more than about 1.8 mm, no more than about 1.7 mm, no more than about 1.6 mm, no more than about 1.5 mm, no more than about 1.4 mm, no more than about 1.3 mm, no more than about 1.2 mm, no more than about 1.1 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , or no more than about 200 μm . Combinations of the above-referenced extension distances *d* are also possible (e.g., at least about 100 μm and no more than about 2 mm or at least about 300 μm and no more than about 1 mm), inclusive of all values and ranges therebetween. In some embodiments, the extension distance *d* can be about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.1 mm, about 1.2 mm, about 1.3 mm, about 1.4 mm, about 1.5 mm, about 1.6 mm, about 1.7 mm, about 1.8 mm, about 1.9 mm, or about 2 mm.

[0374] In FIG. 77A, the cathode **7730** is in proper alignment with the anode **7710**. In FIG. 77B, the cathode **7730** has shifted and is misaligned with the anode **7710**. In FIG. 77C, the interlayer **7760b** has bent, such that the separator **7760b** contacts the cathode **7730**. As shown, the interlayer **7760a** and the separator **7750a** do not extend beyond the separator **7750b**. This configuration provides a short circuit opportunity for the interlayer **7760b** and the cathode **7730** while preventing a short circuit opportunity between the interlayer **7760a** and the cathode **7730**. As shown, the electrochemical cell **7700** is configured such that the interlayer **7760b** short circuits with the cathode **7730**. In some embodiments, the electrochemical cell **7700** can be configured such that the interlayer **7760a** short circuits with the anode **7710**. In other words, the interlayer **7760a** can extend beyond an outside edge of the separator **7750a**.

[0375] FIGS. 78A-78C are illustrations of an electrochemical cell **7800** with multiple interlayers, according to an embodiment. As shown, the electrochemical cell **7800** includes an anode **7810** disposed on an anode current collector **7820**, a cathode **7830** disposed on a cathode current collector **7840**, with a first separator **7850a**, a second separator **7850b**, and a third separator **7850c** (collectively referred to as separators **7850**) disposed between the anode

7810 and the cathode **7830**. A first interlayer **7860a** is disposed between the first separator **7850a** and the second separator **7850b**. A second interlayer **7860b** is disposed between the second separator **7850b** and the third separator **7850c**. In some embodiments, the anode **7810**, the anode current collector **7820**, the cathode **7830**, the cathode current collector **7840**, the first separator **7850a**, the second separator **7850b**, the third separator **7850c**, the first interlayer **7860a**, and the second interlayer **7860b** can be the same or substantially similar to the anode **7710**, the anode current collector **7720**, the cathode **7730**, the cathode current collector **7740**, the first separator **7750a**, the second separator **7750b**, the third separator **7750c**, the first interlayer **7760a**, and the second interlayer **7760b**. Thus, certain aspects of the anode **7810**, the anode current collector **7820**, the cathode **7830**, the cathode current collector **7840**, the first separator **7850a**, the second separator **7850b**, the third separator **7850c**, the first interlayer **7860a**, and the second interlayer **7860b** are not described in greater detail herein.

[0376] As shown, the interlayer **7860a** and the separator **7850a** extend beyond the separator **7850b** by an extension distance *d1*. As shown, the interlayer **7860b** and the separator **7850b** extend beyond the separator **7850c** by an extension distance *d2*. In some embodiments, the extension distances *d1* and/or *d2* can be at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 1.1 mm, at least about 1.2 mm, at least about 1.3 mm, at least about 1.4 mm, at least about 1.5 mm, at least about 1.6 mm, at least about 1.7 mm, at least about 1.8 mm, or at least about 1.9 mm. In some embodiments, the extension distances *d1* and/or *d2* can be no more than about 2 mm, no more than about 1.9 mm, no more than about 1.8 mm, no more than about 1.7 mm, no more than about 1.6 mm, no more than about 1.5 mm, no more than about 1.4 mm, no more than about 1.3 mm, no more than about 1.2 mm, no more than about 1.1 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , or no more than about 200 μm . Combinations of the above-referenced extension distances *d1* and/or *d2* are also possible (e.g., at least about 100 μm and no more than about 2 mm or at least about 300 μm and no more than about 1 mm), inclusive of all values and ranges therebetween. In some embodiments, the extension distances *d1* and/or *d2* can be about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.1 mm, about 1.2 mm, about 1.3 mm, about 1.4 mm, about 1.5 mm, about 1.6 mm, about 1.7 mm, about 1.8 mm, about 1.9 mm, or about 2 mm.

[0377] In FIG. 78A, the cathode **7830** is in proper alignment with the anode **7810**. In FIG. 78B, the cathode **7830** has shifted and is misaligned with the anode **7810**. In FIG. 78C, the separator **7850a**, the interlayer **7860a**, the separator **7850b**, and the interlayer **7860b** have bent, such that the interlayer **7860a** and the interlayer **7860b** contact the cathode **7830**. This configuration provides a short circuit opportunity between the interlayer **7860b** and the cathode **7830** and between the interlayer **7860a** and the cathode **7830**. As shown, the electrochemical cell **7700** is configured such that the interlayer **7860a** and the interlayer **7860b** short circuit

with the cathode **7830**. In some embodiments, the electrochemical cell **7800** can be configured such that the interlayer **7860a** and the interlayer **7860b** short circuit with the cathode **7810**. In other words, the interlayer **7860a** can extend beyond an outside edge of the separator **7850a** and the interlayer **7860b** can extend beyond the outside edge of the separator **7850b**.

[0378] FIGS. **79A-79B** of an electrochemical cell with an interlayer **7960**, according to an embodiment. As shown, the electrochemical cell **7900** includes an anode **7910** disposed on an anode current collector **7920**, a cathode **7930** disposed on a cathode current collector **7940**, with a first separator **7950a** and a second separator **7950b** disposed between the anode **7910** and the cathode **7930**. The interlayer **7960** is disposed between the first separator **7950a** and the second separator **7950b**. As shown, the anode current collector **7920** includes an anode tab **7922** and the cathode current collector includes a cathode tab **7942**. As shown, the electrochemical cell **7900** includes a pouch **7970** formed from films **7970a**, **7970b**, in which each of the other components of the electrochemical cell **7900** are disposed. In some embodiments, the anode **7910**, the anode current collector **7920**, the cathode **7930**, the cathode current collector **7940**, the first separator **7950a**, the second separator **7950b**, and the interlayer **7960** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. Thus, certain aspects of the anode **7910**, the anode current collector **7920**, the cathode **7930**, the cathode current collector **7940**, the first separator **7950a**, the second separator **7950b**, and the interlayer **7960** are not described in greater detail herein.

[0379] FIG. **79A** shows a cross-sectional view of the electrochemical cell **7900**, while FIG. **79B** shows a partially exploded view of the electrochemical cell **7900** with the anode **7910** and associated components separated from the cathode **7930** and associated components. As shown, the interlayer **7960** is coated onto the first separator **7950a**, with the interlayer **7960** and the first separator **7950a** extending beyond the second separator **7950b**. In some embodiments, the interlayer **7960** can include a carbon coating (e.g., CNT, CNF, carbon black, and/or graphene). As shown, the second separator **7950b** extends beyond the anode **7910** by a distance d_a . As shown, the first separator **7950a** and the interlayer **7960** extend beyond the cathode **7930** by a distance d_b . The distance d_b is larger than the distance d_a . This allows for an exposed region of the interlayer **7960** to act as a tab, such that a voltage measurement can be made from the exposed region of the interlayer **7960** without including a tab. As shown, the interlayer has an exposed interlayer length e_{il} .

[0380] In some embodiments, the distance d_a can be at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 2 mm, at least about 3 mm, at least about 4 mm, at least about 5 mm, at least about 6 mm, at least about 7 mm, at least about 8 mm, or at least about 9 mm. In some embodiments, the distance d_a can be no more than about 1 cm, no more than about 9 mm, no more than about 8 mm, no more than about 7 mm, no more than about 6 mm, no more than about

5 mm, no more than about 4 mm, no more than about 3 mm, no more than about 2 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , or no more than about 200 μm . Combinations of the above-referenced distances d_a are also possible (e.g., at least about 100 μm and no more than about 1 cm or at least about 500 μm and no more than about 5 mm), inclusive of all values and ranges therebetween. In some embodiments, the distance d_a can be about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, or about 1 cm.

[0381] In some embodiments, the distance d_c can be at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 2 mm, at least about 3 mm, at least about 4 mm, at least about 5 mm, at least about 6 mm, at least about 7 mm, at least about 8 mm, at least about 9 mm, at least about 1 cm, at least about 2 cm, at least about 3 cm, or at least about 4 cm. In some embodiments, the distance d_c can be no more than about 5 cm, no more than about 4 cm, no more than about 3 cm, no more than about 2 cm, no more than about 1 cm, no more than about 9 mm, no more than about 8 mm, no more than about 7 mm, no more than about 6 mm, no more than about 5 mm, no more than about 4 mm, no more than about 3 mm, no more than about 2 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , or no more than about 600 μm . Combinations of the above-referenced distances d_c are also possible (e.g., at least about 500 μm and no more than about 5 cm or at least about 5 mm and no more than about 3 cm), inclusive of all values and ranges therebetween. In some embodiments, the distance d_c can be about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, about 1 cm, about 2 cm, about 3 cm, about 4 cm, or about 5 cm.

[0382] In some embodiments, the exposed interlayer length e_{il} can be at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 2 mm, at least about 3 mm, at least about 4 mm, at least about 5 mm, at least about 6 mm, at least about 7 mm, at least about 8 mm, at least about 9 mm, at least about 1 cm, at least about 2 cm, or at least about 3 cm. In some embodiments, the exposed interlayer length e_{il} can be no more than about 4 cm, no more than about 3 cm, no more than about 2 cm, no more than about 1 cm, no more than about 9 mm, no more than about 8 mm, no more than about 7 mm, no more than about 6 mm, no more than about 5 mm, no more than about 4 mm, no more than about 3 mm, no more than about 2 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , or no more than about 200 μm . Combinations of the above-referenced exposed interlayer lengths e_{il} are also possible (e.g., at least about 100 μm and

no more than about 4 cm or at least about 1 mm and no more than about 1 cm), inclusive of all values and ranges therebetween. In some embodiments, the exposed interlayer length can be about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, about 1 cm, about 2 cm, about 3 cm, or about 4 cm.

[0383] FIG. 79B shows a partially exploded view of the electrochemical cell 7900. The first separator 7950a, the second separator 7950b, and the interlayer 7960 are shown partially transparent with the interlayer 7960 disposed on the first separator 7950a. The pouch 7970 is shown separate from the other components. In some embodiments, the pouch 7970 can be formed from sheets of laminate film. As shown, the pouch 7970 includes holes 7971a, 7971b, 7971c (collectively referred to as holes 7971). In some embodiments, the holes 7971 can be pre-punched into the pouch 7970. The hole 7971b can allow for contact with the interlayer 7960, as the interlayer 7960 protrudes beyond the second separator 7950b. In some embodiments, the film of the pouch 7970 can be cut after the rest of the electrochemical cell 7900 is assembled. This can allow construction of the electrochemical cell 7900 without any unnecessary cutting of the current collectors. Each of the components shown in FIG. 79B can be brought together to form the electrochemical cell 7900.

[0384] As shown in FIGS. 79A-79B, the first separator 7950a coupled to the cathode 7930 and the interlayer 7960 are both longer than the second separator 7950b. In some embodiments, the second separator 7950b and the interlayer 7960 can be longer than the first separator 7950a. In other words, the positioning of the anode 7910 and the cathode 7930 can be reversed from how they are depicted.

[0385] FIG. 80 is an illustration of an electrochemical cell stack 80000, according to an embodiment. Components shown in FIG. 80 include a common anode tab 80220, a common cathode tab 80420, a common interlayer tab 80630, a thin film resistor 80635, and a casing 8085. For ease of viewing, certain items are excluded from the depiction in FIG. 80, including anodes, anode current collectors, cathode current collectors, anode tabs, cathode tabs, separators, and interlayers. The common anode tab 80220 is electrically coupled to a collection of anode tabs. The common cathode tab 80420 is electrically coupled to a collection of cathode tabs. The thin film resistor 80635 electrically couples the common interlayer tab 80630 and the common cathode tab 80420. The thin film resistor 80635 electrically couples the cathodes of the electrochemical cell stack 80000 and the interlayers of the electrochemical cell stack 80000. In some embodiments, the thin film resistor 8063 electrically couples the common anode tab 80220 and the common interlayer tab 80630. The thin film resistor 80635 can be directly connected between the Cathode and the interlayer allowing for direct pullup. This can be used in systems where BMS modification is not implemented but where protection is desired to prevent dendrites from crossing the separator. Additionally this could be built such that each individual cell has a separate connection from the cathode to the interlayer, allowing the electrochemical cell 80000 to be safe from dendrites from time of manufacture with build in protection. As shown, the resistor would have the same function as in FIG. 3A. It would be possible to implement any of the

aforementioned circuit combinations or other implementations for the connection of the interlayer.

[0386] In some embodiments, the thin film resistor 80635 can have a resistance of at least about 1 Ω , at least about 2 Ω , at least about 3 Ω , at least about 4 Ω , at least about 5 Ω , at least about 6 Ω , at least about 7 Ω , at least about 8 Ω , at least about 9 Ω , at least about 10 Ω , at least about 20 Ω , at least about 30 Ω , at least about 40 Ω , at least about 50 Ω , at least about 60 Ω , at least about 70 Ω , at least about 80 Ω , at least about 90 Ω , at least about 100 Ω , at least about 200 Ω , at least about 300 Ω , at least about 400 Ω , at least about 500 Ω , at least about 600 Ω , at least about 700 Ω , at least about 800 Ω , or at least about 900 Ω . In some embodiments, the thin film resistor 80635 can have a resistance of no more than about 1,000 Ω , no more than about 900 Ω , no more than about 800 Ω , no more than about 700 Ω , no more than about 600 Ω , no more than about 500 Ω , no more than about 400 Ω , no more than about 300 Ω , no more than about 200 Ω , no more than about 100 Ω , no more than about 90 Ω , no more than about 80 Ω , no more than about 70 Ω , no more than about 60 Ω , no more than about 50 Ω , no more than about 40 Ω , no more than about 30 Ω , no more than about 20 Ω , no more than about 10 Ω , no more than about 9 Ω , no more than about 8 Ω , no more than about 7 Ω , no more than about 6 Ω , no more than about 5 Ω , no more than about 4 Ω , no more than about 3 Ω , or no more than about 2 Ω . Combinations of the above-referenced resistances are also possible (e.g., at least about 1 Ω and no more than about 1,000 Ω or at least about 5 (2 and no more than about 500 (2), inclusive of all values and ranges therebetween. In some embodiments, the thin film resistor 80635 can have a resistance of about 1 Ω , about 2 Ω , about 3 Ω , about 4 Ω , about 5 Ω , about 6 Ω , about 7 Ω , about 8 Ω , about 9 Ω , about 10 Ω , about 20 Ω , about 30 Ω , about 40 Ω , about 50 Ω , about 60 Ω , about 70 Ω , about 80 Ω , about 90 Ω , about 100 Ω , about 200 Ω , about 300 Ω , about 400 Ω , about 500 Ω , about 600 Ω , about 700 Ω , about 800 Ω , about 900 Ω , or about 1,000.

[0387] FIGS. 81A-81B are illustrations of a collection of electrochemical cell stacks 81000a, 81000b, 81000c (collectively referred to as electrochemical cell stacks 81000), according to an embodiment. FIG. 81A shows an overhead view of the electrochemical cell stacks 81000, while FIG. 81B shows a side profile view of the electrochemical cell stacks 81000. As shown, the electrochemical cell stacks 81000 are arranged as bicells. As shown in FIG. 81A, two common anode tabs 81220a, 81220b extend from the electrochemical cell stack 81000a. A common cathode tab 81420a and a common interlayer tab 81630a extend from the electrochemical cell stack 81000a. A thin film resistor 81635a electrically couples the common cathode tab 81420a and the common interlayer tab 81630a. A casing 81480a is shown, which houses the components of the electrochemical cells. For ease of viewing, most of the components of the electrochemical cells are not shown, including anodes, anode current collectors, cathode current collectors, anode tabs, cathode tabs, separators, and interlayers.

[0388] FIG. 81B shows a side view of each of the electrochemical cell stacks 81000a, 81000b, 81000c. As shown, the electrochemical cell stack 81000a includes anode tabs 8122a-i, 8122a-ii, 8122a-iii, 8122b-i, 8122b-ii, 8122b-iii (collectively referred to as anode tabs 8122a, b), cathode tabs 8142a-i, 8142a-ii, 8142a-iii (collectively referred to as cathode tabs 8142a), interlayer tabs 8163a-i, 8163a-ii,

8163a-iii (collectively referred to as interlayer tabs **8163a**), and casing **81480a**, disposed in a pre-assembled socket **81005a**. As shown, the electrochemical cell stack **81000b** includes anode tabs **8122c-i**, **8122c-ii**, **8122c-iii**, **8122d-i**, **8122d-ii**, **8122d-iii** (collectively referred to as anode tabs **8122c, d**), cathode tabs **8142b-i**, **8142b-ii**, **8142b-iii** (collectively referred to as cathode tabs **8142b**), interlayer tabs **8163b-i**, **8163b-ii**, **8163b-iii** (collectively referred to as interlayer tabs **8163b**), and casing **81480b**, disposed in a pre-assembled socket **81005b**. As shown, the electrochemical cell stack **81000c** includes anode tabs **8122e-i**, **8122e-ii**, **8122e-iii**, **8122f-i**, **8122f-ii**, **8122f-iii** (collectively referred to as anode tabs **8122e, f**), cathode tabs **8142c-i**, **8142c-ii**, **8142c-iii** (collectively referred to as cathode tabs **8142c**), interlayer tabs **8163c-i**, **8163c-ii**, **8163c-iii** (collectively referred to as interlayer tabs **8163c**), and casing **81480c**, disposed in a pre-assembled socket **81005c**.

[0389] In some embodiments, the interlayer tabs **8163a**, **8163b**, **8163c** (collectively referred to as interlayer tabs **8163**) can be composed of aluminum (e.g., when electrically coupled to cathode tabs **8142**). In some embodiments, the interlayer tabs **8163** can be composed of copper and/or nickel (e.g., when electrically coupled to anode tabs **8122**). In some embodiments, the anode current collectors (not shown) can be coated on a single side. In some embodiments, the anode current collectors can be coated on both sides. In some embodiments, the cathode current collectors (not shown) can be coated on a single side. In some embodiments, the cathode current collectors can be coated on both sides. As shown, the collection of electrochemical cell stacks **81000a**, **81000b**, **81000c** includes 3 cells arranged in parallel, with each parallel arrangement connected in series. In some embodiments, any number of electrochemical cells can be connected in series and/or parallel (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, inclusive of all values and ranges therebetween).

[0390] FIGS. **82A-82B** are illustrations of a collection of electrochemical cell stacks **82000a**, **82000b**, **82000c** (collectively referred to as electrochemical cell stacks **82000**), according to an embodiment. FIG. **82A** shows an overhead view of the electrochemical cell stacks **82000**, while FIG. **82B** shows a side profile view of the electrochemical cell stacks **82000**. As shown, the electrochemical cell stacks **82000** are arranged as single cells. As shown in FIG. **82A**, one common anode tab **8222a** extends from the electrochemical cell stack **82000a**. A common cathode tab **8242a** and a common interlayer tab **8263a** extend from the electrochemical cell stack **82000a**. A thin film resistor **82635a** electrically couples the common cathode tab **8242a** and the common interlayer tab **8263a**. A casing **82480a** is shown, which houses the components of the electrochemical cells. For ease of viewing, most of the components of the electrochemical cells are not shown, including anodes, anode current collectors, cathode current collectors, anode tabs, cathode tabs, separators, and interlayers.

[0391] FIG. **82B** shows a side view of each of the electrochemical cell stacks **82000a**, **82000b**, **82000c**. As shown, the electrochemical cell stack **82000a** includes anode tabs **8222a-i**, **8222a-ii**, **8222a-iii** (collectively referred to as anode tabs **8222a**), cathode tabs **8242a-i**, **8242a-ii**, **8242a-iii** (collectively referred to as cathode tabs **8242a**), interlayer tabs **8263a-i**, **8263a-ii**, **8263a-iii** (collectively referred to as interlayer tabs **8263a**), and casing **82480a**, disposed in a pre-assembled socket **82005a**. As shown, the electrochemi-

cal cell stack **82000b** includes anode tabs **8222b-i**, **8222b-ii**, **8222b-iii** (collectively referred to as anode tabs **8222b**), cathode tabs **8242b-i**, **8242b-ii**, **8242b-iii** (collectively referred to as cathode tabs **8242b**), interlayer tabs **8263b-i**, **8263b-ii**, **8263b-iii** (collectively referred to as interlayer tabs **8263b**), and casing **82480b**, disposed in a pre-assembled socket **82005b**. As shown, the electrochemical cell stack **82000c** includes anode tabs **8222e-i**, **8222e-ii**, **8222e-iii**, **8222f-i**, **8222f-ii**, **8222f-iii** (collectively referred to as anode tabs **8222e, f**), cathode tabs **8242c-i**, **8242c-ii**, **8242c-iii** (collectively referred to as cathode tabs **8242c**), interlayer tabs **8263c-i**, **8263c-ii**, **8263c-iii** (collectively referred to as interlayer tabs **8263c**), and casing **82480c**, disposed in a pre-assembled socket **82005c**.

[0392] In some embodiments, the interlayer tabs **8263a**, **8263b**, **8263c** (collectively referred to as interlayer tabs **8263**) can be composed of aluminum (e.g., when electrically coupled to cathode tabs **8242**). In some embodiments, the interlayer tabs **8263** can be composed of copper and/or nickel (e.g., when electrically coupled to anode tabs **8222**). In some embodiments, the anode current collectors (not shown) can be coated on a single side. In some embodiments, the anode current collectors can be coated on both sides. In some embodiments, the cathode current collectors (not shown) can be coated on a single side. In some embodiments, the cathode current collectors can be coated on both sides. As shown, the collection of electrochemical cell stacks **82000a**, **82000b**, **82000c** includes 3 cells arranged in parallel, with each parallel arrangement connected in series. In some embodiments, any number of electrochemical cells can be connected in series and/or parallel (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, inclusive of all values and ranges therebetween).

[0393] FIGS. **83A-83B** are illustrations of a collection of electrochemical cell stacks **83000a**, **83000b** (collectively referred to as electrochemical cell stacks **83000**), according to an embodiment. FIG. **83A** shows an overhead view of the electrochemical cell stacks **83000**, while FIG. **83B** shows a side profile view of the electrochemical cell stacks **83000**. As shown, the electrochemical cell stacks **83000** are arranged as single cells. As shown in FIG. **83A**, two common anode tabs **8322a**, **8322b** extend from the electrochemical cell stack **83000a** in opposite directions. Two common cathode tabs **8342a**, **8342b** and two common interlayer tabs **8363a**, **8363b** extend from the electrochemical cell stack **83000a**. A thin film resistor **83635a** electrically couples the common cathode tab **8342a** and the common interlayer tab **8363a**. A thin film resistor **83635b** electrically couples the common cathode tab **8342b** and the common interlayer tab **8363b**. A casing **83480a** is shown, which houses the components of the electrochemical cells. For ease of viewing, most of the components of the electrochemical cells are not shown, including anodes, anode current collectors, cathode current collectors, anode tabs, cathode tabs, separators, and interlayers.

[0394] FIG. **83B** shows a side view of each of the electrochemical cell stacks **83000a**, **83000b**. As shown, the electrochemical cell stack **83000a** includes anode tabs **8322a-i**, **8322a-ii**, **8322a-iii** (collectively referred to as anode tabs **8322a**), cathode tabs **8342a-i**, **8342a-ii**, **8342a-iii** (collectively referred to as cathode tabs **8342a**), interlayer tabs **8363a-i**, **8363a-ii**, **8363a-iii** (collectively referred to as interlayer tabs **8363a**), and casing **83480a**, disposed in a pre-assembled socket **83005a**. As shown, the electrochemi-

cal cell stack **83000b** includes anode tabs **8322b-i**, **8322b-ii**, **8322b-iii**, (collectively referred to as anode tabs **8322b**), cathode tabs **8342b-i**, **8342b-ii**, **8342b-iii** (collectively referred to as cathode tabs **8342b**), interlayer tabs **8363b-i**, **8363b-ii**, **8363b-iii** (collectively referred to as interlayer tabs **8363b**), and casing **83480b**, disposed in a pre-assembled socket **83005b**.

[0395] In some embodiments, the interlayer tabs **8363a**, **8363b** (collectively referred to as interlayer tabs **8363**) can be composed of aluminum (e.g., when electrically coupled to cathode tabs **8342**). In some embodiments, the interlayer tabs **8363** can be composed of copper and/or nickel (e.g., when electrically coupled to anode tabs **8322**). In some embodiments, the anode current collectors (not shown) can be coated on a single side. In some embodiments, the anode current collectors can be coated on both sides. In some embodiments, the cathode current collectors (not shown) can be coated on a single side. In some embodiments, the cathode current collectors can be coated on both sides. As shown, the collection of electrochemical cell stacks **83000a**, **83000b** includes 3 cells arranged in parallel, with each parallel arrangement connected in series. In some embodiments, any number of electrochemical cells can be connected in series and/or parallel (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, inclusive of all values and ranges therebetween).

[0396] FIGS. **84A-84B** are illustrations of an electrochemical cell **8400**, according to an embodiment. As shown, the electrochemical cell **8400** includes an anode (not shown) disposed on an anode current collector **8420**, a cathode tab **8422**, a cathode (not shown) disposed on a cathode current collector (not shown), a cathode tab **8442**, a separator **8450**, an interlayer **8460**, an interlayer tab **8463**, and pouch films **8470a**, **8470b** (collectively forming a pouch **8470**). In some embodiments, the anode current collector **8420**, the cathode, the cathode current collector, the separator **8450**, and the interlayer **8460** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the separators **250**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. In some embodiments, the interlayer tab **8463** can be the same or substantially similar to the anode tab **763**, as described above with reference to FIGS. **7A-7B**. In some embodiments, the pouch **8470** can be the same or substantially similar to the pouch **7970**, as described above with reference to FIGS. **79A-79B**. Thus, certain aspects of the anode current collector **8420**, the cathode, the cathode current collector, the separator **8450**, the interlayer **8460**, the interlayer tab **8463**, and the pouch **8470** are not described in greater detail herein.

[0397] FIG. **84A** shows an overhead view of the electrochemical cell **8400**. FIG. **84B** shows a cross section of the electrochemical cell **8400** with some components of the electrochemical cell **8400** obstructed by the interlayer **8460**, the separator **8450**, the interlayer tab **8463**, and the pouch **8470**. As shown, the interlayer tab **8463** extends along approximately the full length of the electrochemical cell **8400**. As shown, a proximal end of the interlayer tab **8463** is recessed from a proximal end of the electrochemical cell by a distance *d*. In some embodiments, the distance *d* can be about 0 μm , about 500 μm , about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, or about 10 mm, inclusive of all values and ranges therebetween. In some embodiments, the

interlayer **8460** can include a carbon coating. In some embodiments, the interlayer tab **8463** can be bonded to the pouch film **8470a**. In some embodiments, the interlayer **8450** can be bonded to the pouch film **8470b**. A long adhesion length between the interlayer tab **8463** and the pouch film **8470a** can provide a strong adhesive bond between the interlayer tab **8463** and the pouch film **8470a**. In other words, a large sealing area provides high structural integrity.

[0398] FIG. **85** is an illustration of an electrochemical cell **8500**, according to an embodiment. As shown, the electrochemical cell **8500** includes an anode **8510** disposed on an anode current collector **8520**, a cathode **8530** disposed on a cathode current collector **8540**, with a first separator **8550a**, a second separator **8550b**, an interlayer **8560** is disposed between the first separator **8550a** and the second separator **8550b**. A first coating layer **8552a** is disposed between the first separator **8550a** and the cathode **8530**. A second coating layer **8552b** is disposed between the second separator **8550b** and the anode **8510**. In some embodiments, the anode **8510**, the anode current collector **8520**, the cathode **8530**, the cathode current collector **8540**, the first separator **8550a**, the second separator **8550b**, and the interlayer **8560** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. Thus, certain aspects of the anode **8510**, the anode current collector **8520**, the cathode **8530**, the cathode current collector **8540**, the first separator **8550a**, the second separator **8550b**, and the interlayer **8560** are not described in greater detail herein.

[0399] In some embodiments, the separators **8550a**, **8550b** (collectively referred to as separators **8550**) can be wetted before or after application of the separators **8550** (e.g., as described above with respect to wettability regarding FIG. **1**). For example, solvents such as ethylene carbonate, propylene, and/or dimethyl carbonate can be used to wet the separators **8550**. In some embodiments, the substance used for wetting can interact with the interlayer **8560**, the anode **8510**, and/or the cathode **8530** can form micropores in the separators **8550**. These micropores can create short circuits between the anode **8510** and the interlayer **8560** and/or between the cathode **8530** and the interlayer **8560**. In some cases, the micropores can be formed via thermal action (i.e., burning). In some cases, the micropores can be formed via dissolving via the wetting solvent. In some cases, a combination of thermal action and dissolving can contribute to the micropore formation. The first coating layer **8552a** and/or the second coating layer **8552b** can prevent the formation of these micropores. In some embodiments, the electrochemical cell **8500** can include the first coating layer **8552a** and not the second coating layer **8552b**. In some embodiments, the electrochemical cell **8500** can include the second coating layer **8552b** and not the first coating layer **8552a**. In some embodiments, the electrochemical cell **8500** can include both the first coating layer **8552a** and the second coating layer **8552b**.

[0400] In some embodiments, the first coating layer **8552a** and/or the second coating layer **8552b** can include alumina, PVDF, boehmite, polyimide, cellulose, cellulosic materials, methyl cellulose, ethyl cellulose, hydroxyethyl cellulose, hydroxypropyl cellulose, hydroxyethyl methyl cellulose, hydroxypropyl methyl cellulose, ethyl hydroxyethyl cellu-

lose, carboxymethyl cellulose, or any combination thereof. In some embodiments, the first coating **8552a** and/or the second coating **8552b** can have a thickness of at least about 1 μm , at least about 1.5 μm , at least about 2 μm , at least about 2.5 μm , at least about 3 μm , at least about 3.5 μm , at least about 4 μm , or at least about 4.5 μm . In some embodiments, the first coating **8552a** and/or the second coating **8552b** can have a thickness of no more than about 5 μm , no more than about 4.5 μm , no more than about 4 μm , no more than about 3.5 μm , no more than about 3 μm , no more than about 2.5 μm , no more than about 2 μm , or no more than about 1.5 μm . Combinations of the above-referenced thicknesses are also possible (e.g., at least about 1 μm and no more than about 5 μm or at least about 2 μm and no more than about 4 μm), inclusive of all values and ranges therebetween. In some embodiments, the first coating **8552a** and/or the second coating **8552b** can have a thickness of about 1 μm , about 1.5 μm , about 2 μm , about 2.5 μm , about 3 μm , about 3.5 μm , about 4 μm , about 4.5 μm , or about 5 μm .

[0401] After the electrochemical cell **8500** has been formed, the conductivity of the separators **8550** can be measured to confirm that micropores have not formed. In some embodiments, this determination can be conducted via a resistance measurement. In some embodiments, the resistance across the combination of the first separator **8550a**, the interlayer **8560**, and the second separator **8550b** (including the first coating layer **8552a** and/or the second coating layer **8552b**) can be at least about 500 k Ω , at least about 600 k Ω , at least about 700 k Ω , at least about 800 k Ω , at least about 900 k Ω , at least about 1 M Ω , at least about 1.1 M Ω , at least about 1.2 M Ω , at least about 1.3 M Ω , at least about 1.4 M Ω , at least about 1.5 M Ω , at least about 1.6 M Ω , at least about 1.7 Ω , at least about 1.8 M Ω , at least about 1.9 M Ω , at least about 2 M Ω , at least about 2.5 M Ω , at least about 3 M Ω , at least about 3.5 M Ω , at least about 4 M Ω , at least about 4.5 M Ω , or at least about 5 M Ω .

[0402] FIG. **86** is an illustration of an electrochemical cell **8600**, according to an embodiment. As shown, the electrochemical cell **8600** includes an anode **8610** disposed on an anode current collector **8620**, a cathode **8630** disposed on a cathode current collector **8640**, with a first separator **8650a**, a second separator **8650b**, an interlayer **8660** is disposed between the first separator **8650a** and the second separator **8650b**. A metal coating layer **8670** is disposed between the first separator on the interlayer **8660**. As shown, the metal coating layer **8670** is disposed on the anode side of the interlayer **8660**. In some embodiments, the metal coating layer **8670** can be disposed on the cathode side of the interlayer **8660**. The metal coating layer **8670** includes a pinhole **8671**. In some embodiments, the anode **8610**, the anode current collector **8620**, the cathode **8630**, the cathode current collector **8640**, the first separator **8650a**, the second separator **8650b**, and the interlayer **8660** can be the same or substantially similar to the anode **210**, the anode current collector **220**, the cathode **230**, the cathode current collector **240**, the first separator **250a**, the second separator **250b**, and the interlayer **260**, as described above with reference to FIGS. **2A-2B**. Thus, certain aspects of the anode **8610**, the anode current collector **8620**, the cathode **8630**, the cathode current collector **8640**, the first separator **8650a**, the second separator **8650b**, and the interlayer **8660** are not described in greater detail herein.

[0403] The metal coating layer **8670** can be coated onto the separator **8650b** (or the separator **8650a**) before the

interlayer **8660** is applied. The metal coating layer **8670** can prevent the risk of local zero-voltage at the interlayer **8660**, as the pinhole **8671** can prevent such occurrences. In some embodiments, the metal coating layer **8670** can have a thickness of at least about 10 μm , at least about 20 μm , at least about 30 μm , at least about 40 μm , at least about 50 μm , at least about 60 μm , at least about 70 μm , at least about 80 μm , at least about 90 μm , at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 1.5 mm, at least about 2 mm, at least about 2.5 mm, at least about 3 mm, at least about 3.5 mm, at least about 4 mm, or at least about 4.5 mm. In some embodiments, the metal coating layer **8670** can have a thickness of no more than about 5 mm, no more than about 4.5 mm, no more than about 4 mm, no more than about 3.5 mm, no more than about 3 mm, no more than about 2.5 mm, no more than about 2 mm, no more than about 1.5 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , no more than about 200 μm , no more than about 100 μm , no more than about 90 μm , no more than about 80 μm , no more than about 70 μm , no more than about 60 μm , no more than about 50 μm , no more than about 40 μm , no more than about 30 μm , or no more than about 20 μm . Combinations of the above-referenced thicknesses are also possible (e.g., at least about 10 μm and no more than about 5 mm or at least about 100 μm and no more than about 600 μm), inclusive of all values and ranges therebetween. In some embodiments, the metal coating layer **8670** can have a thickness of about 10 μm , about 20 μm , about 30 μm , about 40 μm , about 50 μm , about 60 μm , about 70 μm , about 80 μm , about 90 μm , about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.5 mm, about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, or about 5 mm.

[0404] In some embodiments, the pinhole **8671** can have a width of at least about 10 μm , at least about 20 μm , at least about 30 μm , at least about 40 μm , at least about 50 μm , at least about 60 μm , at least about 70 μm , at least about 80 μm , at least about 90 μm , at least about 100 μm , at least about 200 μm , at least about 300 μm , at least about 400 μm , at least about 500 μm , at least about 600 μm , at least about 700 μm , at least about 800 μm , at least about 900 μm , at least about 1 mm, at least about 1.5 mm, at least about 2 mm, at least about 2.5 mm, at least about 3 mm, at least about 3.5 mm, at least about 4 mm, or at least about 4.5 mm. In some embodiments, the pinhole **8671** can have a width of no more than about 5 mm, no more than about 4.5 mm, no more than about 4 mm, no more than about 3.5 mm, no more than about 3 mm, no more than about 2.5 mm, no more than about 2 mm, no more than about 1.5 mm, no more than about 1 mm, no more than about 900 μm , no more than about 800 μm , no more than about 700 μm , no more than about 600 μm , no more than about 500 μm , no more than about 400 μm , no more than about 300 μm , no more than about 200 μm , no more than about 100 μm , no more than about 90 μm , no more than about 80 μm , no more than about 70 μm , no more than

about 60 μm , no more than about 50 μm , no more than about 40 μm , no more than about 30 μm , or no more than about 20 μm .

[0405] Combinations of the above-referenced thicknesses are also possible (e.g., at least about 10 μm and no more than about 5 mm or at least about 100 μm and no more than about 600 μm), inclusive of all values and ranges therebetween. In some embodiments, the pinhole 8671 can have a thickness of about 10 μm , about 20 μm , about 30 μm , about 40 μm , about 50 μm , about 60 μm , about 70 μm , about 80 μm , about 90 μm , about 100 μm , about 200 μm , about 300 μm , about 400 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1 mm, about 1.5 mm, about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, or about 5 mm.

Examples

[0406] FIG. 87 is a graphic representation of monitoring of an interlayer to detect dendritic growth. The electrochemical cell cycled included a lithium metal anode with a thickness of 100 μm , a LFP cathode, a ceramic separator with a thickness of 15 μm on the cathode side, a polyethylene separator with a thickness of 7 μm on the anode side with a 5 μm hard carbon interlayer disposed between the separators. The electrochemical cell included an ether-based electrolyte. A 1 k Ω resistor was placed in a circuit with the cathode current collector and the interlayer. The electrochemical cell was charged and discharged at C/3 under high stack pressure. The green trace represents the voltage potential of an interlayer with respect to the anode. The blue curve represents the voltage of the cathode with respect to the anode. As shown, when the dendrite forms and interfaces with the interlayer (i.e., around 200 h), the voltage of the interlayer is reduced to be similar to the anode potential. In this way, the system that monitors the electrochemical cell is able to detect dendrite growth in real time.

[0407] FIG. 88 is a photograph of dendritic formation at interfaces with the interlayer described above with respect to FIG. 87. The image on the left shows a side of the separator facing the anode, the image in the center includes the anode surface, and the image on the right includes a side of the separator facing the cathode. The cell was contaminated to create dendrite growth. Control cells with this contamination fail within a few cycles. The interlayer was maintained with a passive resistor pull to the potential for the cathode and the life was greatly increased. When the cell failed, it was due to loss of capacity rather than a thermal event. FIG. 89 is a graphic representation of the point in time when the dendrite forms and interfaces with the interlayer (i.e., a close-up view of FIG. 87). The green trace represents the voltage potential of the interlayer with respect to the anode. As shown, the dendrite forms and interfaces with the interlayer (i.e., around 180 h). The voltage of the interlayer has been held up to the cathode potential with a passive resistor.

[0408] FIG. 90 a graphical representation of a prevention of a dendrite from penetrating a separator and creating a localized short circuit. The electrochemical cell cycled included no anode, a LFP cathode, a ceramic separator with a thickness of 15 μm on the cathode side, a polyethylene separator with a thickness of 7 μm on the anode side with a 5 μm hard carbon interlayer disposed between the separators. The electrochemical cell included an ether-based electrolyte. A 1 k Ω resistor was placed in a circuit with the cathode current collector and the interlayer. The electro-

chemical cell was charged at C/10 and discharged at C/3 under high stack pressure. The green trace represents the voltage potential of the interlayer with respect to a copper anode current collector. The blue curve represents the voltage of the cathode with respect to the copper anode current collector. As shown, when the dendrite forms and interfaces with the interlayer (i.e., around 60 h), the voltage of the interlayer increases to be similar to the cathode potential with respect to the anode current collector. The dendrite is dissolved or remediated, and the voltage potential of the interlayer returns to near the voltage potential of the cathode with respect to the anode.

[0409] FIG. 91 is a sample control operating mode state transition diagram. Table 2 shows a sample protocol for the operating mode to follow.

TABLE 2

Summary of algorithm control modules.	
Control State	Purpose
Initial-ization	Executes once at the power-up of the BMS circuit
a. Functions:	
I.	Initialize all control signals OFF.
II.	Initialize all algorithm (ex: PI loop, software filter, etc.) parameters in preparation to be used in the Monitor state.
b. Transitions out:	
I.	A = (IF Initialization = COMPLETE then GOTO Monitor state)
Monitor	Main loop where voltages are read and Detection portion of the algorithm executes
a. Functions:	
I.	Algorithm monitors INTERLAYER(1, 2)_V_SENSE for drop in voltage compared to CELL_V_SENSE.
II.	Controls all switch outputs based on previous state transitions.
III.	Communicates operating mode status, fault status, voltage sense, switch status, data collection, etc. to CAN and/or the master BMS controller.
IV.	Monitors for faults.
b. Transitions out:	
I.	B = (IF Detection algorithm indicates dendrite present the GOTO Remediate state)
II.	L = (IF fault detected then GOTO fault state)
III.	D = (IF remediation failed and transition to Drain state failed in Remediate state OR thermal event probable then GOTO Drain state) NOTE: This will "kill" the pack, only to be performed to protect users and property.
IV.	H = (IF BMS master controller commands powerdown then GOTO Powerdown state)
V.	F = (IF Detection indicates no dendrite growth and algorithm allows, then GOTO Prevention state)
VI.	R = (IF SW4_AIR_TRANSPORT_EN = TRUE then GOTO Air Transport state)

TABLE 2-continued

Summary of algorithm control modules.	
Control State	Purpose
Re-mediate	Voltages are read and Remediation portion of algorithm executes
a. Functions:	
I.	Set SW2_PREVENT(1)_EN = OFF.
II.	Control SW2_PU_REMEDIATE(1)_EN (SW3_PD_REMEDIATE(1)_EN control is TBD) per algorithm.
III.	Algorithm monitors INTERLAYER(1,2)_V_SENSE for recovery rise (indicates dendrite removed) in voltage compared to CELL_V_SENSE.
IV.	Communicates operating mode status, fault status, voltage sense, switch status, data collection, etc. to CAN and/or the master BMS controller.
V.	Monitors for faults.
b. Transitions out:	
I.	C = (IF Remediate algorithm indicates no dendrites present then GOTO Monitor state)
II.	M = (IF fault detected then GOTO fault state)
III.	N = (IF dendrite remediation fails AND thermal event probable then GOTO Drain state) NOTE: This will “kill” the pack, only to be performed to protect users and property.
Drain	Discharges a parallel cell stack to ~0 V to prevent thermal event—only used to protect the users and property.
a. Functions:	
I.	Set SW2_PU_REMEDIATE(1)_EN, SW3_PD_REMEDIATE(1)_EN, SW2_PREVENT(1)_EN = OFF.
II.	Set SW1_DRAIN_EN = ON and monitor CELL_V_SENSE for ~0 V.
III.	Communicates operating mode status, fault status, voltage sense, switch status, data collection, etc. to CAN and/or the master BMS controller.
b. Transitions out:	
I.	E = (IF CELL_V_SENSE = ~0 V then GOTO End)-TBD what effect this will have at the module/pack level . . . will the BMS still be functional/powerd?
Power-down	Commanded by the BMS master controller that the BMS is powering down
a.	Functions:
I.	Set SW2_PU_REMEDIATE(1)_EN, SW3_PD_REMEDIATE(1)_EN, SW2_PREVENT(1)_EN, SW1_DRAIN_EN = OFF.
II.	Communicates operating mode status, fault status, voltage sense, switch status, data collection, etc. to CAN and/or the master BMS controller.
III.	Store any pertinent data to EEPROM.
b.	Transitions out:
I.	I = (IF all powerdown functions complete then GOTO End)

TABLE 2-continued

Summary of algorithm control modules.	
Control State	Purpose
Fault	State entered if a fault was detected by control system
a.	Functions:
I.	Set SW2_PU_REMEDIATE(1)_EN, SW3_PD_REMEDIATE(1)_EN, SW2_PREVENT(1)_EN, SW1_DRAIN_EN = OFF.
II.	Communicates operating mode status, fault status, voltage sense, switch status, data collection, etc. to CAN and/or the master BMS controller.
III.	Check fault status for any faults in the FAILED state (fault maturation). Some faults may be latched for the entire power cycle, some may be allowed to self-clear if the fault condition de-matures (transitions to PASS).
b.	Transitions out:
I.	J = (IF fault self-clearing fault conditions all = PASS then GOTO Monitor state)
II.	P = (IF any latched faults are FAILED then GOTO Powerdown state)
Prevent	Voltages are read and Prevent portion of algorithm execute. This mode is intended to be executed at certain intervals to allow a small potential of current flow the cathode to the interlayer to prevent dendrites from forming.
a.	Functions:
I.	Set SW2_PREVENT(1)_EN = ON.
II.	Set SW2_PU_REMEDIATE(1)_EN, SW3_PD_REMEDIATE(1)_EN = OFF.
III.	Algorithm monitors INTERLAYER(1,2)_V_SENSE for abnormal imbalance in voltage compared to CELL_V_SENSE.
IV.	Communicates operating mode status, fault status, voltage sense, switch status, data collection, etc. to CAN and/or the master BMS controller.
V.	Monitors for faults.
b.	Transitions out:
I.	G = (IF algorithm indicates dendrites present OR Prevent is no longer needed then GOTO Monitor state)
II.	K = (IF fault detected then GOTO fault state)
Air Transport	User commanded via pushbutton or a user interface on the module or pack. This mode will perform same function as Drain state - but exit when the module SOC achieves $\leq 15\%$.
a.	Functions:
II.	Set SW2_PU_REMEDIATE(1)_EN, SW3_PD_REMEDIATE(1)_EN, SW2_PREVENT(1)_EN = OFF.
III.	Set SW1_DRAIN_EN = ON.
b.	Transitions out:
I.	R = (IF module SOC $\leq 15\%$ then GOTO End)

[0410] FIG. 92 is an illustration of the formation of a dual ion system. With a potential applied to an interlayer that is higher than that applied to a cathode, a dual ion system is formed. By holding a voltage in the interlayer that is higher than the voltage of the cathode, cations (e.g., Li^+ ions) are concentrated at the cathode and anion surface, while anions are concentrated near the separator and/or the interlayer. Upon charging and/or discharging, there is little or no ion flux through the interlayer and the separators. However, intercalation and/or lithiation at the cathode and anode is through the local concentrated cation. In some cases, anions (e.g., AlCl_4^- and/or PF_6^-) can also intercalate to the interlayer. Thus, metal migration from the cathode is blocked, which reduces the risk of dendrite formation via metal contamination or metal dissolution. The electrolyte system included can include lithium salt (e.g., LiPF_6 , LiFSI , LiTFSI , LiBF_4 , LiBCN , and/or LiTCB), and solvents, including cyclic carbonates, linear carbonates, and/or ether.

[0411] FIG. 93 is a graphical representation of a cell voltage 9301 as compared to an interlayer voltage 9302 when an external voltage is applied to the interlayer. In such an application, the electrochemical cell can complete the formation of the dual ion system through normal initial cycling.

[0412] Various concepts may be embodied as one or more methods, of which at least one example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments. Put differently, it is to be understood that such features may not necessarily be limited to a particular order of execution, but rather, any number of threads, processes, services, servers, and/or the like that may execute serially, asynchronously, concurrently, in parallel, simultaneously, synchronously, and/or the like in a manner consistent with the disclosure. As such, some of these features may be mutually contradictory, in that they cannot be simultaneously present in a single embodiment. Similarly, some features are applicable to one aspect of the innovations, and inapplicable to others.

[0413] In addition, the disclosure may include other innovations not presently described. Applicant reserves all rights in such innovations, including the right to embodiment such innovations, file additional applications, continuations, continuations-in-part, divisionals, and/or the like thereof. As such, it should be understood that advantages, embodiments, examples, functional, features, logical, operational, organizational, structural, topological, and/or other aspects of the disclosure are not to be considered limitations on the disclosure as defined by the embodiments or limitations on equivalents to the embodiments. Depending on the particular desires and/or characteristics of an individual and/or enterprise user, database configuration and/or relational model, data type, data transmission and/or network framework, syntax structure, and/or the like, various embodiments of the technology disclosed herein may be implemented in a manner that enables a great deal of flexibility and customization as described herein.

[0414] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0415] As used herein, in particular embodiments, the terms “about” or “approximately” when preceding a numerical value indicates the value plus or minus a range of 10%. Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the disclosure. That the upper and lower limits of these smaller ranges can independently be included in the smaller ranges is also encompassed within the disclosure, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the disclosure.

[0416] The phrase “and/or,” as used herein in the specification and in the embodiments, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0417] As used herein in the specification and in the embodiments, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the embodiments, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the embodiments, shall have its ordinary meaning as used in the field of patent law.

[0418] As used herein in the specification and in the embodiments, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to

at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0419] In the embodiments, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

[0420] While specific embodiments of the present disclosure have been outlined above, many alternatives, modifications, and variations will be apparent to those skilled in the art. Accordingly, the embodiments set forth herein are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit and scope of the disclosure. Where methods and steps described above indicate certain events occurring in a certain order, those of ordinary skill in the art having the benefit of this disclosure would recognize that the ordering of certain steps may be modified and such modification are in accordance with the variations of the invention. Additionally, certain of the steps may be performed concurrently in a parallel process when possible, as well as performed sequentially as described above. The embodiments have been particularly shown and described, but it will be understood that various changes in form and details may be made.

1. An electrochemical cell, comprising:
an anode;
a cathode;
a porous material disposed between the anode and the cathode, the porous material configured to electrically isolate the anode from the cathode and allow the passage of ions;
a solid-state electrolyte disposed between the porous material and the anode; and
an interlayer disposed between the porous material and the solid-state electrolyte.
2. The electrochemical cell of claim 1, wherein the interlayer includes carbon.
3. The electrochemical cell of claim 1, wherein at least one of the porous material or the solid-state electrolyte includes a conductive material.
4. The electrochemical cell of claim 1, wherein the anode includes lithium metal.
5. The electrochemical cell of claim 1, wherein the solid-state electrolyte includes a sulfide.
6. The electrochemical cell of claim 1, wherein the porous material and the solid-state electrolyte have porosities of less than about 1%.
7. The electrochemical cell of claim 1, wherein the interlayer includes a binder.
8. The electrochemical cell of claim 1, wherein the interlayer includes at least one of activated carbon, hard carbon, soft carbon, conductive carbon particles, carbon black, graphitic carbon, carbon fibers, carbon microfibers,

vapor-grown carbon fibers (VGCF), fullerenic carbons including “buckyballs”, carbon nanotubes (CNTs), multi-wall carbon nanotubes (MWNTs), single wall carbon nanotubes (SWNTs), graphene, graphene sheets, aggregates of graphene sheets, or materials comprising fullerenic fragments.

9. The electrochemical cell of claim 1, wherein the separator is a first separator and the interlayer is a first interlayer, the electrochemical cell further comprising:

- a second interlayer disposed on the first separator; and
- a second separator disposed on the second interlayer.

10. An electrochemical cell, comprising:

- an anode;
- a cathode;
- a separator disposed on at least one of the anode or the cathode;
- an interlayer disposed on the separator; and
- a solid-state electrolyte coupled to the interlayer, wherein at least one of the separator or the solid-state electrolyte includes a conductive material.

11. The electrochemical cell of claim 10, wherein the solid-state electrolyte contacts the anode.

12. The electrochemical cell of claim 11, wherein the anode includes a lithium metal anode.

13. The electrochemical cell of claim 11, wherein an interface between the solid-state electrolyte and the anode is substantially free of liquid.

14. The electrochemical cell of claim 10, wherein the solid-state electrolyte include a sulfide.

15. The electrochemical cell of claim 10, further comprising:

- a catholyte in contact with the cathode.

16. The electrochemical cell of claim 10, wherein the interlayer includes a binder.

17. The electrochemical cell of claim 10, wherein the separator has a porosity of less than about 1%.

18. The electrochemical cell of claim 10, wherein the interlayer includes at least one of activated carbon, hard carbon, soft carbon, conductive carbon particles, carbon black, graphitic carbon, carbon fibers, carbon microfibers, vapor-grown carbon fibers (VGCF), fullerenic carbons including “buckyballs”, carbon nanotubes (CNTs), multi-wall carbon nanotubes (MWNTs), single wall carbon nanotubes (SWNTs), graphene, graphene sheets or aggregates of graphene sheets, or materials comprising fullerenic fragments.

19. The electrochemical cell of claim 10, wherein the separator includes polyethylene.

20. An electrochemical cell, comprising:

- a first electrode;
- a second electrode;
- a separator coupled to the first electrode;
- an interlayer coupled to the separator, the interlayer including an electronically conductive material; and
- a solid-state electrolyte coupled to the interlayer opposite the separator.

21. The electrochemical cell of claim 20, wherein at least one of the separator or the solid-state electrolyte includes a conductive material.

22. The electrochemical cell of claim 21, wherein the separator includes a non-ionic conductive powder.

23. The electrochemical cell of claim 20, wherein the interlayer includes at least one of activated carbon, hard carbon, soft carbon, conductive carbon particles, carbon

black, graphitic carbon, carbon fibers, carbon microfibers, vapor-grown carbon fibers (VGCF), fullerenic carbons including “buckyballs”, carbon nanotubes (CNTs), multi-wall carbon nanotubes (MWNTs), single wall carbon nanotubes (SWNTs), graphene, graphene sheets, aggregates of graphene sheets, or materials comprising fullerenic fragments.

24. The electrochemical cell of claim **20**, wherein the second electrode includes at least one of graphite, lithium metal, or silicon.

25. The electrochemical cell of claim **24**, wherein the second electrode includes lithium metal.

26. The electrochemical cell of claim **20**, wherein the solid-state electrolyte include a sulfide.

27. The electrochemical cell of claim **20**, wherein the second electrode contacts the solid-state electrolyte.

28. The electrochemical cell of claim **27**, wherein an interface between the second electrode and the solid-state electrolyte is substantially free of liquid.

29. The electrochemical cell of claim **20**, wherein the interlayer includes a binder.

30. The electrochemical cell of claim **20**, wherein the separator has a porosity of less than about 1%.

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